Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 — Unit 2 · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

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· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · UE26MA141A · Unit_02: Unit 2
2. 2 · Introduction to higher order differential equations, complementary function and particular integral
3. 3 · Rules for finding the Complementary Function, Problems
4. 4 · Particular integral of standard functions - Type 1
5. 5 · Particular integral_Type 1 and Type 2
6. 6 · Particular integral of standard functions - Type 2
7. 7 · Particular integral of standard functions - Type 3
8. 8 · Particular integral -Type 3 and Type 4
9. 9 · Particular integral of standard functions - Type 4
10. 10 · Particular Integral when X is any other form
11. 11 · Particular integral when X is any other form_Continued
12. 12 · Legendre differential equation
13. 13 · Legendre differential equations_Continued
14. 14 · Variation of Parameters method_Continued
15. 15 · Variation of Parameters method
16. 16 · Applications of higher-order linear differential equations - RLC Circuits
17. 17 · Application problems on Linear differential equations - Simple harmonic oscillation
18. 18 · Source slide deck: Solutions of Homogeneous DE by introducing Complementary Functions
19. 19 · Historical question-source pages · unit mapping candidates
20. Appendix A · Symbol glossary
21. Appendix B · Sources, coverage and gaps
22. Appendix C · Revision and retrieval practice

M1

Unit_02 · Unit 2

1 UE26MA141A · Unit_02: Unit 2

This unit master organizes source material for Engineering Mathematics - I, Unit_02, officially titled Unit 2. It includes 16 video transcripts and 11 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26MA141A, Unit_02 (Unit 2). Bucket folder present: yes. Videos: 16; slide decks: 11.

EV0000001

2 Introduction to higher order differential equations, complementary function and particular integral

This lesson is indexed from the source video titled Introduction to higher order differential equations, complementary function and particular integral. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000002 · EV0000003

Graphs and diagrams

Figure G000001. Introduction to second and higher order DE with constant coefficients · source slide page 1.

EV0000267

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

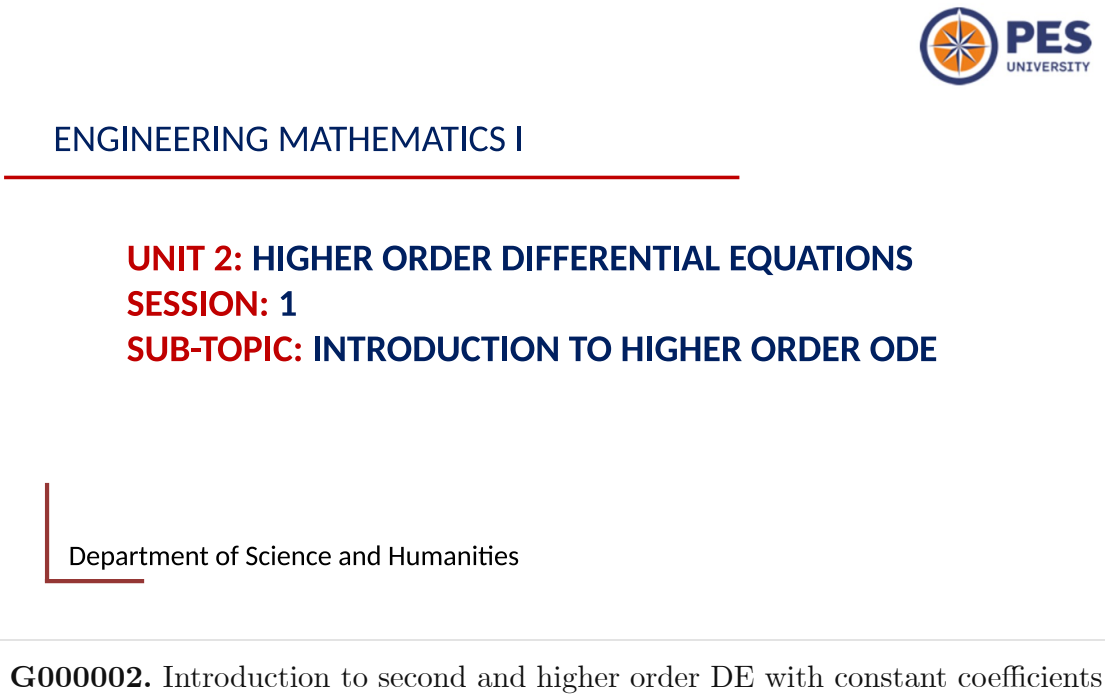

Figure G000002. Introduction to second and higher order DE with constant coefficients · source slide page 2.

EV0000270

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

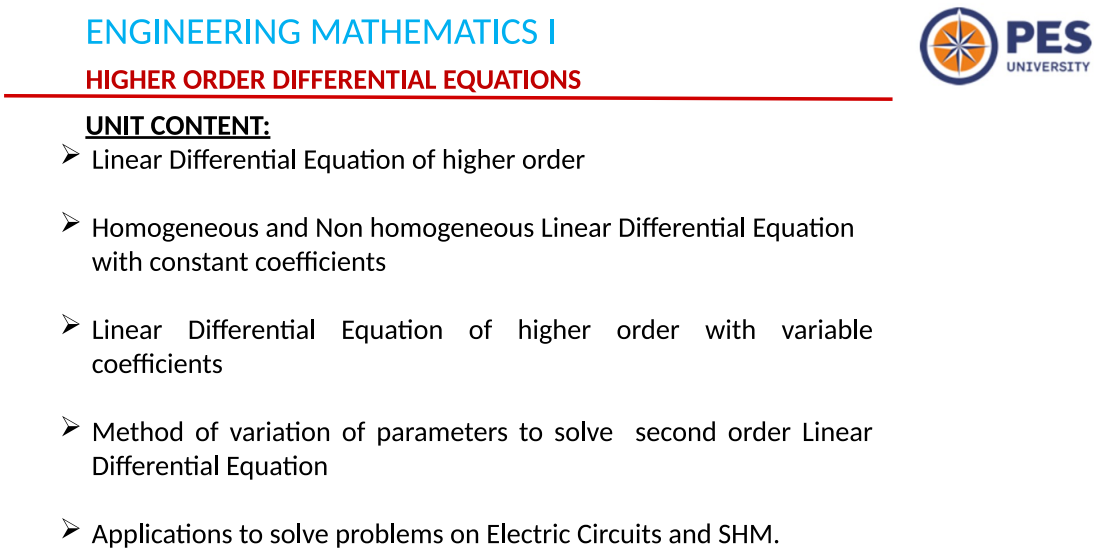

Figure G000003. Introduction to second and higher order DE with constant coefficients · source slide page 3.

EV0000273

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

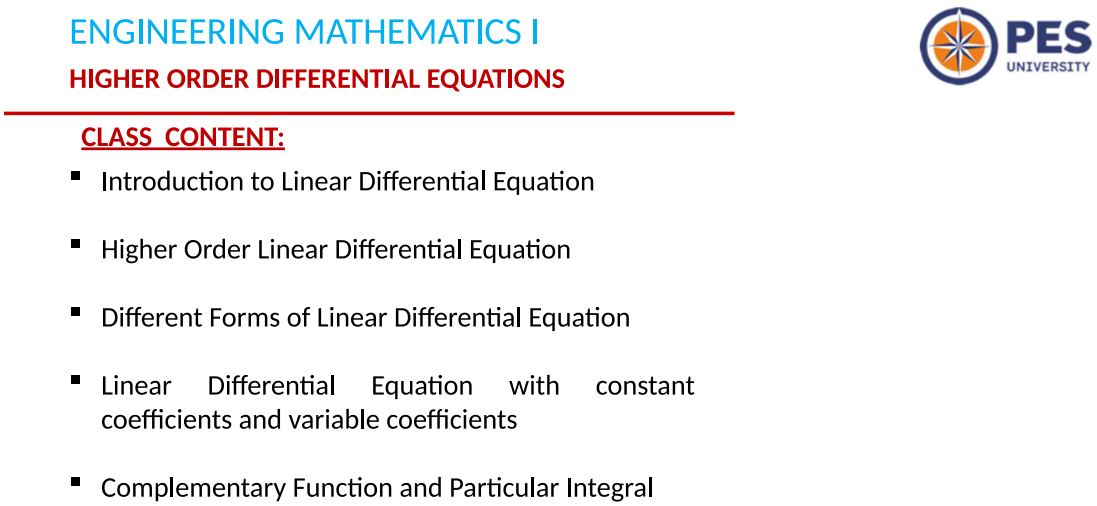

Figure G000004. Introduction to second and higher order DE with constant coefficients · source slide page 4.

EV0000276

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

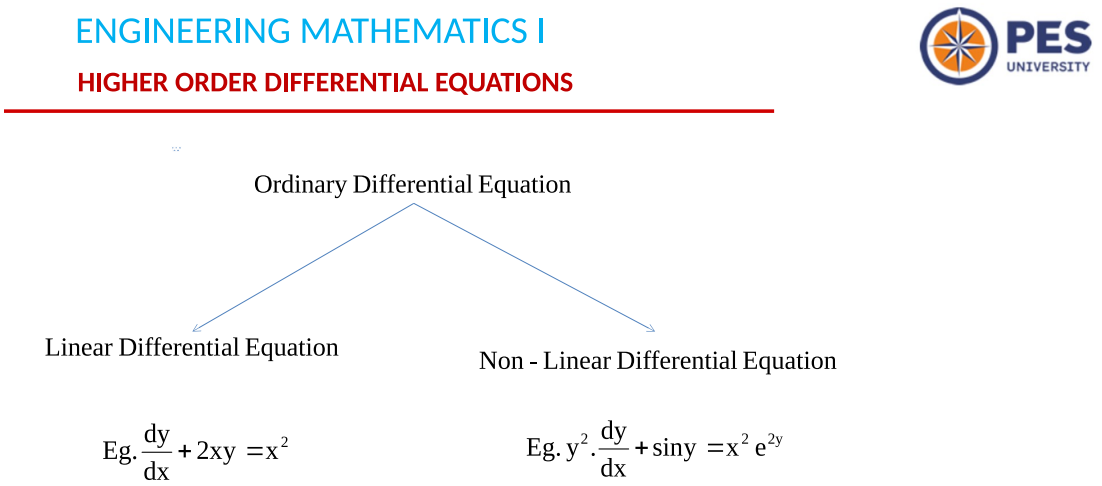

Figure G000005. Introduction to second and higher order DE with constant coefficients · source slide page 5.

EV0000279

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

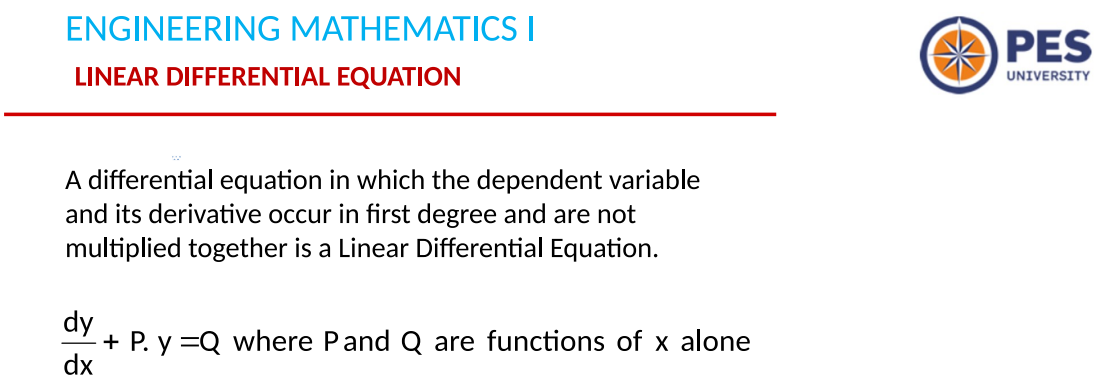

Figure G000006. Introduction to second and higher order DE with constant coefficients · source slide page 6.

EV0000282

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

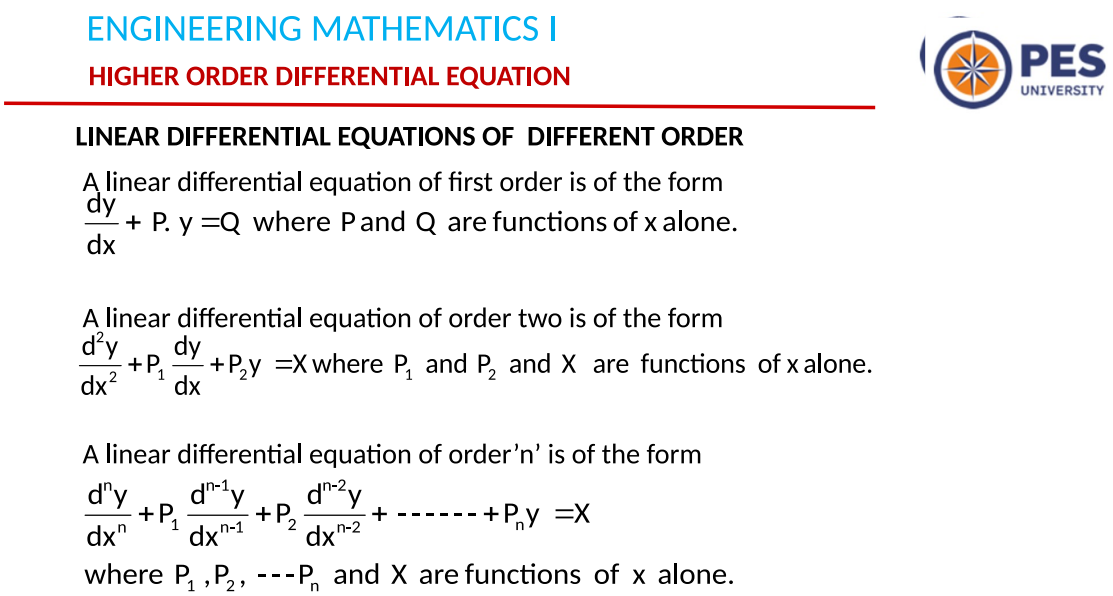

Figure G000007. Introduction to second and higher order DE with constant coefficients · source slide page 7.

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ENGINEERING MATHEMATICS I

HIGHER ORDER LINEAR DIFFERENTIAL EQUATION

A general linear differential equation of order n is of the form

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \text{------} + a_n y = X$$

where $a_1, a_2, \text{---} a_n$ are functions of x or constants

and X is a function of x alone.

Two forms of Higher Order linear differential equation of order n

1. Linear differential equation with constant coefficients

In this type, the coefficient of derivatives are constants.

$$Eg.\ 2\frac{d^3 y}{dx^3} - 5\frac{d^2 y}{dx^2} + 7\frac{dy}{dx} + y = x^2$$

Figure G000008. Introduction to second and higher order DE with constant coefficients · source slide page 8.

EV0000288

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ENGINEERING MATHEMATICS I

HIGHER ORDER LINEAR DIFFERENTIAL EQUATION

Two forms of Higher Order linear differential equation of order n

2. Linear differential equation with variable coefficients

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \ldots\ldots\ldots + a_n y = X$$

In this type, the coefficient of derivatives $a_1, a_2 \text{ -- } a_n$ are functions of x.

$$\text{Eg. } 2x^3 \frac{d^3 y}{dx^3} - (5x^2 + 2)\frac{d^2 y}{dx^2} + x\frac{dy}{dx} + y = x^2 \sin x$$

Figure G000009. Introduction to second and higher order DE with constant coefficients · source slide page 9.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

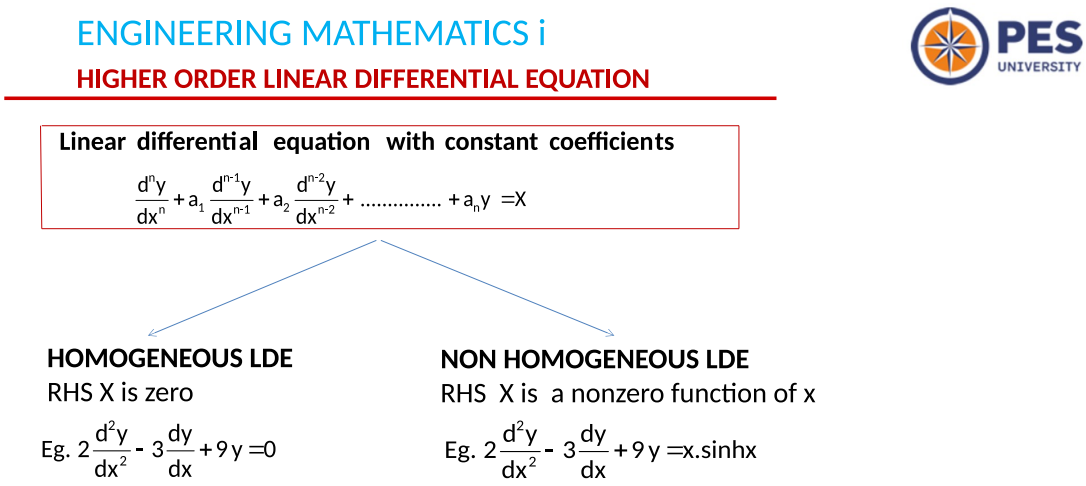

Figure G000010. Introduction to second and higher order DE with constant coefficients · source slide page 10.

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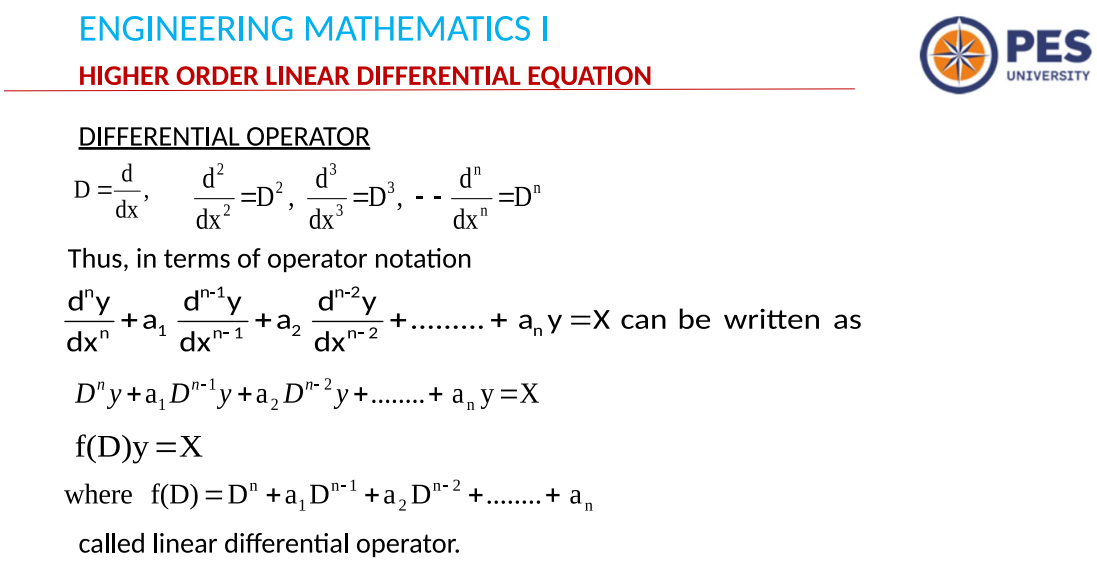

Figure G000011. Introduction to second and higher order DE with constant coefficients · source slide page 11.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

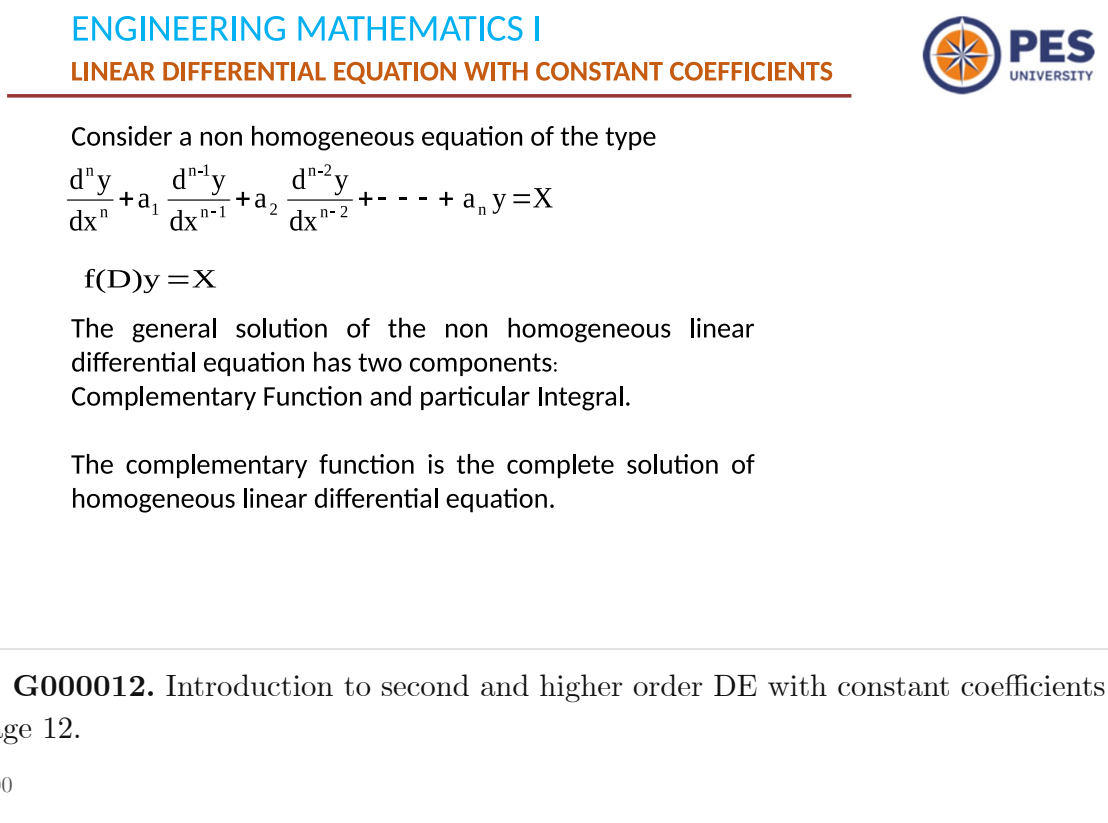

Figure G000012. Introduction to second and higher order DE with constant coefficients · source slide page 12.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

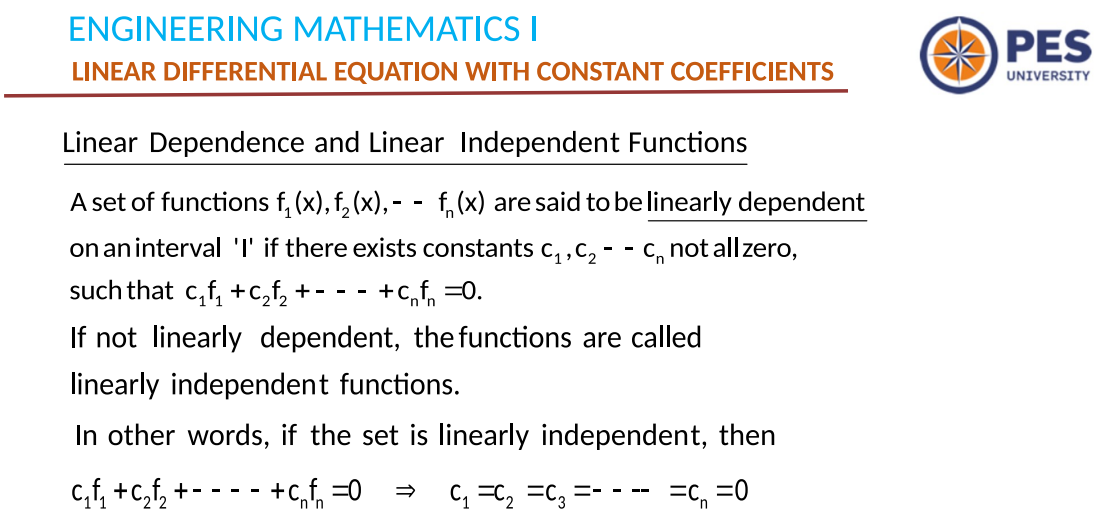

Figure G000013. Introduction to second and higher order DE with constant coefficients · source slide page 13.

EV0000303

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

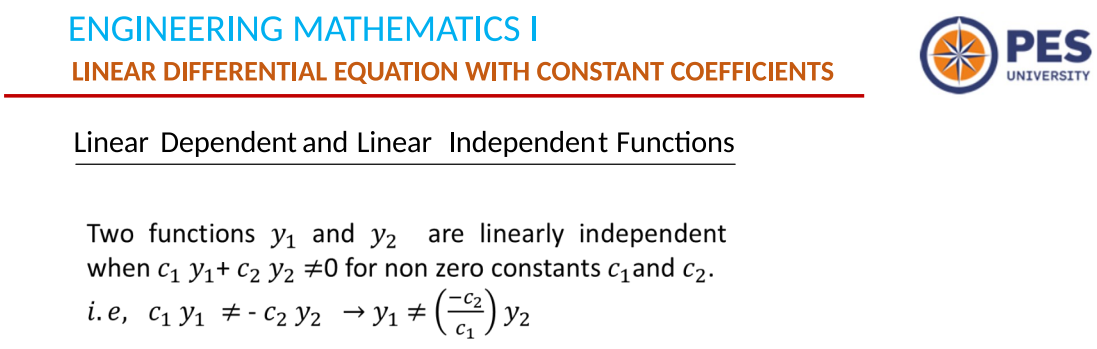

Figure G000014. Introduction to second and higher order DE with constant coefficients · source slide page 14.

EV0000306

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

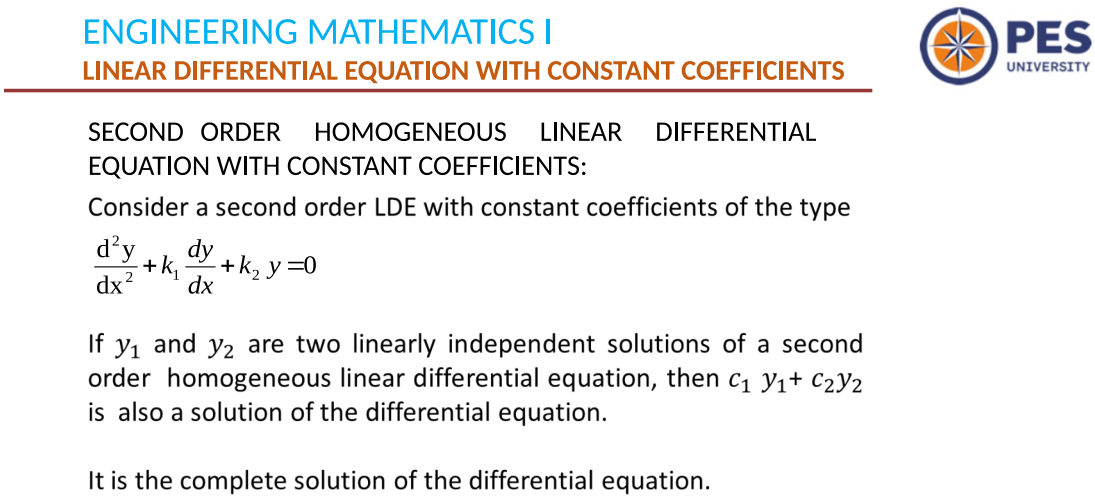

Figure G000015. Introduction to second and higher order DE with constant coefficients · source slide page 15.

EV0000309

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

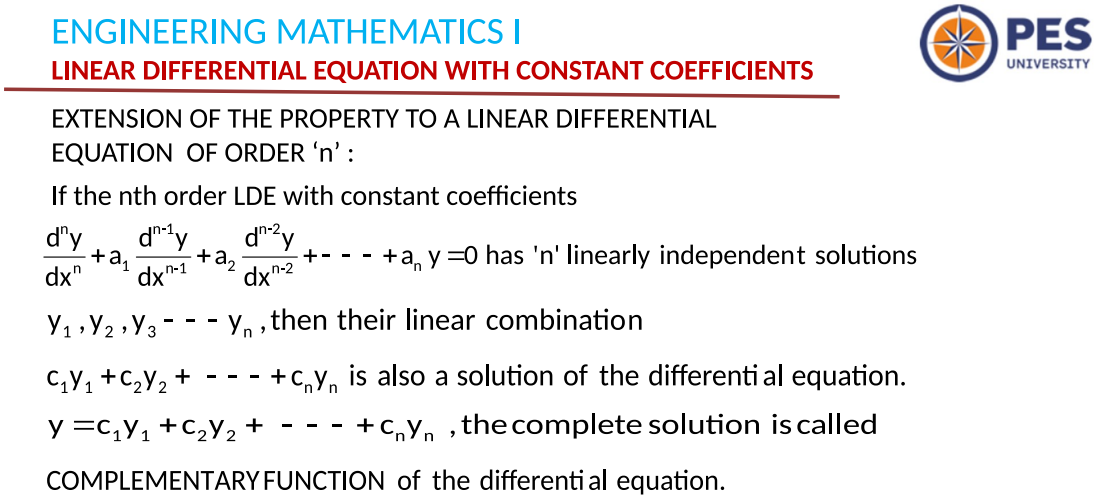

Figure G000016. Introduction to second and higher order DE with constant coefficients · source slide page 16.

EV0000312

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

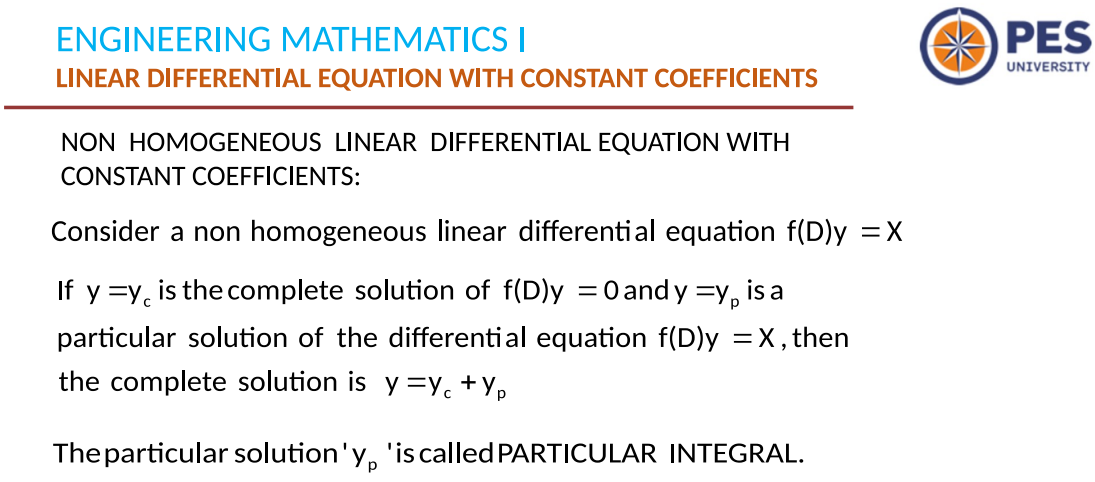

Figure G000017. Introduction to second and higher order DE with constant coefficients · source slide page 17.

EV0000315

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000018. Introduction to second and higher order DE with constant coefficients · source slide page 18.

EV0000318

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000266 · EV0000267 · EV0000268
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 2: HIGHER ORDER DIFFERENTIAL EQUATIONS SESSION: 1 SUB-TOPIC: Introduction to higher order ode Department of Science and Humanities EV0000269 · EV0000270 · EV0000271
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER DIFFERENTIAL EQUATIONS UNIT CONTENT: Linear Differential Equation of higher order Homogeneous and Non homogeneous Linear Differential Equation with constant coefficients Linear Differential Equation of higher order with variable coefficients Method of variation of parameters to solve second order Linear Differential Equation Applications to solve problems on Electric Circuits and SHM. EV0000272 · EV0000273 · EV0000274
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER DIFFERENTIAL EQUATIONS CLASS CONTENT: Introduction to Linear Differential Equation Higher Order Linear Differential Equation Different Forms of Linear Differential Equation Linear Differential Equation with constant coefficients and variable coefficients Complementary Function and Particular Integral EV0000275 · EV0000276 · EV0000277
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER DIFFERENTIAL EQUATIONS EV0000278 · EV0000279 · EV0000280

6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I LINEAR DIFFERENTIAL EQUATION A differential equation in which the dependent variable and its derivative occur in first degree and are not multiplied together is a Linear Differential Equation. EV0000281 · EV0000282 · EV0000283
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER DIFFERENTIAL EQUATION LINEAR DIFFERENTIAL EQUATIONS OF DIFFERENT ORDER A linear differential equation of first order is of the form A linear differential equation of order two is of the form A linear differential equation of order'n' is of the form EV0000284 · EV0000285 · EV0000286
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER LINEAR DIFFERENTIAL EQUATION EV0000287 · EV0000288 · EV0000289
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER LINEAR DIFFERENTIAL EQUATION EV0000290 · EV0000291 · EV0000292
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS i HIGHER ORDER LINEAR DIFFERENTIAL EQUATION HOMOGENEOUS LDE RHS X is zero NON HOMOGENEOUS LDE RHS X is a nonzero function of x EV0000293 · EV0000294 · EV0000295
11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER LINEAR DIFFERENTIAL EQUATION DIFFERENTIAL OPERATOR Thus, in terms of operator notation called linear differential operator. EV0000296 · EV0000297 · EV0000298
12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS I LINEAR DIFFERENTIAL EQUATION WITH CONSTANT COEFFICIENTS Consider a non homogeneous equation of the type The general solution of the non homogeneous linear differential equation has two components: Complementary Function and particular Integral. The complementary function is the complete solution of homogeneous linear differential equation. EV0000299 · EV0000300 · EV0000301
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS I LINEAR DIFFERENTIAL EQUATION WITH CONSTANT COEFFICIENTS EV0000302 · EV0000303 · EV0000304
14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS I LINEAR DIFFERENTIAL EQUATION WITH CONSTANT COEFFICIENTS EV0000305 · EV0000306 · EV0000307
15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS I LINEAR DIFFERENTIAL EQUATION WITH CONSTANT COEFFICIENTS SECOND ORDER HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION WITH CONSTANT COEFFICIENTS: EV0000308 · EV0000309 · EV0000310
16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS I LINEAR DIFFERENTIAL EQUATION WITH CONSTANT COEFFICIENTS EXTENSION OF THE PROPERTY TO A LINEAR DIFFERENTIAL EQUATION OF ORDER 'n' : If the nth order LDE with constant coefficients EV0000311 · EV0000312 · EV0000313
17. Source slide page 17 (extracted text): ENGINEERING MATHEMATICS I LINEAR DIFFERENTIAL EQUATION WITH CONSTANT COEFFICIENTS NON HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION WITH CONSTANT COEFFICIENTS: EV0000314 · EV0000315 · EV0000316

18. Source slide page 18 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000317 · EV0000318 · EV0000319

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.452] Hello, everyone. Welcome to Engineering Mathematics 1. I am Gita TR. I will be discussing with you higher order differential equations. Let us first look at the content of this unit. [00:00:14.272] In this unit, we are discussing about higher order linear differential equation. An introduction to higher order linear differential equation. Next, we will be learning about homogeneous and non-homogeneous linear differential equation with constant. [00:00:32.016] Coefficients. We've been learning methods to solve homogeneous and non-homogeneous LDE linear differential equation with constant coefficients. We are also discussing about linear differential equation of higher order with variance. [00:00:49.584] variable coefficients. In fact, we are discussing two different forms of linear differential equation with variable coefficients and methods to solve the two different forms of linear differential equation with variable coefficients. [00:01:05.344] We are also learning a method called method of variation of parameters, which is used to solve only second order linear differential equation. [00:01:16.896] And we are learning at the end application of higher order linear differential equation to solve problems on electric circuits and simple harmonic motion. Now, in today's session, we are learning the following concepts. [00:01:34.016] First is the definition of linear differential equation. So we are learning the general form of linear differential equation of order one. We are also learning the general form of higher order linear differential equation. [00:01:50.532] We are learning different forms of higher order linear differential equation. Linear differential equation with constant and with variable coefficients. We are learning a very important concept called complementary function and particular integration. [00:02:08.304] So we are discussing these concepts in today's session. Now, introduction. Why do we have to study differential equations? We know that differential equation [00:02:21.472] play a very prominent role in the field of engineering, science, and economics and other disciplines. They serve as mathematical models to solve any real world problem. They help in modeling every physical, biological [00:02:39.120] and technical process. So it is very important to know more about differential equations to learn the methods to solve differential equations. Now if you have an ordinary differential equation [00:02:54.096] Whether the order of the differential equation is 1 or higher than 1, more than 1, ordinary differential equation is classified into two types based on linearity. So one is linear differential equation, and the other one. [00:03:11.728] Non-linear differential equation. So what is the definition of linear differential equation which you have studied before? So a differential equation in which the dependent variable and its derivative do not occur in product form and of degree one. [00:03:29.296] Is a linear differential equation. Now, what is a nonlinear differential equation? A differential equation which is not linear is a nonlinear differential equation. Let us just write an example for linear and nonlinear differential equation. An example for linear [00:03:46.864] differential equation is dyed by dx plus 2x into y is equal to x square and an example for nonlinear differential equation is y square d by dx plus sine y is equal to x square e to the power 2y. You know that this both the equations are of order [00:04:04.432] 1, this is LDE of order 1, and this is nonlinear differential equation of order 1. And in the previous sessions, you learned methods to solve both LDE of order 1 and non-linear differential equation of order 1. But in the coming session, [00:04:22.000] Sessions we will be

talking about only linear differential equation. So we are learning different linear differential equations of different orders, different forms, classification, and the methods to solve higher order linear differential equation. Now [00:04:40.228] Let us consider a differential equation dy by dx plus p into y is equal to q where p and q are functions of x alone. What is x here? x is the independent variable and y the dependent variable which is dependent on x. Now [00:04:58.000] Let us consider a differential equation of this type. Do you call this differential equation a linear differential equation? So, what is the definition of linear differential equation? So, a differential equation in which the dependent variable

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:13.936] and its derivative occur in first degree [00:05:18.608] And are not multiplied together is a linear differential equation. Look at this differential equation. It is y to the power 1, dy by dx to the power 1. So the degree of the dependent variable and its derivative is 1. [00:05:34.352] Also, the equation do not contain the dependent variable and its derivative in product form. So, what do you mean by the equation? Do not have y into dy by dx. So it is a linear differential equation. What is the odd [00:05:51.984] of the differential equation. Order of a differential equation is the order of the highest order derivative present in the differential equation. So what is the order of the differential equation? It is dy by dx. So it is 1. So this is a linear differential equation of order 1. [00:06:09.552] But in the sessions we will be talking about only higher order linear differential equations. So I'll define linear differential equation in a different way, which is more relevant to learn about higher order linear differential equations. [00:06:27.120] Definition goes this way. A differential equation is said to be linear if the coefficient y and its derivatives are functions of x are constants. And [00:06:41.296] The degree of y and its derivatives is 1. So this is a linear differential equation. So a differential equation, if it satisfies these two conditions, then we say that the differential equation is a linear differential equation. [00:06:58.992] Now let us look at linear differential equations of different order. [00:07:04.496] Now, first order linear differential equation. We know it is of the form dy by dx plus p into y is equal to q, where p and q are functions of x alone. Now let us look at linear differential equation of order 2. See how does it look like a different [00:07:22.160] equation the form P naught into D2y by Dx2 plus P1 into Dy by Dx plus P2 into Y is equal to capital X where [00:07:34.320] P0 P1 P2 and capital X are functions of the independent variable X and P0 equal to 0 is a linear differential equation of Albert 2. Why it is a linear different differential equation? [00:07:51.952] What is the definition of linear differential equation? A differential equation in which two conditions. What is the first condition? The coefficient of y and [00:08:02.704] Its derivatives should be functions of x. The coefficient of y is p2, dy by dx is p1, and p naught, they are functions of x. And also the degree of y and all its derivatives is 1 y to the power 1 d y by dx to the power. [00:08:20.432] 1 d2 y by d is 2 to the pi 1. So it is a linear differential equation. Because the order of this differential equation is 2, so it is a second order linear differential equation. Similarly, how does an nth order linear differential equation [00:08:38.000] look like so a differential equation of the form P naught into n third derivative of y with respect to x plus p1 into n minus 1 third derivative of y with respect to x plus pn into y is equal to capital S where [00:08:55.204] P0 P1 P2 P n or functions of X alone is a linear differential equation of R n provided P naught not equal to 0. Now in this differential equation also, you look at [00:09:10.416] The coefficients P naught P1 P2 Pn are functions of x. And also the co and also the degree of y. It's it all its derivatives, it is dn y by dx n to the power 1. [00:09:23.808] n minus one third or derivative of y with respect to x to the

power one dn minus two y by dx n minus two to the power one. So the degree of y and all its derivatives is one, the coefficient of y and all its derivatives are functions of x. So [00:09:41.456] So it is a linear differential equation of order n. [00:09:46.884] Now, n third linear differential equation is of the form, a general n th order differential equation in the form a0 into n third derivative of y with respect to x plus a1 into n minus 1 the derivative of y with respect to x.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.904] Plus E a n into Y is equal to capital X where E naught even into A N are either functions of X or constants. [00:10:11.456] capital X a function of X. Now see based on this there are two different forms of linear differential equations. What are the two different forms? One is linear differential equation with constant coefficients. [00:10:28.560] So what do we mean by it? So the differential equation in which the coefficients a0, a1, a2, a n are just numbers, are constant. [00:10:40.548] So the coefficients in the derivatives are constants. So an example for it, 2d3 by dx3 minus 5, d2y by dx2 plus 7 dy by dx plus y is equal to x square. Now look at this differential equation. So easy to [00:10:58.320] linear differential equation? Yes. Now [00:11:01.924] Look at the coefficients of the derivatives. It is the coefficient of y is 1, dy by dx is 7, d2 y by dx2 is minus p, d3 y by dx3 is 2. So it is a linear differential equation with constant coefficients. What is the order of [00:11:19.696] The differential equation. So the highest order derivative present in the differential equation is 3. So it is a third order linear differential equation with constant coefficients. [00:11:33.200] Now, the second form is linear differential equation with variable coefficients. So in the differential equation, if the coefficients of y and its derivatives are functions of the independent variable x. [00:11:50.832] Then the differential equation is linear differential equation with variable coefficient a01 a2 a n are functions of x. Now, so this is how a differential equation with variable coefficient look like. An example for this is [00:12:08.400] 2x cube into d3y by dx3 minus 5x square plus 2 into d2 y by dx2 plus x into dy by dx plus y is equal to x square into sine x. So it is a linear differential equation of order 3. [00:12:26.768] Now look at the coefficients of the derivatives. So 2x cube minus a 5x square plus 2x and it is 1. 1 can be written as 1 into x to the power 0. So the coefficients of the derivatives are functions of x and the equation is of our del 3. So [00:12:44.464] So it is an example for linear differential equation of order 3 and with variable coefficients. [00:12:55.300] Now we are now discussing about linear differential equation with constant coefficients. So we learn about linear differential equation with constant coefficients. Methods to solve linear differential equation with constant coefficients. Now this [00:13:13.072] LDE with constant coefficients can be classified into two types based on this cap right hand side function capital X. [00:13:22.000] One is called homogeneous linear differential equation. So what is homogeneous linear differential equation? The linear differential equation with constant coefficients in which [00:13:34.448] This capital X right hand side function is zero. And the other one is non-homogeneous differential equation. If RHS function is a non-zero function of the independent variable, then the differential equation is non-homogeneous LD. [00:13:52.144] RHS capital X is 0, RHS capital X is a 10 function of X. An example for a homogeneous linear differential equation with constant coefficients 2 d2y by dx2 minus 3 into dy by dx plus 9 into y is [00:14:09.712] Equal to 0. So this is a linear differential equation of order 2 with constant coefficients. Because RHS is 0, it is a homogeneous LDE. An example for a non-homogeneous LDE. See, I have taken the same [00:14:27.280] Example 2d2 by dx2 minus 3 dy by dx plus 9y is equal to x into sine hyperbolic x. So it is a non-homogeneous linear differential equation. Now you should know one more thing. Every non-homogeneous linear

differential. [00:14:44.848] Equation has a homogeneous linear differential equation associated with it. [00:14:52.740] So the homogeneous differential equation of this non-homogeneous equation is 2d2y by dx2 minus 3 dy by dx plus 9y is equal to 0. Now.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:05.604] Let us learn about the differential operator. We know that the first order differential operator is given by d by ts. So it is given the notation capital D here. [00:15:16.688] capital D is the notation given for the first order differential operator D by Dx. Similarly, the second order differential operator D2 by Dx2 is given the notation D square, capital D square, D3 by DX3 is given the [00:15:34.352] notation d cube and similarly an n third differential operator dn by dxn is given the notation d to the power n. Now if you consider [00:15:46.816] A differential equation. What is this differential equation? A1, a2, a3, a and say they are constants. Now it is. [00:15:54.672] And linear differential equation with constant coefficient, and it is a non-homogeneous linear differential equation. So, this differential equation in this operator notation form, it is written as d to the par n of y. [00:16:11.588] plus a1 d to the power n minus 1 of y plus a2 d to the power n minus 2 of y plus a n into y is equal to capital X. Now you can take y as common factor and write this differential equation as [00:16:29.360] f of d of y is equal to capital X where f of d is d to the power n plus a1 d to the power n minus 1 plus a2 d to the power n minus 2 plus a n called linear differential operator. So this linear differential operator it is a polynomial in D [00:16:49.700] So this linear differential operator behaves like an algebraic expression and it obeys the laws of algebra. You can factorize this f into linear factors, and the linear factors can be written in any order. [00:17:07.472] This can be treated like an algebraic expression and it obeys the laws of algebra. Now you consider a non-homogeneous linear differential equation of this type. [00:17:20.544] Now let us discuss about the solution of this differential equation. So this equation is also written as f of d of i is equal to capital S, where f of d is equal to d to the power n plus a 1, d to the power n minus 1 plus a to d to the power n minus 2 plus a n called. [00:17:38.288] The linear differential operator. Now the general solution of this non-homogeneous LDE with constant coefficients has two components. The first component is called complementary function, and the second component is called particular invariant. [00:17:58.224] And this complementary function is the complete solution of the associated homogeneous linear differential equation. So you should know this. [00:18:11.824] That the solution of the whole non-homogeneous linear differential equation has two components. One is called complementary function, and the other one is called particular integral. So if we are given a non-homogeneous linear differential equation, [00:18:29.488] And asked to solve the non-homogeneous linear differential equation, you need to find the complementary function and the particular integral of the differential equation. Now let us first discuss in detail about this complementary function. [00:18:47.844] So, to learn about complementary function, we need to know a very, very, very important concept called linearly independent and dependent functions. Now, how is it defined? Let us consider a set of n functions. So, how is this linear independent? [00:19:05.616] and dependence is defined for n number of functions. A set of functions f1, f2, f3, fn are said to be linearly dependent on an interval i. [00:19:19.844] If there exists constant c1 and c2, c three, cn, not all zeros such that c1 into f1 plus c2 into f2 plus c n into f n is equal to 0. What do you mean by this? See the [00:19:37.616] linear combination of the functions f1 f2 f n which is c1 f n plus c2 f2 plus c n f n is equal to 0 even for non-zero constant c1 c2 c three c n then the function cf1 f2 f n [00:19:55.184] are said to be linearly dependent functions. [00:19:59.780] What are linearly

independent? If not linearly dependent, the functions are called linearly independent functions. So, in other words, so what do we mean by it? The set of linearly independent. If the set is linearly independent, so what do you mean by it? So this is

EV0000014 · EV0000015 · EV0000016 · EV0000017

Transcript block 5 (source interval 001200-001499 seconds): [00:20:17.552] equal to 0 only when c1 c2 c 3 cn 0. That is linearly independent function. C. The linear combination of the function c 1 f1 plus c2 f2 plus c n f n is equal to 0. It is true only when the arbitrary constant c1 c2 c3 c n [00:20:35.120] are zeros on that is a trivial case otherwise this is not equal to zero for non-zero arbitrary constants c one c2 c three c n this is not equal to zero if f1 f2 f3 fn are leaving [00:20:52.688] Linearly independent of each other. Now let us consider our two functions, say y1 and y2. [00:21:01.508] Now, how do you define these two functions to be linearly independent? See two functions y now y1 of x and y2 of x are said to be linearly independent when the linear combination c1 into y1 plus c2 into y2 is not equal to 0. [00:21:19.280] 0 for non-zero constants c1 and c2. So the what is what is the meaning of it? That is C C1 by 1 is not equal to minus C2 into Y2. That means that y1 not equal to minus C2 by C1 into [00:21:36.848] Y1. R y1 is it not equal to minus c2 by c1 into y2. So what is the meaning? So this is a simple definition we give for [00:21:47.952] Two functions to be linearly independent of each other. So, two functions y 1 and by 2 are said to be linearly independent of each other if [00:22:00.740] Neither of the function is a constant multiple of the other. That is, y1 cannot be written as some constant into y2, or y2 cannot be written as some constant into y1. Then the functions y1 and y2 are said to [00:22:18.512] Be linearly independent functions. So now why do we need linearly independent functions? So you consider a second order homogeneous linear differential equation with constant coefficients. Now let us discuss the solution of homogeneous. [00:22:36.080] Linear differential equation with constant coefficients by first taking an example of now. [00:22:44.048] Let us consider a second order LDE with constant coefficients in the type d2 y by dx2 plus k1 dy by dx plus k2 div is equal to 0. Now how does the general solution of this homogeneous linear differential equation with constant coefficient look? [00:23:01.792] Like see, this is homogeneous because RHS is 0, K1, K2 are constant, so it is a linear differential equation with constant coefficient. How does the general solution of this differential equation? See, now it's very very important if y1 and y2 are two linear. [00:23:19.376] linearly independent solutions of this [00:23:23.024] Second order homogeneous linear differential equation with constant coefficients, then the linear combination C1 by 1 plus C2Y2 is also a solution of the differential equation. See, please understand if y1 and y is [00:23:40.688] 2 are linearly independent, then the linear combination C1 by 1 plus C2Y2 is also a solution. Why C1 by 1 plus C2Y2 is a solution only when y1 and y2 are linearly independent. So what is the definition of linearly independent? [00:23:58.256] independent functions, but if y 1 and y2 are linearly independent, then the linear combination c1 by 1 plus c2 by 2 is not equal to 0 for non-zero constant c1 and c2. So [00:24:11.780] And this is also a solution of the differential equation. In fact, this represents the complete solution of the differential equation. So, what did we learn? Very important concept. The complete solution of homogeneous linear. [00:24:29.552] differential equation with constant coefficient is of the type c1 y 1 plus c2 y2 where [00:24:39.056] Y1 and Y2 are two linearly independent functions. Now, let us extend the same concept to an nth order homogeneous linear differential equation. So consider an n th order in D E with constant coefficients. [00:24:56.752] n third d n y by dxn plus a1 into n minus one third derivative of y with

respect to a x plus a n of y is equal to 0 where a1 a2 a3 a n are constant. Now suppose say this n th order homogeneous linear differential equation

EV0000018 · EV0000019 · EV0000020 · EV0000021

Transcript block 6 (source interval 001500-001799 seconds): [00:25:14.384] has n linearly independent solutions y1 by two y three y ninear combinations c1 by one plus c2 by two plus cn by n also represent a solution of the same [00:25:31.952] differential equation and it is the complete solution of the differential equation. [00:25:38.608] And the complete solution of the differential equation is called complementary function of the differential equation. So what did we learn? The complete solution of linear differential equation, which is homogeneous, and it is of order n will be of the form. [00:25:56.240] is equal to c1 by 1 plus c2 by 2 plus cn by n and the complete solution is called complementary function of the differential equation. So the general solution of homogeneous linear differential equation [00:26:13.808] With constant coefficients is called complementary function of the differential equations. [00:26:22.372] Now, let us look at a non-homogeneous linear differential equation with constant coefficients. F of d of y is equal to capital X, where f of d is the linear differential operator. Now, so the solution of this differential equation, I said it has two components. [00:26:40.144] One is the complementary solution, other one is a particular solution. So y is equal to yc is the complete solution of f of d of y is equal to 0. What do we mean by this first statement? C f of d of y is equal to capital X is the given non-homo. [00:26:57.712] Homogeneous linear differential equation. The associated homogeneous differential equation is f of d of y is equal to 0. So now if y is equal to yc represents the complete solution of the associated homogeneous differential equation and [00:27:16.848] Y is equal to YP is a particular solution. One solution of the non-homogeneous differential equation f of d of y is equal to capital S, then the complete solution of this differential equation is of the form y is equal to yc. [00:27:34.480] C plus YP. Why it is called as the particular solution about which we will be learning later in a different session. So YC plus YP. What is Yc? It is the complete solution of the corresponding homogeneous differential. [00:27:52.048] This particular solution YP is also called particular integral. So this is how the solution of a non-homogeneous linear differential equation looks like. Now in the next session, we will be learning. [00:28:09.616] Methods to find the complementary function of the differential equation. Thank you.

EV0000022 · EV0000023 · EV0000024 · EV0000025

Full transcript: 3831 normalized words, 112 timestamped segments, 1697.045333 seconds duration. Low-confidence segments flagged: 0.

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3 Rules for finding the Complementary Function, Problems

This lesson is indexed from the source video titled Rules for finding the Complementary Function, Problems. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. No slide deck is assigned to this video; candidate slide sources remain separate in the unit master. EV0000026 · EV0000027

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.484] Hello everyone, welcome to the session on higher order differential equations. [00:00:06.048] From now on, we will be discussing about only higher order linear differential equations. So it is more relevant to say higher order linear differential equations. Now, in this session, we are learning the following concepts. We will be solving a higher [00:00:23.792] Order homogeneous linear differential equation with constant coefficients. So we are learning methods to find the general solution of higher order homogeneous linear differential equation with constant coefficients. We know that the general solution of homogeneous [00:00:41.360] LD with constant coefficients is called complementary function. So basically in this session we are learning methods to find complementary function of the given homogeneous linear differential equation. Now [00:00:57.808] Every homogeneous linear differential equation with constant coefficients has an algebraic equation associated with it. The algebraic equation is called auxiliary equation, and the general solution of the [00:01:15.088] Homogeneous linear differential equation depends on the nature of the roots of the auxiliary equation. So in this session we are discussing three different cases. Well [00:01:28.144] The auxiliary equation has real and different rules, real and distinct rules. When the rules of the auxiliary equation are real and equal. And when the rules of the auxiliary equation are imaginary. First let us learn how [00:01:45.360] in detail about auxiliary equation. So what is auxiliary equation? For that let us consider a simple second order homogeneous linear differential equation of the type d2 by b dx2 plus k1 d by dx plus k2 of y is equal to 0. [00:02:04.960] Now let us assume y is equal to e to the power mx. Differentiating this y one time and two times with respect to x, we get y dash is equal to em into e power mx and y double dash is m square into e to the power mx. Now [00:02:24.768] You can substitute, you can plug in these equations into the differential equation. [00:02:31.808] Only when y is equal to e to the power mx is a solution of the differential equation. Now let us assume y is equal to e to the power mx is a solution, and plugging these three equations into the differential equation, we get [00:02:47.904] m square into k plus k1 m plus k2 into a to the power mx is equal to 0. We know that a to the power mx is not equal to 0 for any real value of x. [00:03:01.040] So, which means that m square plus k1m plus k2 equal to 0. So this equation, this algebraic equation is called the auxiliary equation of the second order homogeneous differential equation, capital D square plus. [00:03:18.768] K1 into capital D plus K2 of Y is equal to 0. Now, how do we get the auxiliary equation of the differential equation? Or how are we writing the auxiliary equation of the differential equation? So it's very simple. This equation is f of b. [00:03:36.336] of y is equal to 0. In f of d you just replace b by m [00:03:42.448] And equate the algebraic expression to 0, we get the auxiliary equation. So it is m square plus k1 into m plus k2 equated to 0 is the auxiliary equation of this second order differential equation. Now if you have a second order. [00:04:00.176] Not a homogeneous differential equation of this type. The auxiliary polynomial of this differential equation is f of m. What is f of m? f of d is d squared plus k1 into capital D plus K2. In F of D replace D by M. See every [00:04:17.744] Linear differential operator with constant coefficient f of d has [00:04:23.712] An auxiliary polynomial associated with it, which is F of M. So now if we equate F of M to 0, we get the auxiliary equation. So this auxiliary equation is also called characteristic equation of the given homogeneous linear. [00:04:41.392] Differential equation. Now, very important thing, why do we need this auxiliary equation? Because the general solution of the homogeneous linear differential equation depends on the nature of the

loads of the auxiliary equation. [00:04:58.212] So, to write the general solution of the differential equation, we need to solve the auxiliary equation which is associated with it. Now.

EV0000026 · EV0000027 · EV0000028 · EV0000029

Transcript block 2 (source interval 000300-000599 seconds): [00:05:10.016] If we are discussing three different cases. So what is the first case when the rules of the auxiliary equation m1 and m2 are real and distinct. So because the auxiliary equation is [00:05:24.240] A quadratic equation, it has two roots, m1 and m2. If the roots of the auxiliary equation are real and distinct, then the general solution of this second order homogeneous linear differential equation is given by [00:05:40.208] Y is equal to some constant c1 into e to the power m1x plus c2 into e to the power m2x. So what are m1 and m2? The two real and distinct roots of the auxiliary equation. [00:05:56.336] Which is associated with the given homogeneous linear differential equation with constant coefficients. So it is C1, some constant c1 into e to the power m1x plus c2 e to the power m2 into x. What is x? X is the independent. [00:06:14.000] variable C1 C2 are the arbitrary constant. This is how the general solution of the differential equation look like when the roots are real and distinct. Now just we'll analyze this we'll understand this solution. See what is the order of the differential equation to [00:06:31.236] We have learned in the previous sessions that the general solution of a first order linear diffusion differential equation has one arbitrar constant in its general solution. Similarly, a second order differential equation has [00:06:47.248] Two arbitrary constants and an n th order differential equation has n arbitrary constants in its general solution because here the order is 2, there are two arbitrary constants in the solution. So this is the solution when the roots of the auxiliary equation m1 and m2. [00:07:04.912] are real and distinct. So let us illustrate this concept by taking an example. So solve d2y by dx2 plus 2 dy by dx minus d into y is equal to 0. So now first you need to write the auxiliary equation of this differential equation. How? [00:07:22.480] So you replace this, this is this equation is written as d square plus 2 into capital D minus 3 of y is equal to 0. In f of d replace d by m and equate it to 0. So it is m square plus 2 into m minus 3 is equal to 0. Another [00:07:40.048] simple way of learning to write the auxiliary equation. In this differential equation, you replace y by 1, dy by dx by u. [00:07:49.680] D2 y by dx2 by m square. If the differential equation has d3 y by dx3, replace it by m square. So the equation you get is the auxiliary equation. Solve this auxiliary equation, find out the linear factors and find the roots. The roots are minus 3 and 1. [00:08:07.460] Analyze the roots. The roots are real and they are different. [00:08:11.664] So how would be the solution of the differential equation when the roots are real and distinct? It is c1 e to the power m1x plus c to e to the power m2x. So the general solution of this homogeneous LB with constant coefficient is y is equal to c1 e to the power minus 3. [00:08:29.328] 2x plus c2 e to the power 1 in 2x. So this is the general solution of the second order homogeneous linear differential equation when the roots of the auxiliary equation are real and distinct. Now let us discuss the second case. [00:08:48.560] When the roots of the auxiliary equation are real and equal. M1 is equal to M2 is equal to M C. [00:08:57.024] Then the general solution, the differential equation is y is equal to c1 plus c2 into x, the independent variable x into e power mx. So this represents [00:09:10.256] The general solution of the homogeneous linear differential equation of order 2 if the roots of the auxiliary equation are real and equal. So this is the formula you should remember. Y is equal to C1 plus C2x into E2. [00:09:27.888] the par mx what is m the multiple root the repeated root of the auxiliary equation of this differential equation so now let us take an example solve d2 y by dx2 plus 4 dy by dx plus 4y is equal to 0 simple second order homogeneous [00:09:45.456] Yes, LDE with constant coefficient. So this equation can be written as operator

notation capital D square plus 4d plus 4 of y is equal to 0. And the auxiliary equation is m square plus 4m plus 4 is equal to 0, which is m plus 2 the whole square. So what are the roots?

EV0000030 · EV0000031 · EV0000032 · EV0000033

Transcript block 3 (source interval 000600-000899 seconds): [00:10:03.024] And minus 2. The two rules are real and repeated. So how would be the general solution? It is c1 plus c2 into x into e to the power em x. What is m over here? Minus 2. So it is e to the power minus 2 into. [00:10:20.592] X is the solution of this homogeneous differential equation. So this is case 2 when the roots of the auxiliary equation are equal. Now let us discuss the third case. Third case is when the roots are complex imaginary. So [00:10:38.884] Let the roots be m1 is equal to alpha plus i beta, m2 is alpha minus i beta. We know that the complex roots always occur in pairs, conjugate roots occur in pairs. So the general solution of the differential equation, when the roots are conjugated. [00:10:56.656] complex is y is equal to e to the power alpha is c1 cos beta x plus c2 sine beta is understand this it is e to the power alpha is the real part of the root and beta is the imaginary part of the root so it is e to the [00:11:14.224] alpha into independent variable x. [00:11:17.872] into C1 arbitrar constant C1 cos beta x plus c2 sine beta s. So this is the third case when the roots of the auxiliary equation which is associated with a given homogeneous second order linear differentiation. [00:11:35.536] equation are complex in nature. [00:11:38.960] Now let us consider an example d2y by dx2 minus 4 into dy by dx plus 13y is equal to 0. Now what is the auxiliary equation m square minus 4m plus 13 is equal to 0. That is the auxiliary equation. [00:11:56.592] Using the quadratic formula, find the roots of this equation. So what is that quadratic formula which we had studied before? It is minus b plus r minus square root of b square minus 4ac by 2 into a. [00:12:13.680] A is 1, B is minus 4, and C is 13. Substitute in the quadratic formula. Get the roots. The roots are 2 plus 3i, 2 minus 3i. So what is alpha over here? Alpha is 2 and beta is 2. [00:12:27.536] 3. So how would be the solution? Because the roots of the auxiliary equation are complex. So it is e to the power 2x. C1 cos 3x plus c2 sin 3x. This is the general solution when the roots are complex in nature. So [00:12:45.168] So we discussed three different cases of writing the solution of a second order linear homogeneous differential equation. Now let us extend the same concept. [00:12:58.240] To an n thirder homogeneous linear differential equation. So for that you consider an n thirder homogeneous linear differential equation with constant coefficients of the type d to the power n plus k1 d to the power n by [00:13:15.888] minus one plus k2 t to the power n minus two plus k n of y is equal to zero. Now the [00:13:24.208] auxiliary equation of this differential equation is m to the power n plus k1 m to be power n minus 1 plus k2 m to be power m minus 2 plus kn equated to 0. So this is the auxiliary equation because the degree of this algebraic equation is n [00:13:42.816] How many roots? N roots. So let the roots of this auxiliary equation be m1, m2, comma m three, slash, slash, slash m n. Now let us discuss different cases based on the nature of the n roots of the auxiliary equation. So now let [00:14:00.464] Let us consider different possibilities. Let us consider. So let us say all the N roots M1, M2, M3, Mn are real and distinct. Then [00:14:12.384] The general solution is just extend the same concept which we learned for a second order homogeneous differential equation to an nth order differential equation. So the general solution is y is equal to c1 e to the power em one x plus c2 e to the power m2x. [00:14:29.904] Plus C and E to the power mx. So this represents the general solution when the n roots of the auxiliary equation are real and distinct. Now let us discuss another possibility. What is it when the roots m1, m2? [00:14:47.536] M3 M and all are real and equal. Say equal to M1 equal to M2 equal to M3 equal to M and equal to M. So how would be the general solution? [00:14:58.736] be C1 plus C2X

plus C3 X square plus C n x to the power n minus 1 into e to the power m the equal root mx. So here please look here, see if the order of the differential equation is n.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:15.904] You are stopping at x to the power n minus 1. If the order of the differential equation is 3, say you are stopping at c3 x to the power 3 minus 1, which is x to the power 2. [00:15:28.592] If the order of the differential equation is 2, you are stopping at C2 x to the power 2 minus 1, which is x. [00:15:36.656] So that is C1 plus C2x into e to the parts, which we learned for a second order homogeneous differential equation. So this is for an nth order homogeneous linear differential equation in which the roots of the auxiliary equation are real and equal. [00:15:54.244] Look at the number of arbitrary constants in the general solution. How many arbitrary constant in an nth order differential equation? N constant. See, there are n constant. C1, C2, C3, C. Now let us discuss another possibility. Let us say the first four roots, m1, m2, m1. [00:16:12.016] And the remaining n minus 4 roots are real and distinct. [00:16:19.392] So the first four roots are real and equal equal to same m. Remaining n minus 4 are real and distinct. So I am combining concept 1 and 2, case 1 and case 2, which we discussed for a second order homogeneous differential equation. So how would be the solution of the differential equation? [00:16:37.072] differential equation see for the first four equal roots let us write it is c1 plus c2x plus c3x square plus c 4x to the power 3 because the first four roots are real and equal into e to the power mx. [00:16:51.744] Now you write for the remaining n minus 4 real and distinct roots. So it is C5 to the par m phi x plus c six e to the power m6x plus cn e to the power m nx. So this is one possibility. Let us discuss one more possibility. What is it? [00:17:08.832] If there are two pairs of uh imaginary roots, alpha plus r minus i beta, gamma plus r minus i delta, and the remaining n minus 4 roots are real and distinct. See there are totally n roots. [00:17:23.456] The first two roots are alpha plus i beta and alpha minus pi beta. The next two roots are gamma plus i delta, gamma minus i delta. The remaining n minus 4 are real and distinct. Say. [00:17:36.944] Now the general solution takes this form. What is it? Let us first write for alpha plus r minus i e to beta. E to the power alpha x c1 cos beta s plus c2 sin beta x. Now let us write for the next two. [00:17:53.664] Complex roots e to the power gamma x into c3 cos delta x plus c4 sine delta x and for the remaining n minus 4 roots it is c phi e to the power m phi x [00:18:09.136] Plus C6 e to the power mc plus c n e to the power mnx. Here I should write the constants as C3 and C4. It is written as C and C2. We we have to change this. Now. [00:18:21.380] Let us discuss one more case. What is the other possibility? Let us say this complex roots are repeated in the auxiliary equation. So if a pair of complex roots, say alpha plus r minus i beta, is repeated two times. [00:18:36.256] And the remaining n minus 4 roots say they are real and distinct. So how are you writing? So we are combining actually in this concept 1, 2, 3, 4, all four concepts. So it is the general solution is e to the power alpha x. [00:18:51.536] into C1 plus C2X for the repeated complex load C1 plus C2X into cos beta x plus c three plus c4 x into sine beta x. [00:19:01.696] into for the remaining n minus four roots it is c phi e to the power m phi x plus c six e to the power m c plus cn e to the power m n x. If the complex roots [00:19:14.752] are repeated in the auxiliary equation, then we write this way C1 plus C2x into cos beta x plus c3 plus c4 x into sin beta x. Suppose the complex roots are repeated three times alpha plus r minus i beta alpha [00:19:32.496] plus r minus i beta alpha plus r minus i beta then the general solution is e to the power alpha x c1 plus c2 x plus c three x square cos beta x [00:19:45.760] plus C4 plus C5 x plus C

six x square sine beta s and then you write for the remaining. So this is how we write the general solution of any nth order

EV0000038 · EV0000039 · EV0000040 · EV0000041

Transcript block 5 (source interval 001200-001499 seconds): [00:20:00.400] Homogeneous linear differential equation with constant coefficients. Now let us solve one question. So you consider [00:20:09.796] A third order homogeneous linear differential equation. Dq y by dx q plus 4 d by dx is equal to zero. What is it? It's a homogeneous third order LT with constant coefficients. So what is the [00:20:26.480] General solution of this differential equation called complementary function. So it is D cube plus four into D of Y is equal to zero operator notation. What is the auxiliary equation? F of D equal to when equate. [00:20:41.792] Uh f of m to zero. See this linear differential operator. For every linear differential operator, we can introduce a characteristic polynomial mq plus 4m 4m and the [00:20:56.752] And when mq plus 4m is equated to 0, we get the auxiliary equation. So auxiliary equation is mq plus 4m equal to 0. Let us solve. I have taken a simple cubic equation. So how many roots? 3 roots, because the degree of this equation is 3. So it is m into m square plus 4 equation. [00:21:14.416] to zero m is equal to zero and m is equal to plus or minus divide. So [00:21:20.400] Here C 0 for the real root I'm writing C1 E to the power 0x. Now this is of the form alpha plus R minus I beta, where in alpha is 0, beta is 2. So it is e to the power alpha x, e to the power 0x, c2 cos 2x plus c3. [00:21:38.144] sine to x does the solution of this c one plus c2 cos to x plus c three sin to x. So this way by using these formulae you can find you can write the general solution of any nth order homogeneous linear differential equation with [00:21:55.728] constant coefficients. Thank you.

EV0000042 · EV0000043 · EV0000044 · EV0000045

Full transcript: 3291 normalized words, 90 timestamped segments, 1319.765333 seconds duration. Low-confidence segments flagged: 0.

EV0000026 · EV0000027 · EV0000028 · EV0000029

4 Particular integral of standard functions - Type 1

This lesson is indexed from the source video titled Particular integral of standard functions - Type 1. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000046 · EV0000047

Graphs and diagrams

Figure G000034. Particular integral of Type 1 · source slide page 1.

EV0000366

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

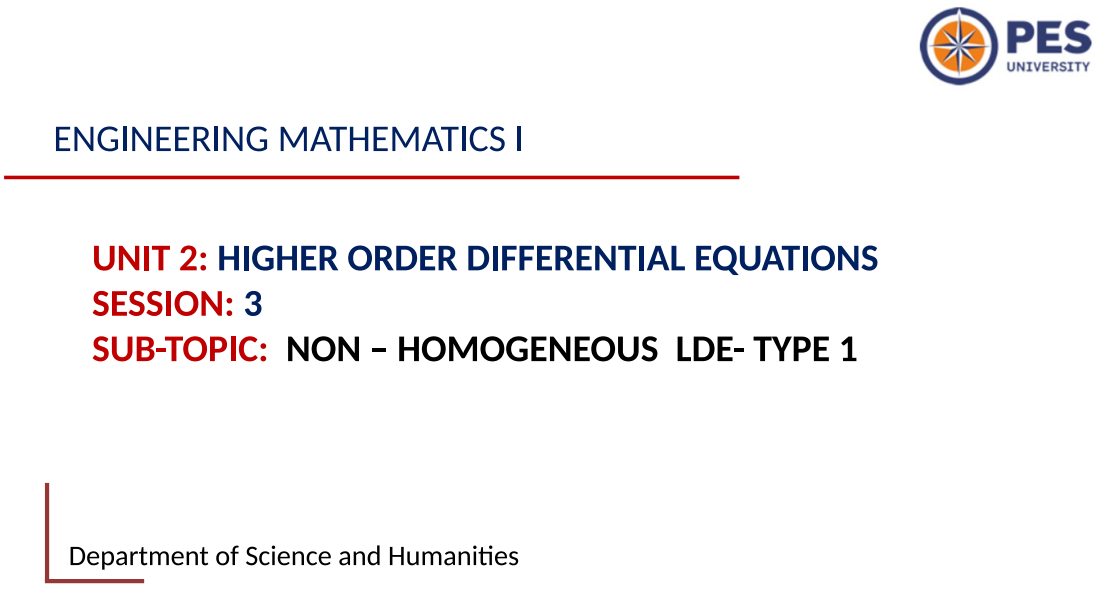

Figure G000035. Particular integral of Type 1 · source slide page 2.

EV0000369

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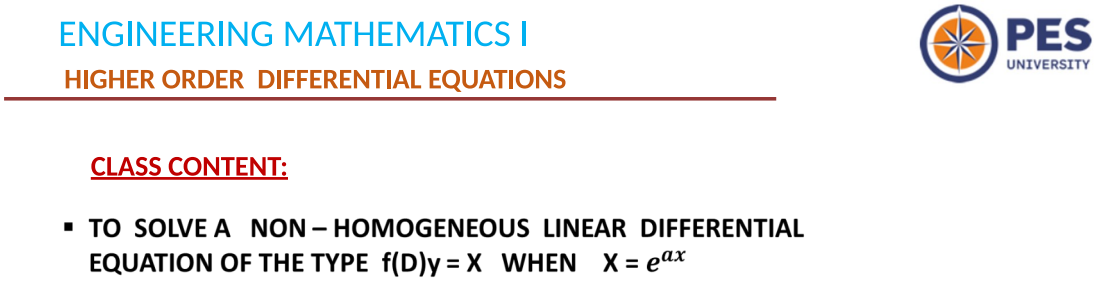

Figure G000036. Particular integral of Type 1 · source slide page 3.

EV0000372

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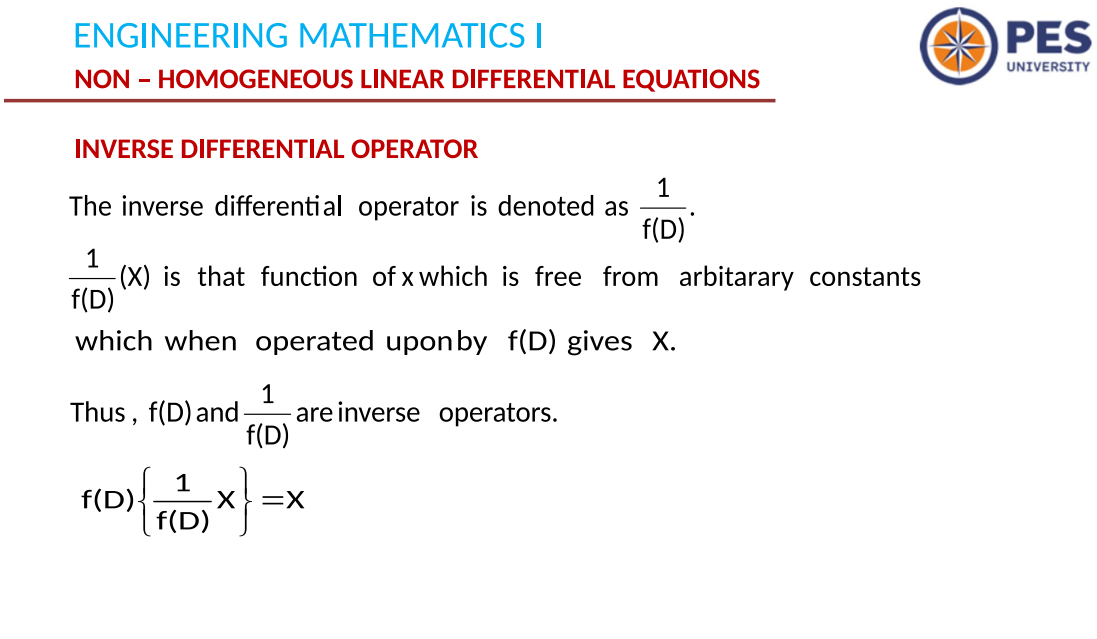

Figure G000037. Particular integral of Type 1 · source slide page 4.

EV0000375

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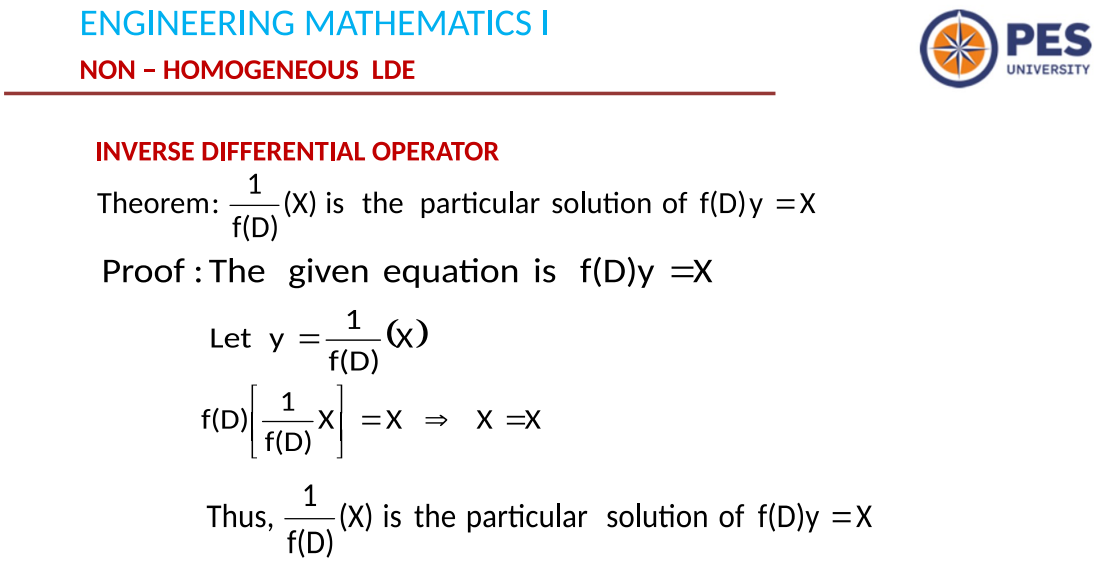

Figure G000038. Particular integral of Type 1 · source slide page 5.

EV0000378

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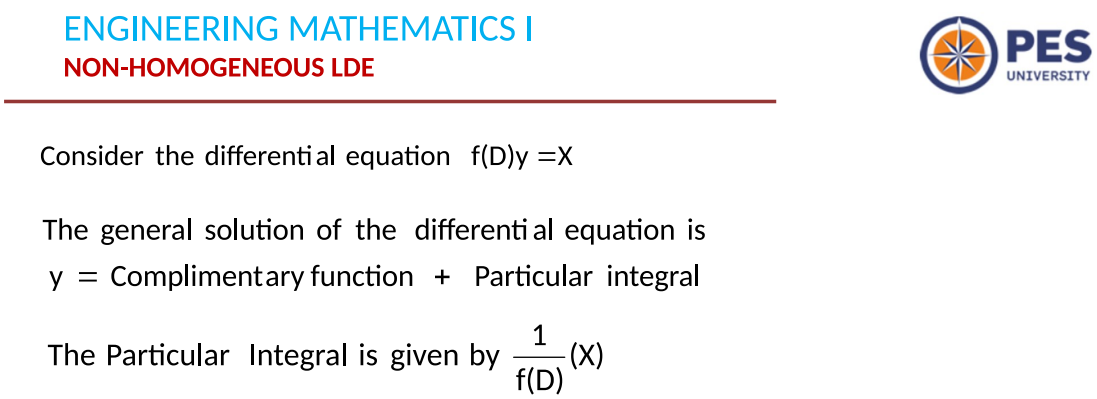

Figure G000039. Particular integral of Type 1 · source slide page 6.

EV0000381

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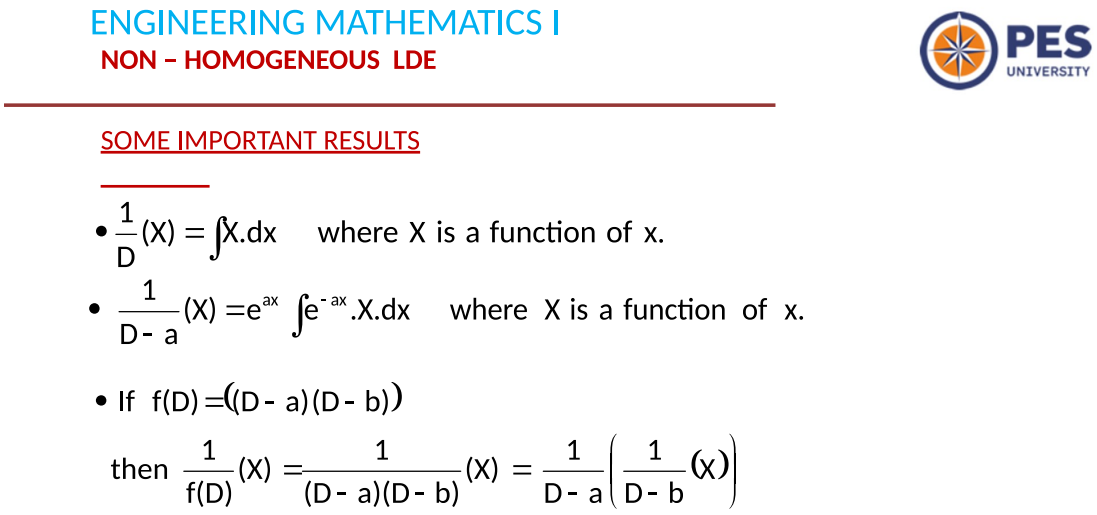

Figure G000040. Particular integral of Type 1 · source slide page 7.

EV0000384

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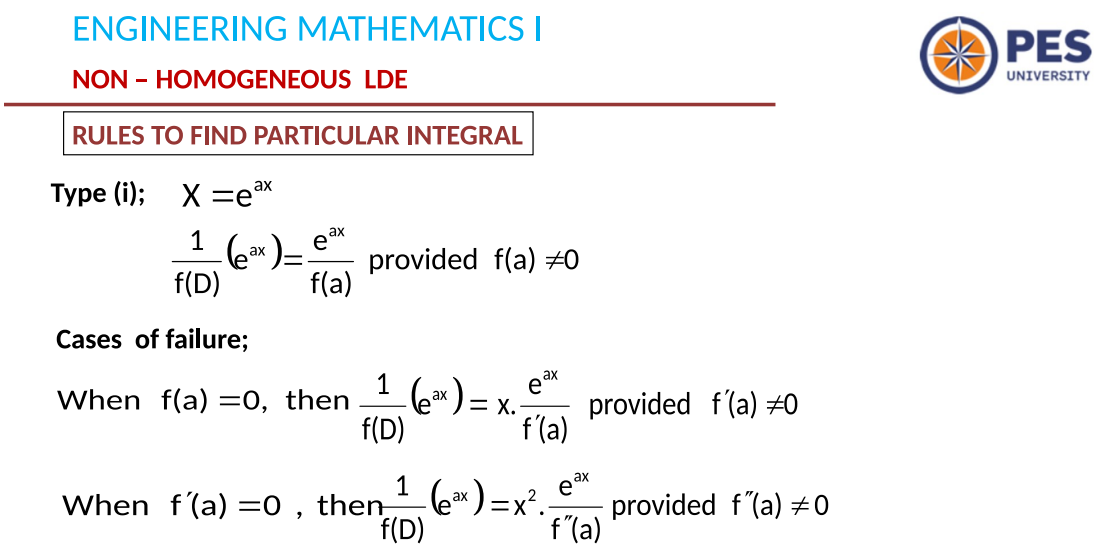

Figure G000041. Particular integral of Type 1 · source slide page 8.

EV0000387

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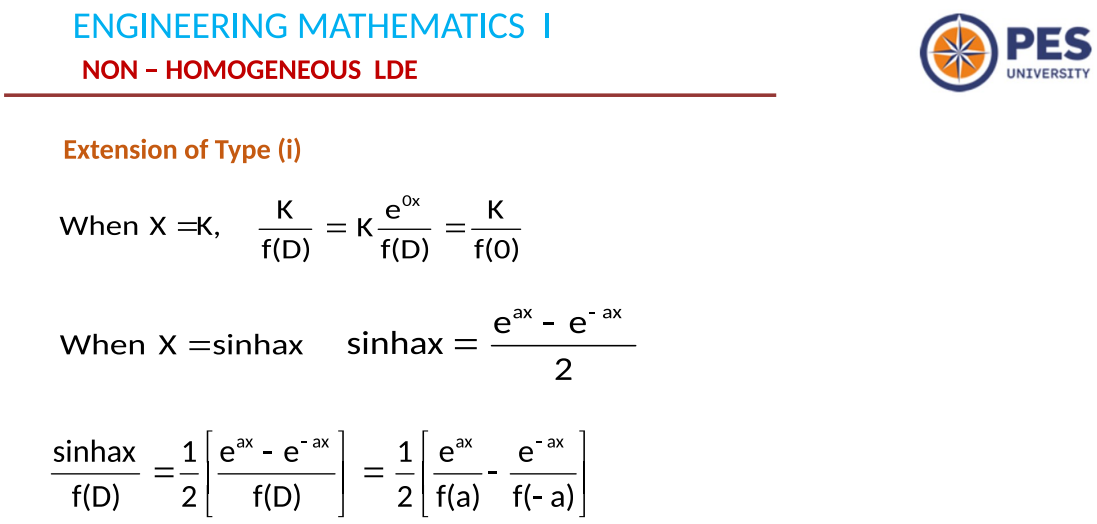

Figure G000042. Particular integral of Type 1 · source slide page 9.

EV0000390

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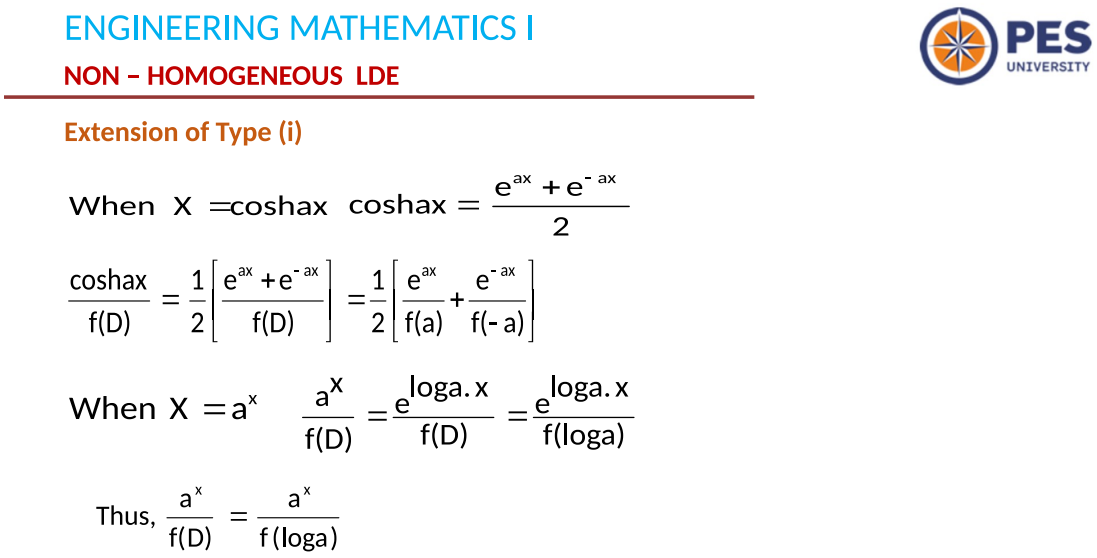

Figure G000043. Particular integral of Type 1 · source slide page 10.

EV0000393

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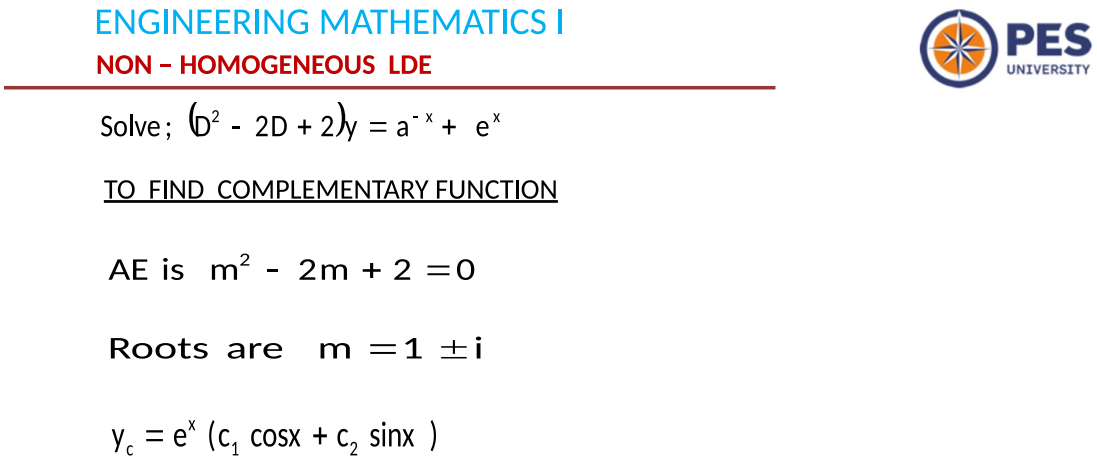

Figure G000044. Particular integral of Type 1 · source slide page 11.

EV0000396

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ENGINEERING MATHEMATICS I

NON – HOMOGENEOUS LDE

TO FIND PARTICULAR INTEGRAL

$$y_p = \frac{a^{-x}}{(D^2 - 2D + 2)} + \frac{e^{x}}{(D^2 - 2D + 2)}$$

$$y_p = \frac{a^{-x}}{((-\log a)^2 - 2(-\log a) + 2)} + \frac{e^{x}}{(1^2 - 2.1 + 2)}$$

Thus, $y = y_c + y_p$

Figure G000045. Particular integral of Type 1 · source slide page 12.

EV0000399

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ENGINEERING MATHEMATICS - I

HIGHER ORDER ODE - Problems

Finding Particular Integral of ODE
Type 1: when $X = e^{ax}$

For the differential equation $f(D)Y = X$, the particular integral is defined as

$$P.I = \frac{1}{f(D)}X$$

Type 1: When $X = e^{ax}$

$$P.I = \frac{e^{ax}}{f(D)} = \frac{e^{ax}}{f(a)}, \quad \text{provided} \quad f(a) \neq 0$$

If $f(a) = 0$, then P.I$= x\frac{e^{ax}}{f'(a)}$, provided $f'(a) \neq 0$
If $f'(a) = 0$, then P.I$= x^2\frac{e^{ax}}{f''(a)}$, provided $f''(a) \neq 0$, similarly, it proceeds on when $f''(a) = 0$.....

Figure G000046. Particular integral of Type 1 · source slide page 13.

EV0000402

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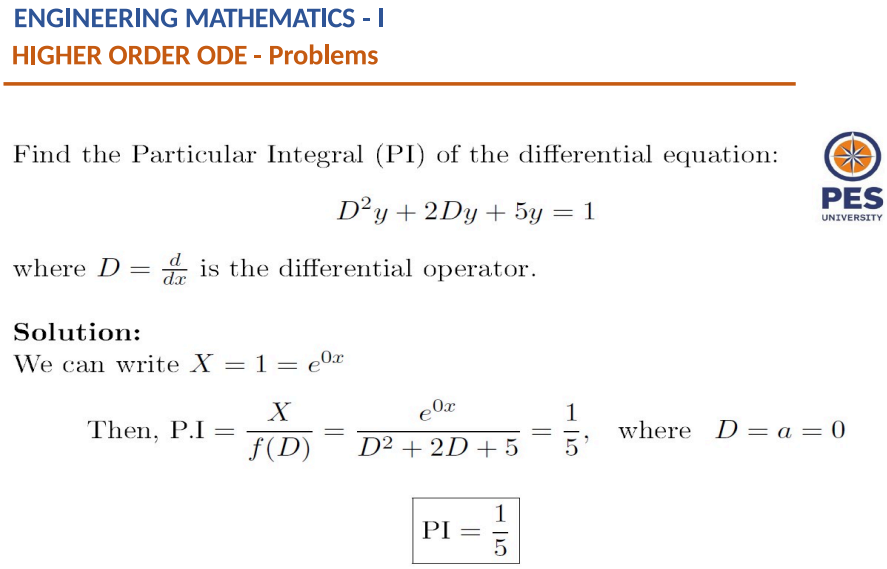

Figure G000047. Particular integral of Type 1 · source slide page 14.

EV0000405

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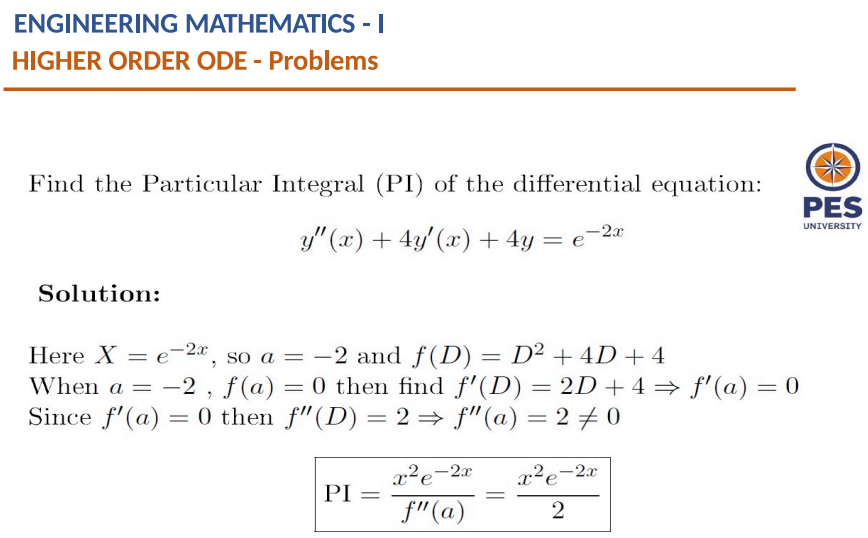

Figure G000048. Particular integral of Type 1 · source slide page 15.

EV0000408

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
HIGHER ORDER ODE - Problems

PES UNIVERSITY

Find the Particular Integral (PI) of the differential equation:

$$D^3y - 3Dy + 2y = 2\sinh x$$

where $D = \frac{d}{dx}$ is the differential operator.

Solution:
We express the right-hand side using the identity:

$$\sinh x = \frac{e^x - e^{-x}}{2} \Rightarrow 2\sinh x = e^x - e^{-x}$$

So the differential equation becomes:

$$f(D) = (D^3 - 3D + 2)y = e^x - e^{-x}$$

$$\text{Then, PI} = \frac{1}{D^3 - 3D + 2}(e^x - e^{-x}) = \text{PI}_1 + \text{PI}_2$$

where:

$$\text{PI}_1 = \frac{1}{D^3 - 3D + 2}e^x, \quad \text{PI}_2 = \frac{-1}{D^3 - 3D + 2}e^{-x}$$

Figure G000049. Particular integral of Type 1 · source slide page 16.

EV0000411

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
HIGHER ORDER ODE - Problems

PES UNIVERSITY

$$\text{In PI}_1, a = 1 \quad \text{and } f(a) = 0, f'(a) = 0 \quad \text{so, find} f''(a)$$

$$f''(D) = 6D = 6, \Rightarrow f''(a) = 6$$

$$\text{PI}_1 = \frac{x^2e^x}{6} \quad \text{then}$$

$$\text{In PI}_2, \quad a = -1 \quad \text{and} \quad f(a) = 4 \quad \text{then, PI}_2 = \frac{e^{-x}}{4}$$

Final Particular Integral is

$$\boxed{\text{PI} = \frac{x^2e^x}{6} - \frac{e^{-x}}{4}}$$

Figure G000050. Particular integral of Type 1 · source slide page 17.

EV0000414

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000051. Particular integral of Type 1 · source slide page 18.

EV0000417

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000365 · EV0000366 · EV0000367
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 2: HIGHER ORDER DIFFERENTIAL EQUATIONS SESSION: 3 SUB-TOPIC: NON – HOMOGENEOUS LDE- TYPE 1 Department of Science and Humanities EV0000368 · EV0000369 · EV0000370
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER DIFFERENTIAL EQUATIONS CLASS CONTENT: EV0000371 · EV0000372 · EV0000373
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LINEAR DIFFERENTIAL EQUATIONS INVERSE DIFFERENTIAL OPERATOR EV0000374 · EV0000375 · EV0000376
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE INVERSE DIFFERENTIAL OPERATOR EV0000377 · EV0000378 · EV0000379
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I NON-HOMOGENEOUS LDE EV0000380 · EV0000381 · EV0000382
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE SOME IMPORTANT RESULTS EV0000383 · EV0000384 · EV0000385

8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE RULES TO FIND PARTICULAR INTEGRAL Type (i); Cases of failure; EV0000386 · EV0000387 · EV0000388
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE Extension of Type (i) EV0000389 · EV0000390 · EV0000391
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE Extension of Type (i) EV0000392 · EV0000393 · EV0000394
11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE TO FIND COMPLEMENTARY FUNCTION EV0000395 · EV0000396 · EV0000397
12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE TO FIND PARTICULAR INTEGRAL EV0000398 · EV0000399 · EV0000400
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Finding Particular Integral of ODE Type 1: when X ac = e [Image text, OCR] For the differential equation f(D)Y = X, the particular integralpES defined as UNIVERSITY P.I = X f(D) Type 1: When X = ear eac eax P.I = provided f(a) ≠ 0 f(D) f(a) If f(a) = 0, then P.I= eax provided f'(a) ≠ 0 f'(a) ; provided f"(a) ≠ O, similarly, it f!"(a); proceeds on when f"(a) = O... EV0000401 · EV0000402 · EV0000403
14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Find the Particular Integral (Pl) of the differential equation: PES D²y + 2Dy + 5y = 1 UNIVERSITY where D = - Solution: We can write X = 1 = eOr X 0c Then, P.I = where D = a = 0 f(D) D2 + 2D + 5 5 1 PI = 5 EV0000404 · EV0000405 · EV0000406
15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Find the Particular Integral (PI) of the differential equation: PES y" (αc) + 4y'(x) + 4y = e-2x UNIVERSITY Solution: Here X = e-2, so α = -2 and f(D) = D2 + 4D + 4 When α = -2 , f(a) = 0 then find f'(D) = 2D + 4 → f'(a) = 0 Since f'(a) = 0 then f"(D) = 2 → f" (a) = 2 ≠ 0 α2e-2x x2e-2x PI = f"(a) 2 EV0000407 · EV0000408 · EV0000409
16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Find the Particular Integral (PI) of the differential equation: D²y - 3Dy + 2y = 2sinh c where D = Solution: We express the right-hand side using the identity: e" -e- sinh = - → 2sinh = e - e-r 2 So the differential equation becomes: f(D) = (D3 - 3D + 2)y = e - e-α 1 Then, PI = (e" - e-") = PI1 + PI2 D3 -3D + 2 where: 1 -1 PI1 = - Pl2 = D3 - 3D + 2 D3 -3D +2 P EV0000410 · EV0000411 · EV0000412
17. Source slide page 17 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] In PIi,a = 1 and f(a) = 0, f'(a) = 0 so, findf"(a) f"(D) = 6D = 6,→ f" (a) = 6 x2ec PI1 = - then 6 In PI2, a = -1 and f(a)= 4 then, PI2 ☰ 4 Final Particular Integral is -C e PI= 6 EV0000413 · EV0000414 · EV0000415
18. Source slide page 18 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000416 · EV0000417 · EV0000418

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.900] Hello, everyone, welcome to the session on higher order linear differential equations. In today's session, we are learning. [00:00:09.664] Methods to solve a non-homogeneous linear differential equation of the type f of b of y is equal to capital X when capital X is equal to E par ax. In the previous session, we learned methods to solve. [00:00:26.288] Homogeneous linear differential equation with constant coefficients. So in today's session, we are learning a method to solve a non-homogeneous differential equation of the type f of d of by is equal to e power ax. [00:00:43.012] We know that the general solution of the non-homogeneous LDE with constant coefficient has two components, complementary function and particular integral. [00:00:55.872] And complementary function is the general solution of the associated homogeneous differential equation. We already learned methods to find complementary function. Basically, in this session, we are finding methods to find [00:01:12.000] The particular integral of the differential equation f of df y is equal to e power ax. [00:01:21.124] First, let us learn a very important concept called inverse differential operator. What is this inverse differential operator? Why do we need this inverse differential operator? The inverse differential operator is denoted as 1 by eight. [00:01:38.896] f of d. Now the definition of 1 over f of d is see 1 over f of d of x is that function of the independent variable x, which is free from arbitrary constants which when [00:01:55.024] operated upon y f of d gives x. So 1 over f of d of capital X is a function of the independent variable X, which do not contain any arbitrary constant in it, but when this 1 over f of b of x [00:02:14.064] is operated upon by f of b gives back x. So that is the inverse differential operator from this it's very clear that this 1 over f of d and f of d are inverse operators. So what is it? 1 over f of d and 1 and f of d [00:02:31.760] are inverse operators and f of d of 1 over f of b of x is equal to x itself. [00:02:40.432] They are inverse operators. You can change the order. You can also write it as 1 over f of d of f f of b of x is equal to capital X. Now this is the definition of the inverse differential operator 1 over f of d. Now why do we need this 1? [00:02:58.064] over E f of D or 1 over Af of B of X. So we have a theorem 1 over E f of D of X represent the particular solution of the non-homogeneous differential equation f of b of y is equal to capital X. Let us prove this [00:03:15.632] It's very important statement. Let us prove this statement. But that consider the given equation. [00:03:22.512] F of B of Y is equal to capital X. It is a non-homogeneous linear differential equation. Just recall what is a non-homogeneous LD? If RHS of the differential equation is a non-zero function of the independent variable, then the differential equation. [00:03:40.176] Is called as a non-homogeneous linear differential equation. Now the given equation is this. Let us assume. Let us just consider y is equal to 1 over f of d of capital X. [00:03:52.368] So let us take y is equal to 1 over f of d of x. Now substitute this in the differential equation. So for y is equal to 1 over f of d of x, it should satisfy the differential equation f of d of y is equal to capital X. Let us check. [00:04:08.944] So put y is equal to 1 over f of d of x in the human differential equation, we get f of d of 1 over f of d of x. [00:04:17.028] Is equal to x itself. Why it is equal to x? Because by definition of 1 over f of d, 1 over f of d of x is that function of x, which when operated upon by f of d gives back x. So it is f of d of 1 over f of d of x is equal to the f [00:04:34.800] equal to capital X, which is equal to X on the right hand side of the differential equation. So y is equal to 1 over F of D of capital X satisfies the equation f of d of y is equal to capital X, which means that y is equal to 1 over F of D of X [00:04:52.368] Is a solution of the differential equation. And this solution is called particular solution of the differential equation f of d of y is equal to capital X because this 1 over F of D of X do not contain any R.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.936] constant. So 1 over f of d of x represents the particular solution of the differential equation f of d of y is equal to capital X. So if you have to find the particular solution of any non-homogeneous linear differentiation [00:05:27.504] equation you are finding one over f of d of capital X now. [00:05:32.240] If you have a non-homogeneous differential equation here of Df y is equal to capital X, we know that the general solution of this non-homogeneous equation has two components. One is called complementary function, which is abbreviated as CF, and the other [00:05:49.968] Component is particular integral which is given the notation PI. So C f plus Pi. So what is Cf complementary function? It is the complete solution of the corresponding homogeneous differential equation. We learn methods to find complementary function and the particular [00:06:07.536] integral is given by 1 over f of d of capital X. Now the next question is how do we find 1 over F of D of capital X? So let us first learn some results, important results wherein f of d takes a simple function of D. [00:06:24.416] Now, what is 1 over D of capital X? What is the meaning of this? It is 1 over F of D of capital X when F of D is equal to simple capital D. So 1 over D of capital X is integration of the function capital S. [00:06:40.896] with respect to x without writing the constant of integration. Suppose example, one over d of x square [00:06:50.884] is x cubed by 3 without writing the constant of integration. Similarly, if you have 1 over d square of capital X, so that is 1 over D, 1 over D of X. So it is [00:07:03.520] 2 times integration of x with respect to x. So 1 over d square of capital X is integral, integral x dx, dx. 2 times integration of x without writing the constant of integration. Now let us consider another one. [00:07:21.168] 1 over D minus Ef capital X, so it is 1 over F of D of capital X, where F of D is the linear factor d minus E. [00:07:29.936] What is 1 over d minus a f capital X? So it is given by this is the formula you should remember it is e to the power ax integral e to the power minus ax into this capital X dx. [00:07:45.572] So it is e power a x integral e power minus a x x dx without writing the constant of integration. Now similarly 1 over d plus a of capital X. Replace a by minus a in the formula. So it is e par minus a. [00:08:03.344] Integral, E power X, X, Dx. Now, [00:08:07.812] If suppose say f of d is the product of two linear factors d minus a and d minus b. So then what is 1 over f of d of capital S? So it is 1 by D minus A into D minus B of capital X. So how do we simplify? [00:08:25.584] this. It can be simplified in two ways. One way is applying the operator one by one. First you apply the operator 1 over d minus b on x. [00:08:35.172] And you then you apply the operator 1 over d minus e on the resultant function of x. So it is 1 over d minus e of 1 over d minus b of x. So to simplify this, use this formula. [00:08:47.972] You get a function fx. Then on the resultant function you apply the operator 1 over d minus a. This is one way of simplifying 1 over d minus a into d minus b of x. [00:09:02.404] The other way is resolve this 1 over d minus e into d minus p into partial fractions. Then use this formula and simplify. So here see we should be very careful. This 1 over f of d is an operator which is acting on the [00:09:20.192] function x. So the operator always should precede the function on which it is operating. So you should write it as 1 over uf of x r x by f of d. [00:09:33.508] It is not x into 1 over f of b because this is an operator which is acting on the function x. Now these are some important resources using which we can find the particular integral if f of d is a simple function d. Now. [00:09:49.744] What if f of d is a cubic equation, cubic expression, then f of d would be product of three linear factors. Then this method is very then the now we are discussing methods to find particular integral when capital X, which is the right hand.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:07.472] Side function of the differential equation is a specific function of x. The first case being [00:10:15.332] When capital X is equal to E par ax. Basically, we are now learning methods to find the particular integral of e par ax, an exponential function a par ax. So the rule, I'll just give you the rule 1 over f of d of e par ax. [00:10:32.388] is e par ax by f of a provided f of a not equal to 0. So what is the meaning of this? It is 1 over f of b of e po is e power a x by f of a. In f of d, just replace d by a to get f of a. [00:10:51.140] Provided f of a not equal to 0, the denominator not equal to 0. Similarly, what is 1 over f of d of e power minus a x? It is e power minus ax by f of minus a in the linear differential operator re. [00:11:08.912] Place d by minus a if you have e par minus a x as the function. So this is the formula which is used to find the particular integral of e po. [00:11:22.304] Now, what if f of a equal to 0? The cases of failure. When is f of a equal to 0? When d minus a is a factor of f of d. [00:11:35.524] Then f of a is equal to 0, then a becomes a root of the equation f of d and f of a is equal to 0. When f of a is equal to 0, the formula is 1 over f of d of e power e x is equal to x into. [00:11:51.360] E par a x by f dash a provided f dash a not equal to 0. So what is this? See x has to be written outside the operator. It is x into 1 over f dash d of e par ax. [00:12:08.448] And in f dash d replace d by a. So what is f dash d? You have f of d differentiate with respect to d to get f dash d. Then in f dash d put d is equal to a to get f dash a. [00:12:22.672] So it is x into e power e x by f dash a provided the denominator f dash a not equal to 0. C please understand x has to be written outside the operator x to the power 1 into e power a x by f dash a. Now [00:12:39.876] What if f dash a equal to 0? Now, when is f dash a equal to 0? When d minus a and d minus a two linear factors of f of b. [00:12:53.060] So d is equal to a, comma a 2 equal roots of f of d is equal to 0. Now when f dash a is 0, the formula is 1 over f of d of e power a x is x square. [00:13:06.608] into e par a is by f double dash a provided the denominator f dash double dash e not equal to zero. So what is f double dash a? You differentiate this f f f of d [00:13:19.808] two times with respect to d and then substitute d is equal to a in the second derivative to get f double dash a so provided this is not equal to 0. What if this f double dash a is equal to 0? The formula is x cube into e power e x by f triple dash a and so [00:13:37.488] So this is the formula to find the particular integral when capital X is equal to e par a x. But the same formula is applicable for the following functions. [00:13:50.244] So, for the functions which can be expressed in terms of exponential functions. So, what is the first one? x is equal to k constant. Now, how do you write constant? In terms of exponential function, k is written as k into e power 0x because e par 0 is [00:14:08.016] anyway one so k by f of d is k into e power 0 x by f of d I am writing k outside the operator because k is just as a constant it can be written outside the operator so it is e par 0 x by f of d now in f of d we place [00:14:25.584] D by 0 it is k by f of 0. So k by f of d is k by f of 0 provided f of 0 is not equal to 0. 0 is not a root of the equation f of d is equal to 0. If 0 is a root, then it is x in 2. [00:14:42.320] K by f dash 0. If f dash 0 is also 0, x square into k by f double dash 0 and so on. Now, so the same formula is applicable to next sine hyperbolic eggs. [00:14:55.680] What is sine hyperbolic A in terms of exponential function? It is sine hyperbolic A is E par Ax minus e power minus e x by 2. Now what is sine hyperbolic ax by f of b? So it is half into right sine hyperbolic x as e par a.

EV0000054 · EV0000055 · EV0000056 · EV0000057

Transcript block 4 (source interval 000900-001199 seconds): [00:15:13.392] minus e power minus a x by f of d. Now you split it into two terms e power a x by f of d. [00:15:19.968] Minus e par minus e x by f of b. In the first term we place d by a and in the second term replaced d by minus a. So sin hyperbolic a x by f of b is half into e power e x by f of a minus e power minus e x by f of minus

[00:15:37.616] A now another function cos hyperbolic x. What is cos hyperbolic x? e par e x plus e par minus e x by 2. And what is cos hyperbolic x by f of d? It is half into e par a x by f of a plus [00:15:54.128] A power minus a is by f of minus a. Now is there any other function which can be written in terms of exponential function? There is one more function. x is equal to a power x. How is a power x written in terms of exponential function? [00:16:11.792] can be written as e par log a par x. It is e par log [00:16:18.916] E per log a par x, that is e per log a into x. So e par log a into x by f of d is so what is the coefficient of x here? It is also a constant which is log a. [00:16:34.660] Now use the same formula. Replace d by log a. So it is e power log a into e x by f of log a when x is equal to a power x. Similarly, when x is equal to a power minus x. [00:16:48.580] So it becomes e bar minus log a into x by f of minus log a. [00:16:54.704] V plus d by minus login if you have a bar minus. So this is how we find the particular integral of exponential function and functions which can be reduced to exponential functions. Now [00:17:12.516] Let us solve a problem. So consider the non-homogeneous differential equation d square minus 2d plus 2 of y is equal to a power minus x plus e power x. So what is it? It is a non-homogeneous LDE. Why it is a non-homogeneous LDE? Because RHS is [00:17:30.288] not equal to 0. What is the order of the differential equation? 2. And this differential equation is a differential equation with constant coefficient. So it is a non-homogeneous linear differential equation of order 2 with constant coefficient. [00:17:47.856] Now the general solution of this differential equation has two components complementary function and particular integral. So [00:17:57.872] Solving this involves two major steps. The first step is to find the complementary function, and the second step is to find the particular integral. First, to find complementary function, I need to write the associated homogeneous differential equation of this differential equation. [00:18:15.504] It is d square minus 2d plus 2 of y is equal to 0. And the auxiliary equation of the homogeneous equation is m square minus 2m plus 2 equal to 0. Now find out the roots of this. [00:18:30.512] So we're using the quadratic formula minus b plus r minus square root of b square minus 4c by 2 a. find out the roots. The roots are 1 plus r minus i because this is a quadratic equation, two roots, one root is 1 plus i and other root is 1 minus i. [00:18:48.144] So the roots here are complex. So how do you write the complementary function? When the roots of the auxiliary equation are complex in nature. So it is alpha plus r minus psi beta recall. So it is e to the power alpha x, c1 cos beta x plus c2 psi beta x. So it is. [00:19:07.664] Y c the complementary function is also given the notation y c it is subscript c y c is equal to e power alpha x that is e to the power 1 into x into [00:19:20.624] C1 cos what is beta here? The coefficient of i is 1. C1 cos 1x plus c2 sin x. The complementary function of the given. [00:19:31.332] Non-homogeneous differential equation is this. Now what is the second step? To find the particular integral. So how is particular integral written? Particular integral is e power minus x. [00:19:44.772] by d square minus 2t plus 2 plus e power x by d square minus 2t plus 2 because the RHS function is a power minus x plus e po x, which I split it into two terms. Now how do you find the second term? It's simple. It is

EV0000058 · EV0000059 · EV0000060 · EV0000061

Transcript block 5 (source interval 001200-001499 seconds): [00:20:01.232] Uh replace d by e to the power a is one. Replace d by one. How? How do you simplify the first term? Replace d by minus log a. [00:20:18.880] So replace b by minus log a here, replace b by 1 here. So it is minus log a the whole square. Minus 2 into minus log a plus 2. Plus e power x in this replace d by 1. So you get y p as this and the complete solution of the differential. [00:20:36.624] equation is Y C plus 1. This is

how we find out the particular integral of the non-homogeneous differential equation when capital X the RHS of the differential equation is an exponential function. Thank you.

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Full transcript: 3347 normalized words, 85 timestamped segments, 1254.634667 seconds duration. Low-confidence segments flagged: 0.

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5 Particular integral_Type 1 and Type 2

This lesson is indexed from the source video titled Particular integral_Type 1 and Type 2. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000066 · EV0000067

Graphs and diagrams

Figure G000052. Particular integral of Type 2 · source slide page 1.

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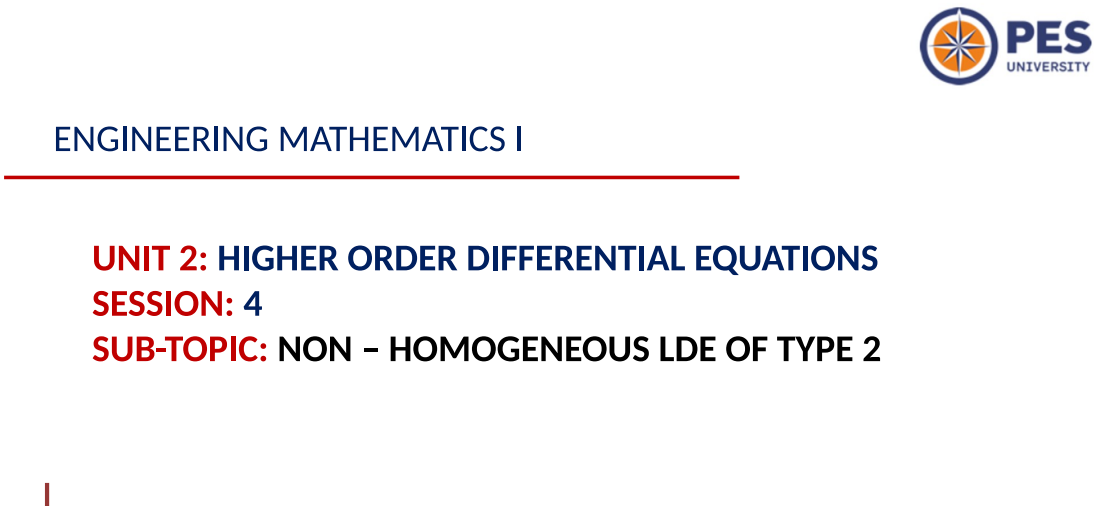

Figure G000053. Particular integral of Type 2 · source slide page 2.

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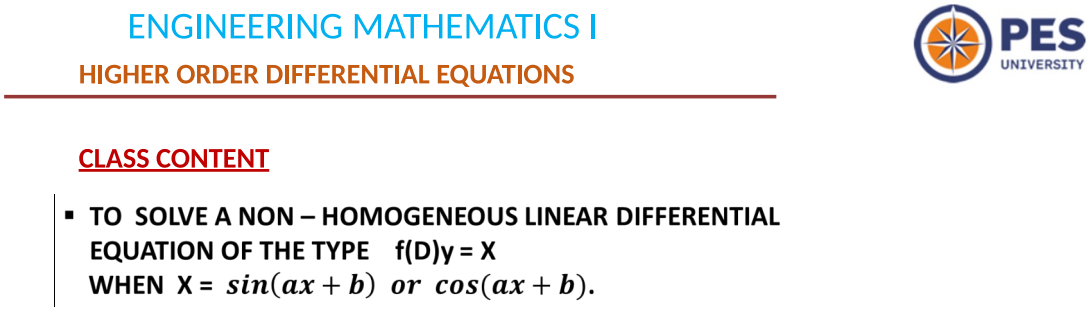

Figure G000054. Particular integral of Type 2 · source slide page 3.

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ENGINEERING MATHEMATICS I

NON – HOMOGENEOUS LDE

RULES TO FIND PARTICULAR INTEGRAL

Type (ii) ; When $X = \sin(ax+b)$ or $\cos(ax+b)$

$$\frac{\sin(ax+b)}{f(D^2)} = \frac{\sin(ax+b)}{f(-a^2)} \quad \text{provided } f(-a^2) \neq 0$$

$$\frac{\cos(ax+b)}{f(D^2)} = \frac{\cos(ax+b)}{f(-a^2)} \quad \text{provided } f(-a^2) \neq 0$$

Figure G000055. Particular integral of Type 2 · source slide page 4.

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ENGINEERING MATHEMATICS I

NON – HOMOGENEOUS LDE

Type (ii) continued,

Case of failure; If $f(-a^2) = 0$, then

$$\frac{\sin(ax+b)}{f(D^2)} = x.\frac{\sin(ax+b)}{f'(-a^2)} \quad \text{provided } f'(-a^2) \neq 0$$

$$\frac{\cos(ax+b)}{f(D^2)} = x.\frac{\cos(ax+b}{f'(-a^2)} \quad \text{provided } f'(-a^2) \neq 0$$

Figure G000056. Particular integral of Type 2 · source slide page 5.

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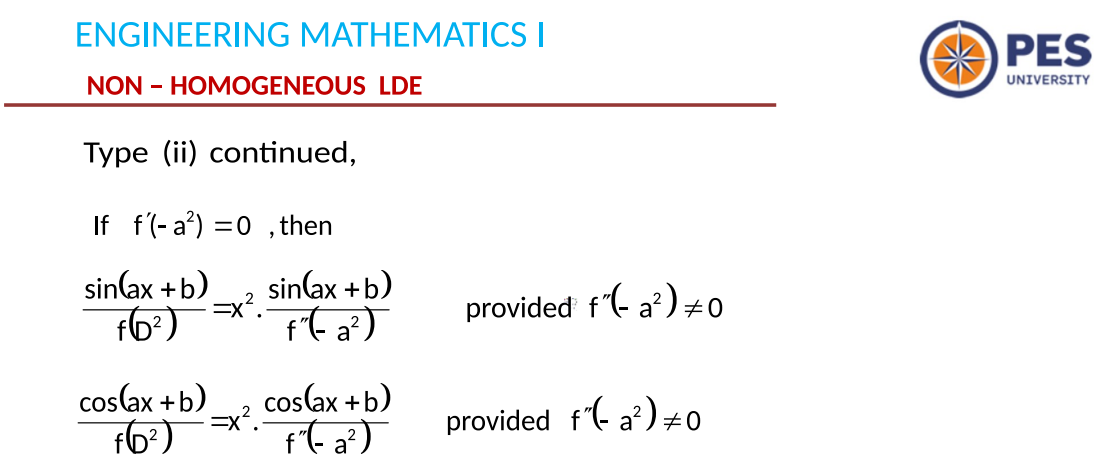

Figure G000057. Particular integral of Type 2 · source slide page 6.

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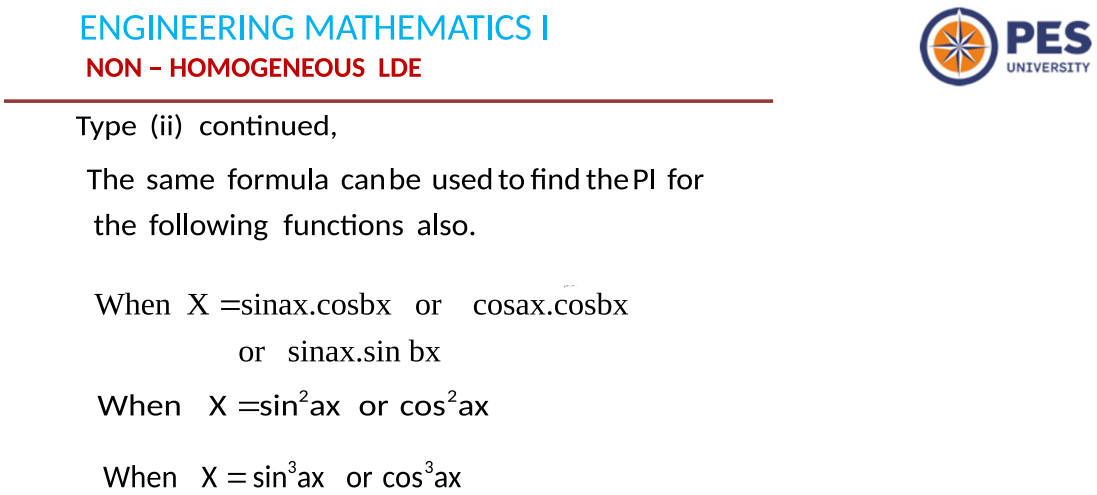

Figure G000058. Particular integral of Type 2 · source slide page 7.

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ENGINEERING MATHEMATICS I

NON – HOMOGENEOUS LDE

Solve ; $(D^3+1)y=\cos(2x+3)$

To Find CF

AE is $m^3+1=0$

$(m+1)(m^2-m+1)=0$

Roots are $m=-1$ and $\frac{1}{2}\pm\frac{\sqrt{3}}{2}i$

$$y_c=c_1e^{-x}+e^{\frac{x}{2}}\left(c_2\cos\left(\frac{\sqrt{3}}{2}x\right)+c_3\sin\left(\frac{\sqrt{3}}{2}x\right)\right)$$

Figure G000059. Particular integral of Type 2 · source slide page 8.

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ENGINEERING MATHEMATICS I

NON – HOMOGENEOUS LDE

To Find PI

$$y_p=\frac{\cos(2x+3)}{D^3+1}=\frac{\cos(2x+3)}{D.D^2+1}=\frac{\cos(2x+3)}{D.(-2^2)+1}$$

$$=\frac{\cos(2x+3)}{(1-4D)}=\frac{(1+4D)\cos(2x+3)}{(1+4D)(1-4D)}=\frac{(1+4D)\cos(2x+3)}{(1-16D^2)}$$

$$=\frac{(1+4D)\cos(2x+3)}{65}=\frac{1}{65}(1+4D)\cos(2x+3)$$

$$=\frac{1}{65}(\cos(2x+3)-8\sin(2x+3))$$

$\therefore y=y_c+y_p$

Figure G000060. Particular integral of Type 2 · source slide page 9.

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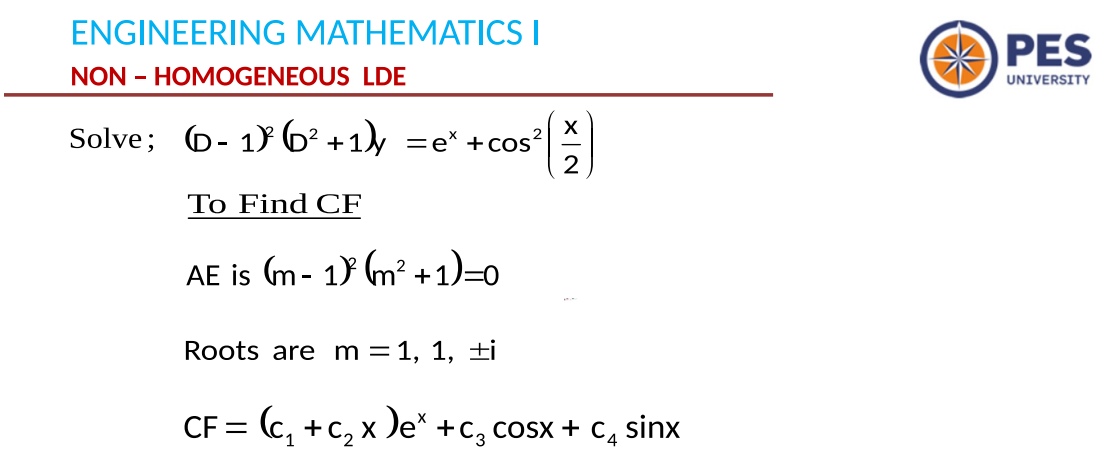

Figure G000061. Particular integral of Type 2 · source slide page 10.

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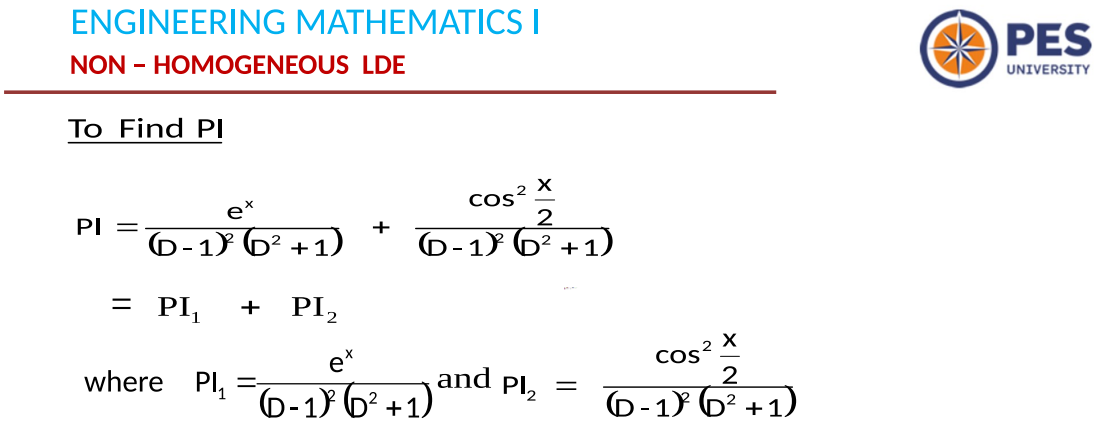

Figure G000062. Particular integral of Type 2 · source slide page 11.

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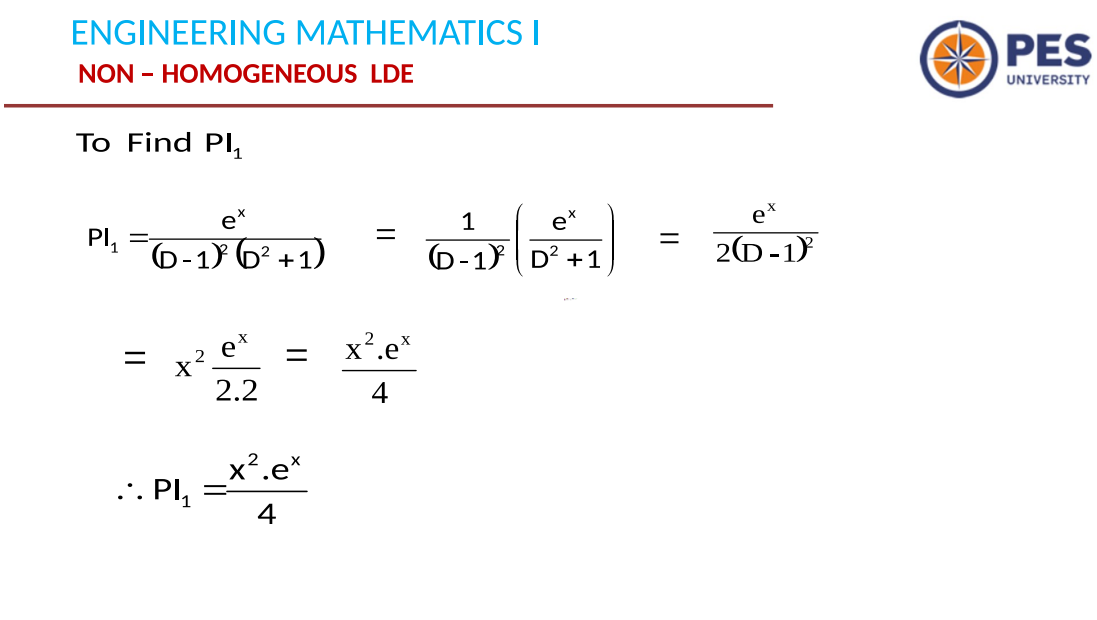

Figure G000063. Particular integral of Type 2 · source slide page 12.

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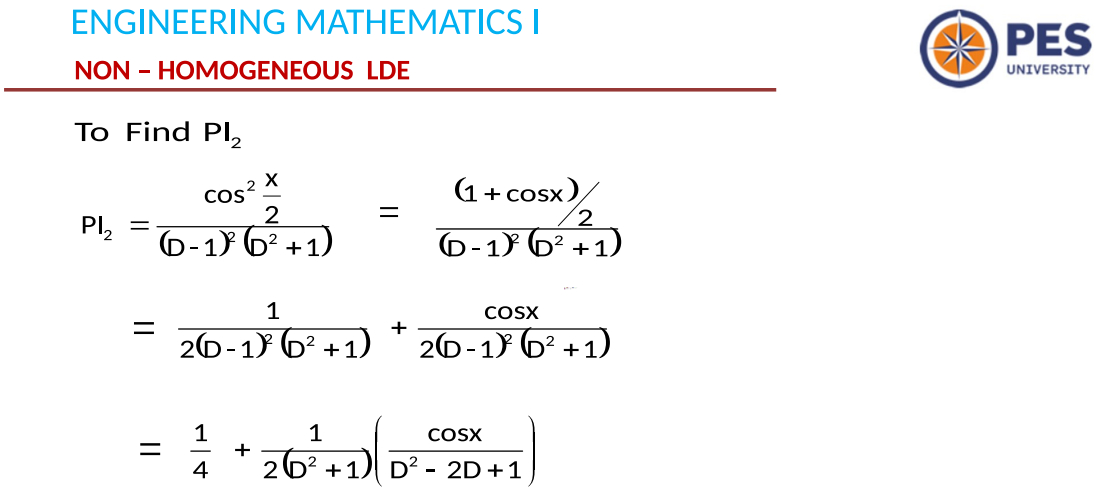

Figure G000064. Particular integral of Type 2 · source slide page 13.

EV0000456

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ENGINEERING MATHEMATICS I

NON – HOMOGENEOUS LDE

PES UNIVERSITY

To Find PI_2

$$PI_2 = \frac{1}{4} + \frac{1}{2(D^2+1)}\left(\frac{\cos x}{-2D}\right) = \frac{1}{4} - \frac{\sin x}{4(D^2+1)}$$

$$= \frac{1}{4} - x\frac{\sin x}{4(2D)} = \frac{1}{4} + x\frac{\cos x}{8}$$

$$\therefore PI_2 = \frac{1}{4} + x\frac{\cos x}{8}$$

Thus, $PI = PI_1 + PI_2$ And the GS is $y = CF + PI$

Figure G000065. Particular integral of Type 2 · source slide page 14.

EV0000459

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ENGINEERING MATHEMATICS - I

HIGHER ORDER ODE - Problems

PES UNIVERSITY

Type 2: When $X = sin(ax+b)/cos(ax+b)$

When $X = sin(ax+b)$

$$P.I = \frac{sin(ax+b)}{f(D^2)} = \frac{sin(ax+b)}{f(-a^2)}, \quad \text{provided} \quad f(-a^2) \neq 0$$

where $D^2 = -a^2$

If $f(-a^2) = 0$, then P.I$= x\frac{sin(ax+b)}{f'(a)}$, provided $f'(-a^2) \neq 0$

If $f'(-a^2) = 0$, then P.I$= x^2\frac{sin(ax+b)}{f''(a)}$, provided $f''(-a^2) \neq 0$,
similarly, it proceeds on when $f''(a) = 0$.....

Figure G000066. Particular integral of Type 2 · source slide page 15.

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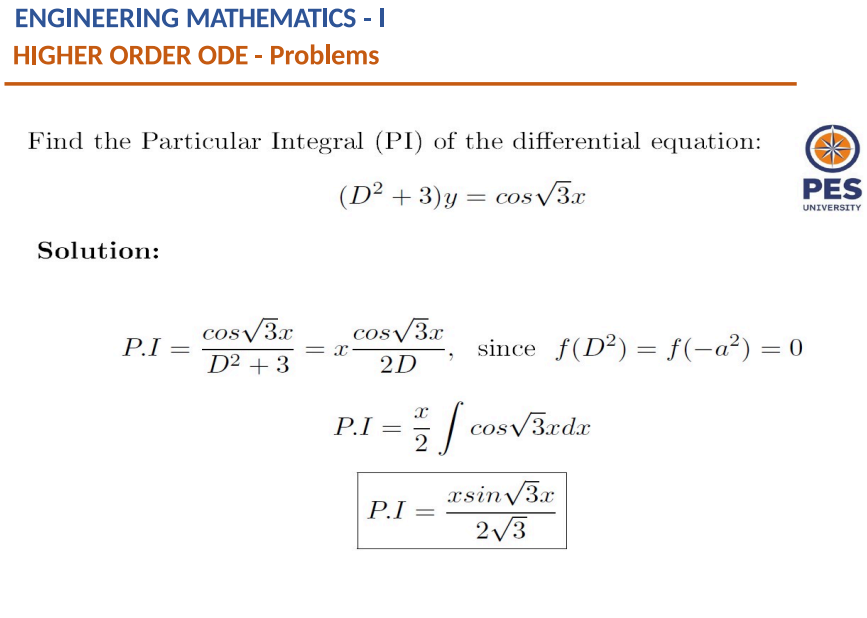

Figure G000067. Particular integral of Type 2 · source slide page 16.

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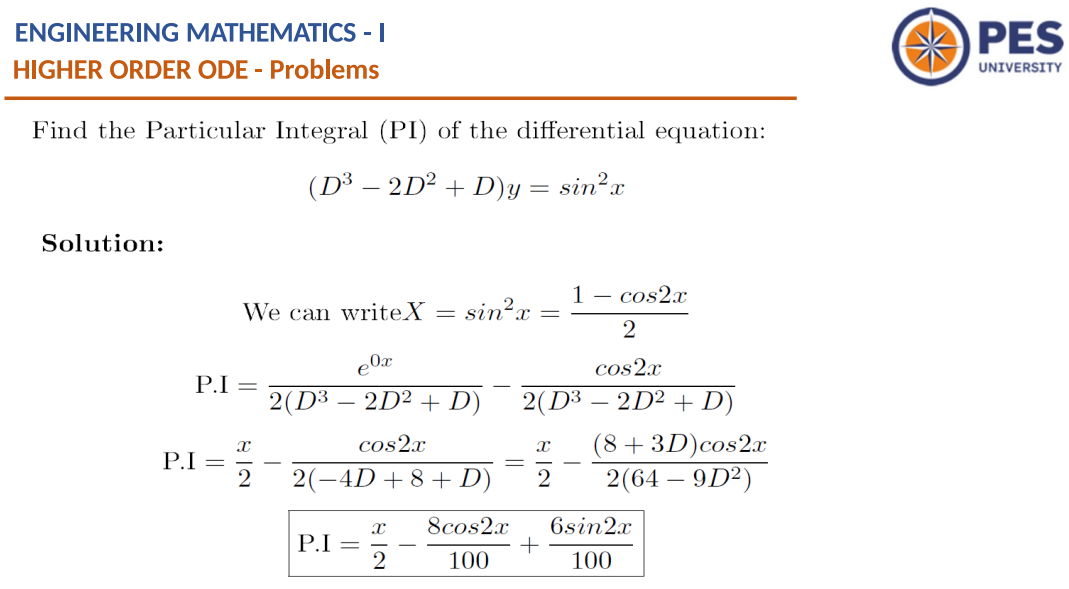

Figure G000068. Particular integral of Type 2 · source slide page 17.

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Figure G000069. Particular integral of Type 2 · source slide page 18.

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000419 · EV0000420 · EV0000421
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 2: HIGHER ORDER DIFFERENTIAL EQUATIONS SESSION: 4 SUB-TOPIC: NON – HOMOGENEOUS LDE OF TYPE 2 Department of Science and Humanities EV0000422 · EV0000423 · EV0000424
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER DIFFERENTIAL EQUATIONS CLASS CONTENT EV0000425 · EV0000426 · EV0000427
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE RULES TO FIND PARTICULAR INTEGRAL EV0000428 · EV0000429 · EV0000430
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000431 · EV0000432 · EV0000433
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000434 · EV0000435 · EV0000436
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000437 · EV0000438 · EV0000439
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000440 · EV0000441 · EV0000442

9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000443 · EV0000444 · EV0000445
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000446 · EV0000447 · EV0000448
11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000449 · EV0000450 · EV0000451
12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000452 · EV0000453 · EV0000454
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000455 · EV0000456 · EV0000457
14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000458 · EV0000459 · EV0000460
15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Type 2: When X = sin(ax + b)/cos(ac + b) [Image text, OCR] When X = sin(ar + b) sin(ac + b) sin(ac + b) P.I = provided f(-a²) ≠ 0 f(D2) f(-a²) where D2 = -α2 sin(ar+b) If f(-a²) = 0, then P.I= provided d f'(-a²) ≠ 0 f'(a) 2 sin(ax+b) If f'(-a²) = 0, then P.I=x provided f"(-a²) ≠ 0, f"'(a) similarly, it proceeds on when f"(a) =0..... EV0000461 · EV0000462 · EV0000463
16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Find the Particular Integral (PI) of the differential equation: (D2 + 3)y = cosV3c PES UNIVERSITY Solution: cos√3c cosV3c P.I = since f(D²) = f(-a²) = 0 D2 +3 Ξ2 2D C P.I = cosV3xdc 2 xsin√3x P.I = 2v3 EV0000464 · EV0000465 · EV0000466
17. Source slide page 17 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Find the Particular Integral (PI) of the differential equation: (D3 - 2D2 + D)y = sin²c Solution: 1 - cos2c We can writeX = sin² = 2 eOx cos2c P.I = 2(D3 - 2D2 + D)2 2(D3 - 2D2 + D) cos2c (8 +3D)cos2c C C P.I = 2 2(-4D + 8 + D) 2 2(64 - 9D2) C 8cos2c 6sin2c P.I = 2 100 100 EV0000467 · EV0000468 · EV0000469
18. Source slide page 18 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000470 · EV0000471 · EV0000472

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.460] Hello everyone. Welcome back to the course on engineering mathematics one. In this session, we solve problem on finding the particular integral when capital X is k power X and X equals [00:00:23.556] Sine of AX plus B or cos of AX plus B. [00:00:34.244] So let's start with case number one when x says e power axe. [00:00:41.700] You have already learned how to find Pi when right hand side capital X is e power AX, where a can be positive, negative, or even zero. So in this case, we write the particular integral in the form e power ax. [00:01:01.604] over f of a it means we will replace d by a in f of d provided f of a has to be not zero in case f of a becomes zero as you know from theory the particular integral is x [00:01:20.548] Expressed in this particular form that is x times e power ax divided by f dash of a. It means we differentiate f of d with respect to d once and again replace d by a again f dash of a should be not zero. [00:01:41.540] In case f dash f a becomes zero, so the particular integrity will be

modified in this particular form. That is x square times e power a x divided by f double dash a and so on. So having understood this concept. [00:02:00.900] Let us solve problems. [00:02:06.756] Take an example of finding the particular integral for this different equation. Observe carefully the right hand side is one. This is nothing but e power 0x where a is taken as 0. [00:02:23.300] So when we use the particular integral formula, we get e po 0x divided by d square plus 2d plus 5, which is coming from here. Replace d by 0, we get 0 plus 0 plus 5, so 1 by 5. [00:02:42.596] So because this is non-zero, we can stop here. So finally 1 by 5 is the particular integral for this particular problem. [00:02:59.716] Let us take another example where the right hand side is a power minus 2x. So a is minus 2 and the polynomial is d square plus 4d plus 4. Let's see what happens when we find out. [00:03:18.564] P particular integral PI write the formula one over [00:03:25.796] D square plus four d plus four times a power minus two x. Observe carefully, we need to replace d by a when a is minus two. So when we do this [00:03:44.068] denominator becomes zero. So we introduce 1x, we differentiate the denominator with respect to d one time. So we get 2d plus 4 a power minus 2x [00:04:02.532] Again we see that denominator is zero when d is minus two. Therefore this is written as x square times one over two d becomes two four becomes zero e power minus two x. [00:04:22.916] So this is the final answer for this particular problem. So the particular integral for this example is h square times e power minus 2x divided by 2 and we can stop here because this is now zero. [00:04:43.972] Let us solve this problem. The right hand side is sine hyperbolic X. So we know very well this is nothing but 2 times a power X minus A power minus X divided by 2.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:00.196] Therefore this is same as e power x minus e power minus x. Let us apply case one two times. So one for p1 and another for p2. [00:05:15.748] As you can see this is P1 and P2. So you can observe carefully here. Now what happens when you apply the procedure to find P1? [00:05:28.868] So when we replace D by 1, we see that f of a b becomes 0. Even in the second stage, f dash f is also zero. Therefore, P becomes x square times e power x divided by 6. Whereas in the [00:05:47.236] Second case denominator is non-zero. So that's why we directly get PI2 to be a power minus x divided by 4. Finally Pi is P1 minus P2. So this is going to be the answer. [00:06:04.964] I hope this concept is clear to you. [00:06:12.580] Let us move on to second case. When x is sine of a x plus b or cos of a plus b. Remember, if we want to apply this case, there are certain constraints. It means [00:06:32.740] F of D should have D square in it. Constraint number two. We replace D square by minus A square. Remember this is not minus A whole square, it's minus A square. [00:06:52.548] So what to do in case the fo of d doesn't have d square? [00:06:59.236] So if F of D doesn't have D square, we use the process called rationalization. [00:07:11.908] to get d square in f of d. So remember these are three constraints what we have in case number two where x is sine of x plus b or cos of x plus b. That is why when you write the particular integral this is not f of t this is f of d square. [00:07:32.196] And we are replacing d square by minus a square, provided this has to be non-zero. In case it becomes zero, we go by case one. Introduce one x differentiate f of d bet. [00:07:48.452] Respect to D once with respect to D1s. Again, that derivative should have d square, then replace d square by minus a square, and so on. [00:08:01.476] So this has a minus a square. Again this is minus a square. [00:08:13.764] Let's say let us take an example. Find the particular integral for this differential equation where the right hand side is cos of row 3x. And you can see a is root three. [00:08:32.292] So this is the formula for particular integral. So when you replace t square by minus a square, it is going to be [00:08:45.284] Minus three whole square. Therefore this is minus three.

Therefore denominator becomes zero. In this case Pi is expressed in the form x into derivative of d squared becomes two d. So one at two d then cost root three x. [00:09:05.380] So we have to use inverse differential operator. Take x by two side integrate cos root 3x with respect to x, then we arrive at the result. X by 2 sine root 3 x divided by root 3. So this is the answer for this particular problem. [00:09:26.276] Let us take another example where the right hand side is sine square x. So we know from trigonometry sine square x is nothing but 1 minus cos of x divided by 2. [00:09:42.596] So this problem should be solved by both case 1 and case 2 because for this one we use e ball 0x and for this we use case number 2. So take half outside, split this into two parts, this is P1 and this is P2.

EV0000070 · EV0000071 · EV0000072 · EV0000073

Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.444] It's very easy to see that when you compute d1, when you replace d by gate, denominator becomes zero. Then we introduce one x. [00:10:14.916] I differentiate with respect to D this 3D square. This is minus 40 and this is 1. Again 3d square becomes 0 and 4d becomes 0, the left out number is 1. Therefore, the answer for this P1 is simply x by 2. How do we solve P2? [00:10:35.108] So timing just ignore this two. So you have a p2 equals cos to x divided by d power three is nothing but d square into d minus [00:10:54.916] to d square plus d replace d square by minus four. So when you do that we get cos to x divided by. So you get the eight minus [00:11:14.180] 3D right replace d square by minus 4. And you observe carefully as per theory part this f of d doesn't have d square. So to get d square in f of d we have to use rationalization. [00:11:29.380] It means multiplying divided by the conjugate of 8 minus 3d. It means 8 plus 3d of cost 2x divided by 8 minus 3d into 8 plus 3d. That's going to be 64. [00:11:48.804] Minus 36 B square. So in the numerator, a T into cost 3x and 3 is constant, B of cost 2x becomes minus 3 sin 2x. So that's why this is 6 sine 2x. In the denominator, [00:12:08.292] When you replace d square by minus 4, you get 100. So when you combine both p1 and p2, this is the answer for p i in this problem. So I hope this concept is also clear to you. So these are the problems on finding PI when the right [00:12:28.196] Sine of Ax plus B or cos of X plus B. Thank you.

EV0000074 · EV0000075 · EV0000076 · EV0000077

Full transcript: 1442 normalized words, 46 timestamped segments, 756.629 seconds duration. Low-confidence segments flagged: 0.

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6 Particular integral of standard functions - Type 2

This lesson is indexed from the source video titled Particular integral of standard functions - Type 2. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000078 · EV0000079

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.740] Hello everyone, welcome back to the session on higher order linear differential equations. In today's sessions, we are learning a method to solve a non-homogeneous linear differential equation of the type f of df is equal to capital X when capital X is trigonometric function of the [00:00:19.632] type sine a x plus b or cos a x plus b. In other words in today's session we are learning a method or a rule to find the particular integral of sine a x plus b or cos a x plus b. Now [00:00:36.452] Well capital X is sine A X plus B or Casi X plus B. [00:00:41.344] So, what is the particular integral of sine ax plus b r cos a x plus b or in general capital X? So it is given by 1 over F of D of capital X. Now how do we find 1 over F of D of sine a x plus b or 1 over f of d of cos e? [00:00:58.992] plus b. So the rule says sin a x plus b by f of d square is equal to sin a x plus b by f of minus a square provided f of minus into a square not equal to 0. So what do we mean by this? [00:01:15.392] So, in the given function of d, which is f of d, you just pick only d square terms and replace d square by minus into a square. [00:01:28.612] If the resultant function still contains any d term, then convert those d terms into d square terms and then replace d square by minus into a square. [00:01:42.948] I'll just give you an example. Suppose say f of d, the given function of d is say d plus d square plus dq. Now, [00:01:53.348] And d write as it is. D square replace it by minus into a square. And d cube write it as d square into d. Replace only d square by minus into a square. [00:02:07.300] So the resultant function will be containing the constants and d terms. Now the d term is converted into d square terms and d square is replaced by minus into a square. Similarly, this formula [00:02:24.816] For cos a x plus b is cos e x plus b by f of d square is cos e x plus b by f of minus into a square provided the denominator f of minus a square not equal to 0. Now the next question is what if the denominator f of minus a [00:02:42.480] into a square equal to 0. Now when is f of minus into a square equal to 0? When b square plus a square [00:02:53.220] is a factor of f of b. Then f of minus a square will be equal to 0. Now the rule says that if f of minus a square is equal to 0, then [00:03:07.716] Sine a x plus b by f of d square is x into sine a x plus b by f dash of minus a square provided the denominator f dash of minus a square not equal to 0. C x has to be written outside the operator. [00:03:26.480] x into sine a x plus b by f dash of minus a square. Now how do we get f dash of minus a square? In the given f of d, differentiate it with respect to d to get f dash d. [00:03:40.612] And in f dash d, you replace only d square terms by minus into a square. [00:03:47.684] If that function still contains any d term, then convert those d terms into d square terms and then replace d square by minus a square, provided the demo denominator not equal to 0. Similarly, for cos e x plus b, [00:04:04.868] It is cos a x plus b by f of d square is x into cos a x plus b by f of minus into a square where the denominator not equal to 0. Now the next question is what if f of minus f dash of minus a square equal to 0? The rule continues if [00:04:23.904] f dash of minus a square equal to zero, then sin a x plus b by f of d square is x square. [00:04:31.300] into sine a x plus b by f double dash of minus a square where in f double dash of minus a square not equal to 0. Now how do we get f double dash of minus a square? Differentiate the given function f of d two times with respect to d and then re [00:04:49.072] replace b square by minus into a square. So now for cos a x plus b it is cos e x plus b by f of d square is x square into cos e x plus b by f double dash of minus into a square provided the denominator not equal to 0. Now the same

EV0000078 · EV0000079 · EV0000080 · EV0000081

Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.640] rule continues if f double if double dash of minus a square is 0 then it is x cube into sin a x plus b by f triple dash of minus a square and so on. But we can use the same rule for the following functions also. [00:05:24.292] Now

if you have sine ax into cos bx or cos a x into cos bx, sine ex into sine b x. So the these functions, the fa the particular integral of these functions can also be determined using the same rule. How? [00:05:42.756] Here you have product of two trigonometric functions. Convert the product of trigonometric functions into sum or difference by using transformation formulae. So what is it? See, we know that sine A into cos B. [00:05:57.540] is 1 by 2 sin a plus b plus sin a minus b. Using the transformation formulae, convert the product of two functions into summary of the trigonometry functions and use the root. [00:06:12.688] Now, if you have sin square ax or cos square ax, the particular integral of these functions can also be determined using the same rule. [00:06:23.536] How? Using the trigonometric identity. What is sin square ax? It is 1 minus cos 2x by 2. Cos square ax 1 plus cos 2x by 2. Using the trigonometric identity, you just write sin square or cos square in terms of cos e. [00:06:41.168] Ax or sine Ax and then use the same rule. This so same rule applicable for sine cube Ax or cos cube Ax using the same using the trigonometric identity. Now let us understand this method in a simple way by taking an example. [00:07:01.572] Consider the differential equation dq plus 1 of y is equal to cos 2x plus 3. So the differential equation is written in operator notation form. Let us first analyze this differential equation. [00:07:14.992] It is a non-homogeneous linear differential equation. Why? Because of cos 2x plus 3. Non-zero function on the right hand side of the differential equation. It is a linear differential equation. What is the order of the differential equation? 3. [00:07:29.968] So it is a non-homogeneous linear differential equation with constant coefficient and of order 3. Now the question is to find the general solution of this differential equation. We know that. [00:07:45.984] The general solution of this non-homogeneous differential equation has two components, the complementary function and the particular integral. So solving this equation has two major steps. [00:07:58.896] What is the first step? Step one is to find the complementary function. So to find complementary function we first have to set up the auxiliary equation which is associated with the given differential equation. So the auxiliary equation is mq plus 1 is equal to the [00:08:16.592] equal to 0 because f of d here is dq plus 1 replace d by m and equate it to 0. mq plus 1 is equal to 0. Now solve this equation to find the roots. So this mq plus 1 can be factorized as [00:08:34.192] M plus 1 into M square minus M plus 1. What is the formula used here? A Q plus B cube. What is AQ plus B cube? A plus B into A square minus A B plus B square. [00:08:50.564] So it is m plus 1 into m square minus m plus 1 equal to 0. Now factor is simplify this m plus 1 is equal to 0, which implies m is equal to minus 1 root. [00:09:03.696] To factorize this to find the roots of this, use the quadratic formula minus b plus or minus square root of b square minus four ac by two a and find the roots. The roots are m is equal to minus one and half plus or minus root three by two into i. [00:09:21.328] The first root is a real root. [00:09:23.824] And the other two roots are complex in nature. Now, how do we write the complementary function when one root is real and the other two roots are imaginary. So the complementary function is given by C1 e to the power minus 1 into x. [00:09:41.988] For the real root. Now for the complex root, just recall alpha plus or minus i into beta. Alpha here is half and beta is root 3 by 2. So what is the root? It is e to the power alpha x into [00:09:58.192] A C2 cos beta x plus c3 psi beta x. So it is c even e to the power minus x plus e to the power 1 by 2 is alpha. 1 by 2 into x into c2 cos root 3 by 2 into x.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:14.656] plus C3 sin root 3 by 2 into x. So this is how we find the complementary function of the given non-homogeneous linear differential equation. Now the second step the major step is to find a particular integral

[00:10:32.336] The particular integral is given by cos of 2x plus 3 divided by dq plus 1 because f of d is dq plus 1. It is 1 over f of d of cos 2x plus 3. Now what does this rule say? So cos a x plus b by f [00:10:49.904] So, D square is cos AX plus B. [00:10:52.944] by f of minus into a square provided f of minus into a square not equal to 0. Now write this d cube as d square into d, replace d square by minus into 2 square. [00:11:10.532] Because a here is 2. So this is equal to cos 2x plus 3 by d into d square plus 1. Now replace this d square by minus into 2 square. So it is [00:11:24.256] Cause 2x plus 3 by C D I have written as it is, and D square is replaced by minus into 2 square. [00:11:33.024] Plus 1. See please note here it is not minus 2 the whole square. It is minus into 2 square, which is minus 4. So this simplifies to cos 2x plus 3 divided by 1 minus 4d. Now [00:11:48.420] See, I told you if there is any D term remaining, D term has to be converted into D square term. How do we convert this 1 minus 4D into D square term? By [00:12:01.216] Multiply the numerator and denominator by its conjugate. What is a conjugate of 1 minus 4d? It is 1 plus 4d. Multiply the numerator and denominator with 1 plus 4d. [00:12:15.472] So we get 1 plus 4d, cos 2 e x plus 3 by 1 plus 4d into 1 minus 4d. See you please look here. See this 1 plus 4d or 1 over, 1 plus 4d are 1 over, 1 minus 4 d are all operators. So you cannot write cos 2x plus 3 to the left. [00:12:34.852] So it should the function always should be to the right of the operator. Now how do we simplify this? So we can do it in different ways. So what I do is [00:12:45.616] I'll write first I'll simplify the denominator. So it becomes one plus four d of cost two x plus three divided by one minus sixteen d square. Now let us simplify this cos two x plus three by one minus sixteen into d square using the rule which we have. [00:13:03.280] learned now. What is that rule? We place d square by minus into 2 square which is minus 4. So this becomes 1 plus 4 d cos 2x plus 3 divided by 1 minus 60 into minus 4. That is minus into minus become plus 64 plus 1 65. [00:13:23.652] Now you apply the operator 1 plus 4d on cost 2x plus 3. So this simplifies to 1 plus 4d cos 2x plus 3. Now this is equal to now multiply term by term 1 into cost 2x plus 3 plus 4 into d of cost. [00:13:41.424] Cas 2x plus 3. What do you mean by d? It is the first order differential operator d by dx. So derivative of cos 2x plus 3. So what is the derivative of cos 2x plus 3? It is minus 2 into sin 2x plus 3. So this expression simplifies to [00:13:58.992] to 1 over 65 into cos 2x plus 3 minus 8 into sin 2 e x plus 3. So the general solution or the complete solution of the differential equation is y is equal to yc plus y p. [00:14:16.548] Now let us consider another problem. So solve d minus 1 the whole square into d square plus 1 of y is equal to e to the power x plus car square x by 2. Just look at this problem. So the equation. So it is [00:14:32.704] A non-homogeneous linear differential equation with constant coefficients and the order of the differential equation is 4 because the equation has d square into d square, d to the power 4. What is d to the power 4? It is fourth order differential operator d4. [00:14:50.352] 4 by dx to the power 4. So the order of this differential equation is 4. [00:14:55.856] Now, because it is a non-homogeneous linear differential equation, we need to find the complementary function and the particular integral. So, solving this involves two major steps. The first step is to find the complementary function.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:13.156] Now, so to find the complementary function, I should write the associated auxiliary equation. So the auxiliary equation is [00:15:22.848] M minus 1 the whole square into m square plus 1 equal to 0. How did I get this? See in the given differential equation f of d is d minus 1 the whole square into d square plus 1. You just have to replace d by m. [00:15:40.132] And equate the expression to 0. That is the auxiliary equation. So it is m minus 1 to whole square into m square plus 1 is equal to 0. Let us solve this

algebraic equation because the degree of this algebraic equation is 4. There are 4 roots and the roots are. [00:15:57.904] 1 comma 1 comma plus r minus i. Why these two 1 and 1 roots? Because m minus 1 the whole square is 0. M minus 1 into m minus 1 0. 1 comma 1 are the two roots here. M square is equal to minus 1. M is equal to square root of minus 1, which is plus r minus i. [00:16:17.924] Now just look at the roots. The first two roots are real and equal, and the next two roots are complex in nature. So how do we write the complementary function? The complementary function is c1 plus c2x into e to the power 1x. This is for the first two equal roots. Next for this [00:16:35.696] complex roots. So I'll I can write plus R minus I as 0 plus R minus I. So which is of the form alpha plus R minus I into beta where alpha is 0 beta is 1. So it is e to the power alpha x [00:16:52.832] e to the power 0x into C3 cos 1x plus C4 sine 1x. So this is the complementary function of the given differential equation. The second step is to find the particular integral. Now I have written [00:17:10.480] particular integral as e power x by f of d plus car square x by 2 by f of d because it is e power x plus car square x by 2 by f of d. Now simplifying these two terms is not very simple it's difficult. So what I do is I [00:17:28.048] I call this as particular integral 1 and particular integral 2 and simplify them separately. Add the two to get the particular integral of the given differential equation. So it is written as [00:17:41.168] Pi 1 plus PI2 where PI1 is a first term and P i2 is a second term cos square x by 2 by d minus 1 the whole square into d square plus 1. Now let us first find out the particular integral 1. [00:17:59.812] To find particular integral and what is it? E power x by d minus 1 the whole square into d square plus 1. How do I simplify this? I'm using rule 1. What is that rule 1? E power e x by f of d is e power a x by f of a. Provided f of a not equal to [00:18:17.584] 0, but it's very clear from this d minus 1 the whole square 1 comma 1 are the two roots. If I directly substitute d is equal to 1, I get 0 in the denominator. So what I do is first I apply the operator 1 over d square plus 1 on e par x. [00:18:37.476] And then I'll apply the operator 1 over d minus 1 the whole square on the resultant function of x. So I write it as 1 over d minus 1 the whole square into e power x by d square plus 1. Let us first simplify this by applying rule 1. Replace b by [00:18:55.248] A. What is a here? 1. So it is e par x by 1 square plus 1. That is 1 by 2. [00:19:02.368] So this becomes 1 by 2 into e power x by d minus 1 the whole square. Now how do I simplify this? How do I find this now if I replace d by 1 I get f of a 0. If I so if I differentiate it again, again I am getting 0. [00:19:20.016] So this is equal to first let me differentiate it one time. It is x into e power x by. [00:19:26.560] 2 into 2 into d minus 1, which is 4 into d minus 1. Why did I write x? The rule says if f of e is 0, e po ax by f of d is x into e par ax by f dash d and in f dash d replace d by 1. [00:19:44.208] Now if I replace D by 1, I again get F dash A0. So what I have to do, I need to differentiate this again with respect to D by writing another X outside the operator. So this becomes. [00:19:59.332] X square into E power x by 4 into D minus 1. Differentiation is 1. So the particular integral 1 is x square e power x by 4. Now, how do I find the other term particular integral 2?

EV0000090 · EV0000091 · EV0000092 · EV0000093

Transcript block 5 (source interval 001200-001499 seconds): [00:20:16.228] It is ca square x by 2 by d minus 1, the whole square by d square plus 1. I don't have any formula for ca square x by 2 by f of d. That is I cannot directly find particular integral of ka square x by 2. [00:20:31.300] I have formula for cos a x or sine x, but not for cos square x by two. [00:20:36.736] So what I do? I use trignometric identity and write cos square in terms of cos ax or cos bx. So what is that identity identity I'm using? cos square x is 1 plus cos 2x by 2. So cos square x by 2 is 1 plus [00:20:56.128] Cos x by 2. So this is equal to 1 plus cos x by 2 by d minus 1 the whole square into d square plus

1. What I do now it here in the numerator it is the sum of two terms. I split it into two terms. It is 1 by 2 into d minus. [00:21:13.776] 1 the whole square into d square plus 1 plus half into cos x by d minus 1 the whole square into d square plus 1. Now how do we simplify this? Which rule I am using? I'm using rule 1. Why? Because 1 can be written as 1 into e power 0x. [00:21:33.028] So now what is e power 0x by f of d? Replace d by 0. So this simplifies to 1 by 2 into 0 minus 1, the whole square into 0 square plus 1, which is 1 by 4. Now coming to the second term. [00:21:50.916] What is that rule? Cos a x by f of d replace d square by minus into a square. So if I replace I get minus 1 square, that is minus 1 plus 1 gets cancelled. So what I do? I apply the operator 1 over d minus 1 the whole square on cos x first. [00:22:10.372] And I'll apply the operator d square plus 1 on the resultant function of x. So it is first term is 1 by 4 plus 1 by 2 into d square plus 1 into cos x by d minus 1 the whole square. What is it? d square minus 2d plus 1. How do I simplify this? [00:22:28.144] replace d square by minus into a square. D term right as it is. So it is minus 1 plus 1 gets cancelled. So I get cos x by minus 2d. [00:22:41.028] So cos x by minus 2d. Now, so how do I simplify this? So what I do is first I simplify the term inside the bracket and then apply the operator 1 over d square plus 1. So what is 1 over d of cos x? It is integration of cos x with respect to x. [00:23:00.000] without writing the constant of integration. So what is integration of cos x? It is sine x. So this simplifies to 1 by 4 minus 1 by 4 cos x by d square plus 1. Now how do I simplify this? How do I calculate this? I use [00:23:17.744] rule sin a x plus b by f of d replace d square by minus into a square I get minus 1 square plus 1 which is 0. So what I do I rate it as 1 by 4 minus 1 by 4 x into [00:23:34.656] sine x by differentiate the denominator. When you are writing x here you have to differentiate the denominator. Differentiation of d square is 2d. [00:23:44.196] Now, how do I simplify this further? So for this I need not need not have to use any rule because 1 over d of sin x, that is integration of sine x with respect to x, which is minus cos x, so it is 1 by 4 plus x into cos x by 8. So this is [00:24:01.968] particular integral 2. Now so what is the particular integral? It is particular integral 1 plus particular integral 2 and [00:24:12.480] The general solution y is equal to the complementary function plus particular integral. See if they ask you to find only the particular integral, you can write up till here. If they ask you to solve the differential equation, you should mention this. Y is equal to complementary. [00:24:30.128] function plus particular integral. So this is how we find the particular integral of sin a x plus b or cos a x plus b. Thank you.

EV0000094 · EV0000095 · EV0000096 · EV0000097

Full transcript: 3942 normalized words, 97 timestamped segments, 1483.989333 seconds duration. Low-confidence segments flagged: 0.

EV0000078 · EV0000079 · EV0000080 · EV0000081

7 Particular integral of standard functions - Type 3

This lesson is indexed from the source video titled Particular integral of standard functions - Type 3. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000098 · EV0000099

Graphs and diagrams

Figure G000070. Particular integral of Type 3 · source slide page 1.

EV0000474

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

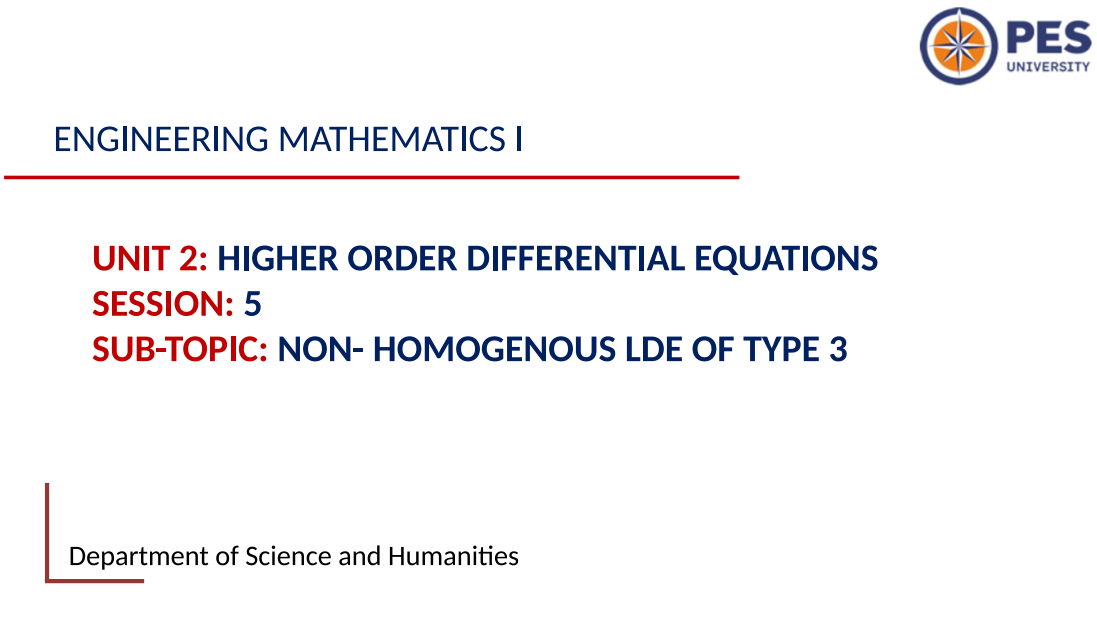

Figure G000071. Particular integral of Type 3 · source slide page 2.

EV0000477

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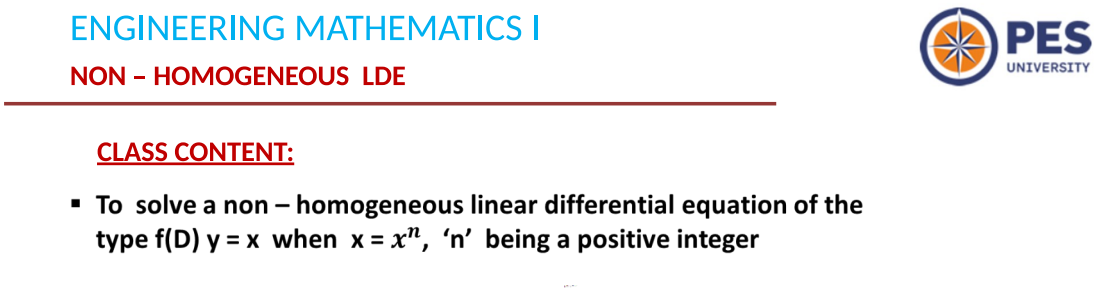

Figure G000072. Particular integral of Type 3 · source slide page 3.

EV0000480

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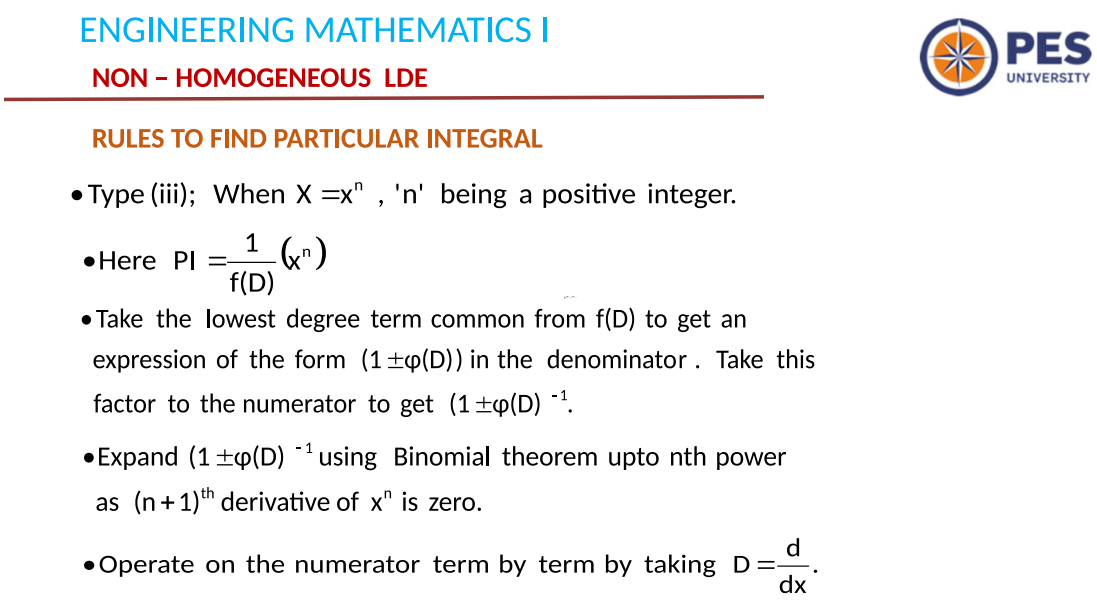

Figure G000073. Particular integral of Type 3 · source slide page 4.

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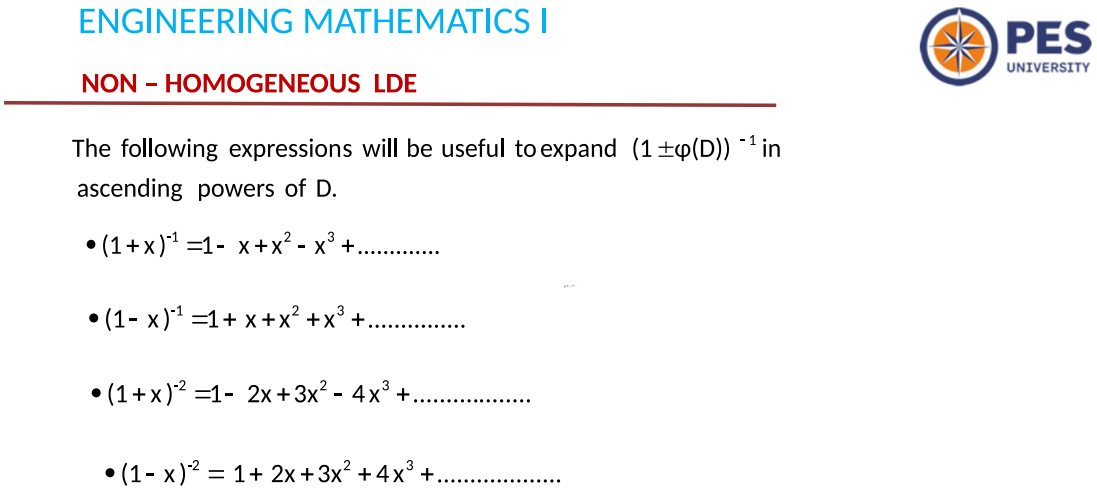

Figure G000074. Particular integral of Type 3 · source slide page 5.

EV0000486

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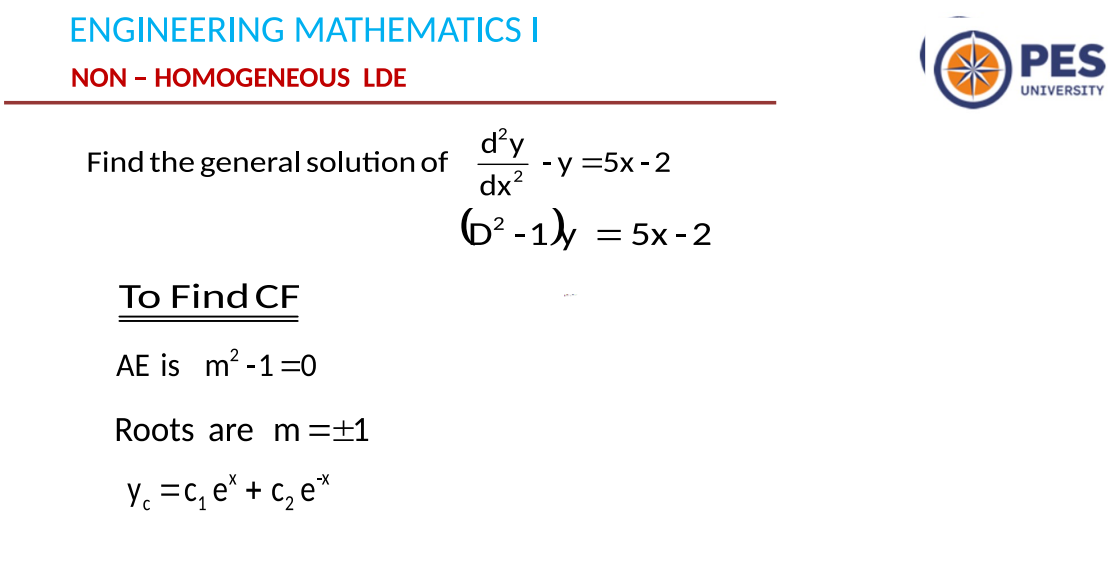

Figure G000075. Particular integral of Type 3 · source slide page 6.

EV0000489

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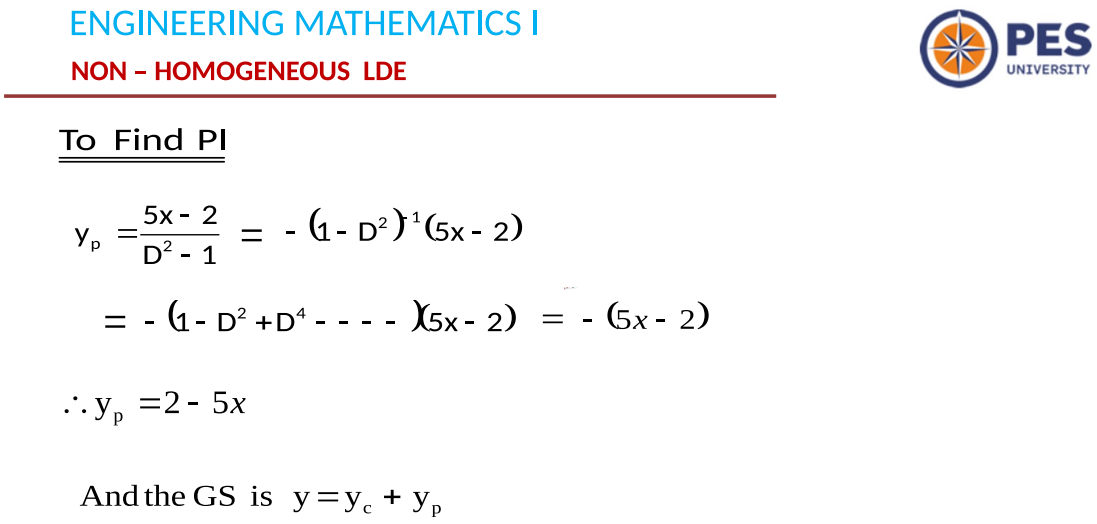

Figure G000076. Particular integral of Type 3 · source slide page 7.

EV0000492

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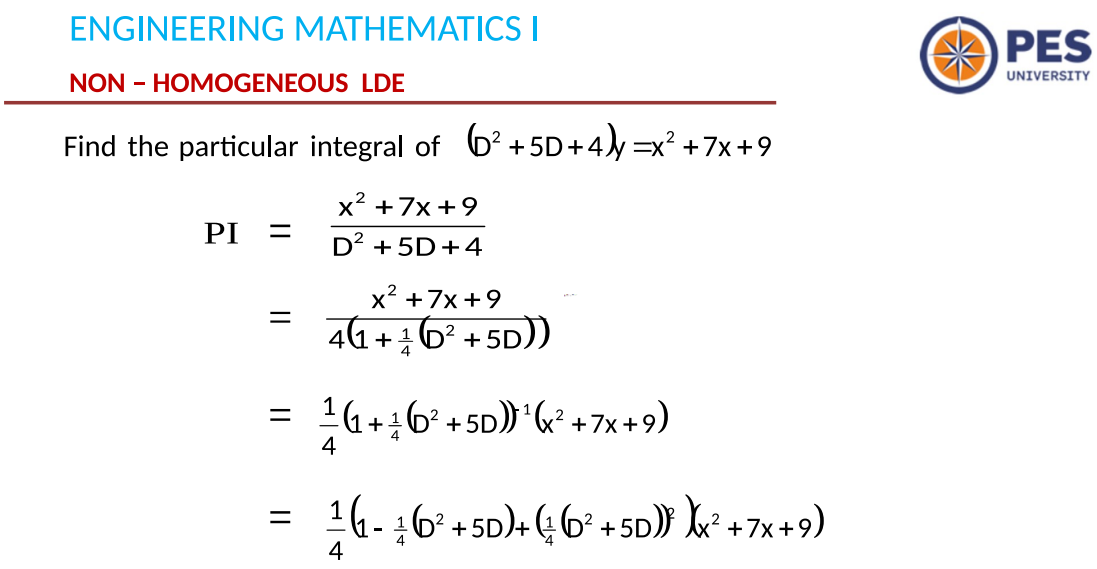

Figure G000077. Particular integral of Type 3 · source slide page 8.

EV0000495

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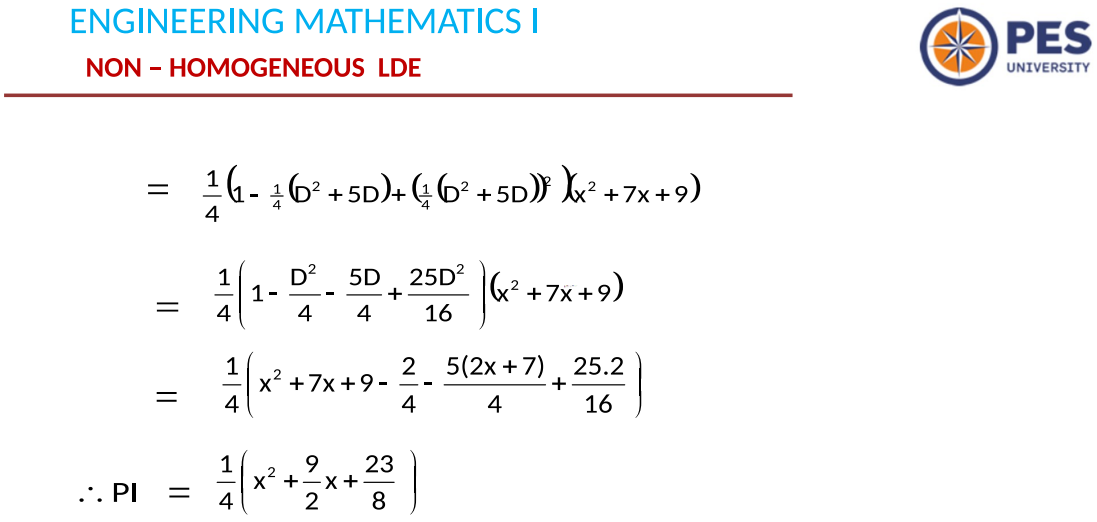

Figure G000078. Particular integral of Type 3 · source slide page 9.

EV0000498

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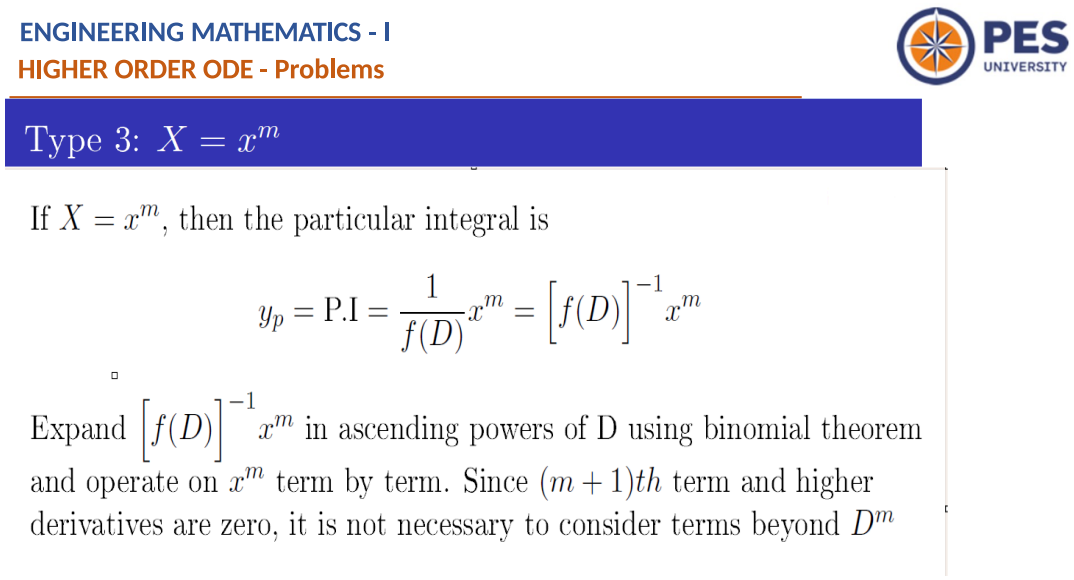

Figure G000079. Particular integral of Type 3 · source slide page 10.

EV0000501

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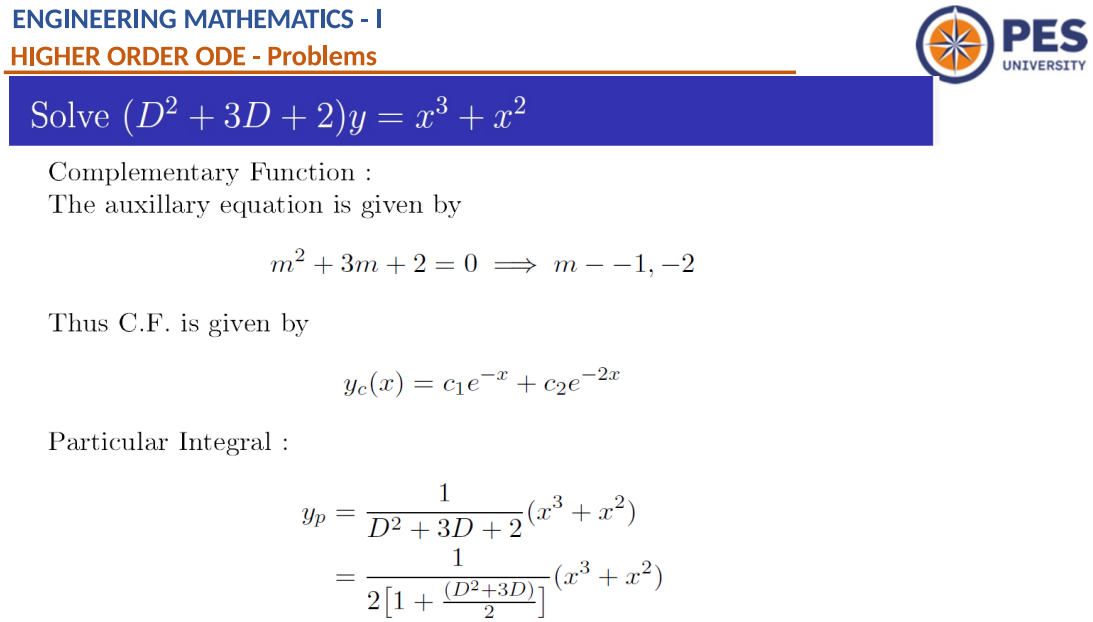

Figure G000080. Particular integral of Type 3 · source slide page 11.

EV0000504

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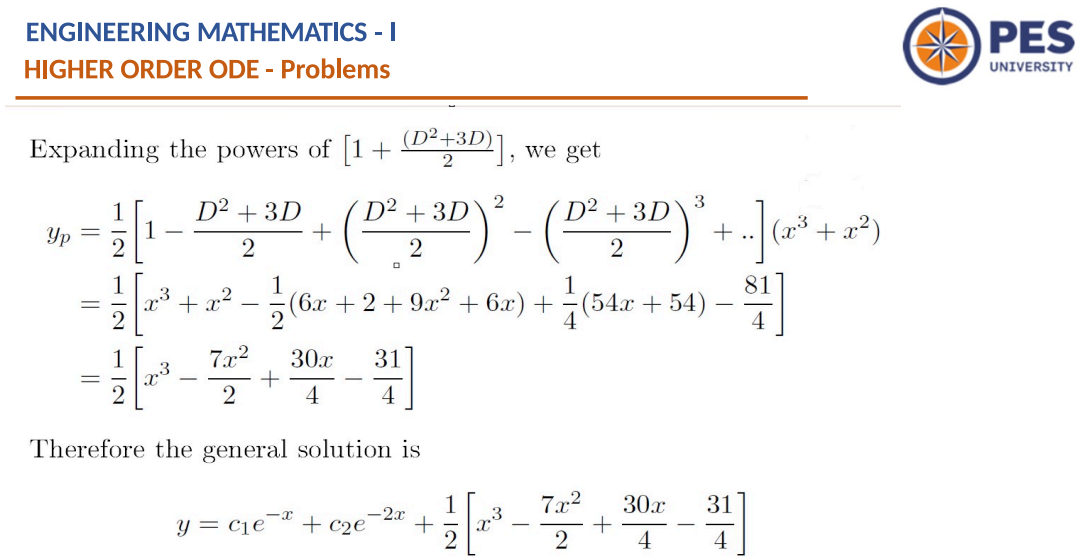

Figure G000081. Particular integral of Type 3 · source slide page 12.

EV0000507

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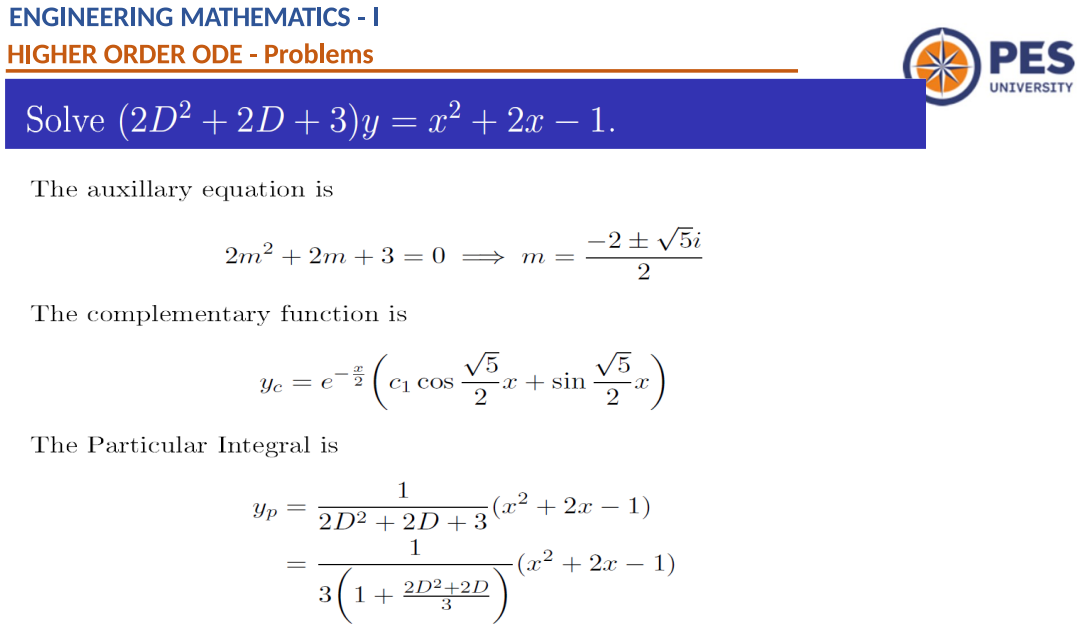

Figure G000082. Particular integral of Type 3 · source slide page 13.

EV0000510

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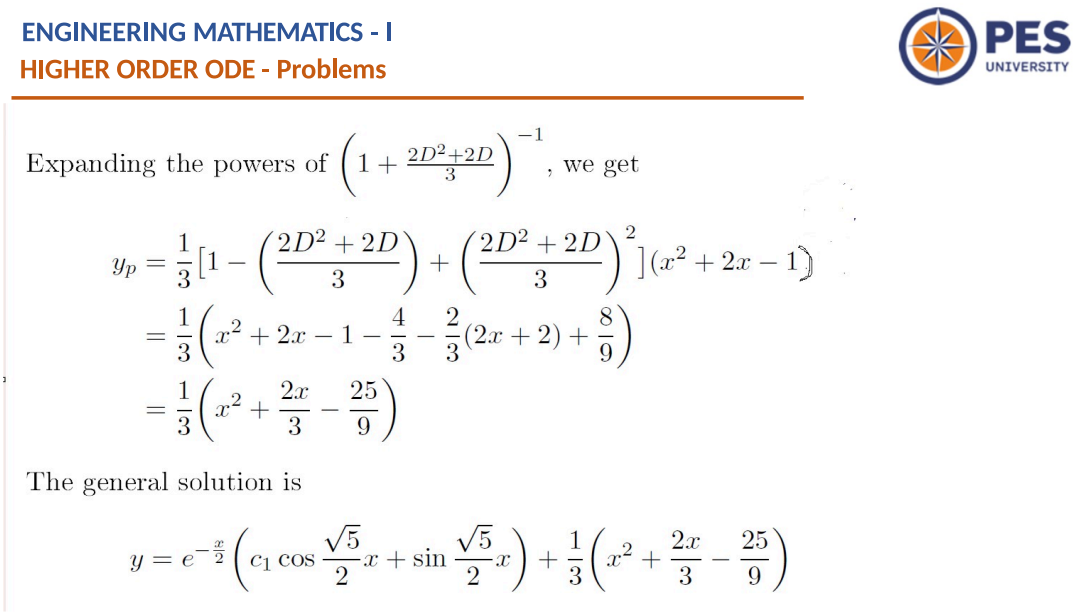

Figure G000083. Particular integral of Type 3 · source slide page 14.

EV0000513

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Figure G000084. Particular integral of Type 3 · source slide page 15.

EV0000516

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000473 · EV0000474 · EV0000475
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 2: HIGHER ORDER DIFFERENTIAL EQUATIONS SESSION: 5 SUB-TOPIC: NON- HOMOGENOUS LDE OF TYPE 3 Department of Science and Humanities EV0000476 · EV0000477 · EV0000478
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE CLASS CONTENT: EV0000479 · EV0000480 · EV0000481
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE RULES TO FIND PARTICULAR INTEGRAL EV0000482 · EV0000483 · EV0000484
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000485 · EV0000486 · EV0000487
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000488 · EV0000489 · EV0000490
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000491 · EV0000492 · EV0000493
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000494 · EV0000495 · EV0000496
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I NON – HOMOGENEOUS LDE EV0000497 · EV0000498 · EV0000499
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] If X = cm, then the particular integral is f(D) yp = P.I = cn f(D) - Expand f(D) in ascending powers of D using g binomial theorem and operate on x"m term by term. Since (m + 1)th term and higher derivatives are zero, it is not necessary to consider terms beyond Dm EV0000500 · EV0000501 · EV0000502
11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Complementary Function : The auxillary equation is given by m² + 3m + 2 = 0 =→ m - -1,-2 Thus C.F. is given by ye(x) = C1e-α + c2e-2c Particular Integral : yp = D2 + 3D + 2 (x²+ x²) 1 (x3 + x2) (D2+3D) 21 + 2 EV0000503 · EV0000504 · EV0000505
12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] ,we get 2 D2 +3D D2 + 3D D²+3D 3 3 +c² 2 2 2 1 81 (6x + 2+ 9x²+ 6x) +÷(54π+54) 2 7x2 30c 31 2 4 4 Therefore the general solution is 7x2 30c 31 y = C1e- + C2e-2x 2 4 4 EV0000506 · EV0000507 · EV0000508
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] The auxillary equation is -2±√5i 2m²+2m+3=0 => m = 2 The complementary function is V5 V5 C1 COS - + sin C 2 2 The Particular Integral is (x² + 2x - 1) yp 2D2 +2D+3 1 (x²+ 2 - 1) 2D2+2D 3 1+ 3 EV0000509 · EV0000510 · EV0000511

14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] 2D2+2D Expanding the powers of 1+ we get 3 2D2+ 2D 2D2 +2D 2](x² + 2c - 1 9p 3 3 3 4 2 8 +2c -1 (2c +2) + 3 3 3 9 1 2c 25 C 3. 3 9 The general solution is V5 2c 25 y=e- C1 COS + sin C 2 2 3 9 EV0000512 · EV0000513 · EV0000514

15. Source slide page 15 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000515 · EV0000516 · EV0000517

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.420] Hello, everyone, welcome back to the session on higher order linear differential equations. In today's session, we are learning a method or a rule to find the particular integral of extra the parian and being a positive integer. In today's session, [00:00:18.192] We are solving a non-homogeneous differential equation of the type f of d of y is equal to capital X, where capital X is equal to x to the power n and is a positive integer. [00:00:31.232] Now, how do we find the particular integral of x to the power n? In the previous sessions, we learned methods or rules to find the particular integral of exponential function e to the power ax or trigonometric functions of the type sine ax plus b. [00:00:50.960] R cos c plus b. Now how do we find the particular integral of x to the power n? First particular integral is given by 1 over f of d of x to the power n. So the step the first step being in f of d [00:01:08.132] Take out the lowest d degree common to get the expression in the form 1 plus some function of d which I call it as phi of d or 1 minus phi of d. [00:01:21.092] So, if you take out the lowest D degree term from f of d, you get an expression of the form 1 plus or minus 5 of d in the denominator. Now take this 1 plus or minus 5 of d to the numerator to get 1 plus or minus 5 of d to the power minus 1. [00:01:39.876] Now, expand this 1 plus or minus phi of d to the power minus 1 in ascending power set using binomial theorem. [00:01:51.236] And you are when you are expanding you are writing the terms up to d to the par n because the d to the power n plus 1 of x to the power n, that is n plus 1 third derivative of x to the po n is anyway 0, n plus 2 third [00:02:09.008] Derivative of x to the power n is 0. So when you are expanding 1 plus phi of d to the power minus 1. [00:02:17.552] Or 1 minus 5 of d to the power minus 1, you write the terms up till d to the power n, neglecting the higher powers of d. Then operate on the numerator term by term by taking capital D is equal to the n [00:02:35.184] to d by dx, d square as d2 by dx to d three d cube as d cube by dx cube and so on. Now [00:02:45.508] The following expressions will be helpful to expand 1 plus phi of d to the power minus 1 or 1 minus phi of d to the power minus 1 in ascending powers of d. What is the first equation? 1 plus x whole to the power minus 1 is 1 minus. [00:03:03.280] Minus x, plus x square, minus x cube, and so on. Here in the series, plus and minus sign alternates. [00:03:12.352] Similarly, what is 1 minus x whole to the power minus 1? We need not have to remember that. So in this series itself, replace plus by minus. So what do you get? 1 minus x whole to the power minus 1 is 1. Minus minus x, which is plus x. [00:03:31.456] Plus minus x the whole square, which is plus x square. Minus into minus x cubed, which is plus x cube, and so on. So in the second series only positive signs. Similarly, the next expression may also be used. [00:03:46.672] Uh to expand this 1 plus 5 d to the power minus 1 or 1 minus 5 d to the power minus 1. 1 plus x whole to the power minus 2, which is 1 minus 2x plus 3x square minus 4x cube. See look here in this series also plus and minus sign alternates in the C. [00:04:04.336] Then 1 minus x whole to the power minus 2. We need not have to remember this. So in this you just

replace x by minus x to get this. So if we remember the first and the third, it is sufficient. So the second and the fourth can be derived from the first and the third. Now [00:04:22.564] Let us consider a question. Find the general solution of d2y by dx2 minus y is equal to 5x minus 2. Look at the right hand side function is an algebraic function, 5x minus 2. [00:04:35.120] So to find the particular integral of 5x minus 2, I am using root 3, which I discussed now. So now because the equation is a non-homogeneous linear differential equation, so the general solution has two components: complementary function and particular integral. So the first [00:04:52.880] step being the to find the complementary function. For that, first let me first write this in operated notation form capital D square minus 1 of y is equal to 5x minus 2. Now to find complementary function I need to set up the corresponding auxiliary equation.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:10.448] So, what is the auxiliary equation? M square minus 1 is equal to 0 because the degree of this equation is 2. 2 roots, the two roots are plus or minus 1. So the two roots of the equation are plus 1 and minus 1. So what is a complementary function when the roots are real and [00:05:28.016] thing so it is c1 e to the power 1x plus c2 e to the power minus 1 e to x. So this is the complementary function of the given non-homogeneous differential equation. Now to find the particular integral. So the particular integral is given by 5x minus 2 [00:05:45.584] by d square minus 1. Now what does the rule say? What is the first step? Take out the lowest d degree term to get the expression in the form 1 plus phi d or 1 minus 5 d. So the lowest d degree term here is minus 1. [00:06:01.936] Why? Because minus 1 can be written as minus 1 into d to the power 0. So take out minus 1 and get d square minus 1 in the form 1 plus 5 d or 1 minus 5 d. So this is equal to [00:06:17.380] Minus of 1 minus d square to the power 1 minus d square. Now take that 1 minus d square to the numerator to get minus of 1 minus d square to the power minus 1 of 5x minus 2. [00:06:32.772] That is the first step. I wrote it in the form 1 minus 5 of d to the power minus 1 of the function 5x minus 2. Now what is the next step? Using binomial theorem, expand this in ascending powers of D. [00:06:48.484] So which expression I'm using? 1 minus x whole power minus 1, which is 1 plus x plus x square plus x cube. Use that formula and expand it in ascending powers of d. I get it as minus of 1 minus d square plus d to the power 4. [00:07:07.556] Now, so it is minus of 1. See this is 1 minus x whole to the power minus 1, which is 1 plus x plus x square plus 1. Before it was 1 minus x, so that is wrong. [00:07:19.952] 1 plus x square plus x to the power 4 plus on because it is 1 minus x whole raised to minus 1. Now [00:07:27.872] When I simplify this, see 1 into 5x minus 2 is 5x minus 2, and d square of 5x minus 2. What is d square of 5x minus 2? 2 times differentiation of 5x minus 2 with respect to x. [00:07:42.724] First time differentiation of this 5x minus 2 itself is a constant. First order derivative of 5x minus 2 is 5. [00:07:49.988] If I differentiate the constant again, I get 0. So d square, d to the power 4, d to the power 6 of 5x minus 2 is 0. So this simplifies to minus into 5x minus 2 because these when I operate these uh d terms. [00:08:07.760] terms on 5x minus 2 I am getting 0. What is particular integral? It is 2 minus 5x. Therefore, the general solution is y is equal to the complementary function which is denoted as yc plus the particular integral y p. [00:08:25.092] Now let us consider another question. Find the particular integral of d square plus phi d plus 4 of y is equal to x square plus 7x plus 9. See in this question I need not have to find the complementary function because the question is only to find the particular integrals. Yeah. [00:08:42.880] Considered here an algebraic function of degree 2. Now how do I find the particular integral of x square plus 7x plus 9? Particular integral is given by x square plus 7x plus 9 divided by d square

plus by d plus 4. Now how do I simplify this? [00:09:02.404] Now, so how do I simplify this? What is the first step? In f of d you take out the lowest d degree term as common factor and get this expression in the form 1 plus 5d or 1 minus 5d. [00:09:16.976] So what is the lowest d degree term over here? 4. You if you take 4 outside, it becomes 4 into 1 plus d square plus 5 d by 4. [00:09:27.364] See this is what we get. Now take this 1 plus 5 d to the numerator to get 1 plus phi d to the power minus 1. So it is 1 by 4 into 1 plus d square plus phi d by 4 whole raise to minus 1. [00:09:43.376] Of x square plus 7x plus 9. Now I need to expand this in ascending power sub b using 1 plus x whole power minus 1 formula, which is 1 minus x plus x square minus x cube and so on. But

EV0000102 · EV0000103 · EV0000104 · EV0000105

Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.060] If you look at the expression over here, algebraic expression, it is of degree 2. So d square of x square plus 7x plus 9 is a constant, and higher derivatives of x square plus 7x plus 9 is anyway 0. Now I am writing it as 1. [00:10:18.032] minus this term that is d square plus phi d by four plus [00:10:22.576] X square that is d square plus y d by 4 the whole square. I am neglecting the higher power sub d because d3, d4, dn of x square plus 7x plus 9 is anyway equal to 0. Now let us simplify this further. So what I to what I have to do? [00:10:42.916] So now I have to expand this. So here see I have to simplify this d square plus phi d. [00:10:49.664] The whole square. First use a plus b the whole square and you simplify. So you get what? 1 minus d square by 4 minus 5 b by 4 plus in this a plus b the whole square. A square plus 2ab plus b square. So when I [00:11:07.184] When I differentiate, I get d square of x square plus 7x plus 9 as constant. Higher powers of d is anyway, when it is operated on x square plus 7x plus 9, I get 0. I have written only d square by 4 the. [00:11:24.164] Sorry, 5 d by 4 the whole square, which is 25 d square by 60. I've neglected d square the whole square, that is d to the power 4 into 2 into 5 d into d, that is [00:11:35.664] 10 d cube that I have neglected and I have considered only 5 d by 4 the whole square which is 25 by 5 25 d square by 16. See d of x square plus 7x plus 9 that is 1 time differentiation 2x plus 7. 2 times differentiation is [00:11:51.728] 2. Higher derivatives will be anyway equal to 0. Now operate on this term by term. So what do I get? 1 by 4 into 1 into x square plus 7x plus 9 minus 1 by 4 into what is d square of this? It is equal to 2. [00:12:07.972] Minus 5 by 4 into d of x square plus 7x plus 9 is 2x plus 7. Plus 20 by 16 into d square is 2. So it simplifies to 1 by 4 into x square plus 9 by 2x plus 23 by 8. [00:12:26.244] This is how we find the particular integraph of integral of algebraic function of the type x to the power n. Thank you.

EV0000106 · EV0000107 · EV0000108 · EV0000109

Full transcript: 2081 normalized words, 50 timestamped segments, 756.864 seconds duration. Low-confidence segments flagged: 0.

EV0000098 · EV0000099 · EV0000100 · EV0000101

8 Particular integral -Type 3 and Type 4

This lesson is indexed from the source video titled Particular integral -Type 3 and Type 4. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text

is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000110 · EV0000111

Graphs and diagrams

Figure G000085. Particular integral of Type 4 · source slide page 1.

EV0000519

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

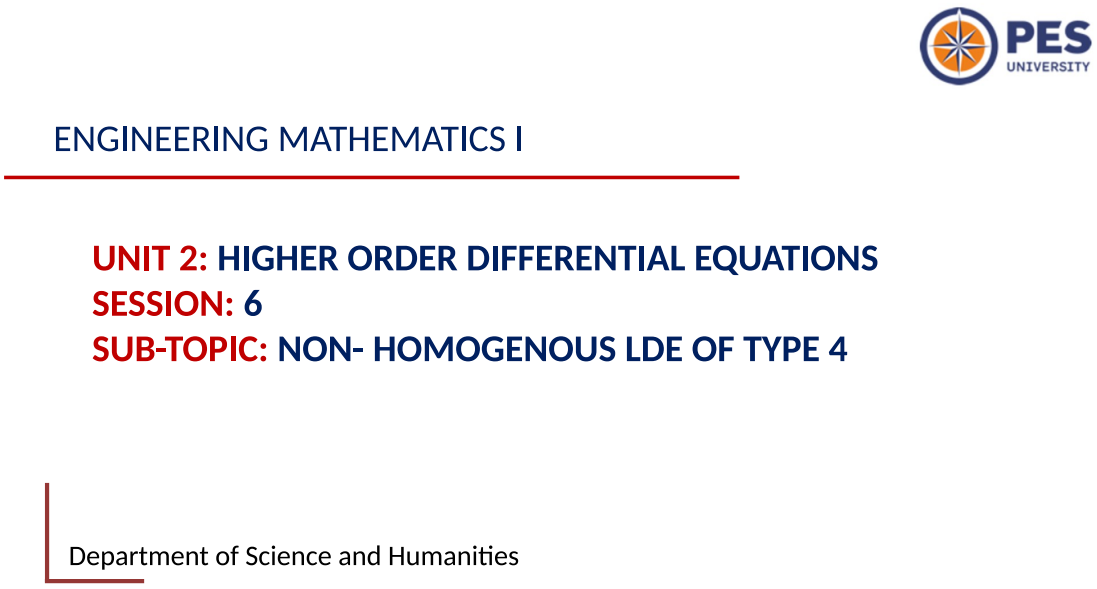

Figure G000086. Particular integral of Type 4 · source slide page 2.

EV0000522

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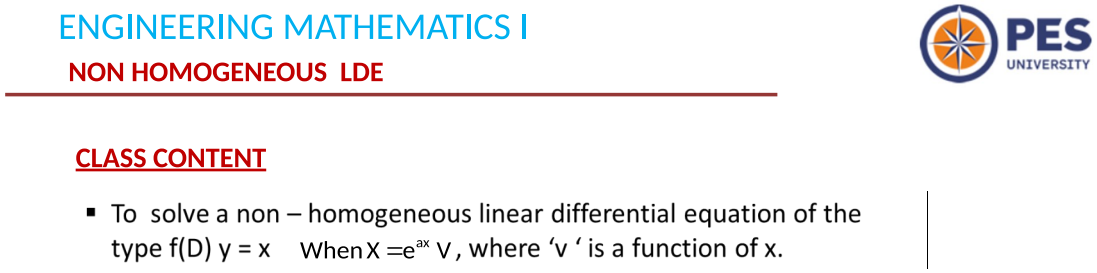

Figure G000087. Particular integral of Type 4 · source slide page 3.

EV0000525

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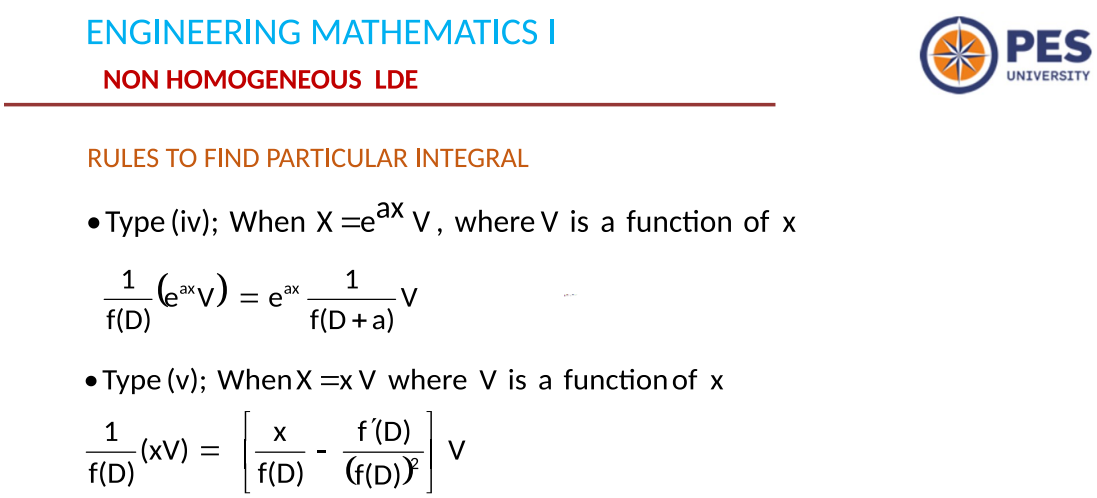

Figure G000088. Particular integral of Type 4 · source slide page 4.

EV0000528

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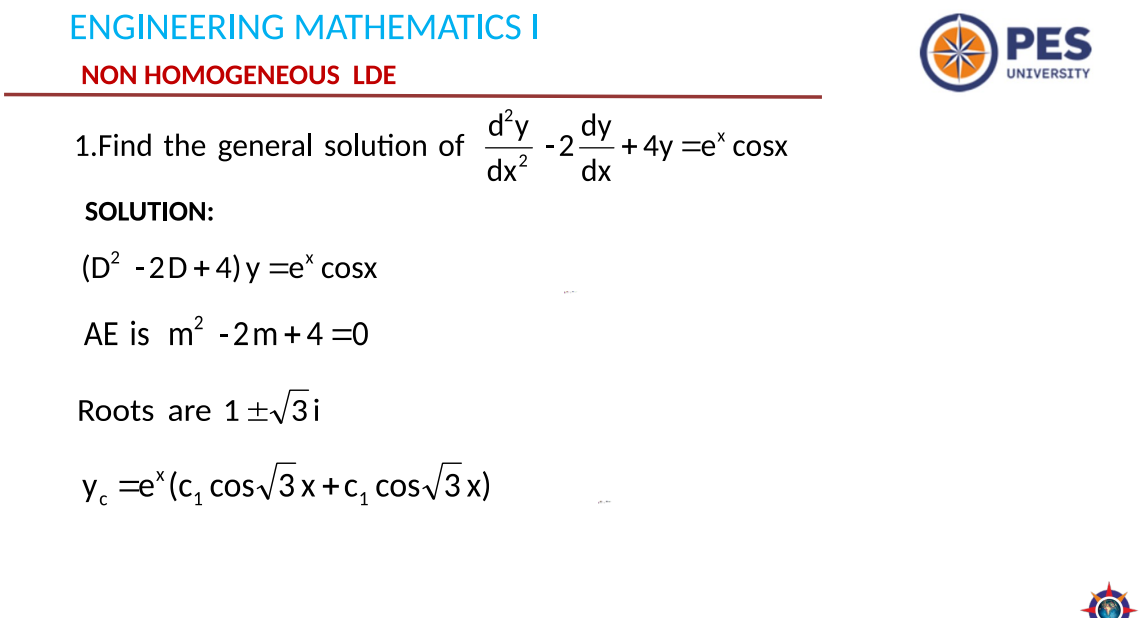

Figure G000089. Particular integral of Type 4 · source slide page 5.

EV0000531

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ENGINEERING MATHEMATICS I

NON HOMOGENEOUS LDE

$$y_p = \frac{e^x \cos x}{D^2 - 2D + 4}$$

$$= e^x \frac{\cos x}{(D+1)^2 - 2(D+1) + 4}$$

$$= e^x . \frac{\cos x}{D^2 + 3} = e^x . \frac{\cos x}{-1^2 + 3}$$

$$\therefore \; y_p = \frac{e^x . \cos x}{2}$$

Thus, $y = y_c + y_p$

Figure G000090. Particular integral of Type 4 · source slide page 6.

EV0000534

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ENGINEERING MATHEMATICS I

NON HOMOGENEOUS LDE

2. Find the general solution of $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = x e^x \cos x$

SOLUTION:

$$(D^2 - 2D + 1)y = x e^x \cos x$$

To find CF

AE is $m^2 - 2m + 1 = 0$

Roots are $m = 1, \; 1$

$$\therefore \; y_c = (c_1 + c_2 x) e^x$$

Figure G000091. Particular integral of Type 4 · source slide page 7.

EV0000537

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

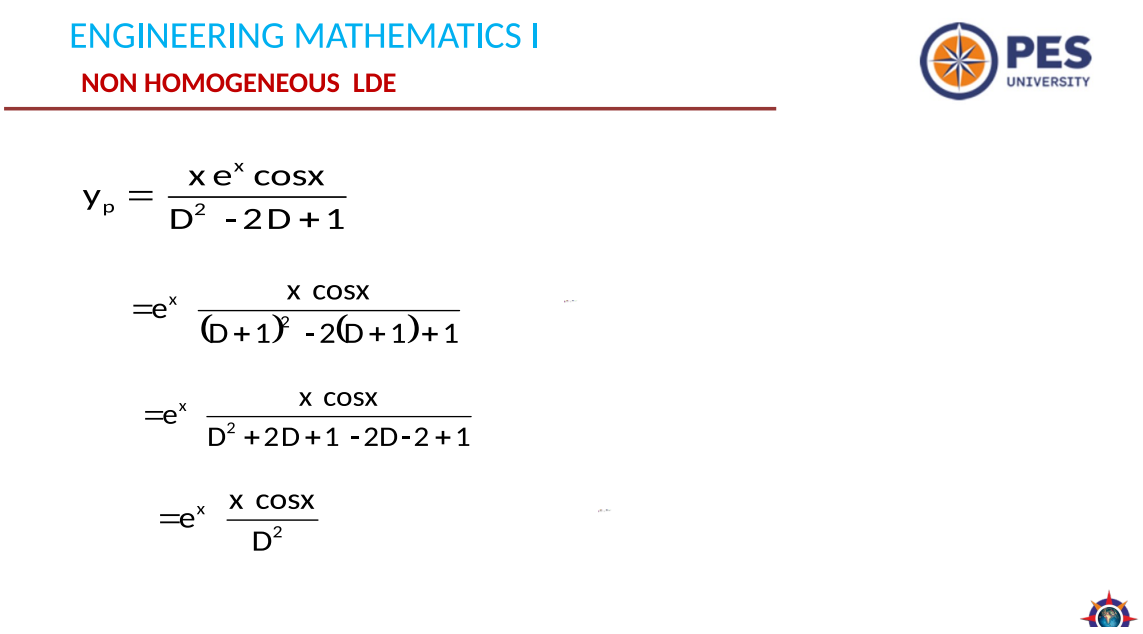

Figure G000092. Particular integral of Type 4 · source slide page 8.

EV0000540

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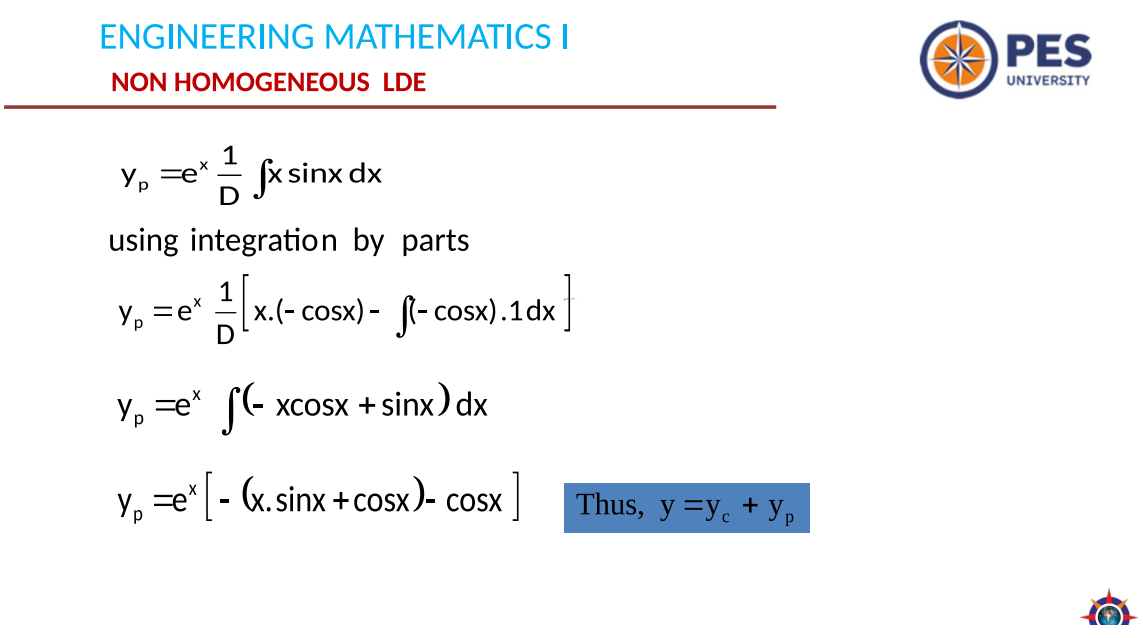

Figure G000093. Particular integral of Type 4 · source slide page 9.

EV0000543

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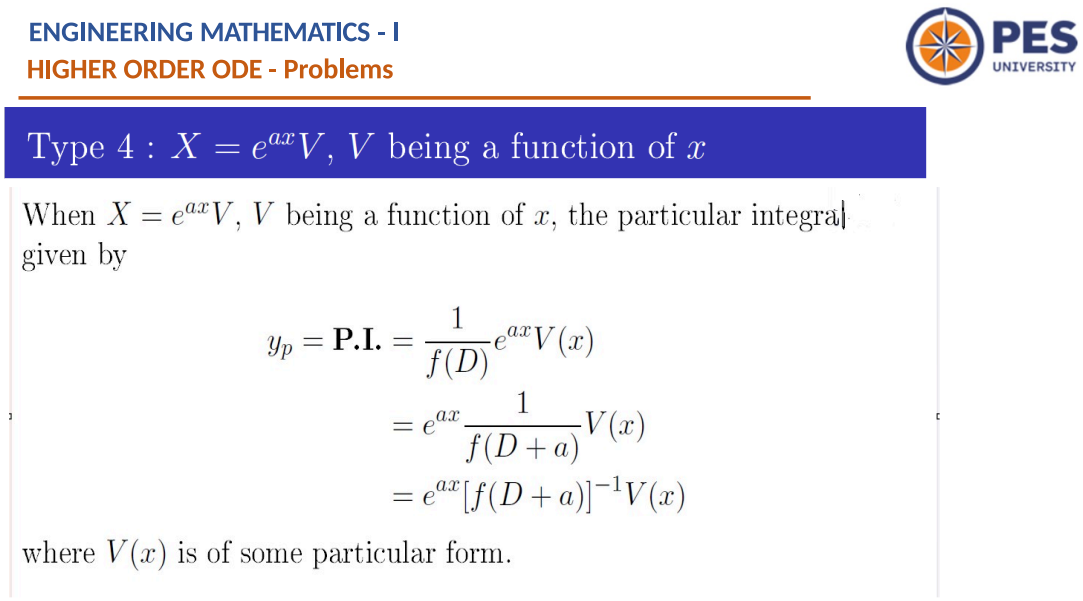

Figure G000094. Particular integral of Type 4 · source slide page 10.

EV0000546

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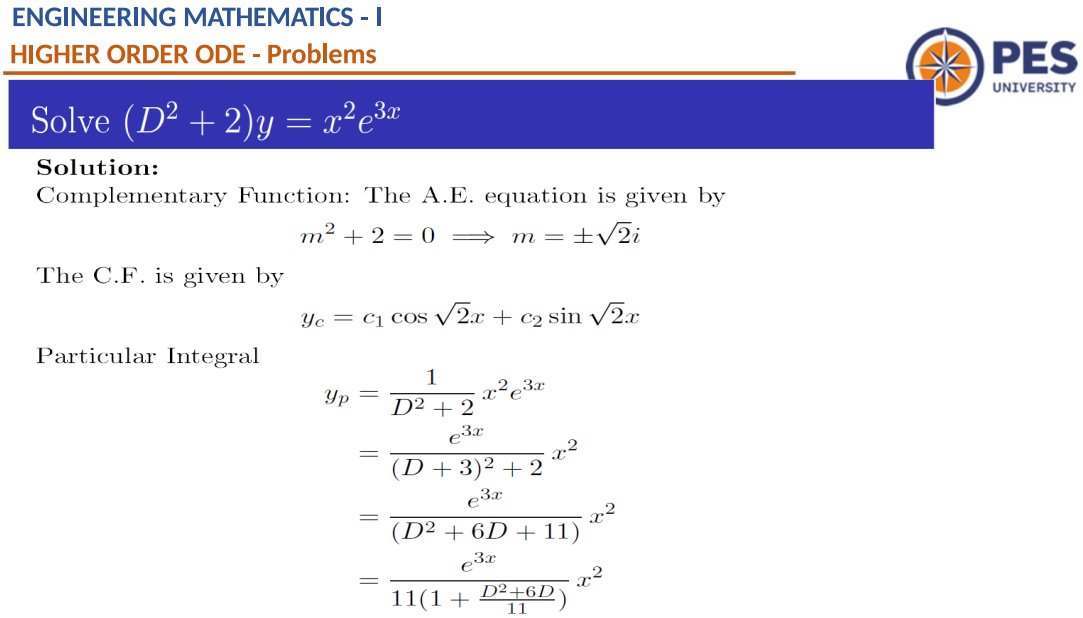

Figure G000095. Particular integral of Type 4 · source slide page 11.

EV0000549

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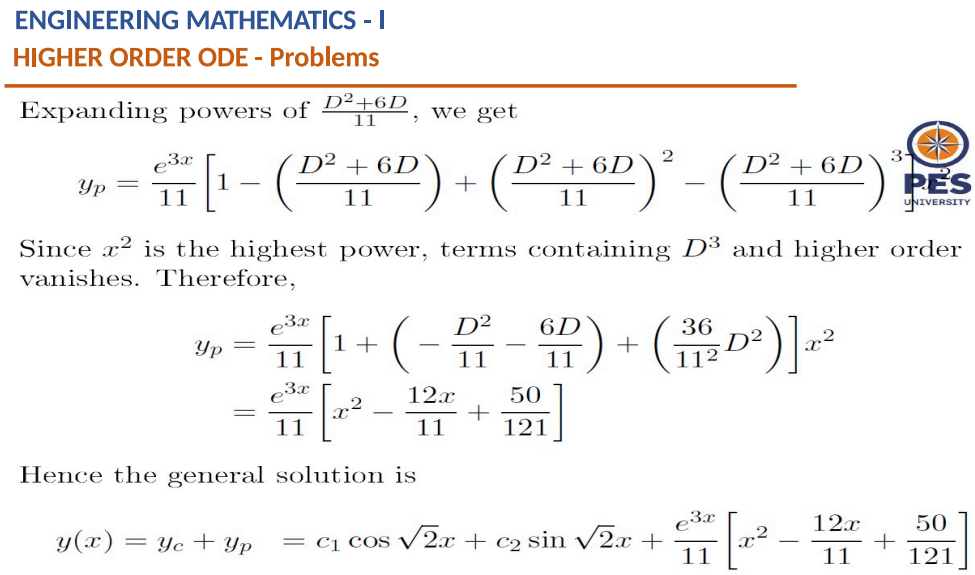

Figure G000096. Particular integral of Type 4 · source slide page 12.

EV0000552

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $(D^2 - 2D + 1)y = e^x sinx$

Solution:

Complementary Function:
The Auxillary equation is

$$m^2 - 2m + 1 = 0 \implies m = 1, 1$$

The C.F. is

$$y_c(x) = (c_1 + c_2 x)e^x$$

Figure G000097. Particular integral of Type 4 · source slide page 13.

EV0000555

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ENGINEERING MATHEMATICS - I
HIGHER ORDER ODE - Problems

PES UNIVERSITY

The particular Integral is

$$\begin{aligned} y_p(x) &= \frac{1}{D^2 - 2D + 1} e^x \sin x \\ &= \frac{1}{(D-1)^2} e^x \sin x \\ &= e^x \frac{1}{(D+1-1)^2} \sin x \\ &= e^x \frac{1}{D^2} sinx \\ &= -e^x \sin x \end{aligned}$$

The general solution is

$$y(x) = (c_1 + c_2 x)e^x - e^x \sin x$$

Figure G000098. Particular integral of Type 4 · source slide page 14.

EV0000558

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Thank you

Department of Science and Humanities

Figure G000099. Particular integral of Type 4 · source slide page 15.

EV0000561

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000518 · EV0000519 · EV0000520
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 2: HIGHER ORDER DIFFERENTIAL EQUATIONS SESSION: 6 SUB-TOPIC: NON- HOMOGENOUS LDE OF TYPE 4 Department of Science and Humanities EV0000521 · EV0000522 · EV0000523
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I NON HOMOGENEOUS LDE CLASS CONTENT EV0000524 · EV0000525 · EV0000526
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I NON HOMOGENEOUS LDE RULES TO FIND PARTICULAR INTEGRAL EV0000527 · EV0000528 · EV0000529
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I NON HOMOGENEOUS LDE SOLUTION: EV0000530 · EV0000531 · EV0000532
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I NON HOMOGENEOUS LDE EV0000533 · EV0000534 · EV0000535
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I NON HOMOGENEOUS LDE SOLUTION: EV0000536 · EV0000537 · EV0000538
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I NON HOMOGENEOUS LDE EV0000539 · EV0000540 · EV0000541
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I NON HOMOGENEOUS LDE EV0000542 · EV0000543 · EV0000544
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Type 4 : X = earV, V being a function of x [Image text, OCR] When X = eaV, V being a function of , the particular integra] given by eaxV(x) Up = P.I. = f(D) 1 ar V(c) f(D + α) = eax[f(D +a)]-1V(x) where V(c) is of some particular form. EV0000545 · EV0000546 · EV0000547
11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Solution: Complementary Function: The A.E. equation is given by m^2 + 2 = 0 =→ m =±√2i The C.F. is given by yc = C1 cos √2x + C2 sin √2c Particular Integral 1 yp = - 3 D2+2 C e3c (D + 3)2 + 2 e3x 2 C (D2 + 6D + 11) 11(1 + D2+6D 11 EV0000548 · EV0000549 · EV0000550
12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Expanding powers of D2+6D, we get 11 3c D2 +6D D2 + 6D D2 +6D 3 PES yp : 11 11 11 UNIVERSITY Since c2 is the highest power, terms containing D3 and higher order vanishes. Therefore, D2 6D 36 D2 yp C 11 112 3c 12c 50 C 11 121 Hence the general solution is 12c 50 y(c) =yc +yp = C1 cos √2x + C2 sin √2 + 11 121 EV0000551 · EV0000552 · EV0000553
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Solution: Complementary Function: The Auxillary equation is m = 1, 1 The C.F. is yc(c) =(c1 + C2x)e" EV0000554 · EV0000555 · EV0000556

14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] The particular Integral is yp(x) = e" sin c D2 - 2D + 1 1 sin D: - 1)2 1 sin (D + 1 - 1)2 1 sinc D2 = -e sin c The general solution is y(x) = (c1 + c2x)e" - e" sinc EV0000557 · EV0000558 · EV0000559

15. Source slide page 15 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000560 · EV0000561 · EV0000562

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.716] Hello everyone. Welcome back to the course on engineering mathematics one. In this session, we will solve problems on finding the particular integral when x is a polynomial, that is x is x power m and [00:00:25.060] When x is epoch a x into v, where v is a function of x. [00:00:36.548] Let us start with case number three when x is x power m. As usual, in this case the particular integral is expressed in the form one over f of d times x power m. Then [00:00:55.812] On all f of b is written as f of b inverse times x power m. Once we do this, we expand this in ascending power sub b using binomial series. [00:01:15.044] Up to the term containing d power m it means the terms beyond d power m are neglected because [00:01:29.732] P power n of x power m is zero for n is greater than m. So what's the meaning of this? Let us take an example. Let us take [00:01:49.220] x square plus seven x plus eight. It means n is three and m is two. And you can clearly see n is better than m and it's going to be zero. Oh [00:02:06.852] When uh n is one, when you differentiate first time, x square become two x, seven x becomes seven. When n is two, two x becomes two, seven becomes zero, when n is three, [00:02:23.140] It becomes zero. It means the nth derivative of x power m is zero when n is m. So because of this condition, it comes beyond b power m or neglected. [00:02:37.764] So having understood this idea, let us solve a problem. Solve the differential equation d square plus 3d plus 2 of y equals x power 3 plus x square. [00:02:56.356] You can observe carefully the value of M is three here. [00:03:03.460] So let's find out the complementary function. The axillary equation is m square plus 3m plus 2 0. The roots are minus 1 and minus 2. So the complementary function will be c1 times e power minus x. [00:03:21.636] plus c 2 times e power minus 2x. Let's move on to the particular integral. It is 1 over the d square plus 3d plus 2 of x square plus x square. [00:03:40.068] You know from theory part, when we apply case number three, it is convenient to express f of d either in the form of 1 plus d or 1 minus d, so that we can apply binomial series. [00:04:00.228] So now our task is to express this polynomial in the form of 1 plus d. So we take 2 outside. It is written as 2 of 1 plus d square plus 3d by 2. Now currently it looks like something like 1 by x or 1 minus x. [00:04:19.076] So typically 1 plus x for this case. Let us bring this to numerator. It becomes 1 plus 3d, 1 plus d square plus 3d by 2, something like minus 1. [00:04:36.964] Apply binomial series which is given by so what is one plus x minus one? You know from logic whenever we have one plus it is alternative sign on the right hand side one minus x plus x square minus x power three and so on. [00:04:56.548] And remember x is nothing but d square plus three d by two. So this is x for the problem. So when we use binomial series, we get this particular line that is half. EV0000110 · EV0000111 · EV0000112 · EV0000113

Transcript block 2 (source interval 000300-000599 seconds): [00:05:14.628] One minus [00:05:17.988] This is x. This is x square. This is x power three and so on. Now what is the meaning of [00:05:30.788] Expanding this using binomial series up to the term containing d power m. Logically

speaking, from x power 3 plus x square. [00:05:45.380] The highest power of x is 3. It means when we use binomial series we should go only up to d power 3. The term beyond d power 3 are neglected, it means d power 4, d power 5, so on they're all neglected because anyway they become zero. So having understood this uh simple technique. [00:06:05.572] So when we expand the third term and the fourth term using A plus B whole square and A my A plus B whole Q, so what terms to be considered? Obviously this is considered, D square is considered, 3D is considered because they are less than 3. [00:06:22.852] When you look at d square plus 3d whole square. We don't want a square. We want b square. We want 9 d square. And we don't want uh yeah we want 2ab that is uh 6d q. What about the fourth term? [00:06:43.588] We don't want a power 3 but we want bq because it's 27 d power 3. So we don't want 3a square b and 3ab square is also not needed. So once you collect all those d's then you should take out this polynomial inside and start applying the differential operator. [00:07:02.852] If you have d of that, differentiate x power three plus x square once with respect to x. If it is d square, differentiate two time. If it is d power three, differentiate three times. So when you do this systematically. [00:07:23.204] The final PI is going to be half times x power 3 minus 7x square by 2 plus 30x by 4 minus 31 by 4. So this is the logic behind solving PI when we have a polynomial case. [00:07:45.124] Let us look at another example. x square plus 2x minus 1. You know very well how to compute the complementary function. The roots are minus 1 plus r minus root phi by 2 into x. [00:08:03.012] So complex rules, so this is the complementary function for this problem. Then what about particular integral? So this is going to be our PI 1 by f of b times the polynomial. Again the task is to [00:08:22.212] reduce this polynomial to one plus x mar let us take three common you get one plus two d square plus two d by three and this part looks like even x. So let us use binomial series [00:08:43.204] we get this this and this now why are we stopping after the second term because highest power of x is two we go only up to t square the terms beyond t square are neglected on it because of zero so [00:09:04.164] based on this logic so one into x square plus two x minus one as usual we want two d square we want two d and what about a plus b whole square form of this we don't want a square but we want b square and we don't want to ab [00:09:22.788] So once you collect all those cities, take this polynomial inside, then differentiate it with respect to x once, twice, and so on. So when you do that, the final answer comes out to be 1 by 3 of x square plus 2x by 3. [00:09:41.508] Minus 25 by 9. So these are the two problems on finding the particular integral when the right hand side is a polynomial. I hope this concept is clear to you. [00:09:55.972] Let us move on to case four. Where b is a function of x. Remember, this is a combination of case four plus case three and case two.

EV0000114 · EV0000115 · EV0000116 · EV0000117

Transcript block 3 (source interval 000600-000899 seconds): [00:10:15.428] Depends on B. [00:10:19.780] So in this particular case, the PA is written as f of b type e bore x into b. This is expressed in the form e po x times we choose a from here, replace a d by d plus c in f of b that is here. [00:10:40.452] then retain v of x as it is. So as you can see from here to here this is case four. From here to here depends on case four it could be case two and it could be case three. [00:10:53.956] If v is sign or cos, we go with case 2. If v of x is a polynomial, we go with case number 3. [00:11:07.076] Okay. Let us find out the solution to this differential equation. D square plus two of y equals x square times a power three x. So you know very well how to find out. [00:11:23.492] Complementary function the roots are plus or minus root two i because alpha zero beta is root two the complementary function becomes C1 cos root two x plus c two sine root two x. So what about bi? [00:11:41.220] You need to uh go by this logic. [00:11:45.828] You have x square

into e power 3x. If x square is not there, case 1. If e power 3x is not there, case 3. [00:11:57.892] So now you should go by case number four because both are available. So e power a x is taken as e power x, v is taken as x square. So this is p i 1 by d square plus 2 x square. [00:12:13.348] times e power 3x. This is written as a power 3x times 1 divided by. Replace d by d plus a. And you can see a values 3. So you must replace b by d plus 3 square plus 2. [00:12:33.412] and rate it be as it is and observe carefully because it's x square you go by case number three so when you simplify you get case [00:12:47.940] plus 6 d is 11x square. Now from here onwards we need to go by case number three. It's a polynomial case. We want 1 plus x form. So we're taking a constant outside to make 1 plus x. Then it is written as a power 3x by 11 [00:13:06.020] Then apply binomial series we get 1 minus x plus x square minus x power 3. Now where do we stop? Because the highest power of x is 2, we need to go up to d square. So after writing binomial series, this has to be considered. [00:13:25.956] This is considered. This is considered. A square is not needed. B square is needed. 2AB is not needed. And in this case, nothing is needed because it is more than 2. So we get this particular step. Then take x square inside. Then start up. [00:13:44.772] Like differential operator, we get this. So finally the answer to this problem is given by this complementary function and this particular integral. [00:14:02.404] Let us move on to another problem where x is e power x times sine x. As usual, the complementary function is going to be c1 plus c2x times e power x because the roots are repeated one and one. And this is a particular integrity. [00:14:22.756] So after this it is something like e power x one divided by d square becomes anyway. This denominator equals d minus the whole square so problem becomes simple. [00:14:42.276] So when I replace d by d plus 1, this is going to be 1 by d square of sine x. So it means we need to apply inverse differential operator integrate sine x to times. So when we do that

EV0000118 · EV0000119 · EV0000120 · EV0000121

Transcript block 4 (source interval 000900-001199 seconds): [00:15:02.372] Answer comes out to be minus a power x times sine x. So this is PI for this problem and this is the general solution to this problem complementary function as well as Pi. So we have learned the problems of finding the particular integral. [00:15:18.596] When x is a polynomial, when x is epo x, epo a x into v, where v is a function of x. Thank you.

EV0000122 · EV0000123 · EV0000124 · EV0000125

Full transcript: 1901 normalized words, 55 timestamped segments, 928.064 seconds duration. Low-confidence segments flagged: 0.

EV0000110 · EV0000111 · EV0000112 · EV0000113

9 Particular integral of standard functions - Type 4

This lesson is indexed from the source video titled Particular integral of standard functions - Type 4. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000126 · EV0000127

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.452] Hello everyone, welcome back to the session on higher order linear differential equations. In this session, we are learning a method or a rule to find the particular integral of function of the type a par ax into v where v is any function of x, that is v. [00:00:18.224] We are learning a method to find the particular integral of product of two functions in which one function is exponential function e bar ax. Now, in the previous sessions, we learned methods to find the particular integral of e bar ex, that is rule 1. [00:00:37.444] Then sine a x plus b or cos e x plus b and algebraic function of the type x to the par n. In today's session we are finding the particular integral of e par a x into b. So it is the formula used. [00:00:52.976] 1 over f of d of e par e x into b is e par x into 1 over f of d plus a of b. [00:01:00.996] So what is the meaning? See e par ax can be taken outside the operator by replacing b by d plus a in f of d. So now you are applying this operator 1 over f of d plus a only on [00:01:18.768] B. What is A here? A is the coefficient of x in the exponential function. So this is the rule used to find the particular integral when. [00:01:28.432] The given function is of the type e par a x into b. Now I'll give you another type wherein capital X is equal to X into B, just the formula 1 over f of b of x into b is x by f of d minus f dash d by f of d the whole square of v. [00:01:47.876] This is the rule used to find the particular integral of product of two functions in which one function is x. [00:01:55.556] But in this session, we are solving problems only on type 4, wherein the function is e par ax into v. Let us consider a problem. Find the general solution of d2y by dx2 minus 2 dy by dx plus 4y is equal to e to the pi. [00:02:13.328] x into cos x because the given equation is a non-homogeneous LD with constant coefficient, I need to find the complementary function and particular integral. Now to find the complementary function let me write the given equation as [00:02:30.896] d square minus 2t plus 4 of y is equal to e to the power x cos 6. [00:02:36.772] Let us first set up the auxiliary equation, which is required to find the complementary function. The auxiliary equation is m square minus 2m plus 4 is equal to 0. Find out the roots of this quadratic equation. By using the quadratic formula, we get the roots as 1 plus R minus. [00:02:54.544] Minus root 3i alpha plus r minus i beta alpha is 1 beta is root 3. So the complementary function is e to the power x that is e to the power alpha x, c even [00:03:06.000] cos root three x plus c2 psi root three x. So this is the complementary function of the given differential equation. Now the second step being finding the particular interval which is given by e par x into cos x by t square minus two t plus four [00:03:23.632] See look at this. This is e par x into b where v is equal to cos x. Now what does that rule say? E power x can be taken outside the operator by replacement by d plus a, a is one. [00:03:41.200] d plus 1 in this e f of d. So that is equal to e par x into into write it to the left into cos 6 pi. Just replace all d'present in f of d by d plus 1. So d plus 1 the whole square minus 2 into d plus 1 plus 4 [00:03:58.768] This simplifies to e power x into cos x by d square plus 2d plus 1 minus 2d minus 2 plus 4. Now plus 2d gets canceled with minus 2d and it simplifies to e power x. [00:04:12.880] into cos x by d square plus 3. Now, how do we simplify this? Which rule are we using? [00:04:20.356] The second one, cos a x by f of d. So this f of d contains only d square terms. So replace d square by minus into 1 square. So it is equal to e power x into cos. [00:04:34.336] Cos x by minus 1 square that is minus 1 plus 3, which is e power x into cos x by 2. So this is how we find the particular integral. This should be by p. [00:04:47.296] Therefore, the general solution is y is equal to yc plus y p. This is y p. So this is how we find the [00:04:55.312] Particular integral when you have product of two functions in which one function is

exponential function. Now let us look at another question. Find the general solution of d2y by dx2 minus 2 dy by dx plus y is equal to x into e power x into cos x. Product of 3

EV0000126 · EV0000127 · EV0000128 · EV0000129

Transcript block 2 (source interval 000300-000599 seconds): [00:05:13.072] functions here one function is e par x so it is e par x into v where v is equal to x into cos x first let us first find out the complementary function write the equation in the operator notation form capital d square minus 2 d plus 1 of y is equal to x into e par [00:05:30.640] x cos x now to find the complementary function so what is the auxiliary equation m square minus 2 m plus 1 very simple [00:05:39.184] to uh simplify so it is 1 comma 1 the roots so the complementary function e c1 plus c2x e to the power 1x why this is the complementary function because the roots are real and repeated [00:05:54.436] Now let us find the particular integral. It is given by x into e power x cos x by d square minus 2d plus 1. [00:06:05.604] So which is now see this is of the type e par x into x cos x. So now take e par x outside the operator by replacing d by d plus 1 in the denominator you are of d. So we get e to d par x into into write it to the left. [00:06:23.136] into x cos x by d plus 1 the whole square minus 2 into d plus 1 plus 1 which simplifies to this is a plus b the whole square d square plus 2 d plus 1 minus 2 d minus 2 plus 1. Now plus 2d minus 2 d gets cancelled. Plus 2 gets cancelled [00:06:40.816] with minus 2 I get a very simple function of D in the denominator x cos x by d square. Now to simplify this I need it need not have to use any rule. I know that 1 over d square of any function of x which I call it as capital X is [00:06:58.912] Two times integration of capital X with respect to X. So how do I simplify this? I need to integrate this X into cos X with respect to X two times without writing the constant of integration, that's very important. [00:07:16.560] without writing the constant of integration. So this simplifies to let me integrate it one time. So it is e par x into 1 by d of 1 over d of x sine x is integration of x sin x. Now how do I integrate this? I use integration by [00:07:34.128] parts what is integration by parts integral of u into b is u integral b minus integral integral b into differential of u which you had studied in your 12th standard so it is [00:07:49.040] e to the power x into 1 over d of let me first simplify this is it is x into integration of sine x is minus cos x minus integral of [00:08:00.240] Integration of sin x is minus cos x into differentiation of x with respect to x is 1 dx. Now let us simplify this further. It simplifies to e to the power x into 1 over d of minus x cos x minus into minus is plus and integration of cos. [00:08:17.936] cos x is just sine x. We are not writing the constant of integration here. Now to simplify this further I need to integrate it. So it is e to the par x into integration of minus x cos x dx plus integration of sine x dx. Now [00:08:37.028] How do I integrate this minus x cos x? Again by using the same integration by parts minus x into integration of cos x is sine x minus integration of cos x is sine x. [00:08:49.796] That sine x when you integrate you get minus cos x. So that is so we get minus of x sine x plus cos x. Integration of this just sine x is minus cos x. So [00:09:01.280] This is the particular integral and the general solution is Yc plus YPC. Please look here to integrate this x is sine x. [00:09:11.232] product of two functions. I have used integration by parts. You have used integration by parts two times. When to integrate x into sin x here and again to integrate this minus of x into cos x. So integration by parts uv rule which you had studied in your 12th standard. [00:09:28.896] So this is how we find out the general solution of the differential equation. So you need to write this. This is must y is equal to yc plus y p. If the question is to find the general solution of the differential equation. Now let us consider one more question. [00:09:45.764] Find the general solution of d2 y by dx to minus three dy by dx plus two

into y is equal to [00:09:53.056] x into e power 3x plus sine 2x here right hand side I have sum of two functions one is sine a x plus b form for the thirst first term I need to use this rule now for

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:09.156] Now let us first write the equation in our notation form, which is d square minus 3d plus 2 of y is equal to x into e power 3x plus sin 2x. Now to find the complementary function, I need to first write the auxiliary equation. This is m square minus 3m plus 2. [00:10:26.928] equal to 0, find the roots. The roots are 1 comma 2. So the complementary function is c 1 e to the power 1x plus c2 e to the power 2x y because the roots are real and distinct. Now to find the particular integral. [00:10:40.576] It is x e to the power 3x plus sin 2x divided by d square minus 3d plus 2. First split this into two terms. X e to the power 3x by f of d plus sine 2x by f of d where f of d is d square minus 3d plus 2. Now [00:10:57.892] I'll call the first term as particular integral 1 and the second term as particular integral 2, where particular integral 1 is the first term and particular integral 2 is the second term sine 2x by t square minus 3t plus 2. Let me first simplify. [00:11:15.072] particular integral one and particular integral two and then substitute back in this now to find particular integral one. So what is it? x into e power three x by d square minus three d plus two. So which rule I am using rule four. What is it? [00:11:31.728] Product of two functions in which one function is e par three x. So what does that rule say? E par three x can be taken outside the operator by replacing b by b plus three in this f of b, so which which is [00:11:49.360] [ASR low confidence; check audio] PES University. PES University. PES University. PES University. [00:11:51.856] x divided by b plus 3 the whole square minus 3 into d plus 3 plus 2. So that's equal to e power 3x into x divided by d square plus 6 d plus 9 minus 3d minus 3 into 39 plus 2. So which simplifies to [00:12:09.488] X divided by D square plus 3d plus 2. Now the next question is how do I simplify this? Which rule? Type 3. So what is the type 3 says? Has many steps. What is the first step? [00:12:25.252] I need to take the lowest d degree term to get the expression in the form 1 plus 5d or 1 minus 5 d. So the lowest d degree term was 2. So I took two common factors, it is e to the power 3x into x by 2 into 1 plus d square plus 3d by 2. So it is of the form 1 plus 5d. [00:12:43.024] of D where phi F D is D square plus three D by two. Now [00:12:48.544] Take this to the numerator and expand using binomial expansion. So it is e to the power 3x by 2. 1 plus d square plus 3d by 2 whole raise to minus 1 of x. [00:13:03.072] Now expand it. So use the formula 1 plus x whole raise to minus 1, which is 1 minus x plus x square minus x cube and so on. So it is 1 minus y I have written only 3d by 2. Though the tur though the expansion has infinite. [00:13:20.720] number of terms because the function on which it has to be operated is x to the power 1. [00:13:27.556] The first order derivative of x to the power 1, that is d of x to the power 1 itself is a constant 1. When you differentiate it 2, 3, n times the derivatives would be 0. So I want only the first derivative d to the power 1. I have neglected [00:13:45.328] All higher degree terms. So it is 1. In fact, it is 1 minus b square plus 3d by 2 to the power 1. [00:13:54.596] Plus d square plus 3d by 2 whole raise to 2. I have not considered those terms. In the second term also, I have not taken d square by 2. I have considered only 3d by 2. So it is 1 minus [00:14:10.096] 3 d by 2 of x. So that is equal to e to the power 3x by 2, 1 into x is x minus 3 by 2 into d of x, which is 1. Now this is particular integral 1. [00:14:26.884] I need to find particular integral 2 also. Now what is that particular integral 2? It's sine 2x divided by d square minus 3d plus 2. Which rule I am using over here? Rule 2, type 2 type. [00:14:40.496] So it is sine ax by f of d square. Replace d square by minus into 2 square that is minus 4. So I get sine 2x by minus 4 minus 3d plus 2. Retain D term as it is. So this simplifies to minus of sine.

[00:14:58.128] And 2x by 3d plus 2. Now, how do I simplify this? I have a D term. I can apply the rule only when I have D square term. Now

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:08.048] Convert this d term into d square term by multiply and dividing by its conjugate. What is the conjugate? 3d minus 2. So it is minus of 3d minus 2 by 3d minus 2 into 3d plus 2 off sine. [00:15:25.680] 2x, which simplifies to 3d minus 2 sin 2x divided by a minus b into a plus b, a square minus b square, 9 d square minus 4. Now how do I simplify it further? [00:15:37.808] Two different ways. See I I can first apply the operator 1 over 9 d square minus 4 unsine 2x and [00:15:45.828] Apply the operator 3d minus 2 on the resultant function of x. R. First itself you apply the operator 3d minus 2 on sine to x and then you apply the operator 1 over 9d square minus 4 on the resultant function. So how how I have done? I have [00:16:01.776] Taken this first 1 over 9 d square minus 4 of sine to x. How do I simplify? Replace d square by minus into a square. What is that minus into a square? Minus into 4. So minus 36 minus 4 is minus 40, and there is one like negative sign over here which becomes plus. [00:16:19.408] It is 1 by 4 t into 3d minus 2 of sine 2x. Multiply 3d of sine 2x minus 2 of sine 2x. What is d of sine 2x? It is 2 cos 2x. So it is 6 cos 2x minus 2 into sine 2x. So this is particular integral 2. Now what is particular integral? [00:16:39.556] It is PI1 plus PI2, and because the question is to find the general solution, y is equal to complementary function plus particular integral. [00:16:51.556] This is how we find the particular integral when the given function is x into b, where b is a function of x. Thank you.

EV0000138 · EV0000139 · EV0000140 · EV0000141

Full transcript: 2839 normalized words, 71 timestamped segments, 1022.378667 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:11:49.360

EV0000126 · EV0000127 · EV0000128 · EV0000129

10 Particular Integral when X is any other form

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Graphs and diagrams

Figure G000100. Particular integral when X is any other form · source slide page 1.

EV0000564

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000101. Particular integral when X is any other form · source slide page 2.

EV0000567

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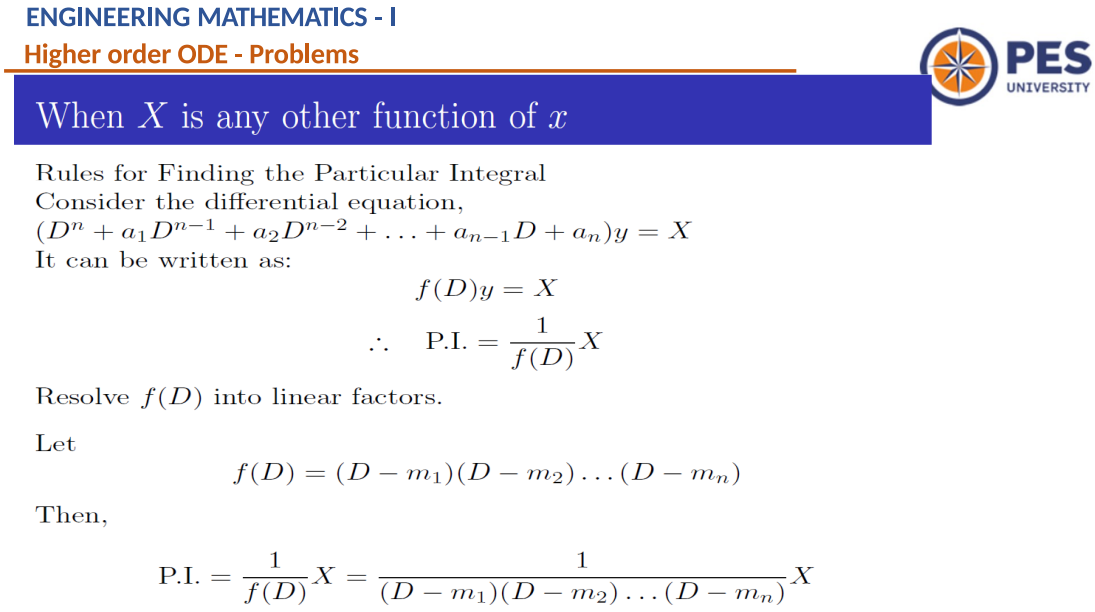

Figure G000102. Particular integral when X is any other form · source slide page 3.

EV0000570

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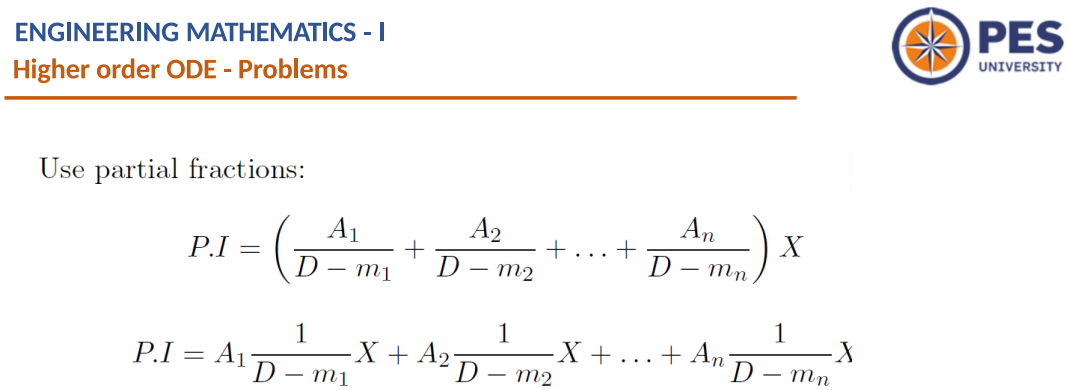

Figure G000103. Particular integral when X is any other form · source slide page 4.

EV0000573

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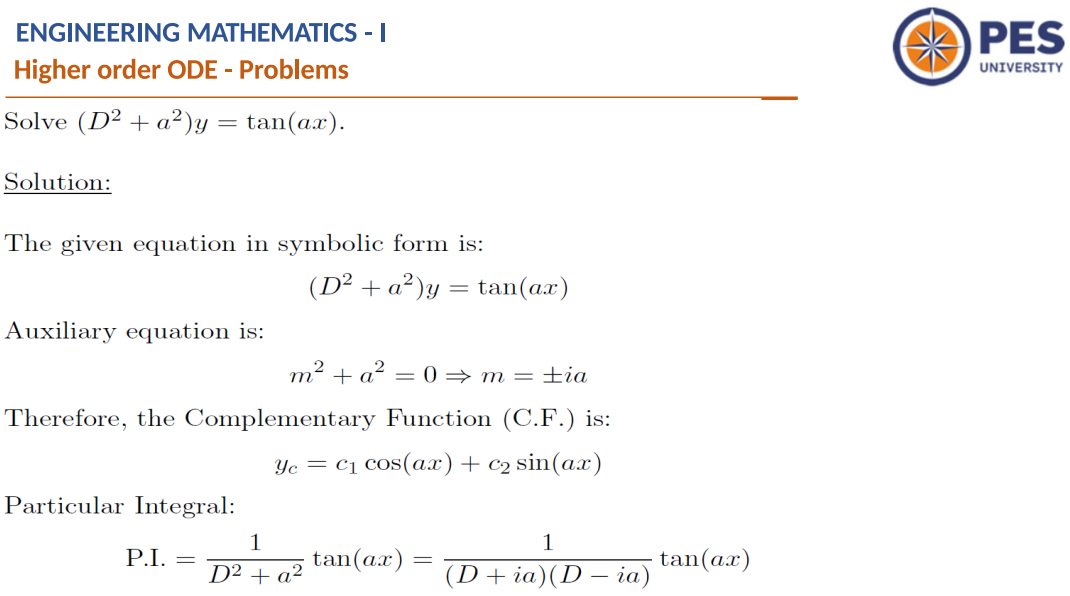

Figure G000104. Particular integral when X is any other form · source slide page 5.

EV0000576

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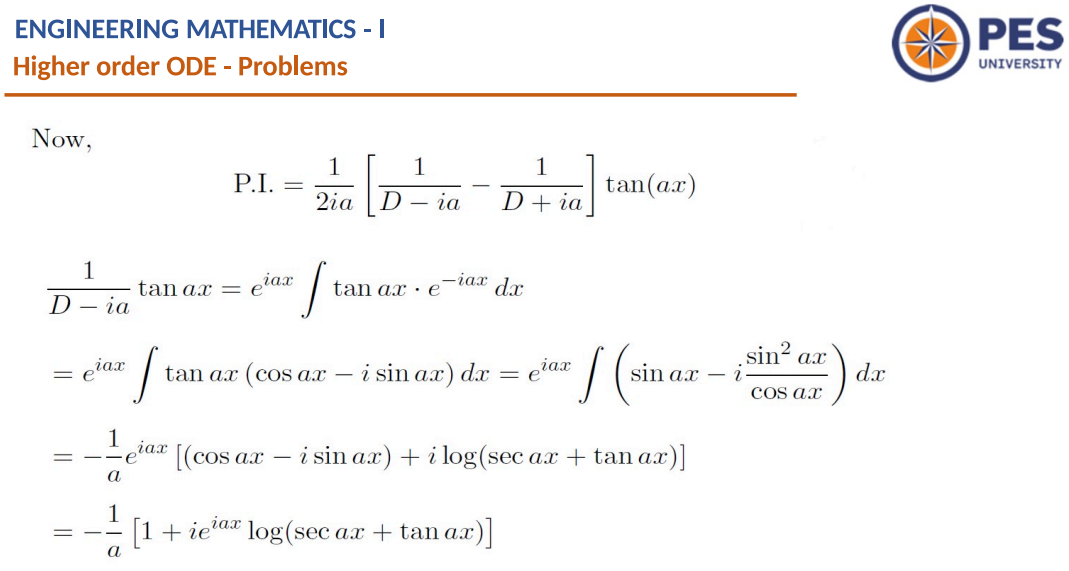

Figure G000105. Particular integral when X is any other form · source slide page 6.

EV0000579

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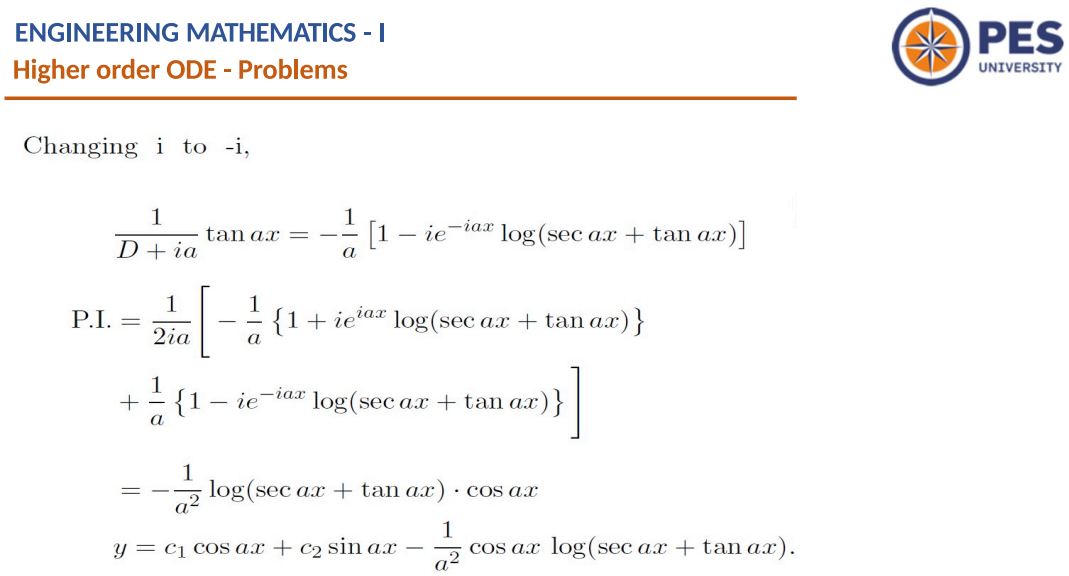

Figure G000106. Particular integral when X is any other form · source slide page 7.

EV0000582

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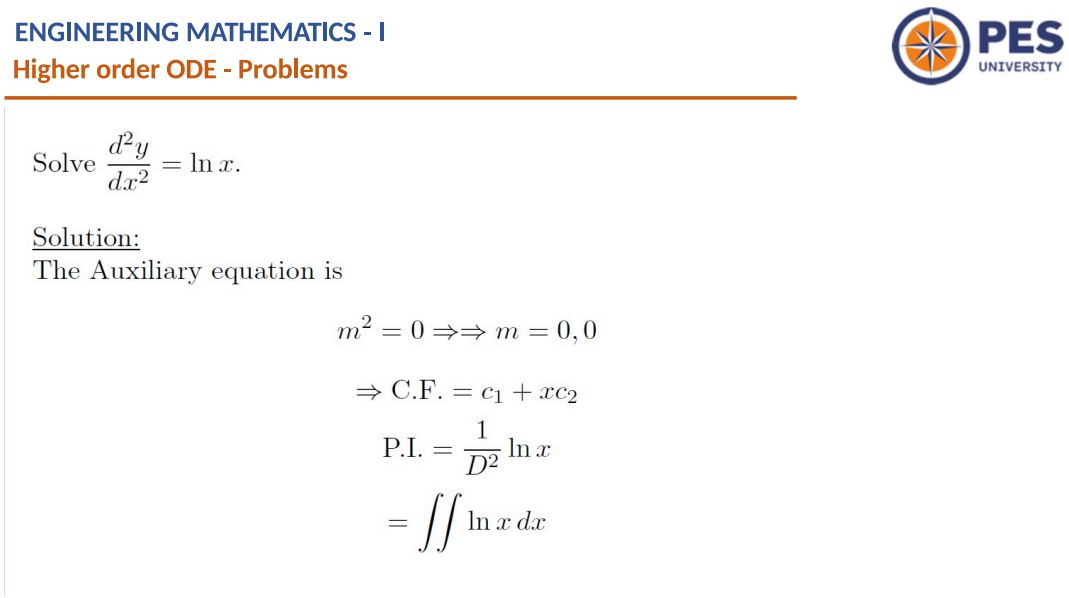

Figure G000107. Particular integral when X is any other form · source slide page 8.

EV0000585

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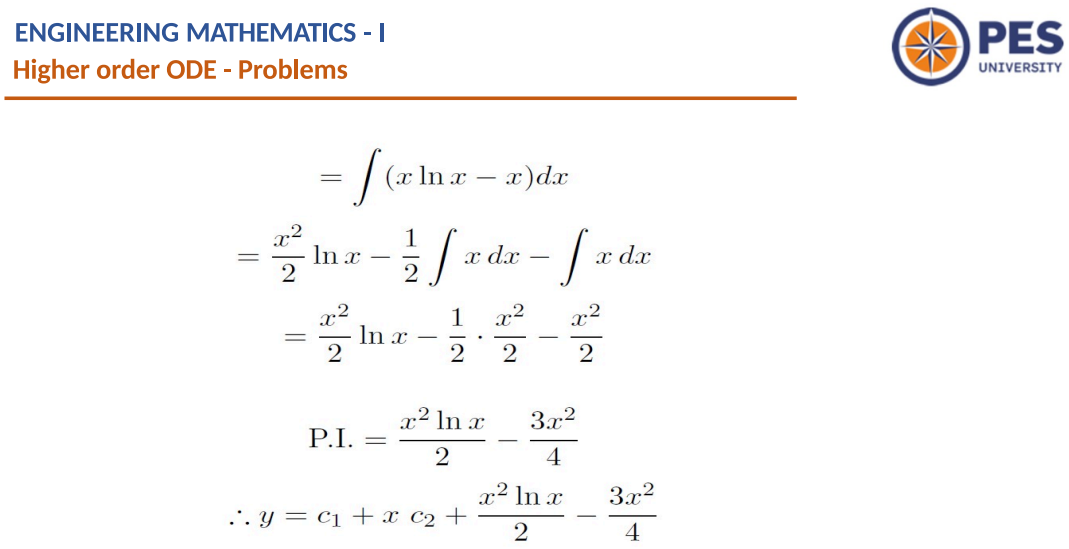

Figure G000108. Particular integral when X is any other form · source slide page 9.

EV0000588

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ENGINEERING MATHEMATICS - I
Higher order ODE - Problems

PES UNIVERSITY

Find the particular integral for the differential equation:
$(D^2 + 3D + 2)y = e^{e^x}$.

Solution:

$$(D^2 + 3D + 2)y = e^{e^x}$$
$$\Rightarrow \frac{1}{(D+2)(D+1)} e^{e^x}$$

Use partial fractions:

$$\frac{1}{(D+2)(D+1)} = \frac{A}{D+2} + \frac{B}{D+1}$$
$$1 = A(D+1) + B(D+2) \Rightarrow A = -1, \quad B = 1$$

So,

$$P.I = \left(\frac{1}{D+1} - \frac{1}{D+2}\right) e^{e^x}$$

Figure G000109. Particular integral when X is any other form · source slide page 10.

EV0000591

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ENGINEERING MATHEMATICS - I
Higher order ODE - Problems

PES UNIVERSITY

$$= e^{-x} \int e^x e^{e^x} dx - e^{-2x} \int e^{2x} e^{e^x} dx$$

Substitute: $e^x = t \Rightarrow e^x dx = dt$

$$= e^{-x} \int e^t dt - e^{-2x} \int t e^t dt$$
$$= e^{-x} e^t - e^{-2x} \left[te^t - \int e^t dt\right]$$
$$= e^{-x} e^{e^x} - e^{-2x} \left[e^x e^{e^x} - e^{e^x}\right]$$
$$= e^{e^x} \left[e^{-x} + e^{-2x} - e^{-x}\right]$$
$$\therefore P.I = e^{-2x} \cdot e^{e^x}$$

Figure G000110. Particular integral when X is any other form · source slide page 11.

EV0000594

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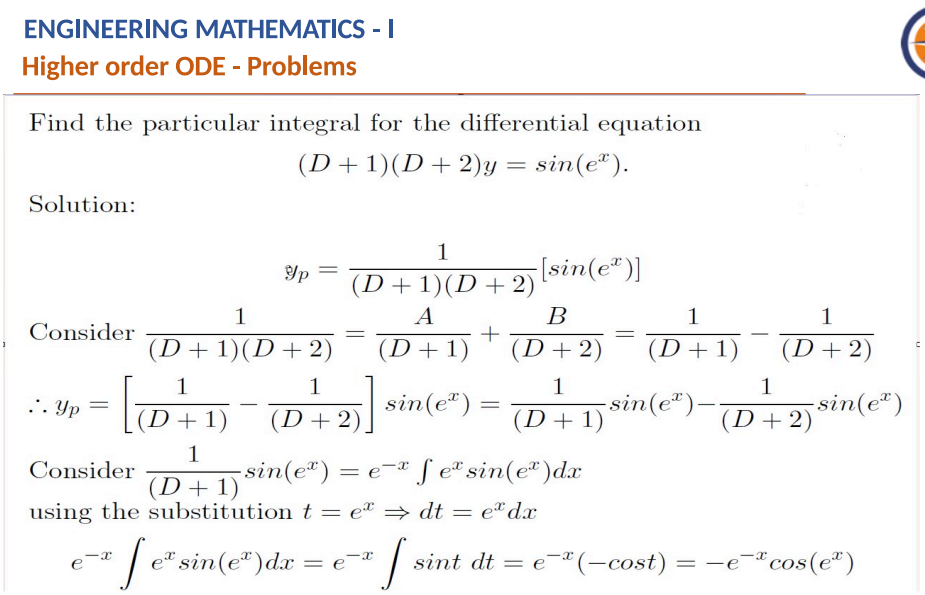

Figure G000111. Particular integral when X is any other form · source slide page 12.

EV0000597

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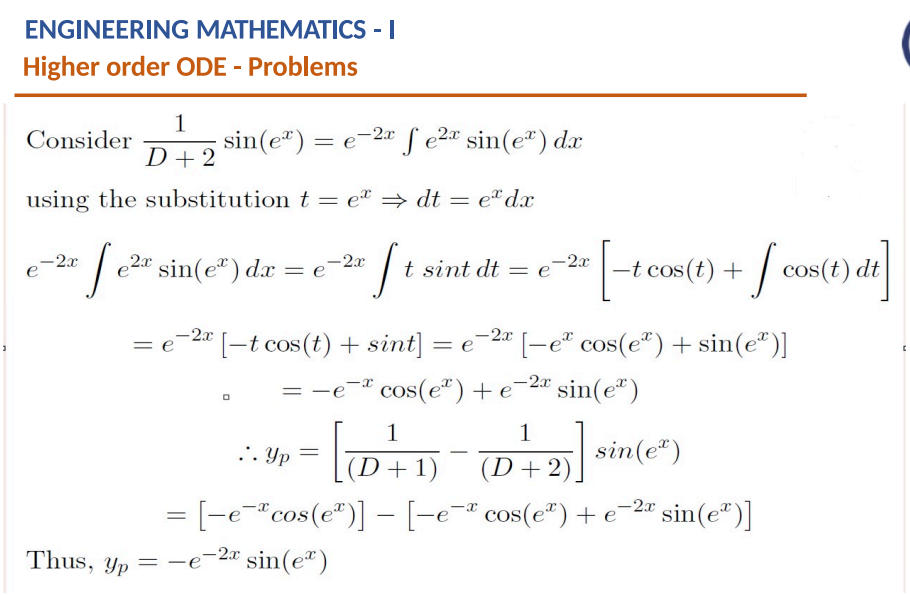

Figure G000112. Particular integral when X is any other form · source slide page 13.

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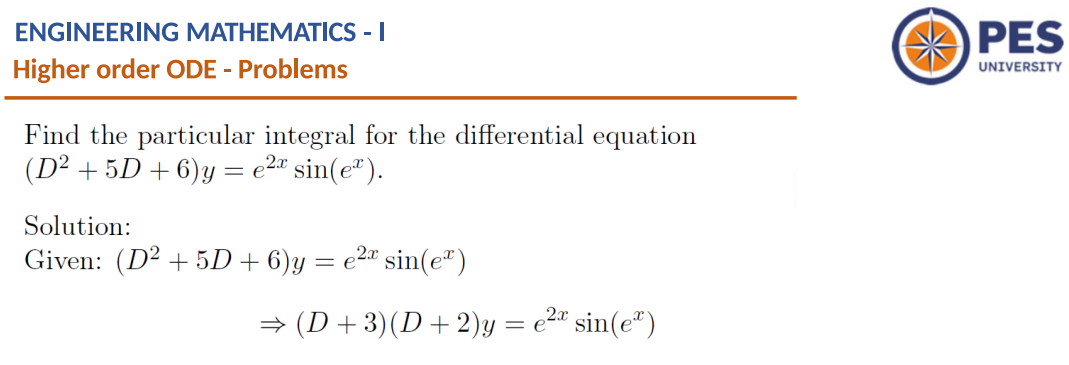

Figure G000113. Particular integral when X is any other form · source slide page 14.

EV0000603

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

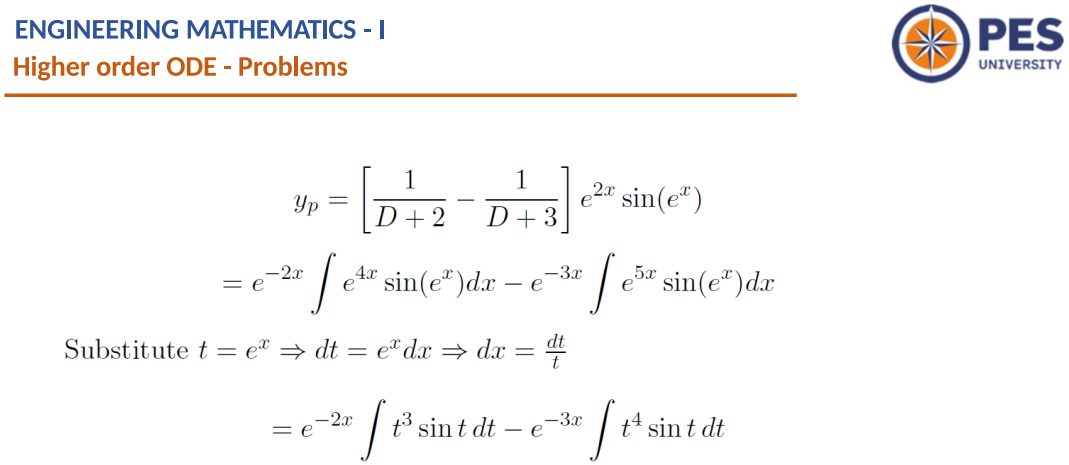

Figure G000114. Particular integral when X is any other form · source slide page 15.

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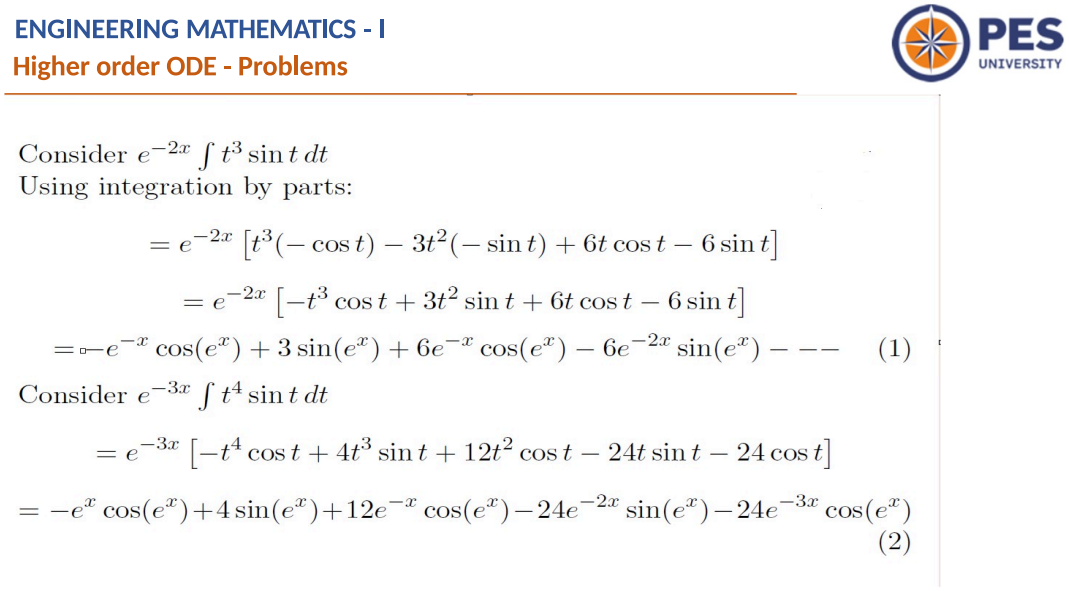

Figure G000115. Particular integral when X is any other form · source slide page 16.

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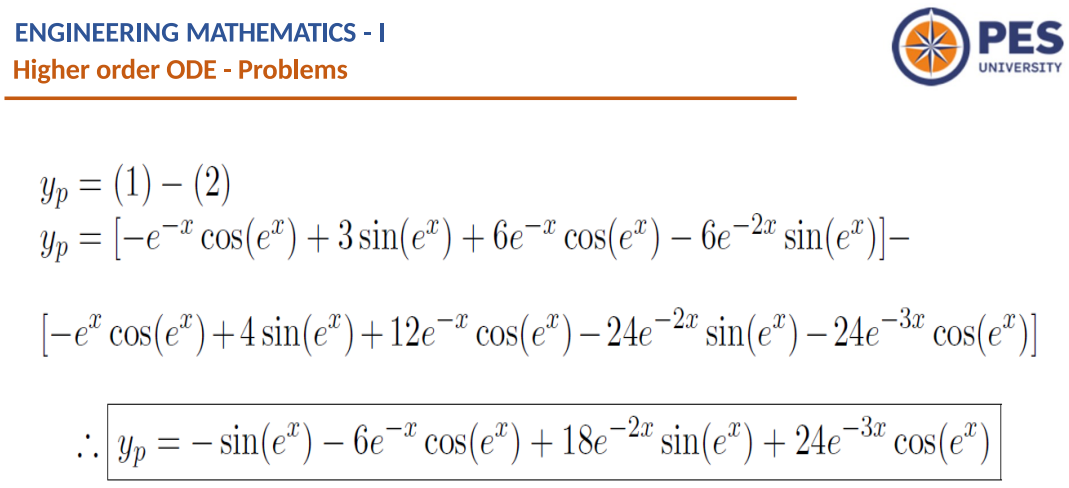

Figure G000116. Particular integral when X is any other form · source slide page 17.

EV0000612

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Figure G000117. Particular integral when X is any other form · source slide page 18.

EV0000615

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000563 · EV0000564 · EV0000565
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 2: HIGHER ORDER DIFFERENTIAL EQUATIONS SESSION: 7 SUB-TOPIC: NON-HOMOGENUOUS LDE WHEN RHS IS ANY OTHER FORM Department of Science and Humanities EV0000566 · EV0000567 · EV0000568
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] Rules for Finding the Particular Integral Consider the differential equation, (Dn + aDn-1 + a2Dn-2 +...+ an-1D+ an)y = X It can be written as: f(D)y = X 1 P.I. = X f(D) Resolve f(D) into linear factors. Let f(D)=(D-m1)(D-m2)...(D-mn) Then, P.I. = X: X f(D) (D -m1)(D -m2)...(D -mn) EV0000569 · EV0000570 · EV0000571
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] Use partial fractions: P.I = +...┼ X - m1 D) - m2 D-mn 1 1 P.I = A1 X + A2 -X+...+ An D D) -m1 - m2 D. - mn P.I = A1em1 f Xe-m1xdc + A2em2x f Xe-m2xdc + ... + Anemnr ↑ Xe-mnxdx EV0000572 · EV0000573 · EV0000574

5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] Solve (D2 + a2)y = tan(ac). Solution: The given equation in symbolic form is: (D2 + a²)y = tan(ax) Auxiliary equation is: m² + α² = 0 → m = ±ia Therefore, the Complementary Function (C.F.) is: yc = C1 cos(ac) + C2 sin(ax) Particular Integral: 1 P.I. = tan(ac) = tan(ax) D2 + a2 (D+ ia)(D -ia) EV0000575 · EV0000576 · EV0000577

6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] Now, P.I. = tan(ax) 2ia D -ia D+ia tan ar = iar D -ia sin? ac iar tan ac (cos ar - i sin ar) dr = é iar sin ar - dc cOs ac iac [(cos ac - i sin ax) + i log(sec ac + tan ax)] a [1 + ieiax log(sec ax + tan ax)] EV0000578 · EV0000579 · EV0000580

7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] Changing i to -i, 1 - ie-iax log(sec ax + tan ax)] -tan ar = D +ia a P.I. = {1 + ieiax log(sec ax + tan ax)} 2ia a {1 - ie-iax log(sec ax + tan ax)} a log(sec ar + tan ac) · cos ac y = C1 cOs ac + c2 sin ac cos ax log(sec ar + tan ax). EV0000581 · EV0000582 · EV0000583

8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] Solve In c. dc2 Solution: The Auxiliary equation is m² = 0 →> m = 0,0 → C.F. = C1 + C2 1 P.I. = ln c D2 In c dc EV0000584 · EV0000585 · EV0000586

9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] (clnc -x)dc 2 C 1 In c x dc x dc 2 2 1 心 穴 In c 2 2 2 2 x² In c 3x2 P.I. = 2 4 x2 ln c 3c2 .. y = C1 + C2 + 2 4 EV0000587 · EV0000588 · EV0000589

10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] Find the particular integral for the differential equation: (D2 + 3D + 2)y = ee Solution: (D² + 3D + 2)y = 1 (D + 2)(D + 1) e Use partial fractions: 1 B (D+ 2)(D+ 1) D+ 2 D+1 1 = A(D+1)+ B(D+2)→ A =-1， B= 1 So, P.I = e e D+1D+2 EV0000590 · EV0000591 · EV0000592

11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] dr - e -2c Substitute: e" = t = e"dr = dt etdt -e -2c tetdt 2m etdt =e-r =e e .. P.I = e-2r .e EV0000593 · EV0000594 · EV0000595

12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] Find the particular integral for the differential equation (D + 1)(D + 2)y = sin(e"). Solution: yp= [sin(e*)] (D+ 1)(D + 2) 1 A B 1 1 Consider (D + 1)(D + 2) (D+ 1) (D+2) (D+1) (D+2) 1 1 1 1 sin(e") = :.9p sin(e")- sin(e") (D+1) (D+2) (D + 1) (D+2) 1 Consider sin(e") = e- f e"sin(e")dc (D+1) using the substitution t = e" => dt = e"dc C esin(e")dc = e-α sint dt = e-"(-cost) = -e-αcos(e") e EV0000596 · EV0000597 · EV0000598

13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] 1 Consider sin(e") = e-2α f e2α sin(e") dc D+2 using the substitution t = e" =→ dt = e*d e2u sin(e") dx = e-22 -2c t sint dt = e-2r -t cos(t) + cos(t) dt = e-2α [-tcos(t) + sint] = e-2x [-e" cos(e") + sin(e")] = -e-α cos(e") + e-2r sin(e") 1 :.yp sin(e") (D+1) (D+2) [-e-"cos(e")] - [-e-α cos(e") + e-2r sin(e")] Thus, Up = -e-2w sin(e") EV0000599 · EV0000600 · EV0000601

14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] Find the particular integral for the differential equation (D2 + 5D + 6)y = e2r sin(e*). Solution: Given: (D2 + 5D + 6)y = e2α sin(e") ≥ (D + 3)(D + 2)y = e2 sin(e") 1 2w sin(e") →yp= ((D + 2)(D + 3) e Using partial fractions: 1 A B (D + 2)(D + 3) D+2 D+3D+2D+3 EV0000602 · EV0000603 · EV0000604

15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] 2c yp sin(D + 2 D + 3 -2c 4c sin(er)dc - e-3r 5c sin(e˜)dc e D e Substitute t = e" → dt = e"dc → dc = dt t -2c sin t dt EV0000605 · EV0000606 · EV0000607

16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] Consider e-2α f t3 sint dt Using integration by parts: = e-2x [t²(- cost) - 3t2(- sint) + 6t cost - 6 sint] = e-2α [-t3 cost + 3t² sint + 6t cost - 6sint] =-e-" cos(e") + 3 sin(e") + 6e- cos(e") - 6e-2α sin(e") - -- (1) Consider e-3α f t4 sint dt [-t4 cost + 4t3 sint + 12t² cost - 24t sint - 24cost] = e-3c = -e" cos(e")+4sin(e")+12e-α cos(e")-24e-2u sin(e")-24e-3α cos(e") (2) EV0000608 · EV0000609 · EV0000610

17. Source slide page 17 (extracted text): ENGINEERING MATHEMATICS - I Higher order ODE - Problems [Image text, OCR] yp=(1)-(2) cos(e*)+3sin(e) + 6e-" cos(e") - 6e-2x sin(e")]- Yp = [-e- c [-e" cos(e")+4 sin(e") + 12e- c cos(e") - 24e-2x sin(e")-24e-3x cos(e")] = -sin(e") -6e- cos(e") + 18e-2r sin(e") + 24e-3r yp EV0000611 · EV0000612 · EV0000613

18. Source slide page 18 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000614 · EV0000615 · EV0000616

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.748] Hello everyone. [00:00:07.300] Welcome back to the course on engineering mathematics 1. In this session we learn how to find the particular integral and the right hand side capital X doesn't belong to any of the four standard forms what we have discussed earlier. [00:00:27.236] It means when x is not e power x, when x is not sine of a x plus b or cos of a x plus b, when x is not a polar case, that is x per m, or when x is not e power a x into v case. [00:00:47.812] So let's look at the procedure how to find out PI in this kind of situation. [00:00:58.212] Let us consider the differential equation in this particular form. So as we know the formula for particular integral is given by 1 over f of d times capital X where f of d is a polynomial in D. [00:01:19.620] So in this case, we first resolve f of b into linear factors in this form. [00:01:32.740] Once we do this, Pi becomes 1 over d minus m1 times d minus m2 and so on, d minus mn. Then we need to resolve this into partial fraction. [00:01:52.388] So when we apply partial fraction, P i reduces to a1 by d minus m1 plus a2 by d minus m2 and so on, a n by d minus mn of capital X. So once we do this. [00:02:11.364] We need to apply a standard formula which is given by one over e minus a of f of x. This is nothing but e power ax integral e power. [00:02:32.772] Minus ax f of x into dx. So we apply this standard formula in this particular case. Once we apply this standard formula for each of these n terms, we get a1 e power m1x integral capital X e [00:02:52.548] P power minus m or dx and so on. [00:02:59.876] So this is a way to find out particular in this case particular integral when x is not any of the standard four forms. [00:03:13.156] Let's take up a problem. Solve the differential equation. T square plus a square of y equals tan ax. Observe carefully. This is not e power x or sine of cos or polynomial or e power x in

to be. [00:03:33.124] So this is solved by this special case. There's no change in computing the complementary function. So the axillary equation for this problem is M square plus a square equals 0. The roots are plus or minus a i. [00:03:53.604] So alpha 0 and beta is A. Therefore, the complementary function becomes C1 cos Ax plus C2 sine ax. [00:04:12.164] Let's concentrate on finding the particular integral PI. As we know that this f of b should be expressed in this particular form, that is linear factors. So based on this idea, these [00:04:31.940] square plus a square can be written as d plus a i times d minus a i it means a plus b into a minus b so once we use partial fraction so this part reduces to one by two ia off [00:04:52.132] of one by d minus i a minus one by d plus i a so probably a and b becomes

EV0000142 · EV0000143 · EV0000144 · EV0000145

Transcript block 2 (source interval 000300-000599 seconds): [00:05:02.916] 1 by 2 IA and minus 1 by 2 IA. [00:05:08.644] Let us apply the power two times. [00:05:13.252] Now consider 1 by t minus i a time tan x. As per the standard formula, this is going to be a power i ax integral tan a x tan a power minus a x into dx. [00:05:33.988] Let's see how to integrate this. And X as it is. And let us use Euler's formula. Which we know very well. What is a power plus or minus Ix? This is cos x. [00:05:52.676] plus or minus i into sin x. Therefore e power minus a x becomes cos a x minus i sine a x. So let us multiply. [00:06:08.292] This tan ax into cos a x. After cancellation of cos and cos we get sin a x. Then multiply tan ax and sin a x. So we get sin square ax divided by cos ax. [00:06:28.996] So in the next step what we do let us keep let me solve only that particular part let us keep e power x as it is integral no change in sine ax then minus i [00:06:46.404] Write sin square x to be one minus cos square ax divided by cos a x. [00:07:03.076] So further this is same as e power i ax integral sine ax minus i one by cos that's going to be second ax plus [00:07:23.652] I into cos a x of dx. So when we do integration it is e power i a x outside [00:07:38.628] So sine ax becomes minus cos ax by a constant here. So this is log of second ax plus. [00:07:57.060] than a x divided by a then the last term is going to be i then it is sine ax divided by a so this integration so let us take minus one by a common [00:08:16.644] Then that leads to this particular step. [00:08:21.572] Okay after this we have to multiply by e power i ax. So when you multiply this by e power iax, you can observe that that leads to car square ax plus sin square ax, which is 1. [00:08:41.764] And there's no change in second. So finally the value of 1 over D minus IA tan ax is going to be minus 1 by a of 1 plus i. [00:08:58.276] E power i x times log of secant a x plus tan a x. Similarly, to find the value of b plus i a of tan a x, we simply replace i2 minus i. Then that leads to this particular step. [00:09:18.500] So finally PI becomes 1 by 2 IA of this is P1 and this is P2. [00:09:28.356] [ASR low confidence; check audio] Then [00:09:32.388] after this particular step let us uh you can simplify like this so 1 by 2i a so minus 1 by a as it is then we have minus uh i by a [00:09:49.540] Then uh e power iax then as usual log of that particular part plus one by a minus i times e power minus iax

EV0000146 · EV0000147 · EV0000148 · EV0000149

Transcript block 3 (source interval 000600-000899 seconds): [00:10:09.892] into log of that particular part. So we can observe minus 1 by a and plus 1 by gets canceled out. So we are left out with 1 by 2 a of you can take minus i by a [00:10:30.276] common even log of that is also we can take as a common so then we get e power ieax plus e power minus i ax into log of that [00:10:48.676] So we can cancel i and i and you know this is same as uh 2 cos a x. So we can also cancel 2 and 2. After doing this simplification, we see that the final PI is going to be [00:11:09.380] Minus 1 by a square log of sec Ax plus than Ax times cos ax. So the general solution is y equals

complementary function plus particular integral. I hope this concept is clear to you. [00:11:30.340] Let us solve another problem. Sol the differential equation d square y by dx square e plus log x as you can see it's a special function as compared to those x for standard forms. [00:11:49.156] So the axlary equation going to be m square equals zero. It means zero appears two times. So the complementary function is going to be c1 plus t2x. So to find out p i we get 1 by d square interlog [00:12:09.892] So unfortunately we can solve this by a normal method. It means log x should be differentiated with respect to x2 times because this is an inverse differential operator. So when integrate log x with respect to x2 times, we get this. [00:12:28.868] So for the first time, as we know, integration of log x. [00:12:37.892] should be written as log x times one into dx. When we use by parts we get this. So for the second time integration the integration of x into log x is going to be [00:12:55.268] x square by 2 log x minus half x px and integration of this going to be x square by 2. Finally after the simplification PI becomes x square log x by 2 minus 3x square by 4 and the general solution is [00:13:12.132] Complementary function plus particular integral. [00:13:23.396] Let us solve this particular differential equation A square plus 3D plus 2 of y equals EE power x. Again observe this. EE power x. So as per the standard procedure, so this is the expression of the linear factors. [00:13:43.108] Then this is the partial fraction. A becomes minus one, we become plus one. Therefore P i becomes 1 by D plus 1 minus 1 by D plus 2 of E power X. So we can take this as P1 minus P2. [00:14:05.956] So this is p1 part and this is p2 part. So when we use substitution method, this part becomes e power t then dt because e power s dt become dt. So that's going to be simply e power t only. [00:14:22.884] So similarly for this particular case we can split e power 2x to be e power x e power x e e power x. So this into t x become dt. So this is going to be e power t dt. [00:14:43.556] So when we use by parts, we get this particular answer. So finally, so Pi becomes e power minus two x time e power after cancellation of e power minus x and this e power minus x.

EV0000150 · EV0000151 · EV0000152 · EV0000153

Transcript block 4 (source interval 000900-001199 seconds): [00:15:01.380] So these are the four standard problems where we computed the particular integral. Thank you.

EV0000154 · EV0000155 · EV0000156 · EV0000157

Full transcript: 1603 normalized words, 55 timestamped segments, 914.133 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:09:28.356

EV0000142 · EV0000143 · EV0000144 · EV0000145

11 Particular integral when X is any other form_Continued

This lesson is indexed from the source video titled Particular integral when X is any other form_Continued. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000158 · EV0000159

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:04.388] Hello everyone. Welcome back to the course on Engineering Mathematics 1. [00:00:13.668] In this session, we will take up problems of finding particular integral when the right hand side as capital X doesn't belong to any of the four standard forms we have discussed earlier. [00:00:32.740] It means when x is not a power ax, when x is not sine of a x plus b as well as of x plus b [00:00:50.660] When x is not a polynomial case, when x is not e power x into v, where v is a function of x. So we use this particular case. It means let us uh learn how to find out particular integral when the right hand side x is not. [00:01:10.596] any one of these four forms. So let us solve a problem on finding the particular integral for the differential equation [00:01:25.124] D plus one times d plus two of y equals sine of v power x. Absorb it. This is not any one of these four forms. So as per clearly part. [00:01:39.940] The particular integral is written as 1 divided by f of b which is nothing but d plus 1 times d plus 2 of capital X. [00:01:52.196] So we first resolve this into partial fraction. So this is written a by d plus one plus b by d plus two. Once we use partial fraction [00:02:08.260] This expression reduces to one divided by d plus one minus one divided by d plus two. It means a value becomes one, b becomes minus one. Therefore, the particular integral is written as [00:02:27.172] 1 divided by d plus 1 minus 1 divided by d plus 2 of this capital X. So let us split this into two parts. So this is going to be 1 divided by d plus 1 of sine of e power x. [00:02:45.156] Minus 1 by d plus 2 of sin of v power x. So we can take this as p1. We can take this as p2. Finally, after computing p1 and p2, we go for p1 minus p2. Okay. Let us compute [00:03:01.764] 1 divided by p plus 1 of sine of p power x. We know a standard formula that is 1 over p minus 1 of f of x equals [00:03:20.068] e power ax integral e power minus ax into f of x to d x. So this standard formula we use in this special case cases. [00:03:36.708] So when you apply this formula for this particular case we get a power minus x because area is minus minus 1. Therefore a power minus a x becomes e power minus x. A power minus ax becomes a power x, then sine of e power x times dx. [00:03:57.540] So we need to integrate this by substitution method. Let us take e power x equal to t so that dt becomes e power dx. Then this expression becomes integral psi t into dt. [00:04:15.236] So integration of sine t becomes minus cos t. Therefore, the value of P1 is going to be minus a power minus x of cos of e power x. In a similar manner, let us compute. [00:04:33.220] One over D plus two of sine of P power X. [00:04:39.620] So in this case a value is minus 2. Therefore, as per the standard formula, we get this particular step. In a similar manner, let us use substitution method, whereas so this is written as a power minus 2x integral. [00:04:57.380] Let us split a power 2x to be a power x times a power x then sine of a power x into dx. So when we use substitution method that is a power x equal to t, so that reduces to a power minus 2x integral.

EV0000158 · EV0000159 · EV0000160 · EV0000161

Transcript block 2 (source interval 000300-000599 seconds): [00:05:16.356] t s t dt. So let us integrate this by parts. So that leads to minus t cos t plus integral cos t dt because you know very well how [00:05:30.884] how to apply integration by parts let's use the technique so finally we get e power minus 2x of minus t cos t and integral of cos t becomes sine t so finally this is same as e power minus 2x times this [00:05:46.884] So when we multiply this by e power minus 2x we get minus e power minus x times cos of e power x plus e power minus 2x times e sine of e power x. So this is our p2. [00:06:04.004] Therefore, the final PI is going to be P1 minus P2. So when we do P1 minus P2 after cancellation of minus e power minus x cos of e power x and e power minus x cos of e power x after cancellation of these two. [00:06:24.068] We get this as the answer. Provided we

have negative sign outside. So therefore the particular integral for this problem is minus e power minus 2x into sine of e power x. I hope this concept is clear to you. [00:06:43.748] In a similar manner, let us solve one more one more problem to get a clarity. [00:06:50.660] Find the particular integral for the differential equation d square plus phi d plus six of y equals e power 2x times sin of e power x. So what we do normally [00:07:08.964] Let us find the factors of this d plus 3 times d plus 2 of y equals right hand side. So this is a formula for particular integral p i 1 divided by f of d times capital X. Similar to last problem. [00:07:28.356] Let us resolve this into partial fraction. So that leads to 1 over d plus 2 minus 1 over d plus 3. [00:07:40.132] Finally, Pi can be written as 1 by D plus 2 minus 1 by D plus 3 of E power 2x times sine of E power x. Let us split this into two parts and apply the standard formula, we get this as well as this. Now both should be solved by. [00:08:00.452] parts so let us take the substitution e power x to be t so in this case what happens we can write e power four x to be power x times e power three x into sine of e power x [00:08:21.124] dx. So when we use the substitution e power x as t so this reduces to e power three x becomes p power three then and e power x dx become dt then we have another sign t term [00:08:37.380] So we can easily find the integration of this either by parts or something called Bernoulli rule. In a similar manner, for this case we get e power x times e power 4x times e power. [00:08:57.092] E power x into dx. After using the same substitution, it reduces to integration of t power four sin t. Even this can be solved by integration by partnoli rule. So finally [00:09:16.580] The integration of first one is going to be this. When we use x t to be e power x, we get this particular step. Similarly, the integration of this is going to be this particular step. [00:09:35.076] when we use T as e polix we get this particular step and hence the particular integral becomes P1 minus P2 which is same as this so these are the two problems we consider [00:09:56.068] To find out PI when the right hand side X doesn't belong to any of the four standard forms we have discussed earlier. Thank you.

EV0000162 · EV0000163 · EV0000164 · EV0000165

Full transcript: 1225 normalized words, 36 timestamped segments, 610.389 seconds duration. Low-confidence segments flagged: 0.

EV0000158 · EV0000159 · EV0000160 · EV0000161

12 Legendre differential equation

This lesson is indexed from the source video titled Legendre differential equation. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000166 · EV0000167

Graphs and diagrams

Figure G000118. Legendre s differential equation · source slide page 1.

EV0000618

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

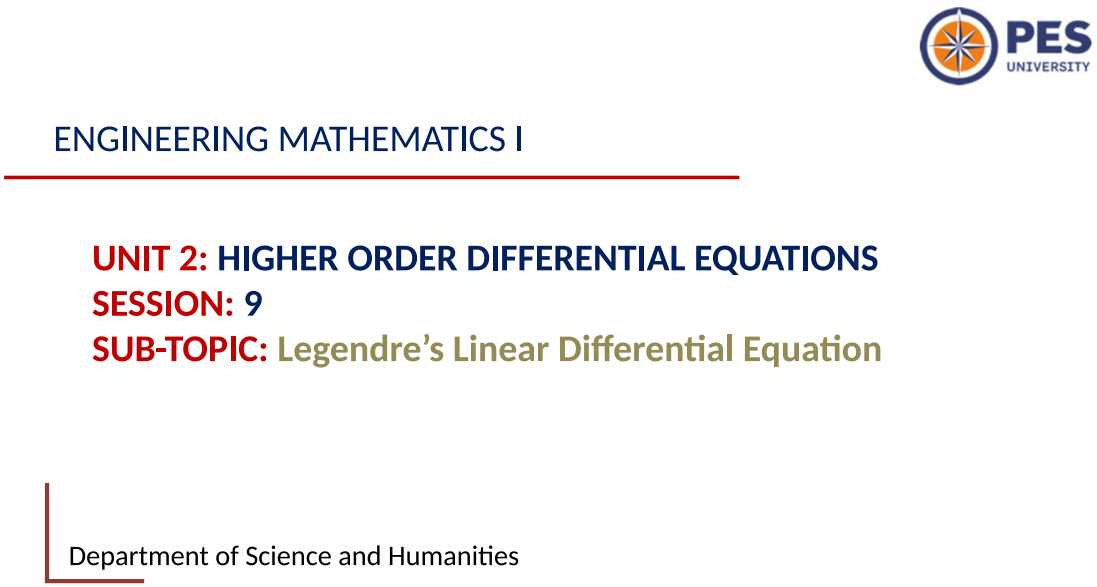

Figure G000119. Legendre s differential equation · source slide page 2.

EV0000621

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

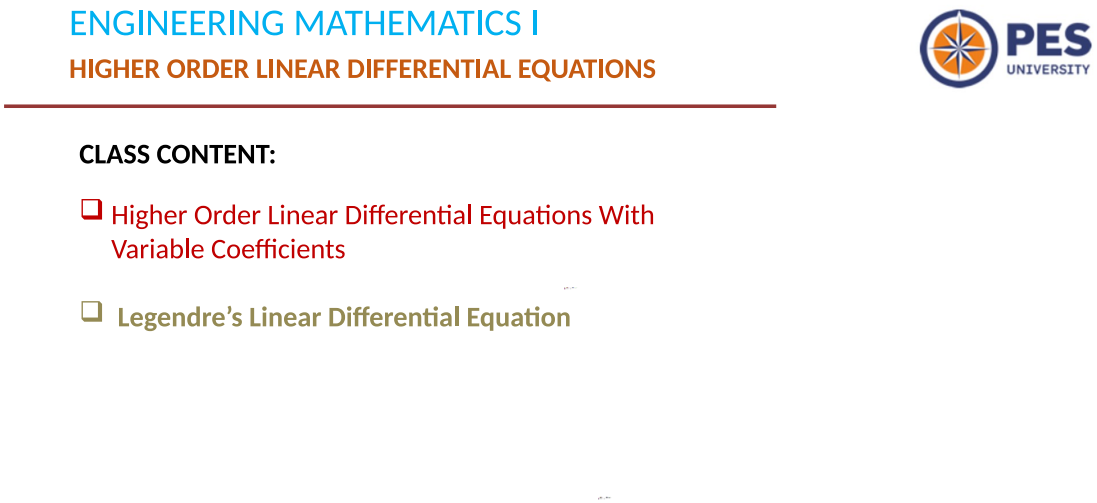

Figure G000120. Legendre s differential equation · source slide page 3.

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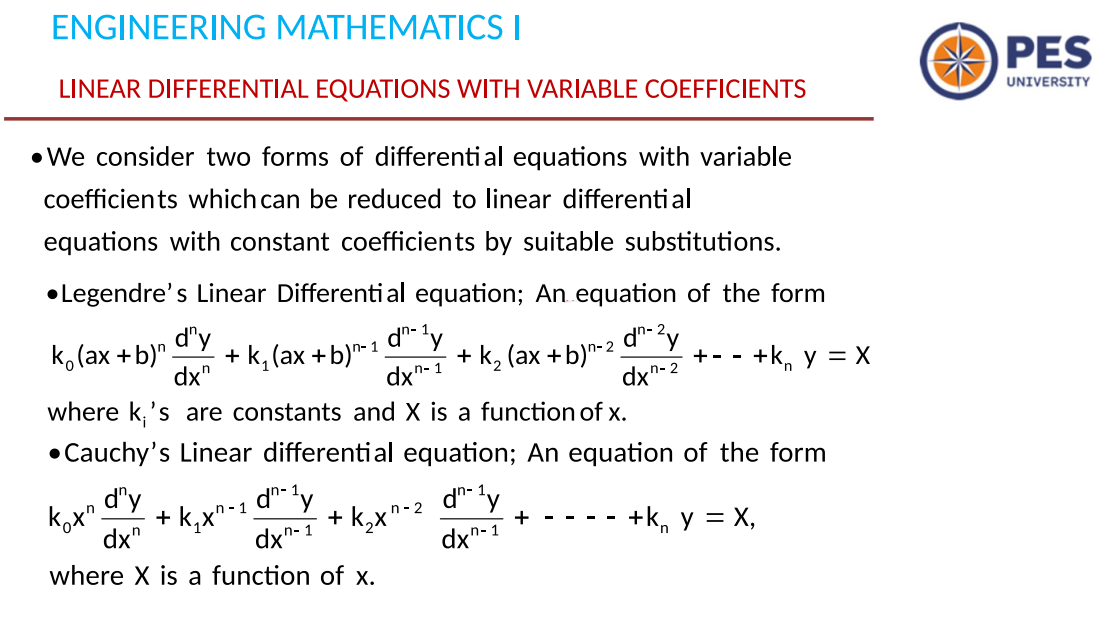

Figure G000121. Legendre s differential equation · source slide page 4.

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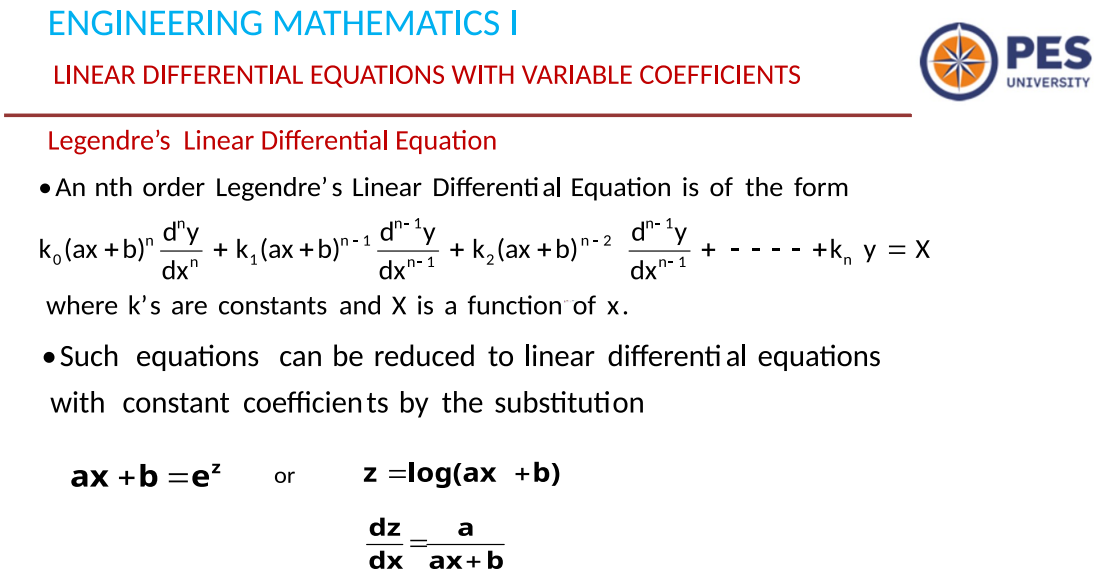

Figure G000122. Legendre s differential equation · source slide page 5.

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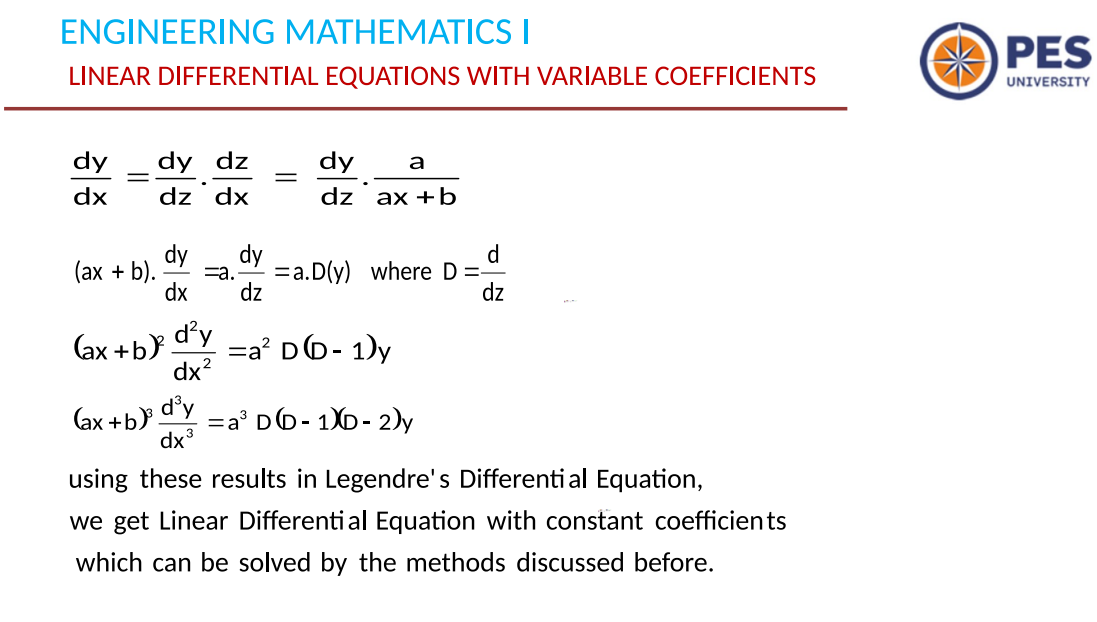

Figure G000123. Legendre s differential equation · source slide page 6.

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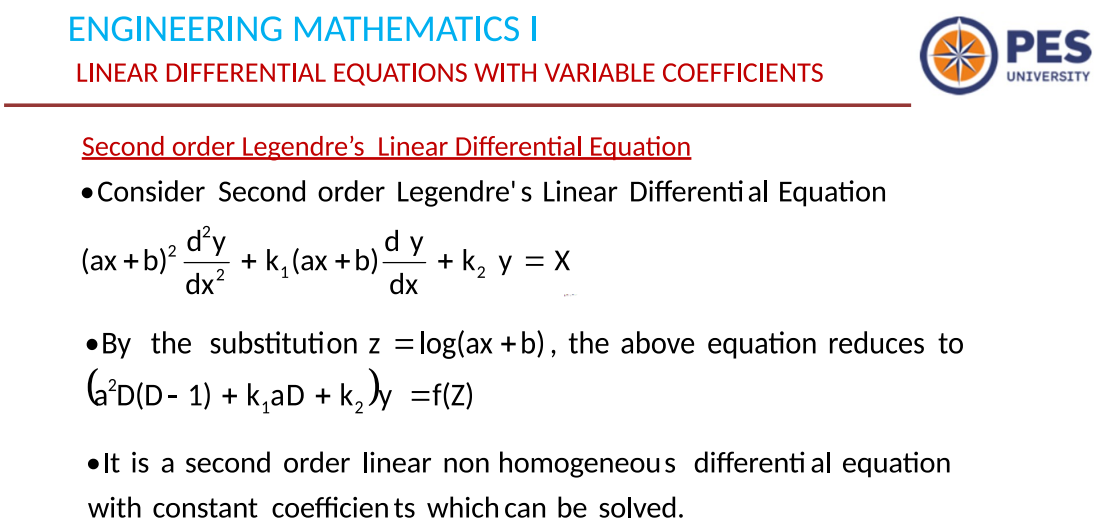

Figure G000124. Legendre s differential equation · source slide page 7.

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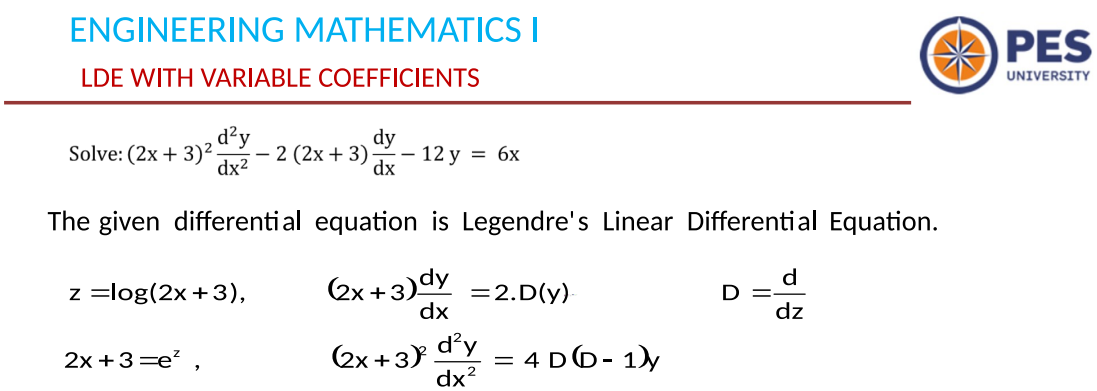

Figure G000125. Legendre s differential equation · source slide page 8.

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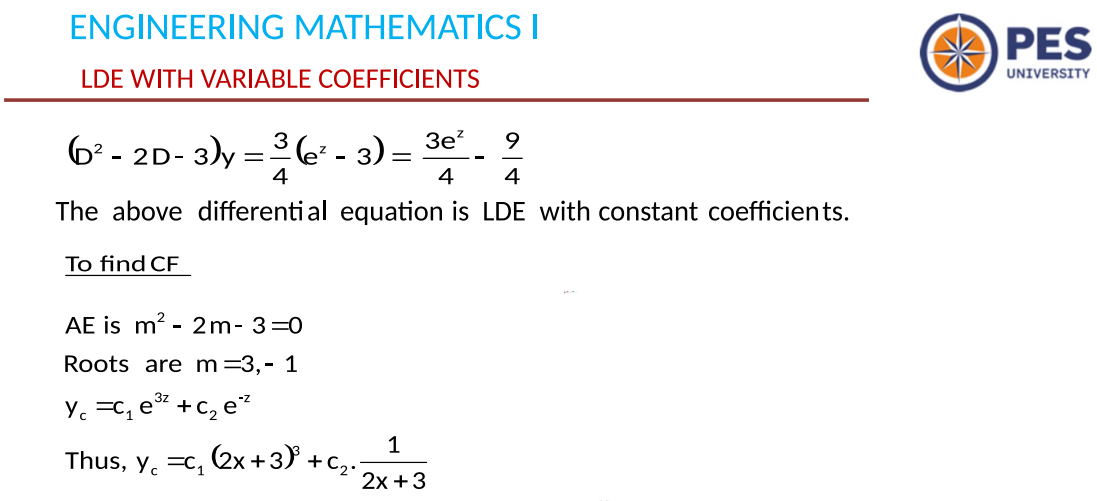

Figure G000126. Legendre s differential equation · source slide page 9.

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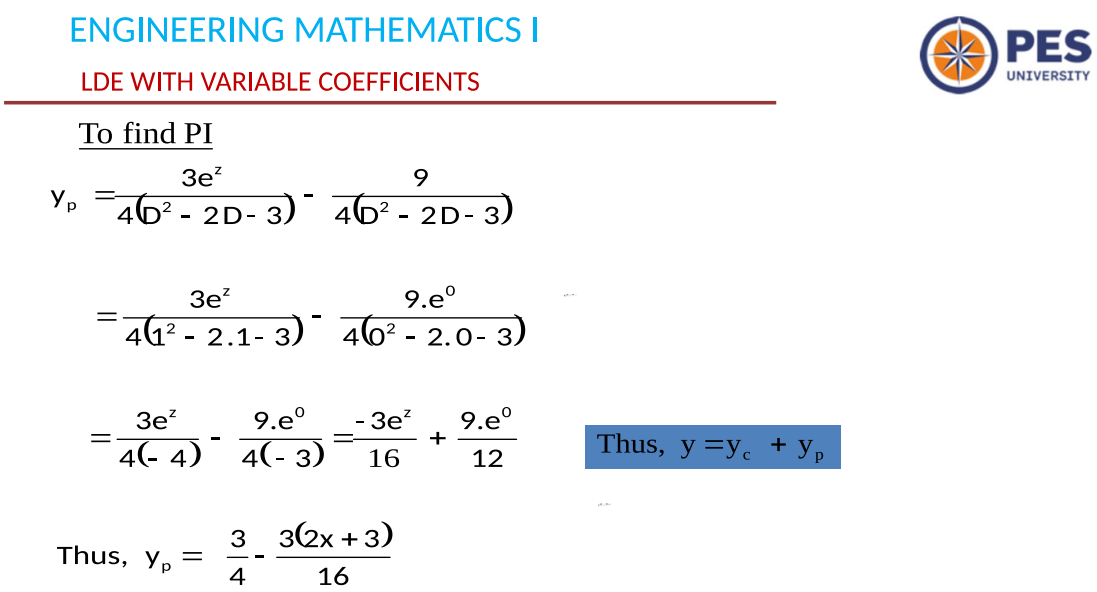

Figure G000127. Legendre s differential equation · source slide page 10.

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Figure G000128. Legendre s differential equation · source slide page 11.

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000617 · EV0000618 · EV0000619
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 2: HIGHER ORDER DIFFERENTIAL EQUATIONS SESSION: 9 SUB-TOPIC: Legendre's Linear Differential Equation Department of Science and Humanities EV0000620 · EV0000621 · EV0000622
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER LINEAR DIFFERENTIAL EQUATIONS CLASS CONTENT: Higher Order Linear Differential Equations With Variable Coefficients Legendre's Linear Differential Equation EV0000623 · EV0000624 · EV0000625
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I LINEAR DIFFERENTIAL EQUATIONS WITH VARIABLE COEFFICIENTS EV0000626 · EV0000627 · EV0000628
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I LINEAR DIFFERENTIAL EQUATIONS WITH VARIABLE COEFFICIENTS Legendre's Linear Differential Equation EV0000629 · EV0000630 · EV0000631
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I LINEAR DIFFERENTIAL EQUATIONS WITH VARIABLE COEFFICIENTS EV0000632 · EV0000633 · EV0000634
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I LINEAR DIFFERENTIAL EQUATIONS WITH VARIABLE COEFFICIENTS Second order Legendre's Linear Differential Equation EV0000635 · EV0000636 · EV0000637
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I LDE WITH VARIABLE COEFFICIENTS EV0000638 · EV0000639 · EV0000640
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I LDE WITH VARIABLE COEFFICIENTS EV0000641 · EV0000642 · EV0000643
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS I LDE WITH VARIABLE COEFFICIENTS EV0000644 · EV0000645 · EV0000646
11. Source slide page 11 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000647 · EV0000648 · EV0000649

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.420] Hello everyone, welcome back to the session on higher order linear differential equations. [00:00:06.948] In today's session, we are discussing about linear differential equations with variable coefficients. We are in fact discussing two different forms of linear differential equation with variable coefficient. The first one being Legendary's linear differential equation. [00:00:24.720] And the second one is Cauchy's linear differential equation. So in today's session, we are learning the general form of legendary linear differential equation and [00:00:36.656] Cauchy's linear differential equation and methods to solve Legendary's and Cauchy's linear differential equation. [00:00:45.152] Now, in the previous sessions, we learned methods to find the solution of linear differential equation with constant coefficients. We

learned methods to find the solution of both homogeneous and non-homogeneous linear differential equation with [00:01:02.896] Constant coefficients, but in today's session, we are discussing two different forms of linear differential equation with variable coefficients, which are also solved by reducing those differential equations into differential equations with variable. [00:01:20.464] Constant coefficient by suitable substitution. So a differential equation with variable coefficient is solved by reducing it to differential equation with constant coefficient. Now what is legendary scenear differential equation? See a differential [00:01:38.032] differential equation of the form. Some constant k naught into a x plus b whole power n into n th order derivative of y with respect to x plus k1 into a x plus b whole power n minus 1 into n minus 1 third derivative of y with respect to x. [00:01:54.768] plus kn into y where k i is k naught k1 k2 k three k n are constants and capital x [00:02:05.648] is a function of x is called legendary linear differential equation of order n provided k not not equal to zero. So [00:02:16.612] The general form of Legendary's differential equation is of this type. Now, in this differential equation, you just put A is equal to 1, B is equal to 0, B get Cauchy's differential equation. So what is Cauchy's differential equation? [00:02:35.776] K naught into a is one and b is equal to zero in legendary's equation. So it is x to the power n into n th order derivative of y with respect to x plus k1 into x to the power n minus one into n minus one third derivative of y with respect to x plus k n into y is the same. [00:02:53.424] equal to capital X where K0 K1 K2 K3 Kn are constants and capital X is a function of X is Cauchy's linear differential equation of order n. Just look at the two differential equations. Why I'm calling them as differential equations [00:03:10.992] with variable coefficient because the coefficients of the derivatives are functions of the independent variable x. [00:03:20.944] So now let us first learn this legendary linear differential equation in detail. So what is legendary's differential equation? This is an nth order legendary linear differential equation. So it is k naught into ax plus b whole par n into [00:03:38.608] n third or derivative of y with respect to x plus k1 into n by a x plus b whole power n minus 1 into n minus 1 third derivative of y with respect to x plus kn. [00:03:50.672] into y is equal to capital X where capital X, the right hand side function, is a function of the independent variable x. K0 K1 K2 K3 Kn are all constants. E is a legendary linear differential equation of order n. [00:04:06.848] Provided this k not not equal to 0. Now just look at the equation. See ax plus b whole par n into n th order derivative of y with respect to x. The degree of the coefficient should be equal to the order of the derivative in all the terms. [00:04:26.628] Look at the second term, ax plus b whole power n minus 1. The degree of the coefficient is n minus 1. And the order of the derivative is n minus 1. Look at the last term, it is k kn into y, that is a x plus b to the power 0 into dy by dx to the power 0, which is y. [00:04:46.532] So, in all the terms, the degree of the coefficient is equal to the order of the derivative. So, such a differential equation with variable coefficient is called legendary linear differential equation.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.168] of order n. Now how do we solve this differential equation with variable coefficients? See the differential equation with variable coefficients is also solved by reducing it to differential equation with constant coefficient. Now how do we reduce [00:05:20.816] Use this differential equation into differential equation with constant coefficient by a suitable substitution. So what is a substitution we do? We take ax plus b is equal to e power z r, z is equal to log a. [00:05:38.384] plus b. So now just look here. I'm introducing another variable z which is taken as a function of the independent variable x only. Z is equal to log ax plus b. This is a substitution I do [00:05:53.584] To convert this differential

equation with variable coefficients into differential equation with constant coefficients. Okay. Now I have z is equal to log ax plus b. Let me differentiate it with respect to x. A simple differentiation. We know what is differentiation of log? [00:06:11.216] It is 1 by. So it is dz by dx is equal to log differentiation is 1 over a x plus b into differentiation of the inner function a x plus b with respect to x, which is a. It is a by ax plus b. We should remember this dz by dx is equal to a by a x plus b. [00:06:28.784] B. How did we get this? I took the substitution z is equal to log ax plus b. Why did I take the substitution z is equal to log ax plus b? Because I want to reduce the differential equation with variable coefficients into differential equation. [00:06:46.352] equation with constant coefficient. So this is dz by dx which is a by a x plus b. Now consider dy by dx. So what is dy by dx? It is derivative of y with respect to x. [00:06:59.152] Because we have introduced another variable z, I can write d by dz as d by dz into dz by dx. How did I write this? Using chain rule. See because z is a function of x, d by dz can be written as d by dz into [00:07:16.816] into dz by dx using the concept over here I get dy by dx as dy by dz into dz by dx. Now substitute for dz by dx which is a by a x plus b. So it is dy by dz into a by a x plus b trans [00:07:34.384] Transposing this a x plus b to the left hand side, I get a x plus b into dy by dz is a into dy by dz [00:07:44.192] which is equal to a into d of y where I have called capital D as D by Dz. Please note down here. See so far I I was using the notation capital D for D by DX. [00:07:59.524] Now I am using the same notation D for D by Dz. I can use a different notation because we are familiar with this notation. Let me use the notation same notation for D by Dz, but please note here the independent variable over here is Z, it is not X, D by Dz. [00:08:19.312] A into d of y. Why this z? Because now I am changing the independent variable from x to z in the differential equation. See that this with the variable coefficient, dy by dx with the variable coefficient. [00:08:35.728] is transformed into the constant coefficient by the substitution z is equal to log ax plus b. Similarly, let me consider d2y by dx2, the second order derivative which is present in the differential equation. [00:08:53.360] d2 y by dx2. What is d2y by dx2? It is the derivative of dy by dx with respect to x, which is d by dx of dy by dx. Now substitute for [00:09:09.540] Dy by dx. So what is dy by dx from this? It is dy by dz. [00:09:14.864] into a by a x plus b. Using the above equation here I get a into x plus b into dy by dz. Now let me simplify this d by dx of a by a x plus b into dy by dz. How how do I simplify this? This is a product of two functions. So I use [00:09:33.136] Product rule. What is product rule in depth in differentiation? U into v. That is u into differentiate differentiation of v plus v into differentiation of u. So now let me use a product rule. So what I do it I consider this as first function, this is the second function. So second [00:09:50.768] function into derivative of first plus first into derivative of second. So that is equal to [00:09:56.228] Ah minus a square. How? I differentiated a by a x plus b with respect to a so what do I get? So I get minus a into a in a divided by a x plus b the whole square retained this d by by dz as it is plus a by a x plus b into d by d by

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:14.000] dx of dy by dz. Now so how do I simplify the second term? Again I use the same chain rule. [00:10:23.104] D by dz can be written as D by dz into dz by DX. So now you write this D by DX as D by dz into dz by DX. So that becomes. [00:10:36.740] Minus a square by x plus b the whole square into dy by d dz plus a by a x plus b into d by dz of d y by dz into dz by dx. Now this simplifies to [00:10:52.868] minus a square by a x plus b the whole square into d y by dz, a by a x plus b [00:10:58.928] into d by dz of dy by dz. What is dz by dx? Which is a by a x plus b. Now you can take a square by a x plus b

the whole square common factor from the two terms. So I get a square by x plus b the whole square [00:11:14.400] I am writing this term first. What is it? D by dz of dy by dz, meaning it is the differentiation of this dy by dz with respect to z, which is d2 y by dz2 minus minus dy by dz. So this is written in notation form. [00:11:32.080] As a square by a x plus b the whole square into this is capital D is the notation used for D by Dz. So D square has to be D2 by Dz 2. So it is D square. [00:11:45.840] Minus D of Y. So a x plus b take this x plus b the whole square to the left hand side. I take a x plus b the whole square into d2y by dx2 is equal to a square into d into d minus 1 of y. See again. [00:12:03.440] With variable coefficient. The derivative with variable coefficient is converted into derivative with constant coefficient, but x is changed to z. So what is d over here? Please remember it is d by dz, but not d by dx. Similarly, [00:12:21.104] We can prove that a ex plus b whole cube into d3 y by dx3 is equal to a cube into d into d minus 1 into d minus 2 of y. [00:12:34.448] So it's easy to remember when you have a x plus b to the power 3, it is a cube into d into d minus 1 into d minus two. Three linear factors. D into d minus 1 into d minus 2 of y. [00:12:48.388] The order remains the same. It is third order. Third order here. D3y by dx3 is d into d minus 1 into d minus 2 of y, where d is equal to d by dz. Now, substituting all these back in the given legendary differential equation, we get. [00:13:09.220] A x plus b dy by dx is a dy, a x plus b the whole square d 2 y by dx2 is a square into d into d minus 1 of y, a x plus b the whole cube d three, y by d x3 is a cube into d into d minus 1 into d minus 2 of y. Using these results in the legendary. [00:13:26.992] General differential equation, we get linear differential equation with constant coefficients, which can be solved by the methods discussed so far. How are we getting differential equation with constant coefficient? Please look here. This is KK. [00:13:43.632] A k naught into x plus b whole power n into n th order derivative of y with respect s become some constant into d into d minus 1 into d minus 2 so far of y. So the entire differential equation gets transformed into differential equation. [00:14:01.360] With constant coefficients, which can be solved by the methods learnt before. So this is how we solve legendary's linear differential equation of any order n. Let us now consider legendary's differential equation of order 2. [00:14:20.420] Second second order legendary scenear differential equation. Consider so it is a x plus b the whole square, d2 y by dx2 plus k1. [00:14:30.352] into a x plus b dy by dx plus k2 of y is equal to capital X. So this differential equation is legendary as linear differential equation of order 2. Why? See here [00:14:42.432] The order of the differential equation is 2. Ax plus B whole square into the second order derivative of y with respect to x. Ax plus b to the power 1 into first order derivative of y with respect to x. So in fact, this is written as a x plus b to the power 0.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:00.080] Into dy by dx to the power 0, which is y. So, in all the three terms, the degree of the coefficient is equal to the order of the derivativity. And the coefficients are pure functions of x. [00:15:16.356] So, and right hand side is a function of it. So, it is a second order legend raise linear differential equation. Now, how do we solve a second order legendary differential equation? Again, by reducing this differential equation into differential equation with constant. [00:15:34.128] coefficients by doing the substitution z is equal to log ax plus b. So what happens to the differential equation? When you put z is equal to log ax plus bc, this a x plus b the whole square d2 y by dx2 becomes a square into d into d minus one of [00:15:51.696] Y plus K1 into A into D of Y plus K2 of Y. [00:15:57.264] is equal to is equal to r here you should note that the right hand side was a function of x in the legendary general form. Now when you are reducing the differential equation into differential equation with constant

coefficient because you are changing the independent variable [00:16:14.896] From X to Z, RHS also should be a function of Z only. So this capital X is written in terms of Z by using the substitution Z is equal to log Ax plus B. So this entire equation is now is a [00:16:32.464] Differential equation in which Z is the independent variable and Y is the dependent variable. Now see the equation to differential equation with constant coefficients and it can be solved by the methods which we learned in the previous sessions. Now [00:16:51.524] Cauchy's linear differential equation. So what is Cauchy's linear differential equation? An n third or Cauchy's linear differential equation is of the form x to the power n into n third derivative of pi with respect to x plus some constant k1 into x to the power n minus one into n minus [00:17:09.296] one third derivative of y with respect to x plus kn into y is equal to capital x where all your constants k1 k2 k three kn are constants and x is a function of x. So this is [00:17:23.920] Cauchy's linear differential equation of order n. See in this differential equation also please note the degree of the coefficient is n and the order of the derivative is n. [00:17:35.204] The degree of the coefficient is n minus 1, order of the delimitative is n minus 1. So with the the final term, the last term. So this is a Cauchy's linear differential equation of order n. But how did we get this Cauchy's linear differential equation? It is a special [00:17:52.976] case of legendaries linear differential equation in which a is equal to 1 and b is equal to 0. The legendary differential equation was a x plus b whole power n into n th order derivative of y with respect to x. Just put a is equal to 1, b is equal to 0 [00:18:10.544] We get Cauchy's linear differential equation. Now see in the differential equation also all the terms the degree of the coefficient is equal to the order of the derivative. Now how do we solve such a linear differential equation with variable coefficient? The same concept is used. [00:18:30.692] So it is reduced to linear differential equation with constant coefficients by the substitution x is equal to e power z, rz is equal to log x. [00:18:42.756] There in Letch and Grace equals Z is equal to log Ax plus B. Here Z is equal to log x. So differentiate it. When you differentiate, you get bz by dx as 1 by x. So similarly, [00:18:54.112] Where if you do the derivation the same way, we get x into dy by dx as dy by dz, that is d of y where capital D is equal to d by dz. [00:19:07.104] Now similarly we get x square into d2 y by dx2. I am not doing the derivation for this because it is it is the same way as we did for legendary's differential equation. So x square into d2 y by dx2 is d into d minus 1. [00:19:24.784] Y where d is equal to d by dz. x cube into d3 y by dx3 would be d into d minus 1 into d minus 2 of y. See the [00:19:36.368] Coefficient, the variable coefficient is converted into a constant coefficient term. Substituting these equations in the [00:19:45.808] Given Cauchy's differential equation, that is the general form of Cauchy's differential equation, the equits reduced to differential equation with constant coefficients, and the equation can be solved by the methods learnt in the previous sessions. Now let us discuss uh

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:03.472] Cautious differential equation of order two. [00:20:06.628] How does it look like? So it is of the type x square into d2 y by d is 2 plus k1 into x into d y by dx plus k2 into y is equal to capital X where capital X is a pure function of the independent variable. See this is the order of the differential equation is 2. [00:20:24.400] and it it is with variable coefficient and it is of the form x square into d2 y by dx2 x to the power n 1 into dy by d x to the power 1 x to the power 0 into dy by dx to the power 0. So it is [00:20:39.776] Cautious linear differential equation of order 2. Now how do we solve such a differential equation? The same way we do the substitution, z is equal to log x. If you do the substitution, z is equal to log x, x into dy by dx becomes d of y where [00:20:57.424] D is equal to

d by dz, x square into d2 y by dx2 becomes d into d minus 1 of y. If you substitute in the differential equation, the differential equation gets converted into differential equation with constant coefficient. See, please note here. [00:21:15.008] This function of x also should be transformed to function of z using this substitution. Z is equal to log x. Now. [00:21:24.196] This is a simple second order linear differential equation with constant coefficients, which can be solved by the methods which we learnt in the previous session. So, this is about Legendary's and Cauchy's linear differential equation of order n and of order 2. [00:21:41.680] So we learned method to solve both Legendre's and Cauchy's linear differential equation in today's session. Thank you.

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Full transcript: 3428 normalized words, 87 timestamped segments, 1311.317333 seconds duration. Low-confidence segments flagged: 0.

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13 Legendre differential equations_Continued

This lesson is indexed from the source video titled Legendre differential equations_Continued. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000186 · EV0000187

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.292] Hello everyone, welcome back to the session on higher order linear differential equations. In this session, we will be solving problems on Legendre's linear differential equation and Cauchy's linear differential equation. [00:00:16.112] In the previous session, we discussed about two different forms of linear differential equation with variable coefficients. One is legendary's linear differential equation, and the other one is [00:00:30.224] Cauchy's linear differential equation. We also learn method to solve Cauchy's and Legendary's linear differential equation. Now because in this session we are solving problems on Legendary's LDE and Cauchy's LDE, let us have [00:00:47.856] A quick recap of the concept studied in the previous session. What is this legendary linear differential equation? Our differential equation of the form k naught into a x plus b whole par n into nth order derivative of y with respect to x plus. [00:01:05.200] K1 into a x plus b whole power n minus 1 into n minus 1 for the derivative of y with respect to x plus so on. [00:01:13.904] Plus kn of y is equal to capital X where capital X is a function of the independent variable X, K0, K1, K2, K3, Kn are constants is legendary's linear differential equation of order n. [00:01:31.536] See here, the degree of the coefficient is equal to the order of the derivative in all the terms. Now, how do we solve such a differential equation? So the differential equation with variable coefficient is transformed into differential equation with variable. [00:01:49.104] constant coefficients by the substitution z is equal to log a x plus b or e to the power z is equal to a x plus b or e to the power z minus b by a is equal to x. Now [00:02:04.932] In the previous session, we derived that a x plus b into dy by dx is equal

to a into d of y by capital. D stands for d by d dz. It is the differential operator with respect to z because [00:02:22.704] By using this substitution, we are changing the independent variable from X to Z. [00:02:31.460] We also derived a x plus b the whole square into d2 y by dx2 is a square into d into d minus 1 of y, and a x plus b the whole cube into d3 y by dx3 is a cube into d into d minus 1 into d minus 2 of y. [00:02:49.248] Using these equations back in the given legendary linear differential equation. The differential equets transformed into LDE with constant coefficients, and the differential equation is solved by the methods which we learnt in the previous. [00:03:06.832] Sessions. Now, what is the second form of LDE with variable coefficient? We learned that is Cauchy's linear differential equation. Now, what is this Cauchy's linear differential equation? We learned that Cauchy's linear differential equation is a special case of [00:03:24.400] Legendary C wherein a is equal to 1 and B is equal to 0. So the differential equation of the type k naught into x par n into n third derivative of y with respect to x plus k1 into x to the power n minus 1 into n minus 1 third. [00:03:41.968] derivative of y with respect to x plus kn into y is equal to capital X where K0 K1 K2 K3 Kn are constant and K0 not equal to 0 and the right hand side function is a pure function of X Cauchy's LDE of all [00:03:59.536] Now, how do we solve this differential equation? The same method is used. So the differential equation. [00:04:06.736] With variable coefficient is reduced to differential equation with constant coefficients by using the substitution z is equal to log x or x is equal to e power z and x into dy by dx is equal to d of. [00:04:24.368] X square where capital D stands for d by dz. X square into d2 y by d2 y by dx2 is d into d minus 1 of y. X cube d3 y by dx3 is d into d minus 1 into d minus 2 of y. [00:04:41.936] Using these equations back in the differential equation, the equation gets transformed into LDE with constant coefficients, and the equation can be solved by the methods which we learned in the previous sessions. Now let us apply the method which is learned to solve problems on [00:04:59.520] and legend raised linear differential equation. Now let us consider a question x cube into d3 y by dx3 plus two x square into d2 y by dx2 plus two y is equal to 10 into x plus 1 by x.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:14.096] So what is this differential equation? It is LDE with variable coefficients. And we learned only two different forms of LDE with variable coefficients. So what is this type? This is Cauchy's linear differential equation of order 3. Now, how do we solve this differential equation? [00:05:31.760] equation by doing the substitution z is equal to log x r x is equal to e power z. We also derived x into dy by dx is equal to df y x square capital D stands for d by dz, x square into d2 y by dx2 [00:05:49.328] D into d minus 1 of y x cube d3 y by dx is d into d minus 1 into d minus 2 of y. Now using these equations back in the given causation d of order 3, we get d into d minus 1 into d minus 2 of [00:06:06.896] plus 2 into d into d minus 1 of y plus 2 of y is equal to 10 into r. Here RHS also has to be converted into a function of z because we are changing the independent variable from x to z. [00:06:23.216] So what is x? X is e power z. What is 1 by x? It is e power minus z. So this is [00:06:29.648] A differential equation in which y is the dependent variable and z is the independent variable. Now let us simplify this differential equation. So we get see there are three linear factors multiply you get dq minus three d square plus two d plus. [00:06:46.112] Multiply these two terms 2 d square minus 2 d plus 2 of y is equal to 10 into e power z plus e power minus z. Simplifying it further, I get dq minus d square plus two of y is equal to 10 into e power z plus e power minus z. What is this differential equation? [00:07:03.760] It is LD of order 3 with constant coefficient. Now, because RHS of the differential equation is not equal to 0, it is a non-homogeneous LD with constant

coefficients and of order 3. Now. [00:07:22.736] How do I solve this differential equation? I know that the non-homogeneous LDE with constant coefficients has two components. One is the complementary function, and the second component is the particular integral. Now I need to find the complementary function. [00:07:40.368] function and particular integral of this differential equation. Now next to find complementary function. So how do I find complementary function? I need to write the associated auxiliary equation. So auxiliary equation is mq minus m square plus 2 is equal to 0. Is it [00:07:57.936] cubic equation three roots now how do I find the roots see just by infect inspection I I I can find that m is equal to minus one is one root m is equal to minus one because it is minus one [00:08:12.336] And this is minus 1 minus 2 plus 2 equal to 0. Once you get one root, the other two roots are found by factorizing. So just divide this entire expression mq minus m square plus 2 with m plus 1, you get the quadratic equation. So all the quadratic equations. [00:08:29.968] The roots are m is equal to minus 1, comma, 1 plus r minus i. So the three roots, just analyze the roots. One is a real root and the other two roots are complex. Now how do you write the complementary function for this? One real root and two complex roots. [00:08:47.536] The complementary function y c is equal to c 1 e to the power minus 1 into z. Ah, here it is not x because the independent variable in the differential equation is z. [00:09:00.644] C1 E power minus Z plus. For this now I'm writing P par Z into C2 cos 1 Z plus C3 psi 1 Z, but [00:09:11.504] The complementary function has to be written in terms of x itself. So by use using that substitution, so what is e power minus z? 1 by x. What is e power plus z? It is x. So it is c1 by x plus e x into c2 cos of log x plus c3 c. [00:09:29.200] Sign of log x. So this represents the complementary function of the given Cauchy's Lia differential equation. See here you should note that the final answer should be in terms of x itself. [00:09:43.396] So use the substitution, get the expression in terms of X and report that as the answer. [00:09:49.712] Now this is just the complementary function. Next is to find the particular integral. So what is particular integral? It is 10 into e power z plus e power minus z divided by d q minus d square plus 2.

EV0000190 · EV0000191 · EV0000192 · EV0000193

Transcript block 3 (source interval 000600-000899 seconds): [00:10:04.128] Now let us split it into two types. It is 10 into e power z by d cube minus d square plus 2 plus 10 into e power minus z by d cube minus d square plus 2. [00:10:15.012] See, I know that d is equal to minus 1 is a root of dq minus d square plus 2. So even if I substitute d is equal to minus 1 over here, second term, I get the denominator 0. So I'm using the [00:10:29.792] So if f of a is equal to 0, what I need to do? I need to differentiate this, put 1x over here. Let us consider the first return. See here. We know that d is equal to 1 is not a root, so the denominator is definitely not equal to 0. So when you put a is 1 here, when you substitute a is equal to 1, you get the equation. [00:10:47.472] 1 minus 1 plus 2, which is source 2. So this simplifies to 10 e to the power z by 2. Now for the second term because the denominator is 0, what I need to do. So the first term is 10 e power z by 1 cube minus 1 square plus 2. Here I am writing z in 2. [00:11:07.268] 10 e power minus z by dq. This dq minus d square plus 2 has to be differentiated with respect to d. It is 3d square minus 2d. See it is z into. Please note that it is not x into. [00:11:21.156] Because the independent variable here is z, but not x. So now how do I proceed further to simplify this? I'll do the substitution d is equal to minus 1. So what do I get? I get 10 e power z by 2 plus z into 10 e power minus z by 3. [00:11:38.928] into d is equal to minus 1 minus 1 square minus 2 into minus 1. The denominator is not equal to 0's. Now it simplifies to 10 e power z by 2 which is 5 plus 5 into 10 by 52 into e power minus z. Now we know that the particular [00:11:56.496] integral also has to be written in terms of x. So use the

transformation z is equal to log x and e par z is equal to x. So I get particular integral as 5x plus 2 by x into log x. So now what is the [00:12:12.240] Complete solution of the differential equation it is C F plus PI. So when you are reporting the final answer it the function should be in terms of X itself. It is it should not be in terms of Z. [00:12:26.660] Okay, now let us consider another question. Solve 2x plus 3 the whole square, d2y by dx2 minus 2 into 2x plus 3, dy by dx minus 12y is equal to 6x. [00:12:39.044] So it is a second order LDE with variable coefficient and it is legendary cell DE afforded to. [00:12:47.396] 2x plus 3 the whole square into second order derivative minus 2 into 2x plus 3 to the power 1 into dy by dx minus 12 into y is equal to 6x is the independent variable. Now how do I solve this legendary linear difference? [00:13:05.168] differential equation. So I need to do the substitution z is equal to log 2x plus 3 r 2x plus 3 is equal to e power z. See log to the base e. So this is 2x plus 3 is equal to e power z or x is equal to e power z minus 3 by 2. [00:13:23.664] We also derived a 2x plus 3, that is a x plus b into dy by dx is e into d of y. So 2x plus 3 into dy by dx is equal to 2 into d of y. I need the second one, 2. [00:13:37.424] Where capital D stands for D by Dz. This has to be written where D is equal to D by C D Z. It is not D by DX. Now 2x plus 3 the whole square into D2 Y by Dx2 is 4 into D into D minus 1. Now why? [00:13:55.120] Now using all these equations in the given legendary linear differential equation of order 2, we get 4 into d. What is this? 4 into d into d minus 1 of y minus 2 into 2. [00:14:09.264] The into d of y minus 12 of y is equal to 6 into r. x has to be written in terms of z using this equation. [00:14:17.584] E power z minus 3 by 2. Now this simplifies to 4 t square minus 4 d minus 4 d is minus 8 t minus 12 of by is equal to 3 into e to the power minus z minus 3. What is this differential equation? It is a linear [00:14:35.216] differential equation with constant coefficients. Now how do you solve this? So I need to find the complementary function and particular integral. [00:14:46.596] Now I am splitting the term on the right hand side as 3 par z by 4 minus 9 by 4. Now this is a differential equation with constant coefficients. To find complementary function, I need to write the associated auxiliary equation, which is m square minus 2m minus 3 is equal to 0.

EV0000194 · EV0000195 · EV0000196 · EV0000197

Transcript block 4 (source interval 000900-001199 seconds): [00:15:06.980] To find the roots, the roots are 3 comma minus 1. So the complementary function is c 1 e to the power 3z plus c 2 to the power minus z. It is in terms of z. But I need to reuse this equation 2x plus 3 is equal to e. [00:15:21.616] e power z and write it in terms of x r x is equal to e power z minus three by two. So use that it is c1 into e to the power three z is two e x plus three the whole cube plus c2 into e to the power minus z is one by two e x plus three. This is the complementary function of the given less [00:15:39.312] generalized linear differential equation next is to find the particular integral. [00:15:44.288] So the particular integral is 3 power z by 4 into d square minus 2d minus 3 minus 9 divided by 4 into d square minus 2d minus 3. How did I write this? The numerator 3 to the power z minus 9 is split it is split into two terms. [00:16:02.032] Now, how do I simplify the first one? It's very simple. I just have to replace D by 1. So it is. [00:16:11.776] Because it is e to the power a is a is 1, just put in the place of d you just put 1. So I get 1 square minus 2 minus 3, which is definitely not equal to 0. Now how do I simplify the second term? Constant by f of d. [00:16:27.428] So the constant can be expressed in terms of exponential function as what? 9 into e power 0z. [00:16:35.236] divided by 4 into d square minus 2 d minus 3. Replace D by 0. So if you have constant k by f of d it is k by f of 0 provided f of 0 not equal to 0. So this is 3 power z divided by see here I have read [00:16:53.008] D by 1 in the second term I have replaced D by 0. So this is because 1 is written as e to d by 0 and here because it is 0 into Z, so it is I have to just replace D

by 0. [00:17:09.220] So this is actually I could have written this as e to the power 0 z, which is again equal to 1 itself. So it is 3 e power z by 4 into minus 4 minus 9 by 4 into minus 3, which simplifies to 3 by 4 minus 3 by 16 e to the power z. [00:17:27.488] But you need to write epar z in terms of s, which is 2x plus 3. So this is the particular integral of the given differential equation 3 by 4 minus 3 by 16 into [00:17:39.152] Uh into 2x plus 3 is the particular integral. So the general solution would be y is equal to yc plus y p. Please note that when you are reporting the final solution, the complementary function and particular integral should be a function of x, which is the independent. [00:17:56.816] Variable of the given differential equation. Okay, now let us consider another question. Find the general solution. Now how do you read this? It is x into fourth order derivative of y with respect to x plus 10 into third order derivative of y with respect to x is equal to 0. [00:18:16.516] By just looking at the differential equation, we say that it's a homogeneous differential equation. And but this differential equation doesn't resemble [00:18:25.792] I uh either Cauchy's or Legendary's linear differential equation, but this equation can be brought to Cauchy's linear differential equation. [00:18:37.796] By multiplying the entire differential equation with x to the power 3 y. See it is x into 4th order derivative of y with respect to x. I want because the order of the derivative here is 4, I want x to the power 4 here. [00:18:55.364] And it is 10 into x to the power 0 into third order derivative of y with respect to x. Because it is third order derivative, the coefficient of the [00:19:05.952] Uh the degree of the coefficient should be 3. So if I multiply the entire equation with x cube, it gets converted into different Cauchy's linear differential equation. See multiplying the entire equation with x cube, it becomes x to the power 4 into d4 y by d. [00:19:23.600] X4 plus 10 x cube into third order derivative of y with respect to x is equal to 0. So this is Cauchy's linear differential equation of order 4. Now how do I solve this Cauchy's linear differential equation? Again by doing the substitution. [00:19:42.820] x is equal to e to the power z, or z is equal to log x. [00:19:47.008] Now I need to write this x cube d3 y by dx3 x to the power d4 d y by dx4 into what? X into d y by dx is equal to d of y where d is equal to this has to be mentioned where capital D is equal to d by dz.

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:04.656] X square into D2 Y by Dx2 is D into D minus 1 of Y. I need up to this four therivative. X cube into D3 Y by Dx3 is D into D minus 1 into D minus 2 of Y and X to the power 4 into 4th order derivative of y with respect to X is D into D. [00:20:22.224] into d minus 1 into d minus 2 into d minus 3 of y. Now I need to use all these equations in the given differential equation to convert the given differential equation into differential equation with constant coefficients. [00:20:39.044] So what do I get? I get this term d into d minus 1 into d minus 2 into d minus 3 of y plus 10 into d into d minus 1 into d minus 2 of y is equal to 0. No need to multiply the linear factor. Let us take a common factor from the two terms. So what do I get? What are the common factors? [00:20:56.816] fact is d into d minus 1 into d minus 2 can be taken as common fact. So I get d into d minus 1 into d minus 2 what is remaining the first term d minus 3 plus 10 [00:21:08.224] Of y is equal to 0. Simplification would give d into d minus 1 into d minus 2 into d plus 7 of y is equal to 0. Now very simple. Now I need to uh this because this is a homogeneous LDE with constant coefficient and [00:21:25.872] I do I need not have to find the particular integral. I need to find only the [00:21:32.432] Complementary function and I know that the complementary function represents the complete solution of the homogeneous linear differential equation. So I need to set up the auxiliary equation. The auxiliary equation is m into m minus 1 into m minus 2. [00:21:50.064] To m plus 7 is equal to

0. The roots are 0, comma 1, comma 2, comma minus 7. So all the four roots are real and distinct. So let me write the [00:22:02.976] complementary function which itself is the solution of the differential equation. So I have written I have not written y c I have written y because it is a homogeneous differential equation. So y is equal to c 1 e to the power 0 z plus c2 to the power 1 z plus c three to the power 2 z plus c [00:22:20.656] C4 e to the power minus Z, but the final solution has to be in terms of X itself. So write it as the general solution is C1 e to the power 0 z is 1 plus c2 into x plus c3 into x square plus c4 x to the [00:22:38.224] This is how we find the general solution of Cauchy's and Legendary's linear differential equation. [00:22:50.276] So, we learn method to find the general solution of Cauchy's and Legendary's linear differential equation. Thank you.

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Full transcript: 3793 normalized words, 92 timestamped segments, 1378.282667 seconds duration. Low-confidence segments flagged: 0.

EV0000186 · EV0000187 · EV0000188 · EV0000189

14 Variation of Parameters method_Continued

This lesson is indexed from the source video titled Variation of Parameters method_Continued. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000206 · EV0000207

Graphs and diagrams

Figure G000129. Variation of Parameters method · source slide page 1.

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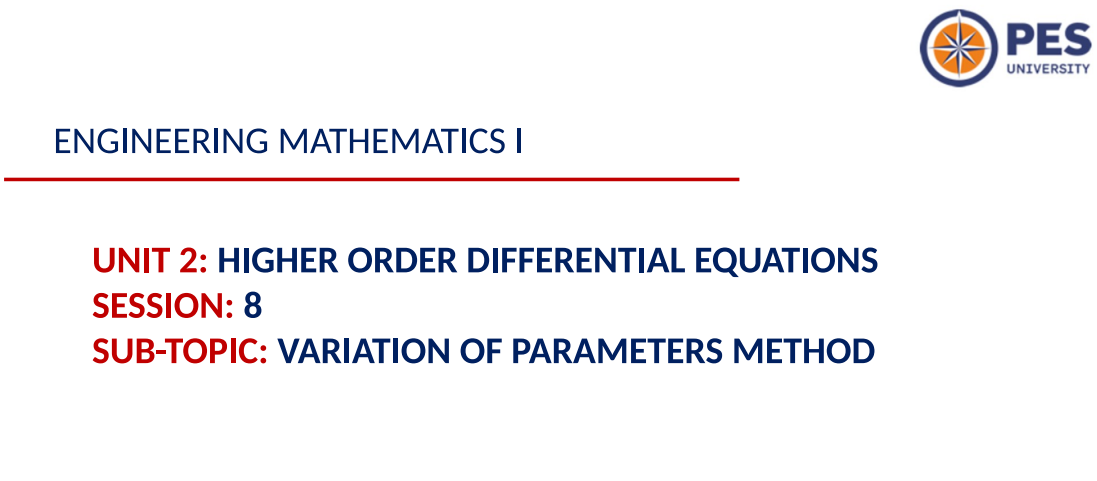

Figure G000130. Variation of Parameters method · source slide page 2.

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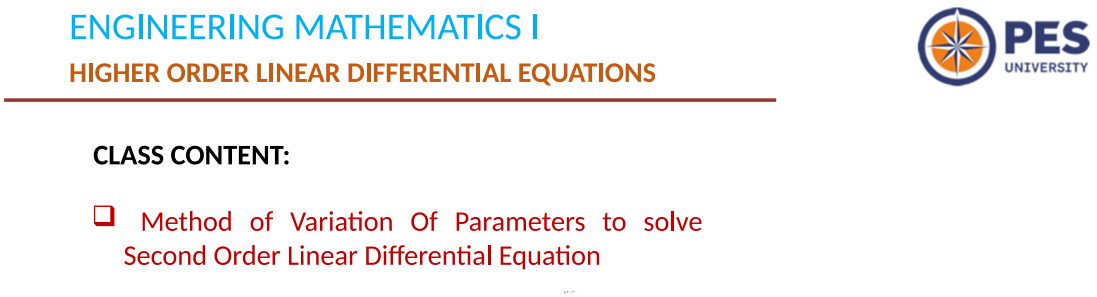

Figure G000131. Variation of Parameters method · source slide page 3.

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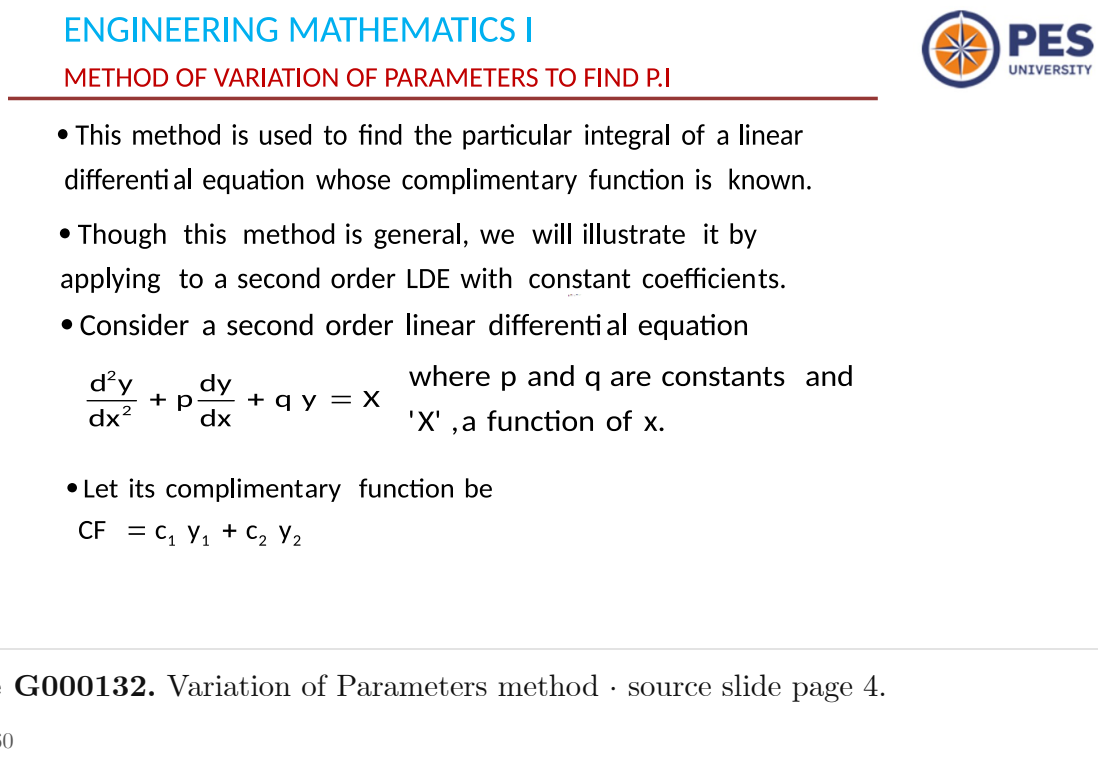

Figure G000132. Variation of Parameters method · source slide page 4.

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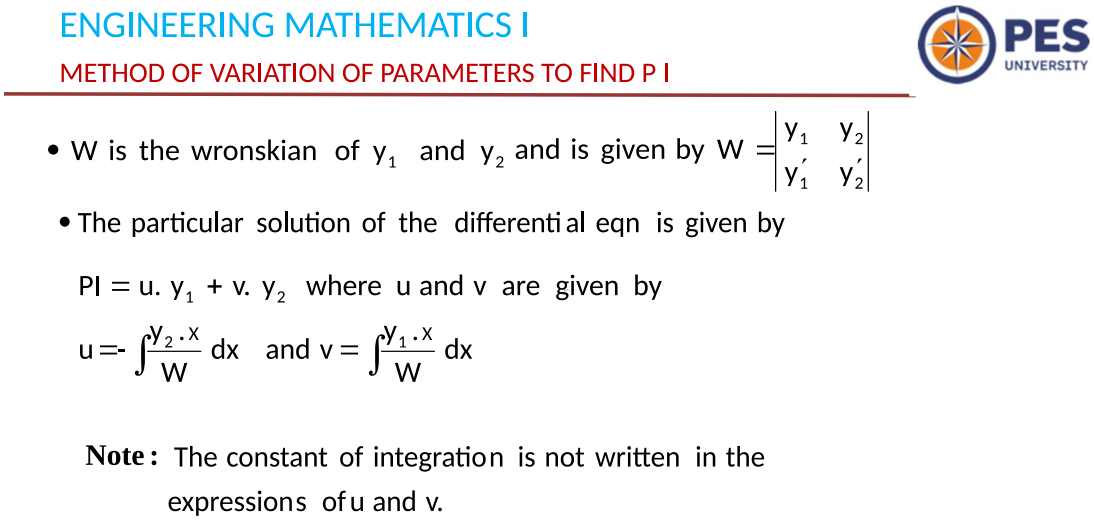

Figure G000133. Variation of Parameters method · source slide page 5.

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ENGINEERING MATHEMATICS I

METHOD OF VARIATION OF PARAMETERS TO FIND P I

- This method can also be used on differential equation with variable coefficient provided we know to find the general solution of the associated homogeneous equation.

- The coefficient of $\frac{d^2y}{dx^2}$ in the differential equation must be unity.If it is not unity ,make it one by dividing the differential equation by the coefficient.

Figure G000134. Variation of Parameters method · source slide page 6.

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ENGINEERING MATHEMATICS I

METHOD OF VARIATION OF PARAMETERS TO FIND P I

Find the particular integral by the method of variation of parameters :

$y'' - 2y' - 3y = x.e^{-x}$

$(D^2 - 2D - 3)y = x.e^{-x}$

STEP 1 To Find CF

AE is $m^2 - 2m - 3 = 0$

Roots are $3, -1$

$y_c = c_1 e^{3x} + c_2 e^{-x}$

$y_1 = e^{3x}, \quad y_2 = e^{-x}$

Figure G000135. Variation of Parameters method · source slide page 7.

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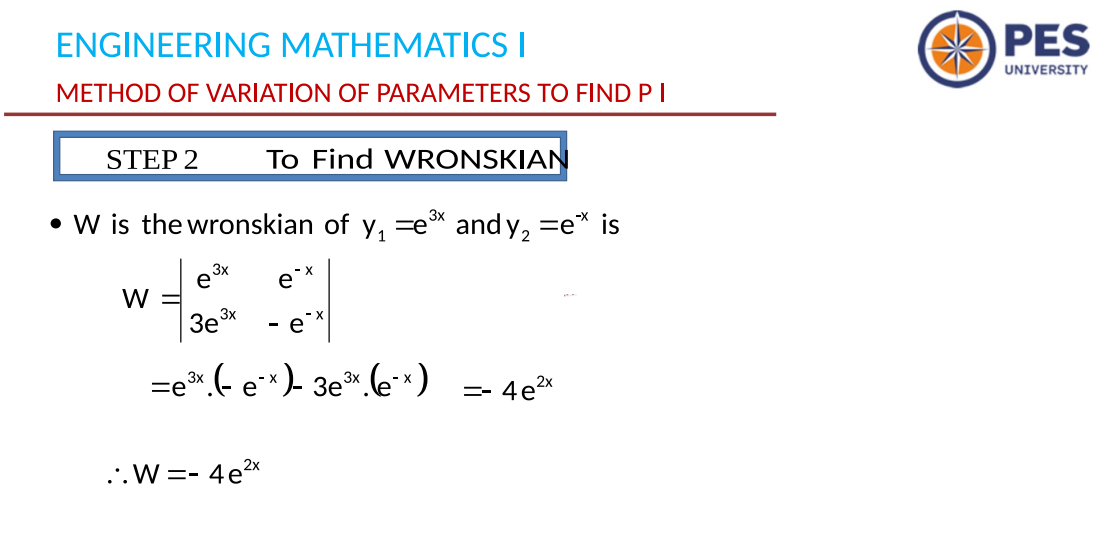

Figure G000136. Variation of Parameters method · source slide page 8.

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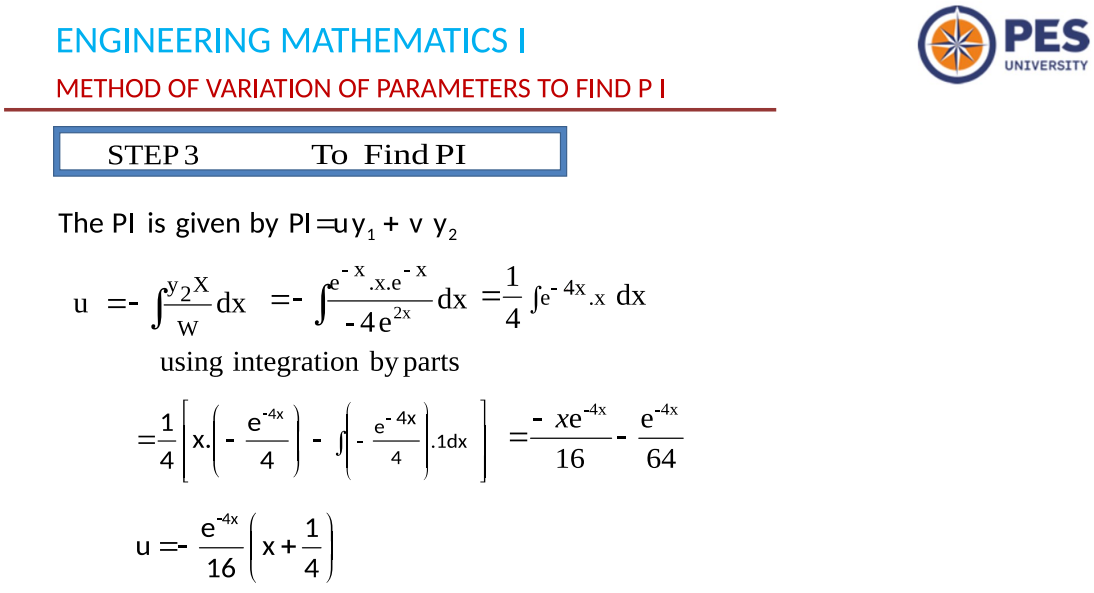

Figure G000137. Variation of Parameters method · source slide page 9.

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ENGINEERING MATHEMATICS I

METHOD OF VARIATION OF PARAMETERS TO FIND P I

STEP 3 To Find PI

$$v = \int \frac{y_1 X}{W} dx = \int \frac{e^{3x}.x.e^{-x}}{-4e^{2x}} dx = -\frac{1}{4}\int x\, dx$$

$$= -\frac{1}{4}.\frac{x^2}{2} \qquad v = -\frac{x^2}{8}$$

$$PI = -\frac{e^{-4x}}{16}\left[x + \frac{1}{4}\right].e^{3x} + e^{-x}\left(-\frac{x^2}{8}\right)$$

$$PI = -\frac{e^{-x}}{16}\left[x + \frac{1}{4}\right] - e^{-x}\left(\frac{x^2}{8}\right)$$

Figure G000138. Variation of Parameters method · source slide page 10.

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ENGINEERING MATHEMATICS I

Problems on Method Of Variation Of Parameters

1. Find the Particular Integral by the method of variation of parameters :

$(D^2 + 4)y = \tan 2x$

SOLUTION:

Step 1 : To Find y_1 and y_2

AE is $m^2 + 4 = 0$

Roots are $m = \pm 2i$

$y_c = c_1 \cos 2x + c_2 \sin 2x$

Here, $y_1 = \cos 2x$ and $y_2 = \sin 2x$

Step 2 : To Find Wronskian

Wronskian is given by

$$W = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix}$$

$$W = 2(\cos^2 2x + \sin^2 2x)$$

$$W = 2$$

Figure G000139. Variation of Parameters method · source slide page 11.

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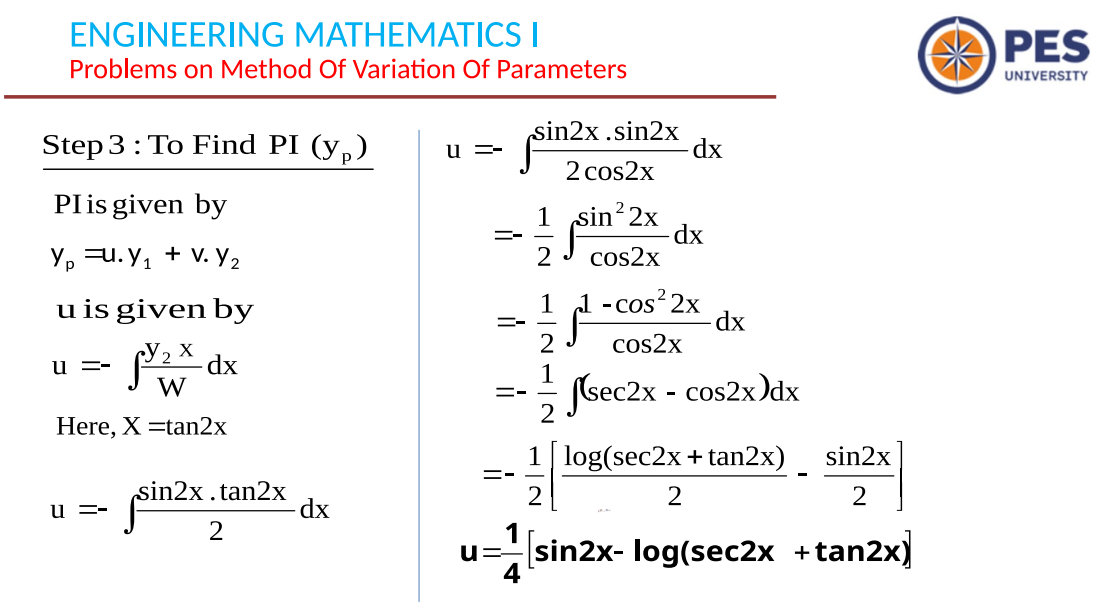

Figure G000140. Variation of Parameters method · source slide page 12.

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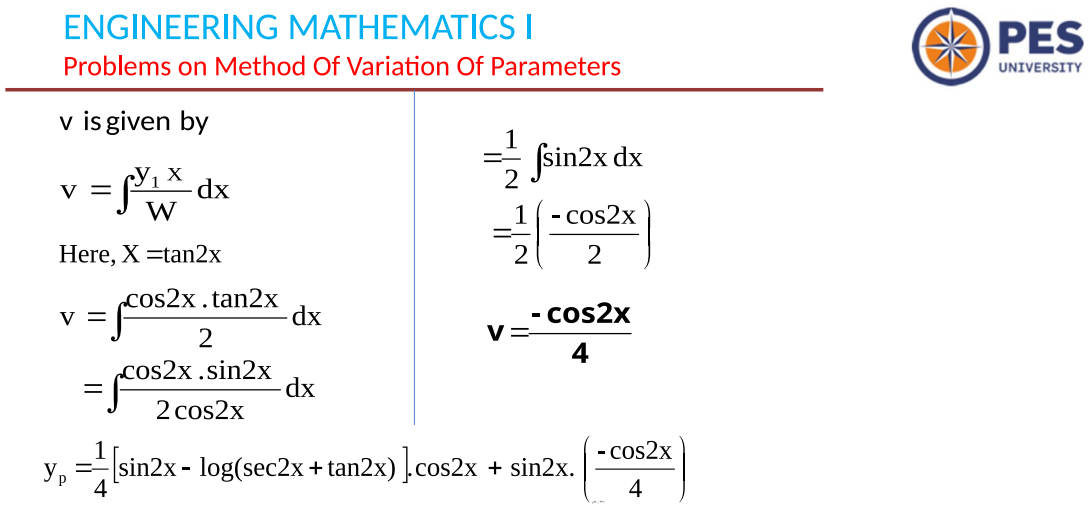

Figure G000141. Variation of Parameters method · source slide page 13.

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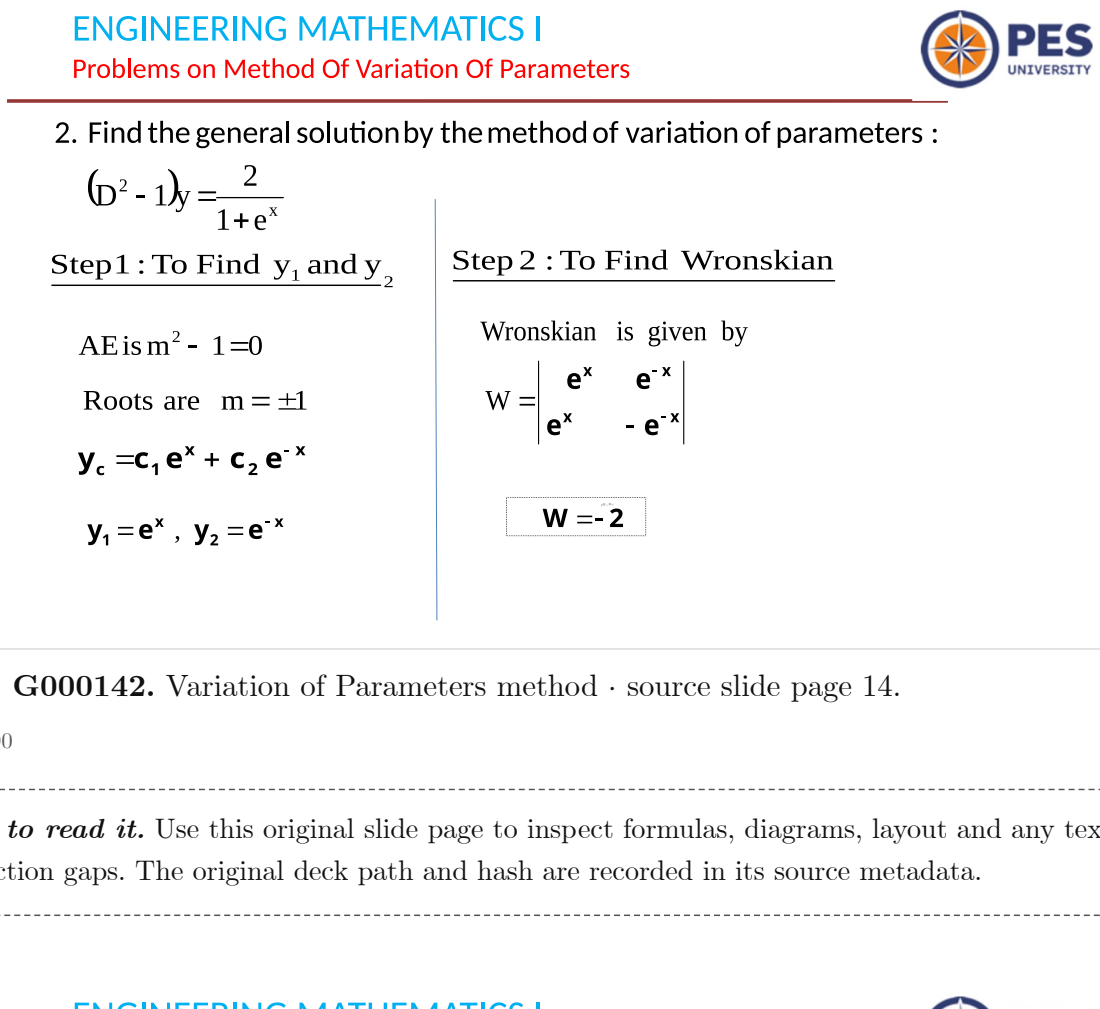

Figure G000142. Variation of Parameters method · source slide page 14.

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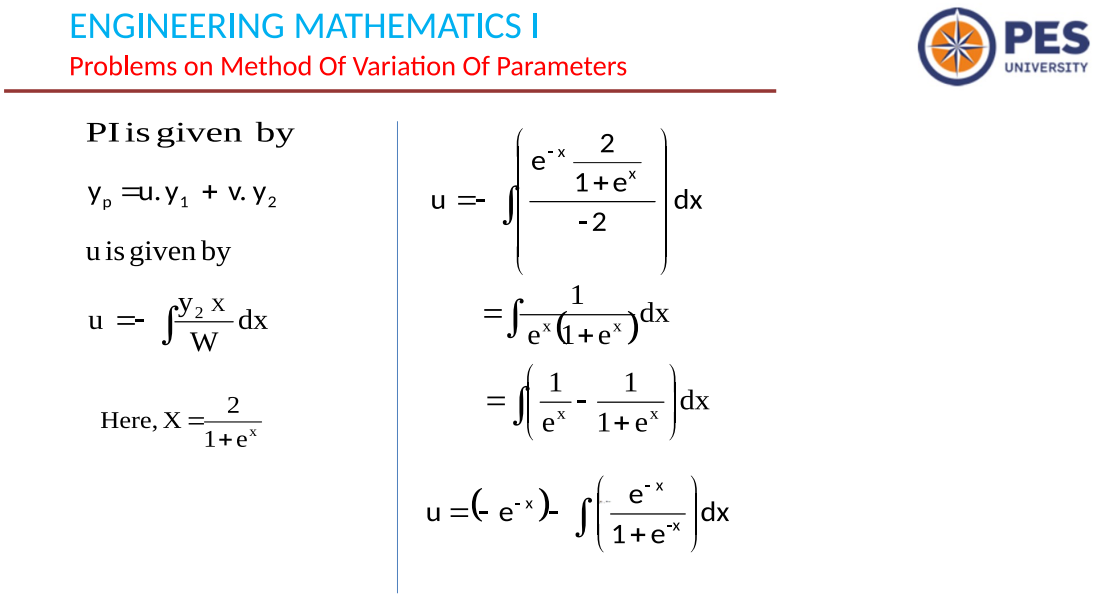

Figure G000143. Variation of Parameters method · source slide page 15.

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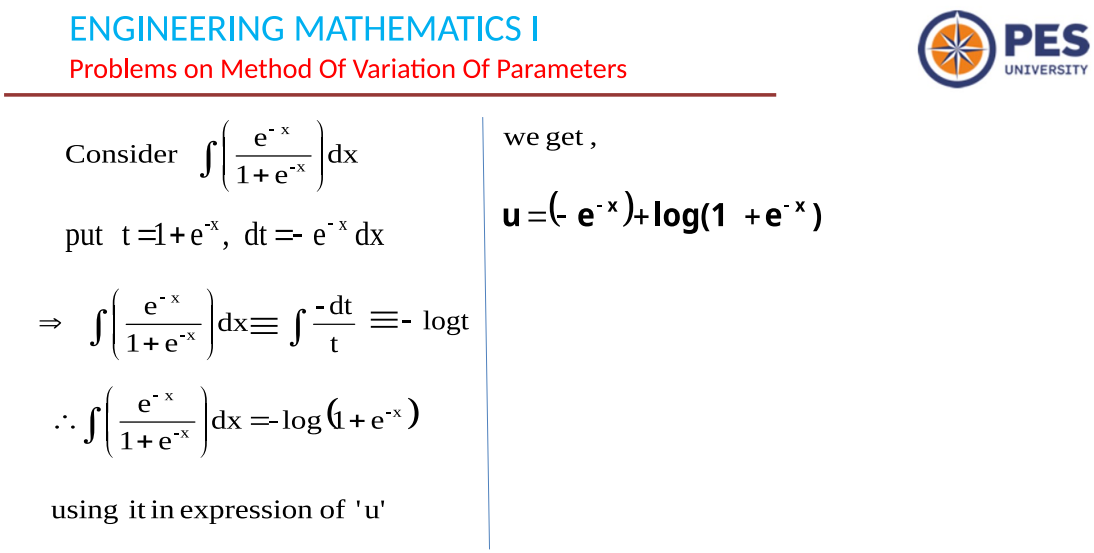

Figure G000144. Variation of Parameters method · source slide page 16.

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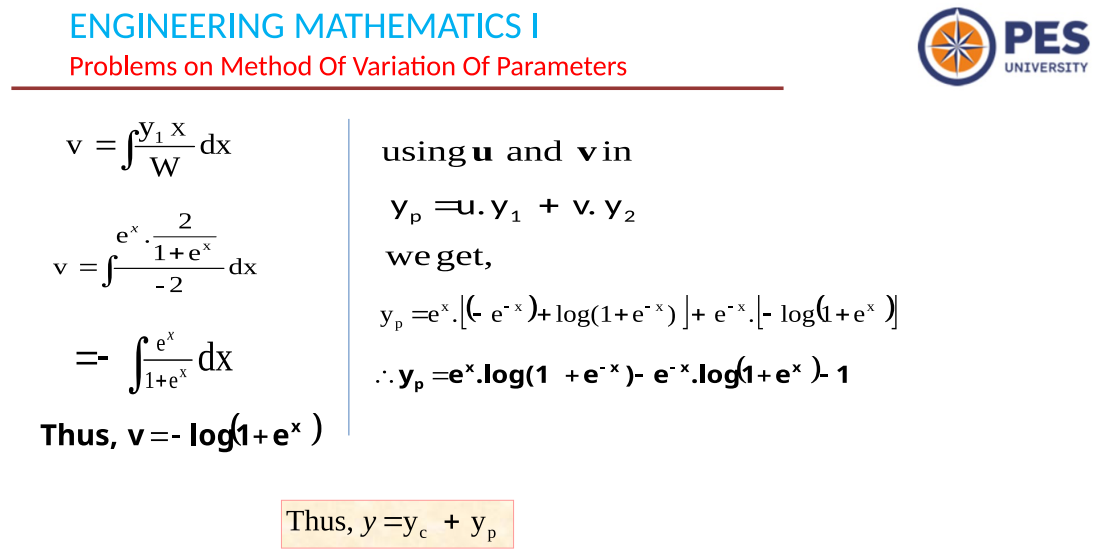

Figure G000145. Variation of Parameters method · source slide page 17.

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Figure G000146. Variation of Parameters method · source slide page 18.

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000650 · EV0000651 · EV0000652
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 2: HIGHER ORDER DIFFERENTIAL EQUATIONS SESSION: 8 SUB-TOPIC: VARIATION OF PARAMETERS METHOD Department of Science and Humanities EV0000653 · EV0000654 · EV0000655
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER LINEAR DIFFERENTIAL EQUATIONS CLASS CONTENT: Method of Variation Of Parameters to solve Second Order Linear Differential Equation EV0000656 · EV0000657 · EV0000658
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I METHOD OF VARIATION OF PARAMETERS TO FIND P.I EV0000659 · EV0000660 · EV0000661
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I METHOD OF VARIATION OF PARAMETERS TO FIND P I EV0000662 · EV0000663 · EV0000664
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I METHOD OF VARIATION OF PARAMETERS TO FIND P I EV0000665 · EV0000666 · EV0000667
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I METHOD OF VARIATION OF PARAMETERS TO FIND P I EV0000668 · EV0000669 · EV0000670
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I METHOD OF VARIATION OF PARAMETERS TO FIND P I EV0000671 · EV0000672 · EV0000673

9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I METHOD OF VARIATION OF PARAMETERS TO FIND P I EV0000674 · EV0000675 · EV0000676
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS I METHOD OF VARIATION OF PARAMETERS TO FIND P I EV0000677 · EV0000678 · EV0000679
11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS I Problems on Method Of Variation Of Parameters SOLUTION: EV0000680 · EV0000681 · EV0000682
12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS I Problems on Method Of Variation Of Parameters EV0000683 · EV0000684 · EV0000685
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS I Problems on Method Of Variation Of Parameters EV0000686 · EV0000687 · EV0000688
14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS I Problems on Method Of Variation Of Parameters EV0000689 · EV0000690 · EV0000691
15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS I Problems on Method Of Variation Of Parameters EV0000692 · EV0000693 · EV0000694
16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS I Problems on Method Of Variation Of Parameters EV0000695 · EV0000696 · EV0000697
17. Source slide page 17 (extracted text): ENGINEERING MATHEMATICS I Problems on Method Of Variation Of Parameters EV0000698 · EV0000699 · EV0000700
18. Source slide page 18 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000701 · EV0000702 · EV0000703

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.452] Hello, everyone, welcome back to the session on higher order linear differential equations. In this session, we are solving problems on method of variation of parameters. In the previous session, we learned a technique. [00:00:16.208] Called method of variation of parameters to solve second order linear differential equations. [00:00:23.840] With constant or with variable coefficients. Now let us apply the method which is learnt in the previous session to solve problems on second order linear differential equations. Second order linear differential equations are very important in science and engineering. [00:00:41.520] They have historically been the most thoroughly studied class of differential equations. Now let us apply the method to solve second order differential equations. [00:00:53.764] You consider a question. Find the particular integral by the method of variation of parameters. See, we learnt in the previous session that this method of variation of parameters is used to find the particular integral of a differential equation. [00:01:12.208] Provided we know the complementary function of the differential equation. So even to find the particular integral, we need to first find out the complementary function of the differential equation. So this method involves three major steps. [00:01:29.860] The first step being finding the complementary function. So we need to first find out y1 and y2 of the given differential equation. So that is the first step. [00:01:42.340] to find by 1 and by 2. So you first set up the auxiliary equation of the given non-homogeneous linear differential equation with constant coefficient. So what is the auxiliary equation? [00:01:55.136] M square plus 4 is equal to 0. So what are the roots? Plus r minus 2i. The

two roots are complex in nature. So how do we write the complementary function? The complementary function is given by see this is of the form alpha plus r minus i into beta. [00:02:12.784] Where alpha is 0 and beta is 2. So the complementary function is given by C1 cos 2x plus C2 psi. [00:02:21.600] 2x. This is this equation is of the form c1 by 1 plus c2 by 2 where y 1 is equal to cos 2x and y2 is equal to sine 2x. This is the first step to find y 1 and y2. [00:02:36.932] The second step being finding the Ron scale of y 1 and by 2. So we know that the rons scale of y1 and by 2 is given by the formula determinant y1 by 2. Y1 dash y2 dash. [00:02:54.160] Here y1 is cos to x and y2 is sine to x. So the run scale is determined cos to x sine to x. That is the first row. [00:03:03.104] And the second row is differentiation of cos to x is minus 2 sine 2x and differentiation of sine 2x is 2 cos 2x. Now expand the determinant, it is 2 cos square to x plus 2 sin square 2x, which is equal to 2, because cos square 2x plus sin squared 2x. [00:03:20.848] This is 2. The Ran scale of y 1 by 2 is 2. The third step being finding the particular integral. So the particular integral is given by u into y 1 plus b into y2. [00:03:36.004] where u is given by minus integral y 2 into x divided by w dx. So what is y what are y 1 by 2 which we found out. [00:03:49.088] y1 is cos to x and y2 is sine to x. W is the r scale and x is the rhs function of the given differential equation. Now [00:04:02.212] Here in this in the given differential equation, this capital X is tan 2x. So using y2x w in the integral we get u is equal to minus integral y2 is sin to x, capital X is tan 2x, and W we found out in [00:04:19.984] It is two, the Roll scan of I1 by two. So now how do we evaluate the integral? See the major disadvantage or drawback of this method is this method requires evaluating the integrals. How do you evaluate this? [00:04:36.256] See I can write this as minus sine two x by two. [00:04:40.592] Tan 2x is written as sine 2x by cos 2x. So which simplifies to minus half into integral. Sin into sine is sin square 2x by cos 2x dx. Now how do we evaluate this integral? So what I do is I use the trigonometric [00:04:58.256] identity and write sin square 2x as 1 minus car square 2x.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:04.292] divided by cos 2x dx. Now I split the term inside the integral into two terms, which is 1 by cos 2x minus cos square 2x by cos to x. So that is equal to minus half. Integral 1 by cos is second [00:05:21.472] secant 2x minus cos 2x dx. Now I know to evaluate second 2x and cos 2x. What is integration of secant 2x? It is log of [00:05:31.984] secant 2x plus tan 2x divided by 2. And what is integration of cos 2x? It is just sine 2x. So this is minus half into integration of sine secant 2x is log of secant 2x plus tan 2x divided by 2. Minus [00:05:51.428] Integration of cos 2x is sin 2x by 2. So this simplifies to u is equal to 1 by 4 into sin 2x minus log of secant 2x plus stand 2x. So we found out u, which is required here. U. [00:06:08.324] To write the particular integral, we need to find another function called v. How do we find v? So v is given by it is given by the formula integral y1 into capital X by W dx. [00:06:24.228] What is y1? Y1 is cos 2x. W is 2 and capital X is stand to X. So using it we get integral cos 2x. [00:06:34.480] Tan to x by 2. This is a simple integral to evaluate because this is cos to x by 2 and tan is written as sin by cos. Cas and cos gets cancelled, so it is half integral sine to x. So what is integration of sine to x? [00:06:49.808] Minus cos 2x divided by 2. So it is half into minus cos 2x divided by 2. B is equal to minus cos 2x by 2. See, I had told you when we were discussing this method of variation of parameters, when you evaluate the integrals. [00:07:06.576] To find u and v, we are not writing the constant of integration. So this is v. So what is a particular integral? So it is given by u, this entire expression is u into y1 plus v into [00:07:22.304] y2. So the particular integral is given by this. But to find the particular integral, first thing is we need to

know the complementary function of the differential equation. The other thing is we need to evaluate the integrals to find the particular integral. [00:07:39.984] So now let us apply this method to solve another question. Find the general solution by the method of variation of parameters. D square minus 1 of y is equal to 2 divided by 1 plus e power x. How is this question different from the previous one? [00:07:57.892] This is again a second order linear differential equation with constant coefficient. [00:08:03.172] Because here RHS function is a little complex function, we need to evaluate the integral. So let us use the method to find the general solution of this differential equation by this method of variation of parameters. Okay. So what are the three steps involved? [00:08:20.944] First step is to find the complementary function, find out by one by two, second step is to find the Ronskian. [00:08:28.800] And third step is to find the particular integral. And to find particular integral, we need to find out the functions u of x and v of x by evaluating the integrals. Now the first step is to find y and byte. [00:08:44.708] The auxiliary equation is m square minus 1 is equal to 0, m square is equal to 1, m is equal to plus r minus square root of 1, which is plus r minus 1, 1 root is plus 1, and the other root is minus 1. So because the roots are real and distinct, the complementary function is written as C [00:09:02.480] e to the power 1x plus c2 to the power minus x. So this equation is of the form c1 by 1 plus c2 by 2 where y 1 is equal to e to the power x, y2 is equal to e to the power minus x. It is not that you have to write in this order only. You can also write c1 e to the power minus x plus c [00:09:20.048] e to e to the power plus x. Then y1 becomes e to the power minus x and y2 is equal to e to the power x. It doesn't matter. So here I have taken y1 as e to the power x y2 is equal to e to the power minus x. The final solution of the differential equation will not change if you interchange y1 [00:09:37.616] and by 2. Now, once you find by 1 and by 2, second step is to find the Rans scan of this e to the power x and e to the power minus x. So the rans scan is given by determinant [00:09:50.464] E to the power x, e to the power minus x, that is the first. The second step is the derivatives. E to the power x or differentiation gives e to the power x. E to the power minus x or differentiation gives minus e to the power minus x. Now simplify this, expand this. So it is minus 1 minus 1, which is minus 2.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:11.076] W is equal to minus 2 because e to the power x into e to the power minus x is 1 into minus 1. So it is minus 1, minus 1, which is 2. Ron's can we found out. What is the third step? Third step is to find the particular integral. [00:10:28.080] So the particular integral is given by u into y 1 plus v into by 2 where u is given by minus integral y 2 into x by w ts. So that is equal to x in this in this equation is 2 by 1 plus c po x. [00:10:48.228] So u becomes minus integral. Y2, I have taken it as e to the power minus x. Capital X is 2 by 1 plus E power X divided by W, which we found out it is minus 2 dx. So if you simplify, minus and minus becomes plus 2 and 2 gets cancelled. [00:11:06.016] 1 by e power x into 1 plus e power x dx. Now the next question is how do we evaluate this integral? What I do is I just resolve this into partial fractions. See this can be written as 1 by e power x. [00:11:23.488] Minus 1 by e 1 plus e power x. How? See you take L Cm and check E par x into 1 plus e power x is the L Cm. So when you simplify numerator becomes 1 plus E power x minus e power x, which is equal to 1. [00:11:37.828] So I resolve this into partial fractions. It is 1 by e power x minus 1 by 1 plus e power x dx. Now the first term is easy to integrate. It is e to the power minus dx. Integration of e to the power minus x is minus c to the power minus x. Now [00:11:55.600] evaluating the second term is little difficult. So the first term I have integrated, the second one I have written as it is minus integral. Now what I have done, how did I get the C power

minus x? I can take the common factor [00:12:10.096] E power x in the denominator of this. If I take e power x common factor, what do I get? 1 by e power x, which is e power minus x plus 1. [00:12:20.928] And if I take this e power x to the numerator, it becomes e power minus x by 1 plus e power minus x. R I can understand this in a different way. Multiply both numerator and denominator by e power minus x. [00:12:38.276] So e power minus x into 1 is e power minus x. Denominator if you consider 1 into e power minus x is e power minus x plus e power x into e power minus x is 5. Now why did I take this e power x common factor you will understand now. [00:12:55.200] So now you consider this integral. Let us solve this integral. How do I solve this integral? By substitution. So what is that substitution? I'll take t is equal to 1 plus e power minus x. So what is dt? [00:13:08.864] Minus e power minus x dx. Differentiate this t with respect to x. I get dt by dx as minus e power minus x. Now I'm changing the variable from x to t by this substitution. T is equal to 1 plus e power minus x. So this integral. [00:13:26.512] integral becomes its equivalent to it is equivalent to integral. So what is this? E power minus x dx can be replaced by minus dt. Divided by the denominator 1 plus e power minus x is written as t. [00:13:42.820] So I have changed the variable from x to t by using the substitution t is equal to 1 plus e power minus x. Because this integral is of the form some dt by t form. Differentiation of the denominator. [00:13:59.152] We'll give the numerator. Now what is integration of 1 over t? It is log p. So it is minus of log p. Don't write the constant of integration. So but t is not the variable, the variable is it's. So just put back the variable x, we get t. [00:14:16.208] The value of this integral e power minus x by 1 plus e power minus x dx is minus log of 1 plus e power minus x. [00:14:26.468] So this is how we are evaluating the second integral and we wanted e power minus x in the numerator for the same reason. So that I can evaluate the integral by the substitution method. [00:14:39.664] Now using this expression in u, we get u is equal to minus e power minus x plus log of there was one minus sign over there, minus u to minus become plus log of 1 plus e power minus x. So this is a we find out u of x, which is required. [00:14:57.424] required to find the particular integral of the given differential equation. Now how do we find B?

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:04.880] V for this again I need to evaluate another integral. Integral y1 into capital X by W dx. So this is integral y 1 is e power x, capital X is 2 by 1 plus E po x, and W is minus 2 dx. So that is equal to [00:15:22.576] Minus 2 and 2 gets cancelled e power x by 1 plus e power dx. This is a very simple integrality valid. This is again of the form dt by t where t is equal to 1 plus e power x. [00:15:34.820] So what is integration of dt by t? It is log t. So it is mine and v is equal to minus log of 1 plus e power x. Now using this u and v in the equation. [00:15:48.164] y p is equal to u into y 1 plus v into y2, we get y p is equal to u e c par x. [00:15:56.452] Sorry, y 1 is e power x and u is the function which we found out minus e power x plus log of 1 plus e power minus x. Plus b is e power minus x into y 2 is minus log of 1 plus e power x. So this is how we find the particular integral. [00:16:16.144] And the general solution is equal to yc plus yp because the question was to find the general solution, not just to find the particular integral. So we need to write this. Now see this method. The major drawback of this method is we need to evaluate the integrals. [00:16:33.824] To find the particular integral of the differential equation. But see, we apply this method to solve differential equation with constant coefficients. Now let us apply the method to solve second order linear differential equation with variable coefficients. [00:16:53.284] Consider a question x square y double dash minus 3xy dash plus 3y is equal to 12x to the power 4. Just look at the look at the question. It is a differential equation

with variable coefficients. [00:17:11.236] We know this form of differential equation with variable coefficient. So, what is this? This is Cauchy's linear differential equation of order 2. I know a method to find the general solution of this differential equation, but because here the question is to find the general solution. [00:17:29.008] By the method of variation of parameters, let us apply method of variation of parameters to find the general solution of this differential equation. But [00:17:40.288] To apply this method, I first have to find out the complementary function. So to find complementary function we know that the differential equation with variable coefficients has to be transformed to differential equity with constant coefficients. Now [00:17:57.968] How do you transform this differential equation into differential equation with constant coefficients? By substitution. What is the substitution we do? Z is equal to log x r x is equal to e power z. Now we also know that. [00:18:15.536] X into dy by dx is equal to capital D of Y where capital D stands for D by Dz and X square into D2 Y by D x2 is D into D minus 1 of Y. So this is very important where capital D differential operator stands for D by D. [00:18:33.232] Said now using these expressions in the given second order LDE with variable coefficients we get d into d minus 1 minus 3 into d plus 3 off by is equal to 12 into r this is very important. [00:18:50.800] and x to the power 4 is written as e to the power 4 z. So now this simplifies to d square minus d minus 3 d is minus 4 d plus 3 of y is equal to 12 e power 4z. Now [00:19:07.376] The question is to find see the same question can be written this way because this differential equation is equivalent of the given differential equation wherein z is equal to log x. Now I need to find out the complementary function of [00:19:25.008] This differential equation to apply this method of variation of my parameters to solve it. Now, how do I find the complementary function? I need to write the associated auxiliary equation. [00:19:39.664] The auxiliary equation is m square minus 4 m plus 3 is equal to 0. Simple quadratic equation. What are the roots? 3 comma 1 because m minus 3 into m minus 1 are the factors. 3 and 1 are the roots. The roots are real and distinct. [00:19:55.392] So what is the complementary function c1 e to the power 3z. Please remember z here, 3z plus c2, e to the power z. Now this is of the form c one, y1 plus c2, y2, because I am applying the method.

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:11.744] To uh solve this differential equation, let me continue with the same z itself. So it is y1 is equal to e to the power 3z, y2 is equal to e to the power z. At the end we will write back x. This z we will write back x. Y1 and y2. Next you find the [00:20:31.716] X is equal to 12 e to the power 4z because I am applying the method to solve this differential equation. Capital X is 12a to the power 4z. Y1 is e power 3z and y2 is e power z. So this is the first step to find y1 y 2. Now what is the second step to find ron skin. So what is ron skin? [00:20:49.488] is given by determinant e power 3 z, e power z derivatives 3 e to the power 3 z, e power z. So you just cross multiply, simplify, you get w is equal to minus 2 into e to the power 4z [00:21:05.188] Please remember I am applying this method to solve this differential equation, which is equivalent of the given differential equation. So I found out y1 by two w the last step b finding the particular integral. [00:21:21.040] Which is given by u into y 1 plus v into y 2 where u is given by minus integral y 2 into capital X by W DS. So this is equal to minus y2 is e power z. Capital X is twelve e power 4z divided by w is minus [00:21:38.672] 2 into e to the power 4z. So e to the power 4z gets cancelled. So this is equal to minus into minus by minus is plus 12 by 2 is 6. [00:21:47.968] integral e to the power z d is d z. So what is integration of e to the power z? It is e to the power z itself. So it is 6 into e to the power z. Now b is given by integral by 1 into x by w dx [00:22:03.808] So this is interval e bar three z capital X is twelve E bar four Z.

[00:22:09.380] Same thing I have written divided by W is minus 2 to the power 4z. This gets canceled, so it is minus 6 integration of e to the power 3z dz. So what is this e to the power 3z integration with respect to z. It is e to the power 3 z divided by [00:22:27.152] by 3 into the 6 so it becomes minus 2 into e to the power 3z. Now y p is is when you substitute u y 1 b by 2 we get y p as this. [00:22:41.220] So this is equal to 4 into e to the power 4z. See here it is 6 e to the power 4z minus 2 e to the power 4z, which is 4 e to the power 4z. Now [00:22:53.924] Y c is equal to c 1 e to the power 3 z to c2 to the power z. We found out y c and y p [00:23:01.828] For the transform differential equations, wherein y was a function of z. But the give in the given question, y is a function of x. So we should write back x. So writing both y p and yc in terms of x. [00:23:18.912] You get Y C is equal to C1, E par three Z is X cube, plus C two e to the C two. [00:23:25.408] into x e power z is x and y p is equal to 4x to the power 4. So what is the general solution of the given differential equation? So that is equal to what this y c plus y p in terms of x c1x cube plus c2x plus 4x to the power 4. [00:23:47.140] So we learned we used this method to solve linear differential equation with constant as well as with variable coefficients. So we learned different methods, different solution techniques to solve linear differential equation. [00:24:04.944] of different forms linear differential equations with constant coefficients, with variable coefficients, homogeneous and non-homogeneous linear differential equations in the sessions. In the next session, we will apply the methods which we learn, the concept which we learn [00:24:22.704] Two real world problems. Thank you.

EV0000222 · EV0000223 · EV0000224 · EV0000225

Full transcript: 3923 normalized words, 98 timestamped segments, 1467.114667 seconds duration. Low-confidence segments flagged: 0.

EV0000206 · EV0000207 · EV0000208 · EV0000209

15 Variation of Parameters method

This lesson is indexed from the source video titled Variation of Parameters method. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000226 · EV0000227

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.676] Hello, everyone, welcome back to the session on higher order linear differential equations. In today's session, we are learning or discussing a method called method of variation or parameters to solve a second order linear differential equation. [00:00:16.132] This method is used to find the particular integral of a linear differential equation whose complementary function is already known. [00:00:25.476] So, this method is used to find only the particular integral of the differential equation. To find the particular integral of the differential equation, the complementary function must be known. Though this method is very general, we will apply this method. [00:00:43.472] To solve a second order linear differential equation with constant coefficient. Now let us consider a second order linear differential

equation with the type d2y by dx2 plus p into dy by dx plus q into y is equal to capital X where [00:01:01.936] P and Q are constants and capital X is a function of X. See, I have considered a simple second order non-homogeneous linear differential equation with constant coefficients. I know to find the complementary function of this differential. [00:01:19.600] equation. So let the complementary function of this differential equation be C1 by 1 plus C2 by 2. So we know that complementary function [00:01:31.044] is the general solution of the associated homogeneous differential equation. And we know that this c1 by 1 plus c2 y2 represents the complementary function of the differential equation only when y1 and y2 are two [00:01:48.816] Two linearly independent solutions of the associated homogeneous linear differential equation. Now we learnt in the previous session that what do you mean by the linearly independent solution? When do we say that y 1 and by 2 are two linearly independent? [00:02:06.384] function. So y 1 and by 2 are said to be linearly independent if the linear combination of the function c1 by 1 plus c2 by 2 is not equal to 0. [00:02:17.488] Now, because this c1 y1 plus c2 y2, we have taken it as the complementary function, y1 and y2 have to be the linearly independent functions. Now, how do you check whether the functions are linearly independent or not? [00:02:35.216] Let us introduce another term called Ronskeian. So let W be the Ronskean of by 1 and by 2, and it is given by W is equal to determinant [00:02:47.568] Y1 by 2 and its derivative y 1 dash by 2 dash. That is w is equal to y1 into y2 dash minus y2 into y1 dash. Now how do you check the two functions are linearly independent or not? So the runs here of the functions y1 and [00:03:05.232] Y2 is not equal to 0 if y1 and y2 are two linearly independent functions. So because here y1 and y2 are linearly independent, so w is not equal to 0. Now the particular solution. [00:03:22.800] Now the differential equation is given by [00:03:26.112] P i is equal to some u of u a function of x, u of x into y1 plus v of x into y2, where u and v are given by u is equal to minus integral y2 into x by w d and u v is equal to the equal to equal [00:03:43.760] equal to integral y1 into x by w dx. So what is this w? w is the long scale. What is capital X? That is the right hand side function of the given differential equation. What are y1 and y2? Two linearly independent solutions of the corresponding [00:04:01.328] Homogeneous differential equation. So u is given by this expression and b is given by this expression. So to find the particular integral anyway by 1 and by 2, I know I just have to find out u and b. I need to evaluate these two. [00:04:18.896] To find the particular integral by this method. But when I am writing the expression for u and b, I am not writing the constant of integration because the constant of integration gets observed with the [00:04:35.716] constants of the complementary function when I'm writing the general solution of the differential equation. So no need to write the constant of integration when you are writing the expressions for u and b. So this method basically involves finding out [00:04:53.488] are evaluating the integrals u and b and finding out this Ronski and W. Now, so let us discuss more about this uh method. So this method I illustrated this method by taking the example of a second order

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.424] non-homogeneous linear differential equation with constant coefficient. This method can also be used to find the particular integral of [00:05:21.348] Our differential equation with variable coefficient. If we know the general solution of the associated homogeneous equation, in the sense this method can also be used to find the particular integral of LTE with variable coefficient provided. [00:05:39.120] We know the complementary function of the differential equation, and this method we illustrated by taking the example of second order differential equation. This method can be applied to solve a differential

equation of any order that is [00:05:56.688] and th order linear differential equation. But we are learning in this session m uh to solve only LTE of order 2. Now [00:06:06.576] When we are writing, when we are finding the using this method to find the solution of the differential equation, we should note down this. The coefficient of d2y by dx2 in the given differential equation has to be 1. [00:06:21.316] If it is not equal to one any other constant number, divide by the coefficient and make the coefficient of d2 y by dx to one. So now [00:06:31.236] What are the demerits of this method? See what is the main demerit of this method? This method is used to find only the particular integral of the differential equation and [00:06:42.496] To find the particular integral also we need to know the complementary function of the differential equation. So the complementary function is absolutely required to apply this method. [00:06:55.360] And this method involves evaluation of some integrals, which are sometimes very difficult to evaluate or solve. So these are the main two demerits of this method. Now let us apply this method to solve. [00:07:10.384] Linear differential equation with constant coefficient and with variable coefficient, but we are applying this method to find the solution of only second order LBE. Now let us consider a question. Find the particular integral by the method of variation of parameters. [00:07:30.500] So the question is y double dash minus two y dash minus three y is equal to x into e power minus x. So let me write this equation in operator notation form. It is capital D square minus two d minus three of y is equal to x into e to the power minus x. So here this d is equal to [00:07:48.272] D by dz or it is not d by dz. We have we are discussing a different method here, variation of parameter method. So it is because the independent variable here is x, so d is d by dx. Now [00:08:02.052] We know to solve this differential equation. We know to find the complementary function and also the particular integral of this differential equation which we learnt in the previous session. How do you find the particular integral of this equation? Which type? This is type 4. [00:08:19.296] It is e to the power ax into v type where v is a function of x. So it is e to the power minus 1 into x into x where v is equal to x. So using that method I can find the particular integral. But here the question is to find the particular integral using the method of variation of parameters. So [00:08:39.280] To find the particular integral of this differential equation, by using method of variation of parameter, I need to write the complementary function of the differential equation. If the question was to find the particular integral, [00:08:55.376] To find the particular integral by any method, I need not have to find the complementary function because it is by the method of variation of parameters I need to first find out the complementary function of the differential equation. So this method involves three steps. The first [00:09:13.040] step being uh writing the complementary function and finding out by one and by two. The second step is to find the RAN scale and the third step is to find u and b. So let us discuss the first step to find complementary function. [00:09:28.548] So to find complementary function, I should set up the auxiliary equation which is associated with the responding homogeneous differential equation. So it is m square minus 2m minus 3 is equal to 0. So the auxiliary equation, the roots are 3n minus 1, so the complementary. [00:09:46.320] function is c 1 e to the power 3x plus c2 to the power minus x. So this is of the form c 1 into y1 plus c2 into y2 where y1 is equal to e to the power 3x and y2 is equal to e to the power minus x. C e to the power 3x and e to the power minus x are 2

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:03.920] linearly independent functions. Now here in the given differential equation capital X is x into e power minus x. Now the second step being to find the Rand scheme. So what is W of y1 by 2? [00:10:22.736] The Ronskiana

by one by two is given by determinant e to the power three x e to the power minus x that is the first row. Derivatives of it three into e to the power three x minus e to the power minus x is the second row. So you simplify this, it is e minus e to the power two. [00:10:40.400] two x minus three e to the five. [00:10:43.652] e to the power 3x into minus e to the power minus x minus half 3 e to the power 3x into e to the power minus x which simplifies to minus 4 into e to the power 2x. So the answer is minus 4 into e to the power 2x. Now [00:11:00.868] The particular next the last step is to find the particular integral. The particular integral is given by u into y 1 plus v into y2, where u is e minus integral y2 into x by w dx and b is [00:11:19.940] And v is given by integral y 1 into x by wtx. Let us first simplify this u, which is minus integral y2 into x by wtx. So what is so this is equal to [00:11:36.320] Minus what is y 2? e to the power minus x. Capital X is the RHS of the given differential equation, which is X into E to the power minus X. By W which we found out it is minus 4 into E to the power 2x. So minus sign minus sign gets cancelled and e to the power 2x. [00:11:54.032] Here you have e to the power minus 2s, which simplifies to 1 by 4 e to the power minus 4s into xpx. [00:12:03.268] 1 by 4 into C. I have taken x as the first function and I'm integrating e to the power minus 4x. So it is x into integration of e to the power minus 4x is minus e to the power minus 4x by 4 minus. [00:12:18.896] integral of integration of this is this minus e to the power minus four x by four into differentiation of this x. [00:12:28.384] with respect to x is 1 dx. So let's simplify this further. So this simplifies to so it is 1 by 4 into minus x e to the power minus 4x by 4 which is minus x e to the power minus 4x by 16. Now this is minus into minus plus [00:12:45.860] 1 by 4 into integration of e to the power minus 4x is again e to the power minus 4x by minus 4, which becomes minus 1 by 64 into e to the power minus 4x. [00:12:59.376] So if you take e to the power minus 4x by 16 common factor, you get e to the power minus 4x into 16 into x plus 1 by 4. This is just one integral evaluation to find u. Similarly, we need to find the other. [00:13:16.208] uh function b to find the particular integral of the differential equation. Now, how do we find B? V is given by the formula [00:13:26.288] Integration y 1 into x by w dx. So what is y 1? e to the power 3x. What is x? x into e to the power minus x. W the wrong scale which we found out it is minus 4 into e to the power 2x. So this is very simple. E to the power 2x gets canceled with e to the equation. [00:13:43.920] by 2x it is minus 1 by 4 into integration of x dx. So what is integration of x dx? It is x square by 2. See I told you when you are calculating u and v no need to write the constant of integration. And v is equal to x square by 8 [00:14:03.716] Minus x square by 8 and particular integral is given by y1 into u plus y2 into. [00:14:11.600] B. See this is the particular solution. Particular solution should not contain any constant. If the question was to find the complete solution, then you should write the complete solution. Y is equal to C plus PI. So [00:14:28.304] We learn this method to find the particular integral of a second order differential equation with constant coefficients, but this method can be applied to solve linear differential equation of any order with constant coefficient or with variable coefficients. [00:14:45.936] So we will solve more problems on this method in the next session. Thank you.

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Full transcript: 2287 normalized words, 59 timestamped segments, 893.205333 seconds duration. Low-confidence segments flagged: 0.

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16 Applications of higher-order linear differential equations - RLC Circuits

This lesson is indexed from the source video titled Applications of higher-order linear differential equations - RLC Circuits. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence.

EV0000238 · EV0000239

Graphs and diagrams

Figure G000147. Applications of DE LCR circuit · source slide page 1.

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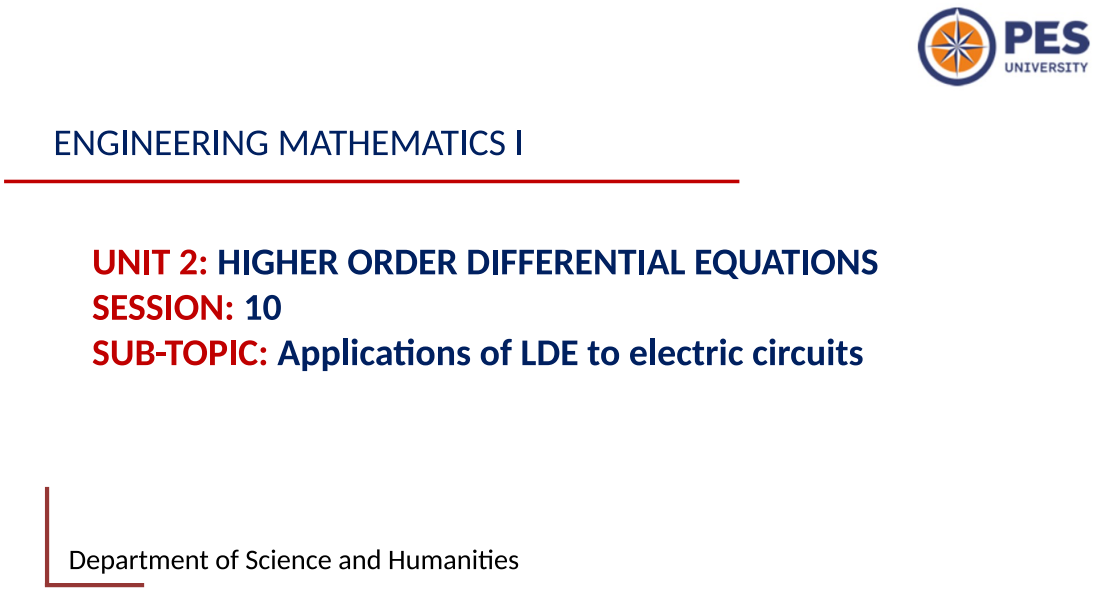

Figure G000148. Applications of DE LCR circuit · source slide page 2.

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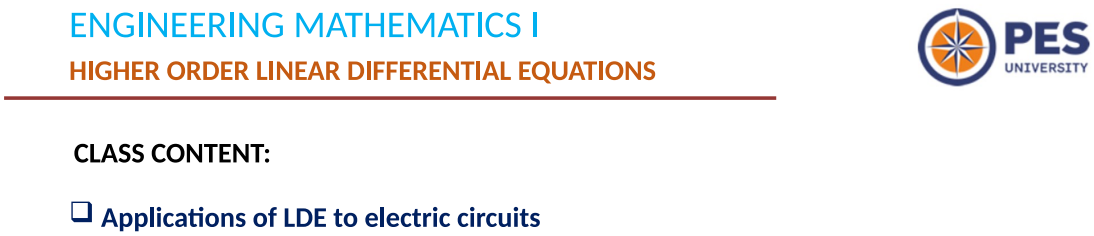

Figure G000149. Applications of DE LCR circuit · source slide page 3.

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ENGINEERING MATHEMATICS I
LCR Circuits

RLC Circuit

A simple RLC circuit consists of an inductor of inductance 'L' (in henries), a Capacitor of capacitance 'C'(in farads) and a resistor of resistance 'R' (in ohms) and a source of electromotive force E (t) (in volts) are connected in series.
The current 'I' through the circuit is measured in amperes and the charge 'Q' on the capacitor is measured in coulombs.

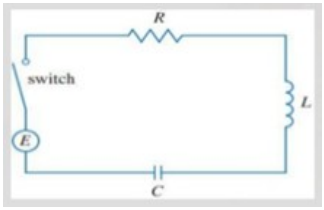

Figure G000150. Applications of DE LCR circuit · source slide page 4.

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ENGINEERING MATHEMATICS I
LCR Circuits

The equation for the current I(t) in the RLC circuit is obtained by considering the voltage drops across three components.

The voltage drop across the inductor is given by $E_L = L\frac{dI}{dt}$

The voltage drop across the resistor is given by $E_R = R\,I$

The voltage drop across the capacitor is given by $E_c = \frac{Q}{C}$

From Kirchoff's Voltage Law,
The voltage supplied by the battery = The voltage drop across the circuit components
(the sum of these voltage drops is equal to the electromotive force)

Figure G000151. Applications of DE LCR circuit · source slide page 5.

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ENGINEERING MATHEMATICS I
LCR Circuits

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i.e, $E(t) = V_L + V_R + V_c$

$$E(t) = L\frac{dI}{dt} + R\,I + \frac{Q}{C}$$

'I' is the current in the entire circuit and 'Q' is the charge on the capacitor.
We know that, $I = \frac{dQ}{dt}$

$$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = E(t)$$

This is the differential equation which describes LCR – Circuit.

Figure G000152. Applications of DE LCR circuit · source slide page 6.

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ENGINEERING MATHEMATICS I
LCR Circuits

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$$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = E(t)$$

It is a second order linear differential equation with constant coefficients. If the charge and the current are known at zero time, we can get an expression for Q(t) and I(t) .

Figure G000153. Applications of DE LCR circuit · source slide page 7.

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ENGINEERING MATHEMATICS I
RLC - Circuits

Example: Find an expression for I(t) in RLC - circuit with R = 8 ohms , L = 2 henry, C = 0.1 farad, which is connected to a source of voltage E(t) = 10 volts , assuming zero charge and zero current when t = 0.

Solution: The differential equation of RLC – circuit is

$$L\frac{d^2Q}{dt^2}+R\frac{dQ}{dt}+\frac{Q}{C}=E(t)$$

Using the values of R, L, C and E from the given problem, we get

$$2\frac{d^2Q}{dt^2}+8\frac{dQ}{dt}+\frac{Q}{0.1}=10$$

Figure G000154. Applications of DE LCR circuit · source slide page 8.

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ENGINEERING MATHEMATICS I
LCR Circuits

$$2\frac{d^2Q}{dt^2}+8\frac{dQ}{dt}+\frac{Q}{0.1}=10$$

$$\Rightarrow \frac{d^2Q}{dt^2}+4\frac{dQ}{dt}+5Q=5$$

The above equation is a non homogeneous LDE of order 2 with constant coefficients.

To find CF :

Auxiliary Equation is

$$m^2+4m+5=0$$

The roots are $m=-2\pm i$

$$CF=e^{-2t}(c_1\cos t+c_2\sin t)$$

Figure G000155. Applications of DE LCR circuit · source slide page 9.

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ENGINEERING MATHEMATICS I
LCR Circuits

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To find PI

PI is given by,

$$PI = \frac{5}{D^2 + 4D + 5} = 1$$

$\therefore$ The general solution is $Q(t) = CF + PI$

$\therefore Q(t) = e^{-2t}(c_1 \cos t + c_2 \sin t) + 1$ --------(1)

Figure G000156. Applications of DE LCR circuit · source slide page 10.

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ENGINEERING MATHEMATICS I
LCR Circuits

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To find c_1 and c_2

Using the given initial conditions, I(0) = 0 and Q(0) = 0 eqn (1)

$0 = e^{-2.0}(c_1 \cos 0 + c_2 \sin 0) + 1$

$0 = 1.(c_1.1 + c_2.0) + 1$

$c_1 = -1$ -------(2)

To find C_2

differentiate eqn (1) wrt 't' and then use the given initial conditions.

$$\therefore \frac{dQ}{dt} = e^{-2t}(c_1(-\sin t) + c_2(\cos t)) + (-2)e^{-2t}(c_1(\cos t) + c_2(\sin t))$$

Figure G000157. Applications of DE LCR circuit · source slide page 11.

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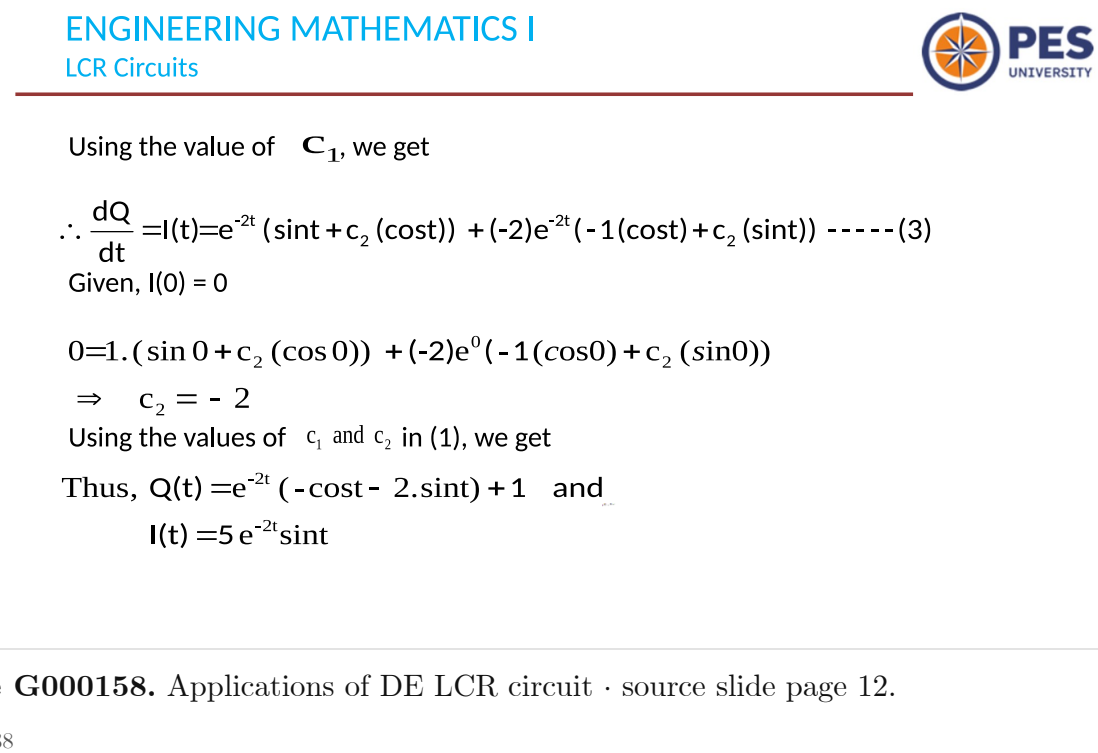

Figure G000158. Applications of DE LCR circuit · source slide page 12.

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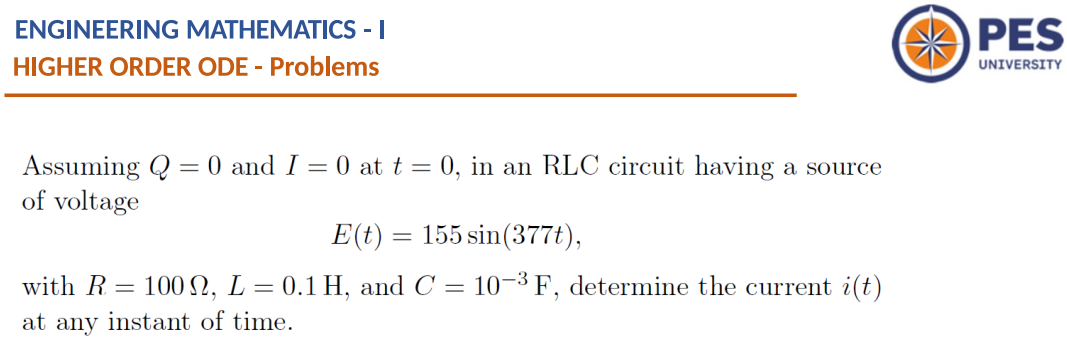

Figure G000159. Applications of DE LCR circuit · source slide page 13.

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ENGINEERING MATHEMATICS - I
HIGHER ORDER ODE - Problems

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For a series RLC circuit with source

$$E(t) = 155\,\sin(377t),$$

and

$$R = 100\ \Omega, \quad L = 0.1\ \text{H}, \quad C = 10^{-3}\ \text{F},$$

Kirchhoff's voltage law in terms of the charge $q(t)$ gives

$$L\,\ddot{q} + R\,\dot{q} + \frac{1}{C}\,q \;=\; E(t), \qquad i(t) = \dot{q}(t).$$

Substituting the numerical values:

$$0.1\,\ddot{q} + 100\,\dot{q} + 1000\,q = 155\,\sin(377t).$$

Dividing through by 0.1:

$$\ddot{q} + 1000\,\dot{q} + 10000\,q = 1550\,\sin(377t),$$

with initial conditions

$$q(0) = 0, \quad \dot{q}(0) = i(0) = 0.$$

Figure G000160. Applications of DE LCR circuit · source slide page 14.

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ENGINEERING MATHEMATICS - I
HIGHER ORDER ODE - Problems

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Solve $r^2 + 1000\,r + 10000 = 0 \;\Rightarrow r_{1,2} = -500 \pm 200\sqrt{6}$
$r_1 \approx -10.1021,\ r_2 \approx -989.8979$).

Assume

$$q_p(t) = A\,\sin(377t) + B\,\cos(377t).$$

Substituting into the ODE yields

$$A \approx -0.00128331, \quad B \approx -0.00366164.$$

Hence

$$i_p(t) = \dot{q}_p(t) = 377A\,\cos(377t) - 377B\,\sin(377t)$$
$$\Rightarrow i_p(t) = -0.483808\,\cos(377t) + 1.38044\,\sin(377t).$$

Figure G000161. Applications of DE LCR circuit · source slide page 15.

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ENGINEERING MATHEMATICS - I
HIGHER ORDER ODE - Problems

The general solution is

$$i(t) = i_h(t) + i_p(t) = C_1\, e^{-10.1021\,t} + C_2\, e^{-989.898\,t} - 0.483808\cos(377t) + 1.38044\sin(377t)$$

Using initial conditions, we find

$$C_1 = -0.0423597, \quad C_2 = +0.526168.$$

Therefore,

$$i(t) = -0.0423597\, e^{-10.1021\,t} + 0.526168\, e^{-989.898\,t} - 0.483808\cos(377t) + 1.38044\sin(377t)$$

Figure G000162. Applications of DE LCR circuit · source slide page 16.

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Thank you

Department of Science and Humanities

Figure G000163. Applications of DE LCR circuit · source slide page 17.

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000704 · EV0000705 · EV0000706
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 2: HIGHER ORDER DIFFERENTIAL EQUATIONS SESSION: 10 SUB-TOPIC: Applications of LDE to electric circuits Department of Science and Humanities EV0000707 · EV0000708 · EV0000709
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER LINEAR DIFFERENTIAL EQUATIONS CLASS CONTENT: Applications of LDE to electric circuits EV0000710 · EV0000711 · EV0000712
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I LCR Circuits RLC Circuit A simple RLC circuit consists of an inductor of inductance 'L' (in henries), a Capacitor of capacitance 'C'(in farads) and a resistor of resistance 'R' (in ohms) and a source of electromotive force E (t) (in volts) are connected in series. The current 'I' through the circuit is measured in amperes and the charge 'Q' on the capacitor is measured in coulombs. EV0000713 · EV0000714 · EV0000715
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I LCR Circuits The equation for the current I(t) in the RLC circuit is obtained by considering the voltage drops across three components. The voltage drop across the inductor is given by The voltage drop across the resistor is given by The voltage drop across the capacitor is given by From Kirchoff's Voltage Law, The voltage supplied = The voltage drop across the by the battery circuit components (the sum of these voltage drops is equal to the electromotive force) EV0000716 · EV0000717 · EV0000718
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I LCR Circuits i.e, 'I' is the current in the entire circuit and 'Q' is the charge on the capacitor. We know that, This is the differential equation which describes LCR – Circuit. EV0000719 · EV0000720 · EV0000721
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I LCR Circuits It is a second order linear differential equation with constant coefficients. If the charge and the current are known at zero time, we can get an expression for Q(t) and I(t) . EV0000722 · EV0000723 · EV0000724
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I RLC - Circuits Example: Find an expression for I(t) in RLC - circuit with R = 8 ohms , L = 2 henry, C = 0.1 farad, which is connected to a source of voltage E(t) = 10 volts , assuming zero charge and zero current when t = 0. Solution: The differential equation of RLC – circuit is Using the values of R, L, C and E from the given problem, we get EV0000725 · EV0000726 · EV0000727
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I LCR Circuits The above equation is a non homogeneous LDE of order 2 with constant coefficients. To find CF : Auxiliary Equation is EV0000728 · EV0000729 · EV0000730
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS I LCR Circuits To find PI PI is given by, EV0000731 · EV0000732 · EV0000733
11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS I LCR Circuits Using the given initial conditions, I(0) = 0 and Q(0) = 0 eqn (1) To find differentiate eqn (1) wrt 't' and then use the given initial conditions. EV0000734 · EV0000735 · EV0000736

12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS I LCR Circuits Using the value of , we get Given, I(0) = 0 Using the values of in (1), we get EV0000737 · EV0000738 · EV0000739
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Assuming 2 = 0 and I = 0 at t = 0, in an F RLC circuit having asource of voltage E(t) = 155 sin(377t), with R = 100 Ω, L = 0.1H, and C = 10-3 F, determine the current i(t) at any i instant of time. EV0000740 · EV0000741 · EV0000742
14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] For a series RLC circuit with source E(t) = 155 sin(377t), and R = 100 Ω, L = 0.1 H， C = 10-3 F, Kirchhoff's voltage law in terms of the charge q(t) gives 1 i(t) = i(t). Substituting the numerical values: 0.1+ 100 q + 1000 q = 155 sin(377t). Dividing through by 0.1: + 1000q + 10000 q = 1550 sin(377t), with initial conditions q(0) = 0， (0) =i(0) = 0. EV0000743 · EV0000744 · EV0000745
15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Solve r2 + 1000r + 10000 = 0 → r1,2 = -500± 200√6 r1 -10.1021, r2 -989.8979). Assume ap(t) = A sin(377t) + B cos(377t). Substituting into the ODE yields A -0.00128331, B -0.00366164. Hence ip(t) = áp(t) = 377A cos(377t) - 377B sin(377t) → ip(t) = -0.483808 cos(377t) + 1.38044 sin(377t). EV0000746 · EV0000747 · EV0000748
16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] The general solution is - 0.483808 c0s(377t) + 1.38044 sin(377t) Using initial conditions, we find C1 = -0.0423597, C2 = +0.526168. Therefore, -0.483808 c0s(377t) + 1.38044 sin(377t) EV0000749 · EV0000750 · EV0000751
17. Source slide page 17 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000752 · EV0000753 · EV0000754

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.220] Hello, everyone. Welcome back to the session on Higher Order Linear Differential Equations. In today's session, we will be discussing an application of linear differential equations to electric circulates. [00:00:13.536] So let us consider a simple L CR circuit or R L C circuit. What is this L Cr circuit? It consists of an inductor of inductance CL [00:00:23.856] capacitor of capacitance C, resistor of resistance are in series with the source of EMF which is given by ESP. So it is a series circuit containing these three electrical components. Now the resistance is measured in poses [00:00:42.048] inductance is measured in Henries and the capacitance is measured in fat. Now the current which is flowing through the circuit is given by IFD and the charge and the capacity is given by Q of D. [00:00:54.704] IFP is measured in the current is measured in amperes and the charge and the capacitor is measured in holes. Now let us derive a differential equation which describes this LCR circuit. [00:01:08.516] So to derive a differential equity for the LCI C2, we need to first consider the voltage drops across the three components. [00:01:18.724] So the equation the LCS are food is obtained by considering the voltage drop across the three components. So first one is the voltage drop across the inductor, which I call it as EL, which is given by L into DIB. [00:01:34.560] Similarly, the voltage drop across the resistor, this is from Ohm's law. So it is er is equal to R into I. R is the resistance and I is the current. E R is equal to R into I. Similarly, the voltage drop

across the C fifth is given by [00:01:51.088] E C is equal to Q by C. Now we have from K chop's voltage law the total voltage drop. The voltage drop across these three components is equal to the EMF supplied by the battery. So what is it? [00:02:05.808] voltage supplied by the battery is equal to voltage log across the three component that is the sum of these three voltage log is equal to electromotte force. So we are using KCO's voltage log to write the differential equation [00:02:21.072] which represents this R and C circuit now. So that is the the M of the circuit CFP is equal to the voltage across the inductor plus voltage across the resistor plus voltage across the capacitor. So what is this mean? [00:02:38.400] VL is l into d by dt, VR is i into i, vc is equal to q by c, where i of p is the current which is flowing to the subdued and q is the charge of the capacity. And also we know that i is equal to dq by dt. Rate of change in the charge is equal to current. [00:02:58.596] Putting this back in the differential equation, the differential equation becomes n into d square q by dt square plus r into d q by dt plus q by c is equal to e for t. Here's inductance resistance and capacitance icon. [00:03:15.792] constant values. This EFT, the EMF which is supplied, is a function of P. [00:03:21.956] So now what is this differential equation? So this is the differential equation which describes the LCI circuit. So the differential equation which describes the LCI circuit is a simple second order non-homogeneous linear differential equation with constant. [00:03:41.732] So we write the differential equation and linear differential equation, which represents this LCR set. Now to find an expression for Q of P and I of P, I need to know the initial conditions. I need to know, we need to know the charge. [00:04:01.380] The current at time t is equal to zero. If we know the charge and current at time t is equal to zero, we get an expression for q of p and m of p. So this is the differential equation which represents the LCR sub 2. [00:04:16.228] we learned in the previous session solving this differential equation now let us apply this concept which we learned to an example so let us solve a problem on NCR side so consider it [00:04:29.348] Find an expression for IFP. If R and C circuit with R the resistance, uh of the resistor which is present in the circuit is eight ohms, the inductance is two h capacitance is 0.15, which is connected. [00:04:46.512] a source of voltage E of T is equal to 10 volts, assuming zero charge and zero current when T is equal to zero. This is the initial condition which is given over here. [00:04:57.124] Let us first write the differential equation, the R and C circuit. What is the differential equation which represents the R and C circuit? Which is L into D square Q by D T square plus R into D Q by DT plus Q by C is equal to E of T. So what is Q? L is equal to 2 degrees.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:17.156] R is equal to 8 ohms. And what else is given? The capacitance is equal to 01 0.15. And E of P is equal to it is also a constant, 10 ohms. Now using the given data in the differential equation, so it becomes 2 into [00:05:36.004] B square Q by DP square plus A T to D Q by D P plus Q by 0.1 is equal to 10. Simplify this further. Nine point one has one by 10. [00:05:46.916] Simplifying the equation further. So we get D square 2 by D square. Divide the entire equation by 2. So you get 8 by 2 which is 4. So this is actually 10Q. 10 Q by 2 is 5Q is equal to 10 by 2. [00:06:03.076] is this differential equation it's a non-homogeneous second order linear differential equation with constant coefficients we learn methods to find the solution of a non-homogeneous LB constant coefficients so what is it we know that [00:06:19.904] The complete solution of the non-homogeneous L V has two components. One is complementary function, and the other one is the particular. So we need to find out the complimentary function. [00:06:31.108] function and particular integral of this non-

homogeneous linear differential equation which ref which is actually which represents the L Cr circuit which is given in the post genetic. Now so uh to find the complementary function I need to [00:06:48.132] auxiliary equation. So what is the auxiliary equation? m square plus 4m plus 5 is equal to 0. Solve the quadratic equation. The roots are minus 2 plus i minus 2 minus n. The two roots are complex in nature. [00:07:03.300] Because there will be self-completed. So how do we write the complementary function? It is e to the power alpha t into c1 cos beta t plus c2. Sin beta t in this case alpha is minus 2 and beta is 1. So it is e to the top. [00:07:18.500] Minus 2t, C1 cos t plus C2 sin t. So this is the complementary function. This doesn't alone represent the complete solution of the differential equation because the even differential equation is a non-homogeneous L D L D E. Now it has another component. [00:07:37.860] is a component particular integral. Particular integral is given by phi by d square plus 4 d plus phi. Now how do you find the particular integral when R's function is a constant function. [00:07:52.676] So just replace D by 0Y because 5 can be written as 5 into E to the part 0 t. Replace D by 0. So what do you get? 5 by. [00:08:02.852] which is equal to 1. So the complete solution is q of t is equal to cm plus p i. So what is it? Minus t 2 t into c1 plus t plus e2 sin t. This is the complementary function plus particular integral is 1. So this is an expression for q of t. But [00:08:21.924] Now to find C1 and C2, we'll use the initial condition which is given. So what is that? It is given. Q of T and E of T is equal to 0, R T is equal to 0. That is F0 is 0. Q of 0 is equal to 0. Use it in equation 1. So what do we get? So it is 0. [00:08:41.028] is equal to e to the power minus 2 into 0 plus c 1 cos 0 plus c2 sine 0 plus 1 sine 0 is 0. C is 1. So c one is equal to I mean 0 is equal to 1 into c1 plus c2 plus 1. So that is equal to now I mean that implies c one is equal to [00:09:00.196] it is c1 plus 1 is equal to 0 and c1 is equal to minus 1. P found out c1. Now to find the other constant c2. So the given expression of q of p differentiate with respect to t. We'll get dq by t t as e to the power minus 2 t into [00:09:19.556] Because it was e to the power minus 2t into C1 cos t plus c2 sin t. Product of two functions. You must product of t to the power minus 2t. Differentiate the second term c1 into cos t differentiation is minus sine t plus c2 into differentiation of sine is cos t. [00:09:37.572] Plus differentiate the first term. P to the power minus 2p differentiation is minus 2 into e to the power minus 2t. Ret in the second term as it is. C1 cos t plus c2 sin d. But what is dq by dt? Which is equal to If P. [00:09:55.876] So I have used the value of C1 also. DQ by DT which is equal to I of T is equal to in the place of C1, I have written the value of C. So simplify this further. Now use the second condition which is given. I of 0 is equal to

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:12.452] zero. So what is it? I of zero zero is equal to e to the power minus zero is one. Sin zero is zero plus c two into power zero plus minus two into e to the power zero minus one power zero plus c to sine zero. Simplify this you will get the value of c two as minus two. [00:10:31.268] So C2 is equal to minus 2. So put this back C1 and C2, we get Q of P is equal to e to the by minus 2t into minus cos t minus 2 sin p plus 1 and i of p which is the derivative of q of t. Substitute for c1 and c2. [00:10:49.444] If t is equal to 5 into e to the power minus 2t sine t. So using differential equation concept, we derive an expression for q of t the charge and the capacity and the current which is flowing through the LCR circuit ISB. So this is how we can apply. [00:11:08.708] linear differential equation to solve problems in physics and engineering, in science and engineering. So we can apply uh this concept of differential equations to [00:11:20.580] model different situations in science and engineering. So we learn

methods to solve the differential equations. So by solving those differential equations we can analyze the compete.

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Full transcript: 1872 normalized words, 44 timestamped segments, 696.064 seconds duration. Low-confidence segments flagged: 0.

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17 Application problems on Linear differential equations - Simple harmonic oscillation

This lesson is indexed from the source video titled Application problems on Linear differential equations - Simple harmonic oscillation. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000250 · EV0000251

Graphs and diagrams

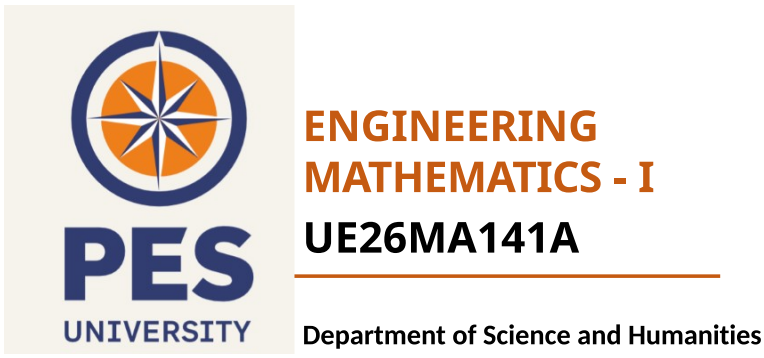

Figure G000164. Applications of DE SHM · source slide page 1.

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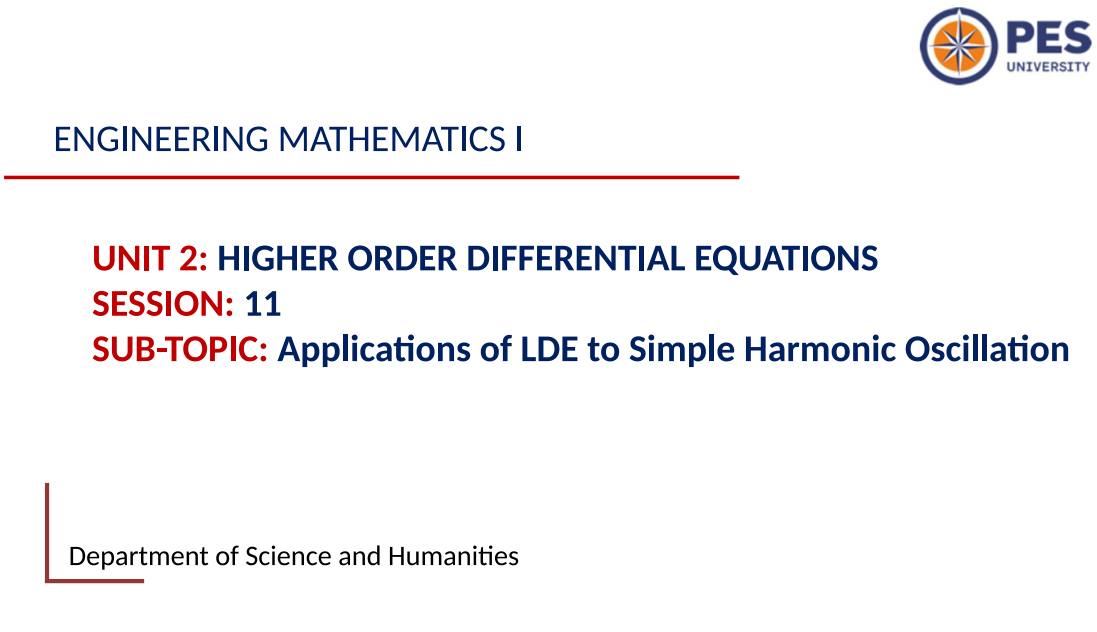

Figure G000165. Applications of DE SHM · source slide page 2.

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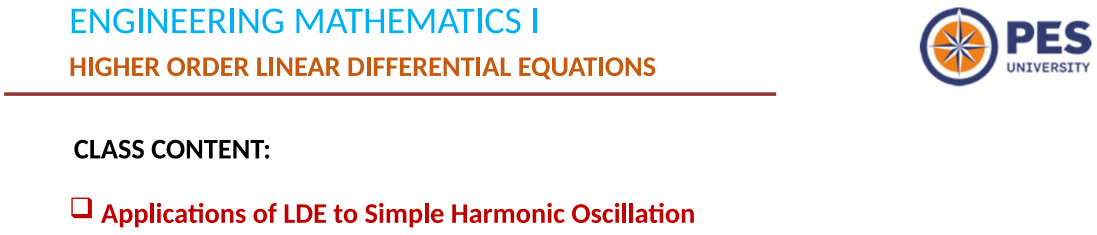

Figure G000166. Applications of DE SHM · source slide page 3.

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APPLICATIONS OF LINEAR DIFFERENTIAL EQUATIONS

- Second order linear differential equations are used to model many situations in physics and engineering.
- Situations like an object with a mass attached to a vertical string and an electrical circuit containing a resistor , an inductor and a capacitor in series can be represented by a linear differential equation.
- Differential equations are useful for modelling and simulating phenomenon and understanding how they operate.

Figure G000167. Applications of DE SHM · source slide page 4.

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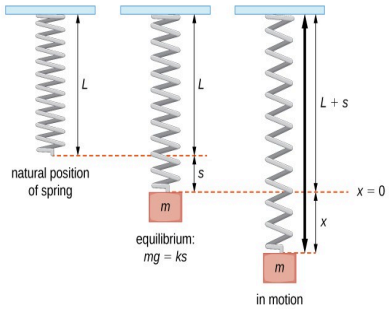
ENGINEERING MATHEMATICS I

SIMPLE HARMONIC OSCILLATION

Consider a mass suspended from a spring which is attached to a support. Gravity is pulling the mass downward and the restoring force of the spring is pulling the spring upwards. When both the forces are equal, we say that the spring is in equilibrium position. If the mass is displaced from equilibrium , the mass oscillates up and down. This motion can be modelled by a second order LDE.



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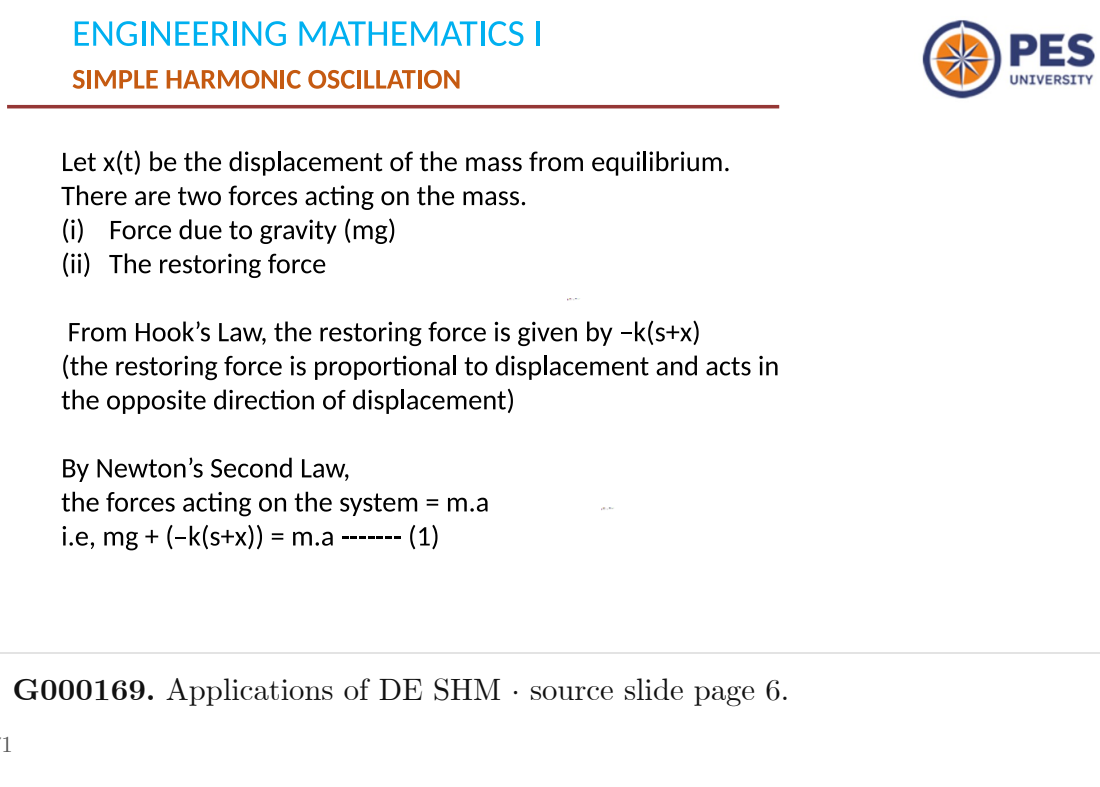

Figure G000169. Applications of DE SHM · source slide page 6.

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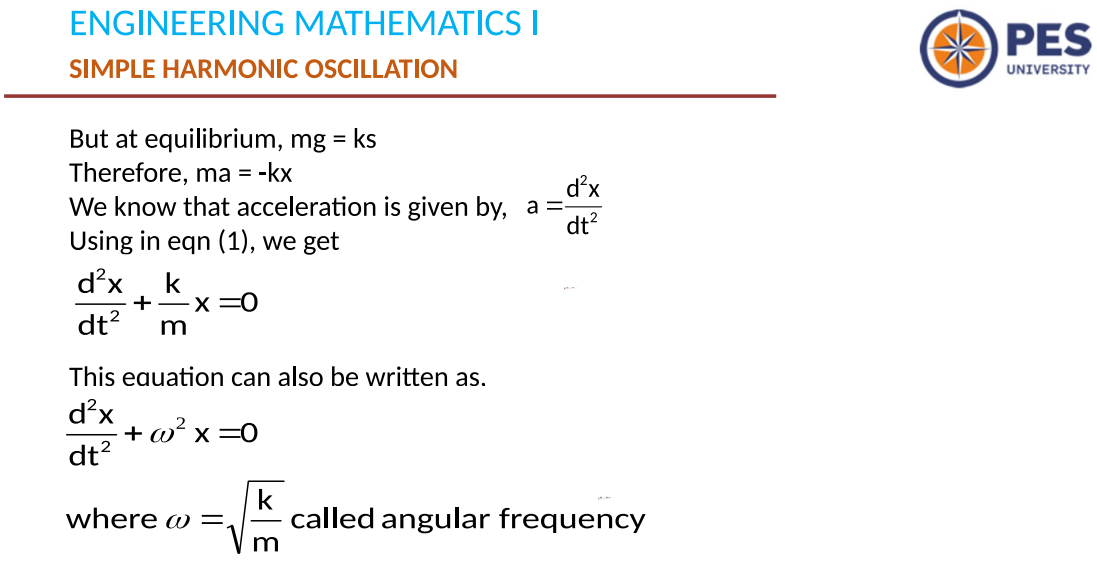

Figure G000170. Applications of DE SHM · source slide page 7.

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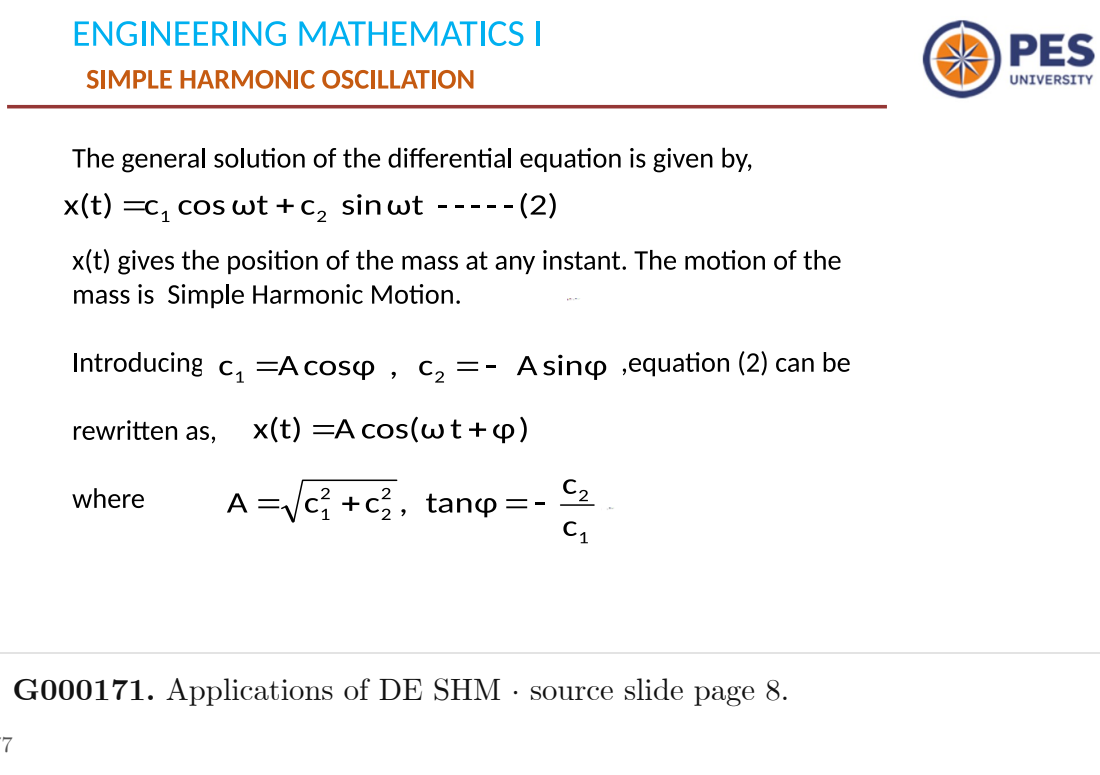

Figure G000171. Applications of DE SHM · source slide page 8.

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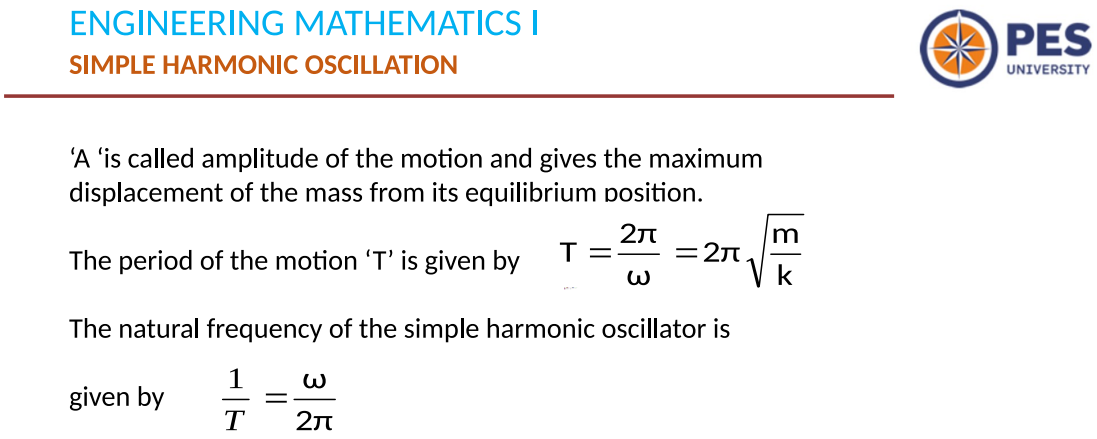

Figure G000172. Applications of DE SHM · source slide page 9.

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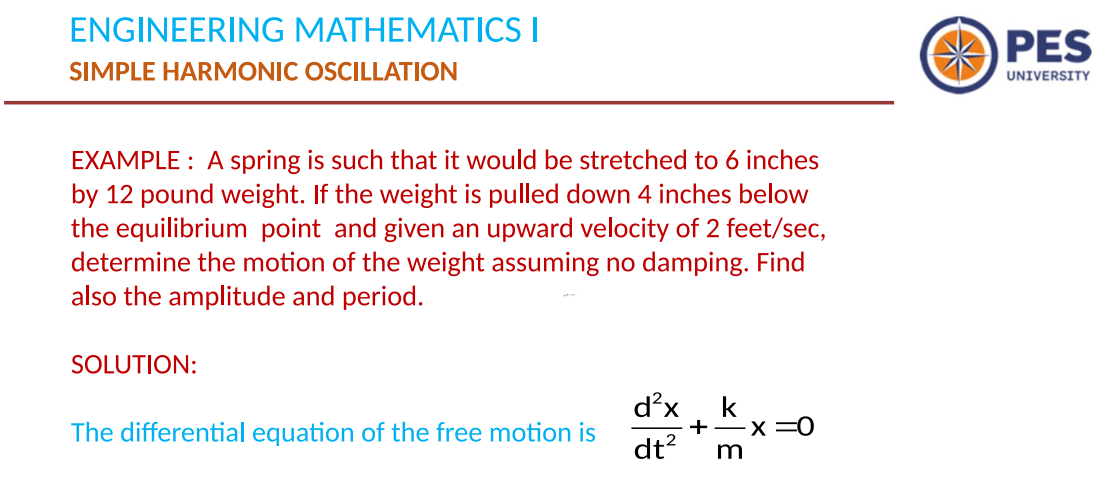

Figure G000173. Applications of DE SHM · source slide page 10.

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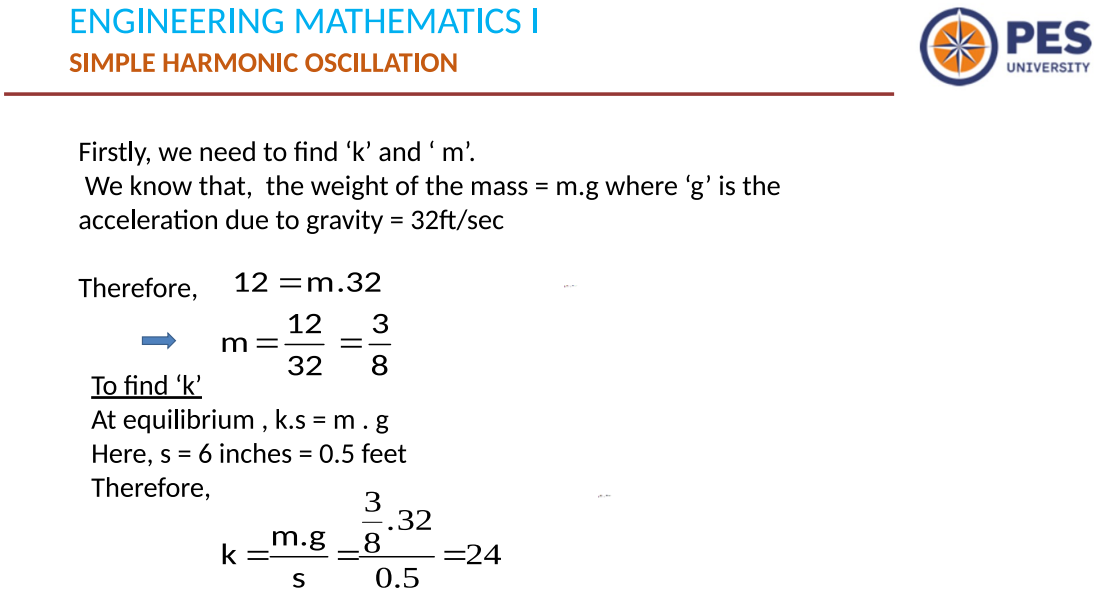

Figure G000174. Applications of DE SHM · source slide page 11.

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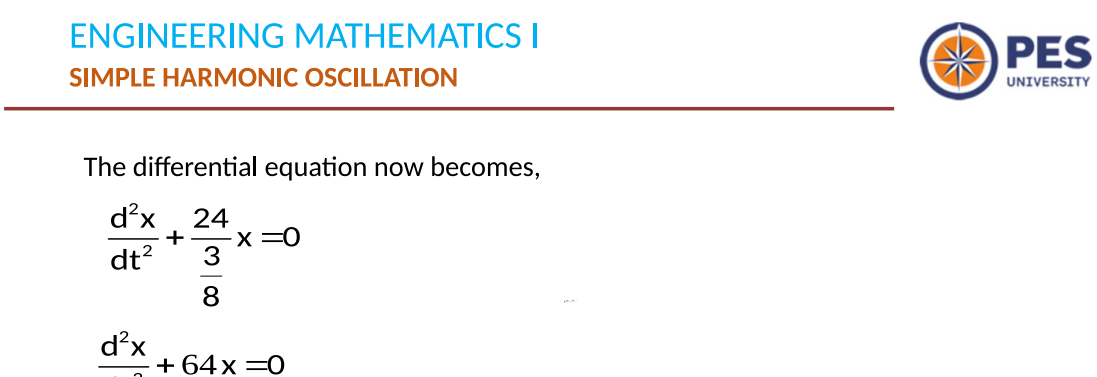

Figure G000175. Applications of DE SHM · source slide page 12.

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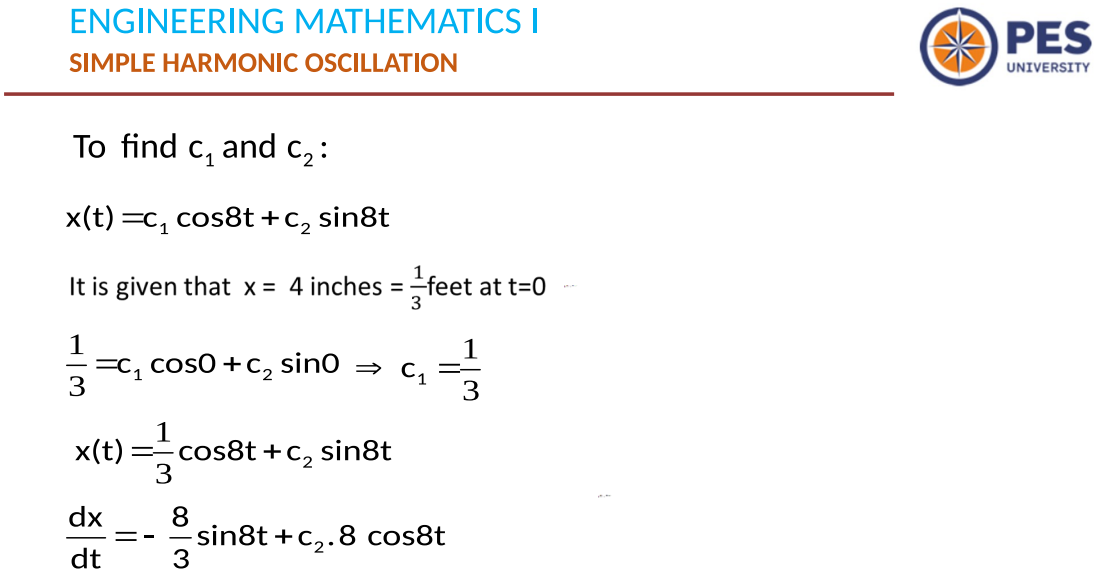

Figure G000176. Applications of DE SHM · source slide page 13.

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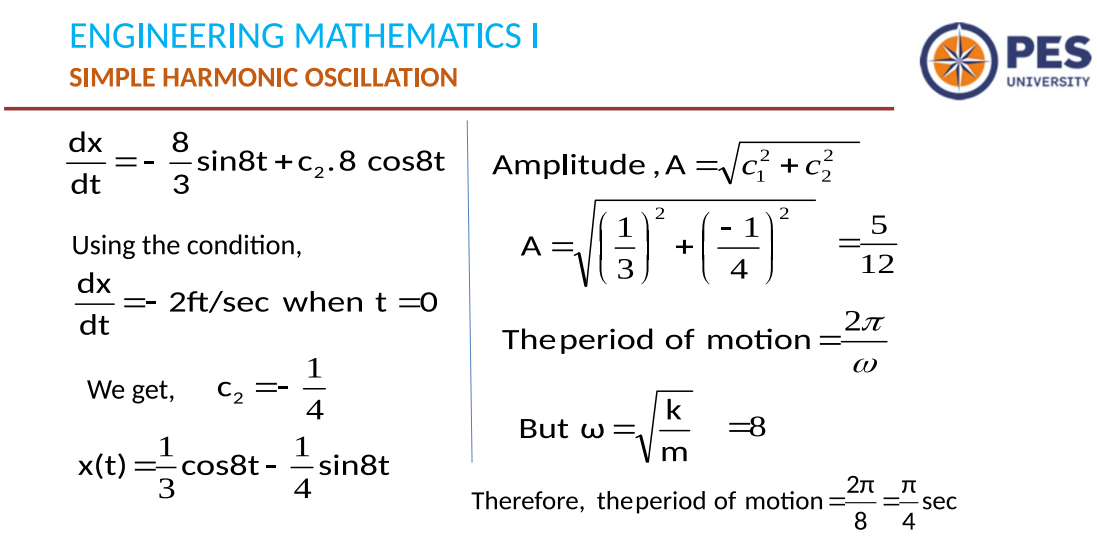

Figure G000177. Applications of DE SHM · source slide page 14.

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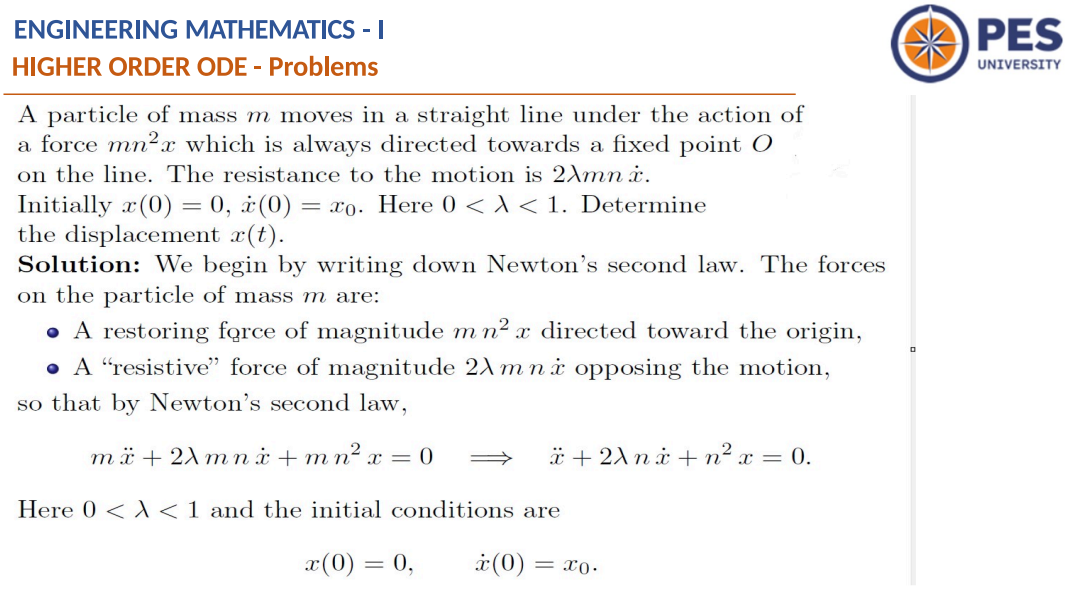

Figure G000178. Applications of DE SHM · source slide page 15.

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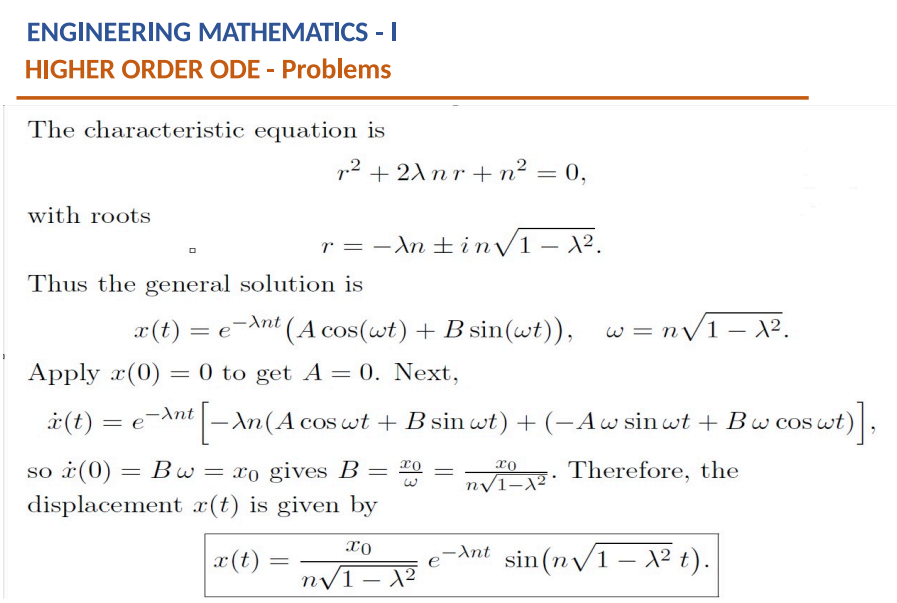

Figure G000179. Applications of DE SHM · source slide page 16.

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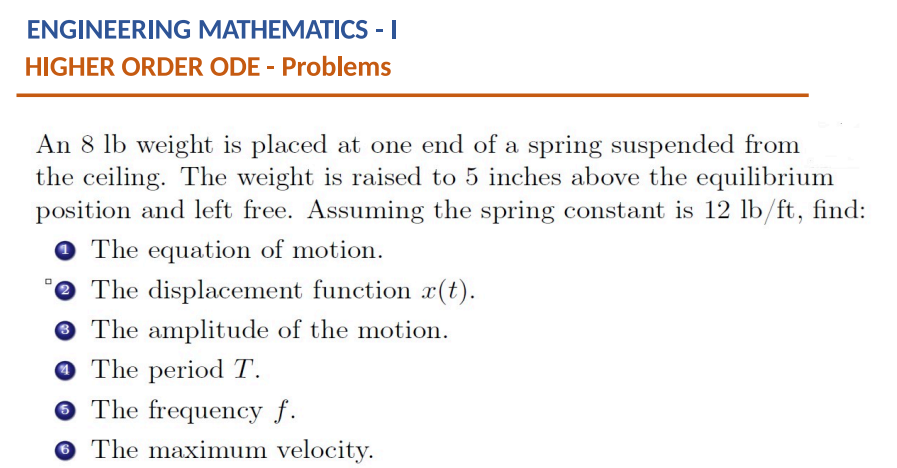

Figure G000180. Applications of DE SHM · source slide page 17.

EV0000804

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ENGINEERING MATHEMATICS - I
HIGHER ORDER ODE - Problems

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Answer: We consider an 8 lb weight suspended from a spring of constant $k = 12$ lb/ft, with no damping. Let $x(t)$ be the displacement (in feet) measured downward from the equilibrium position.

1. Compute the mass.

$$m = \frac{\text{weight}}{g} = \frac{8 \text{ lb}}{32 \text{ ft/s}^2} = \tfrac{1}{4} \text{ slug}.$$

2. Write the equation of motion.
By Newton's law,

$$m\ddot{x} + \underbrace{kx}_{\text{spring force}} = 0 \implies \tfrac{1}{4}\ddot{x} + 12x = 0.$$

Multiply both sides by 4:

$$\ddot{x} + 48x = 0.$$

Figure G000181. Applications of DE SHM · source slide page 18.

EV0000807

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ENGINEERING MATHEMATICS - I
HIGHER ORDER ODE - Problems

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3. Form the characteristic (auxiliary) equation.
Replace $\ddot{x}$ by r^2:

$$r^2 + 48 = 0.$$

4. Solve for the roots r.

$$r^2 = -48 \implies r = \pm\sqrt{-48} = \pm 4i\sqrt{3}.$$

Thus the angular frequency is

$$\omega = 4\sqrt{3}.$$

5. Write the general solution. For complex-conjugate roots $r = \pm i\omega$, the solution is

$$x(t) = C_1 \cos(\omega t) + C_2 \sin(\omega t), \qquad \omega = 4\sqrt{3}.$$

Figure G000182. Applications of DE SHM · source slide page 19.

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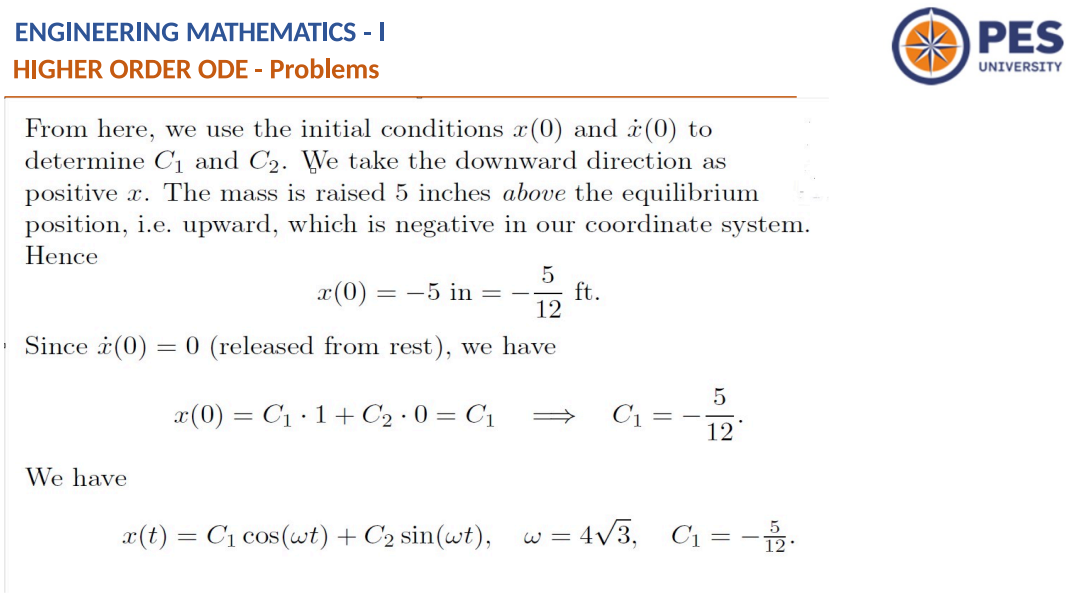

Figure G000183. Applications of DE SHM · source slide page 20.

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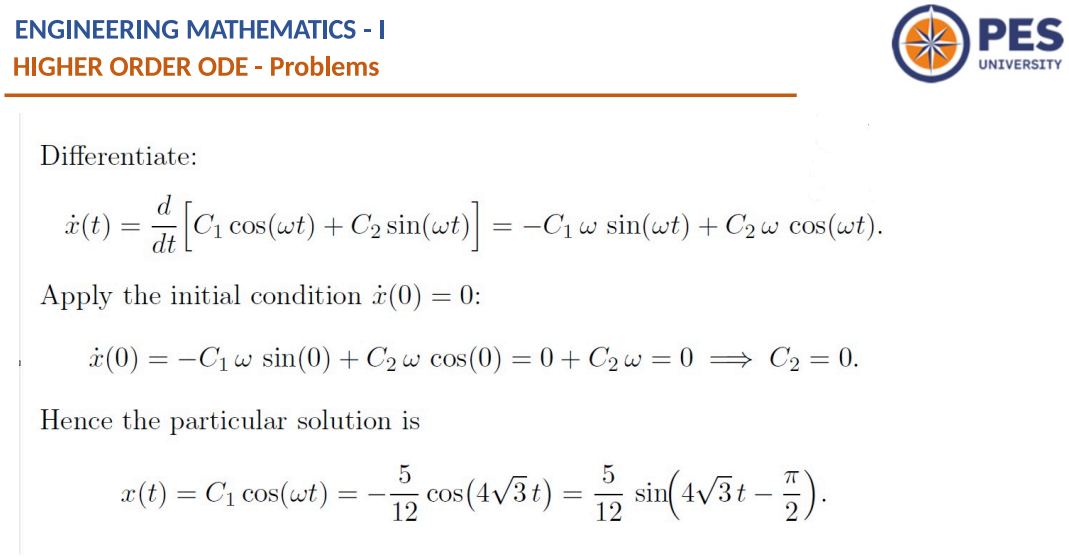

Figure G000184. Applications of DE SHM · source slide page 21.

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ENGINEERING MATHEMATICS - I
HIGHER ORDER ODE - Problems

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Also,

$$\text{Amplitude } A = \frac{5}{12},\ T = \frac{2\pi}{\omega} = \frac{\pi}{2\sqrt{3}},\ f = \frac{1}{T} = \frac{2\sqrt{3}}{\pi},\ v_{\max} = A\,\omega = \frac{5}{\sqrt{3}}.$$

Figure G000185. Applications of DE SHM · source slide page 22.

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Thank you

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Figure G000186. Applications of DE SHM · source slide page 23.

EV0000822

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000755 · EV0000756 · EV0000757
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 2: HIGHER ORDER DIFFERENTIAL EQUATIONS SESSION: 11 SUB-TOPIC: Applications of LDE to Simple Harmonic Oscillation Department of Science and Humanities EV0000758 · EV0000759 · EV0000760
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER LINEAR DIFFERENTIAL EQUATIONS CLASS CONTENT: Applications of LDE to Simple Harmonic Oscillation EV0000761 · EV0000762 · EV0000763
4. Source slide page 4 (extracted text): APPLICATIONS OF LINEAR DIFFERENTIAL EQUATIONS Second order linear differential equations are used to model many situations in physics and engineering. Situations like an object with a mass attached to a vertical string and an electrical circuit containing a resistor , an inductor and a capacitor in series can be represented by a linear differential equation. Differential equations are useful for modelling and simulating phenomenon and understanding how they operate. EV0000764 · EV0000765 · EV0000766
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I SIMPLE HARMONIC OSCILLATION Consider a mass suspended from a spring which is attached to a support. Gravity is pulling the mass downward and the restoring force of the spring is pulling the spring upwards. When both the forces are equal, we say that the spring is in equilibrium position. If the mass is displaced from equilibrium , the mass oscillates up and down. This motion can be modelled by a second order LDE. [Image text, OCR] L+S natural position of spring x=0 m X equilibrium: mg=ks m inmotion EV0000767 · EV0000768 · EV0000769
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I SIMPLE HARMONIC OSCILLATION Let x(t) be the displacement of the mass from equilibrium. There are two forces acting on the mass. Force due to gravity (mg) The restoring force From Hook's Law, the restoring force is given by –k(s+x) (the restoring force is proportional to displacement and acts in the opposite direction of displacement) By Newton's Second Law, the forces acting on the system = m.a i.e, mg + (–k(s+x)) = m.a ------- (1) EV0000770 · EV0000771 · EV0000772
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I SIMPLE HARMONIC OSCILLATION But at equilibrium, mg = ks Therefore, ma = -kx We know that acceleration is given by, Using in eqn (1), we get This equation can also be written as, EV0000773 · EV0000774 · EV0000775
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I SIMPLE HARMONIC OSCILLATION The general solution of the differential equation is given by, x(t) gives the position of the mass at any instant. The motion of the mass is Simple Harmonic Motion. Introducing ,equation (2) can be rewritten as, where EV0000776 · EV0000777 · EV0000778
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I SIMPLE HARMONIC OSCILLATION 'A 'is called amplitude of the motion and gives the maximum displacement of the mass from its equilibrium position. The period of the motion 'T' is given by The natural frequency of the simple harmonic oscillator is given by EV0000779 · EV0000780 · EV0000781

10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS I SIMPLE HARMONIC OSCILLATION EXAMPLE : A spring is such that it would be stretched to 6 inches by 12 pound weight. If the weight is pulled down 4 inches below the equilibrium point and given an upward velocity of 2 feet/sec, determine the motion of the weight assuming no damping. Find also the amplitude and period. SOLUTION: The differential equation of the free motion is EV0000782 · EV0000783 · EV0000784
11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS I SIMPLE HARMONIC OSCILLATION Firstly, we need to find ‘k’ and ‘ m’. We know that, the weight of the mass = m.g where ‘g’ is the acceleration due to gravity = 32ft/sec Therefore, To find ‘k’ At equilibrium , k.s = m . g Here, s = 6 inches = 0.5 feet Therefore, EV0000785 · EV0000786 · EV0000787
12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS I SIMPLE HARMONIC OSCILLATION The differential equation now becomes, The general solution of the differential equation is , EV0000788 · EV0000789 · EV0000790
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS I SIMPLE HARMONIC OSCILLATION EV0000791 · EV0000792 · EV0000793
14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS I SIMPLE HARMONIC OSCILLATION Using the condition, We get, EV0000794 · EV0000795 · EV0000796
15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] A particle of mass m moves in a straight line under the action of a force mn? which is always directed towards a fixed point O on the line. The resistance to the motion is 2入mn . Initially c(0) = 0, (0) = co. Here 0 < 入< 1. Determine the displacement c(t). Solution: We begin by writing down Newton's second law. The forces on the particle of mass m are: A restoring force of magnitude m n? directed toward the origin, ● A “resistive” force of magnitude 2入 m n opposing the motion, so that by Newton's second law, m+2入mnc+mn²x=0 +2入n+n² = 0. Here O < 入 < 1 and the initial conditions are c(0) = 0, c(0) = c0. EV0000797 · EV0000798 · EV0000799
16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] The characteristic equation is r2 + 2入nr + n² = 0, with roots r = -Xn ±inV1 - X2. Thus the general solution is x(t) = e->nt(Acos(wt) + Bsin(wt), w = nV1 - x2. Apply c(0) = 0 to get A = 0. Next, i(t) = e˜Ant [_ -入n(A cos wt + B sin wt) + (-Aw sinwt + Bw cos wt) so (O) = Bw = co gives B = c0 displacement c(t) is given by CO α(t)= -Xnt sin(n/1 - X2 t). nV1- x2 EV0000800 · EV0000801 · EV0000802
17. Source slide page 17 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] An 8 lb weight is placed at one end of a spring suspended from the ceiling. The weight is raised to 5 inches above the equilibrium position and left free. Assuming the spring constant is 12 lb/ft, find: The equation of motion. The displacement function x(t). The amplitude of the motion. The period T. The frequency f. The maximum velocity. EV0000803 · EV0000804 · EV0000805

18. Source slide page 18 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Answer: We consider an 8 lb weight suspended from a spring of constant k = 12 lb/ft, with no damping. Let c(t) be the displacement (in feet) measured downward from the equilibrium position. 1. Compute the mass. weight 8 1b m = slug. 9 32 ft/s2 2. Write the equation of motion. By Newton's law, m+ =0 ↓+12= 0. spring force Multiply both sides by 4: +48 = 0. EV0000806 · EV0000807 · EV0000808

19. Source slide page 19 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] 3. Form the characteristic (auxiliary) equation. Replace by r2: r2 + 48 = 0. 4. Solve for the roots r. r2 = -48 => r = ±√-48 = ±4iv3. Thus the angular frequency is W = 4v3. 5. Write the general solution. For complex-conjugate roots r = ±iw, the solution is c(t) = C1 cos(wt) + C2 sin(wt), W = 4v3. EV0000809 · EV0000810 · EV0000811

20. Source slide page 20 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] From here, we use the initial conditions c(O) and (O) to determine Ci and C2. We take the downward direction as positive c. The mass is raised 5 inches above the equilibrium position, i.e. upward, which is negative in our coordinate system. Hence 5 c(0) = -5 in = ft. 12 Since (O) = 0 (released from rest), we have 5 c(0) = C1 · 1 + C2 · 0 = C1 We have x(t) = C1 cos(wt) + C2 sin(wt), w = 4v3, EV0000812 · EV0000813 · EV0000814

21. Source slide page 21 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Differentiate: c(t) = 1 cos(wt) + C2 sin(wt) = -C1 w sin(wt) + C2 w cos(wt). 1+ Apply the initial condition c(O) = 0: c(0) = -C1 w sin(0) + C2 w cos(0) = 0 + C2w = 0 =→ C2 = 0. Hence the particular solution is 5 5 si(4v√t-) (t) = C1 cos(wt) = cos(4v3t) 二 12 12 EV0000815 · EV0000816 · EV0000817

22. Source slide page 22 (extracted text): ENGINEERING MATHEMATICS - I HIGHER ORDER ODE - Problems [Image text, OCR] Also, 5 2π 1 2V3 TT 5 Amplitude A = T f= Umax =A= 12 2√3 T V3 TT EV0000818 · EV0000819 · EV0000820

23. Source slide page 23 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000821 · EV0000822 · EV0000823

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.708] Hello everyone, welcome back to the session on higher order linear differential equations. In today's session, we will be discussing about an application of linear differential equation to simple harmonic oscillator. [00:00:14.628] Linear differential equations have a wide variety of applications in science and engineering. They are used to model different situations, different phenomena in physics and engineering. Now let us consider an oscillator. Consider a mechanical oscillator and an electrical oscillator. [00:00:32.400] A object with a mass mass attached to a vertical string which is oscillating to and fro. And consider an electrical circuit containing an inductor, capacitor, and resistor in series with a source of EMF. Let us see how [00:00:49.968] How these two situations can be modeled by second order linear differential equations. So let us derive differential equations which represent these two phenomena. [00:01:01.092] We know methods to solve second order linear differential equation. We solve the differential equation and we will learn more about the two different phenomena. So, in general, differential equations are useful for modeling and simulating

the phenomena and understanding how exactly operate. [00:01:21.188] Now, in this session, we will be discussing about a mechanical oscillator. So now let us consider a mass suspended from a spring which is attached to a support. So the gravity of the gravity is pulling the mass downwards. [00:01:38.500] And the restoring force of the spring is pushing it upwards. When the two forces are equal, the spring comes to an equilibrium position. Now, if the mass is pulled downwards, [00:01:51.940] Or if it is pushed upwards, it starts oscillating up and down. So this oscillation can be modeled by a second order linear differential equation. Now let us consider these three diagrams. In the first diagram, there is nothing happening, a spring is just attached to a support. [00:02:11.364] So this is the natural position of the spring. In the second diagram, an object with mass m is attached to a spring which is hung from a level surface. So because of the gravity, the mass is pulled downwards, but the restoring force is pushing it upwards and it comes from the same. [00:02:29.136] To equilibrium position. So at equilibrium, the restoring force of the spring is equal to the gravitational force which is given by Mg. So Mg is equal to K into S. So what is this Mg? This is force due to gravity. This is because of the mass M. [00:02:46.736] And K into S. K is the force constant. We know it is spring constant. K into S is the displacement of the spring from the natural position. So Mg is equal to Kx at equilibrium position. So S is the displacement. [00:03:04.336] Because of the mass m mg is equal to case. Now if the mass is pulled downwards by distance A x of T [00:03:13.796] So the mass starts oscillating up and down. So what is this x? It represents the displacement of the mass from the equilibrium position. So x of t it is a function of t. So it is oscillating up and down. Now let us see how this oscillates. [00:03:31.568] regulatory motion can be represented by a second order linear differential equation. [00:03:37.988] So XFT represents the displacement of the mass from equilibrium. Now there are two forces acting on the mass. So what are the two forces? One is the force due to gravity, and the second one is the restoring force. We know Hook's law. What is from Hooke's Law? Hook's Law states that there [00:03:55.760] Restoring force of the spring is proportional to the displacement. A restoring force is equal to some constant into S S plus X. So what is this S plus X here? It is a total displacement, and K is the proportionality constant, which is called as the force. [00:04:13.328] Constant or the spring constant. And negative sign here indicates that the restoring force is in the opposite direction of the displacement. We know this. It is minus k into s plus x. The total the fourth I mean the restoring force acting on the system. [00:04:30.896] system is minus k into s plus x. Now we have from Newton's second law. What is Newton seconds law? The force acting on the system is equal to m into a where a stands for the acceleration. M into a now [00:04:46.500] Equating the two, the total forces acting on the system is minus force due to gravity, and the second one is the restoring force. So it is mg which is the force due to gravity plus the restoring force is minus k into s plus x is equal to m into a.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.940] A is the acceleration. Now we know that at equilibrium mg is equal to k. So what happens? This mg and k s gets cancelled and it becomes. [00:05:15.652] M A is equal to minus k into x. But we know that the acceleration is a second order derivative of displacement with respect to time t. So a is equal to d square x by dt square. Putting this back in the equation and simplifying the equation becomes d square x by dt square. [00:05:33.424] square plus k by m into x is equal to 0. Now this equation can also be written as d square x by dt square plus omega square x is equal to 0 where omega square is equal to k by m or omega is equal to square root of k by m call as angular frequency of the motion.

[00:05:53.156] Now, so just look at this differential equation. It is a homogeneous differential equation, linear differential equation with constant coefficients. So it is a homogeneous linear differential equation with constant coefficients, and we learn methods to find the general solution of the [00:06:10.928] Homogeneous LDE. So we learned that the general solution of homogeneous LDE has only one component, which is the complementary function. Now how do we find the complementary function of this? Right? The auxiliary equation, which is associated with the differential. [00:06:28.496] equation find out the roots and write the complementary function. So what are the roots here? It is actually square root of omega square minus omega square which is plus r minus i omega. The two roots are complex. So what is the general solution? It is C1 cos omega T. [00:06:46.064] plus C2 sin omega T. X of T is equal to C1 cos omega T plus C2 sin omega T. What is X of T? It is the displacement of the mass from its equilibrium position and this equents a simple harmonic oscillation, simple harmonic motion. Now [00:07:05.988] Introducing another variable phi. So let us take c1 is equal to a cos phi and c2 is equal to minus a sin phi. Put this back and simplify, we get x of t is equal to a into cos of omega t plus phi. [00:07:22.724] Where a is equal to square root of C1 square by C2 square and tan phi is equal to minus C2 by C1. So what is this a? A represent A which is equal to square root of C1 square by C2 square represent the amplitude of the harm. [00:07:40.496] motion and it gives the maximum displacement of the mass from its equilibrium position. Now the period of the motion so what is the period? It is a time taken for one complete oscillation. So it is given by 2 pi by omega and omega is equal to square root of k by [00:07:58.064] So it simplifies to 2 pi into square root of m by k. So you should remember all these formulae. The period of this oscillatory motion is given by 2 pi into square root of m by k. And we'll find out the natural frequency of the harmonic oscillator, which is a reciprocal. [00:08:15.632] Of the period. So which is omega by 2 pi. So what is omega? It is square root of k by m. So 1 by 2. So this is how we can find out the amplitude, frequency, and the period of the oscillatory motion by solving the second order linear difference. [00:08:33.200] Equation which we derive for this simple harmonic motion. Now let us consider a problem. Consider see a big question. A spring is such that it would be stretched to 6 inches by 12 pound weight. [00:08:48.164] If the weight is pulled down 4 inches below the equilibrium point and given an upward velocity of 2 feet per second, determine the motion of the weight assuming no damping, no forces, external forces acting on the system. Find also the amplitude and the period. [00:09:05.936] So, how do we let us analyze the problem? So, what is this? It's given the 12 pound weight. The weight of the object is 12 pound. So, because of the weight, it is stretched to 6 inches. So, what is S over here? S is equal to 6 inches, and W which is equal to Mg is equal to 12 pound. [00:09:25.028] Okay, so using this we can find out the uh we can find we can write the differential equation and we can we can solve the differential equation and get an expression for x of t. But this data which is given over here, the later data is used to find the arbitrary constant c. [00:09:42.800] Okay. So now what is the differential equation of the free motion? We know the differential equation. It is d square x by dt square plus k by m x is equal to 0. So this is a differential equation. Now first we need to find out this k the force constant k and the

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:00.368] Pass of the object M. So how do we find K? K K and M. [00:10:05.488] Now we need to find out K and M. So the weight of the mass, weight of the mass we know it is equal to m into g, where g is the acceleration due to gravity because here it is given in feed. Let us write the gravitational uh mean acceleration due to

gravity. So which is 32 feet per second. Now [00:10:25.764] We have MG is equal to MG is equal to [00:10:30.128] Then this is the weight of the object 12 pounds. So 12 is equal to m into g is 32. So what is the mass of the object? m is equal to 12 by 32, which is equal to 3 by 8. So we found out m. Now to find k I told you we are using the equation k s is equal to [00:10:47.856] M into g s is known, s is given from the question. So it is k s is equal to m into g and k is equal to now what is s is equal to six inches because it is in feet over here, let us convert this into feet. So we know that 12 inches is equal to 1 feet. [00:11:05.424] or 6 inches is equal to 0.5 feet or 1 by 2 feet. So now K is equal to what is K? It is equal to Mg by S. So [00:11:15.568] Uh which what is m m we have derived it is 3 by 8 into this is acceleration due to gravity 32 divided by the displacement of the mass because of the weight uh which is equal to 6. [00:11:30.624] which is equal to how much? 12 pounds. So it is 3 by 8 into 32 divided by ESS is equal to 0.5 feet. So the constant this force constant is equal to 24. We found out Kn. Now let us put this back in the differential equation. [00:11:48.144] So the differential equation becomes 24, I mean d square x by dt square plus 24 by 3 by 8 into x is equal to 0, which simplifies to d square x by dt square plus 64x is equal to 0. So what is omega square over here? 64. Omega is equal to 8. So what is the [00:12:05.872] General solution C1 cos omega t plus C2 sin omega T. So the general solution is C1 cos 8t plus C2 sin t. Did you understand how we got this? Because omega square is equal to 64, omega is equal to 8, and the general solution we wrote. [00:12:23.440] It is C1 cos omega t plus C2 sin omega T. So it is C even cos A T plus C2 sin A T. Now the next question is how do we find the constant C1 and C2? To find the constant C1 and C2, we need to use the data. [00:12:41.008] which is given over there the initial conditions which are given. Now we need to find C1 and C2. So it's given XFT is I mean we know that X of T is equal to C1 cos 8 T plus C2 sine A T. Now [00:12:56.304] To find C1 and C2, we will use the initial condition. What is given at T is equal to 0, X is equal to 4 inches, R1 by 3 feet, that is it is stretched to a distance of 4 inches. So we will take that as T is that position as T is equal to 0. [00:13:13.328] So 4 inches you convert it into feet, which is 1 by 3 feet, put this back in this, we get 1 by 3 is equal to c1 cos 0 plus c2 sine 0. Sine 0 is 0, cos 0 is 1, c1 is equal to 1 by 3. So put this back in the equation, the equation becomes 1 by. [00:13:31.056] Now to find c2, because the velocity is given at t is equal to 0, we need to write an expression for velocity. What is velocity? dx by dt. So differentiate this equation with respect to t. [00:13:48.480] We get dx by dt is equal to 1 by 3 into cos differentiation is minus sine into 8 t differentiation is 8. So it is minus 8 by 3 sin eight t plus c two into 8 into cos 80. So now use the other data which is given. So what is it? [00:14:07.716] Use the condition, it's given dx by dt is equal to 2 feet per second when t is equal to 0. But the negative sign, why is this negative sign? It's it says that it gives a it is given an upward velocity of 2 feet per second. So upward velocity which is in the [00:14:24.688] Opposite direction of the displacement. So this is the convention which is used for upward velocity we use the sign minus. If it is downward velocity, we use the sign plus. Similarly for displacement also. Downward displacement it is given positive sign. Upward displacement [00:14:42.320] We'll give positive negative sign. So dx by dt is equal to minus 2 feet per second when t is equal to 0. So put this back, get the value of c2. C2 is equal to minus 1 by 4. How? Say dx by dt is minus 2 and t is equal to 0. This term by [00:14:59.888] and you get C2 into A t is equal to now minus 2. So C2 is equal to minus 4. So what is X of T? It is 1 by 3 cos C8 minus 1 by 4 sin A T. So this is an expression for the displacement

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:16.580] X of T, it is a function of T time. So you can find out this X at any instant T. Now, so they have asked you to find the amplitude also. The amplitude is given by this is the formula square root of C1 square plus C2 square. So that is C1. [00:15:34.352] is 1 by 3 and C2 is minus 1 by 4, simplify amplitude is equal to 5 by 12. And also what is the next question? Period the motion which is 2 pi by omega. But what is omega? Omega is equal to uh we have derived omega is equal to 8. So it is [00:15:52.612] eight so period of the motion is two pi by eight which is pi by four second [00:16:00.516] This is how we find the period of the motion amplitude by writing the equation for the displacement XFP. So this is how we apply a second order linear differential equation to model a simple harmonic oscillator. Thank you.

EV0000262 · EV0000263 · EV0000264 · EV0000265

Full transcript: 2716 normalized words, 60 timestamped segments, 980.8 seconds duration. Low-confidence segments flagged: 0.

EV0000250 · EV0000251 · EV0000252 · EV0000253

18 Source slide deck: Solutions of Homogeneous DE by introducing Complementary Functions

This entry preserves the complete slide deck titled Solutions of Homogeneous DE by introducing Complementary Functions as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_02, to avoid an unsupported match. EV0000320 · EV0000321

Graphs and diagrams

Figure G000019. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 1.

EV0000321

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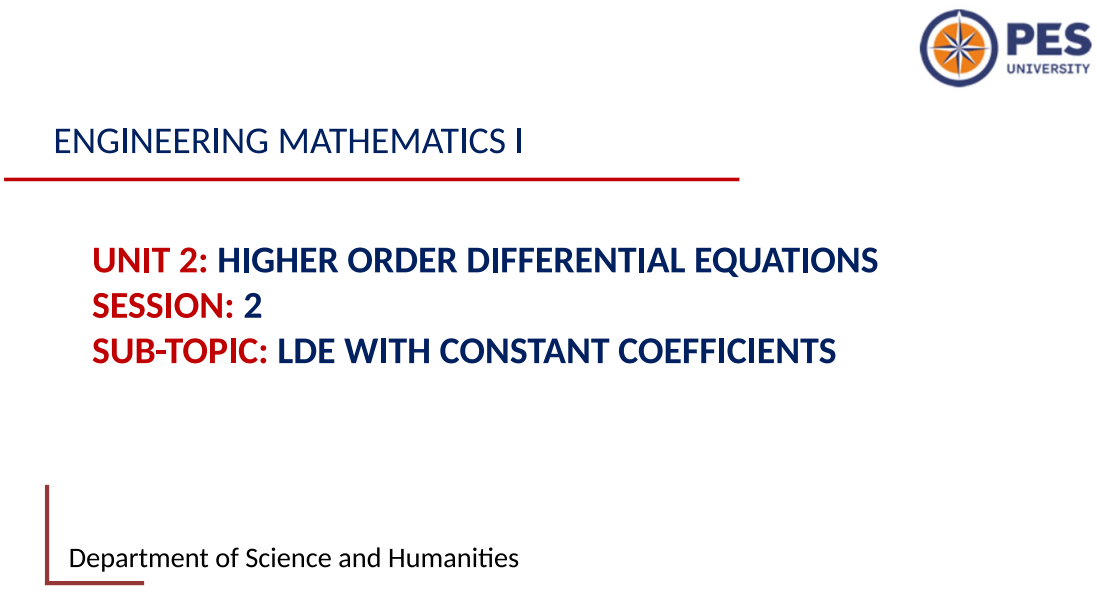

Figure G000020. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 2.

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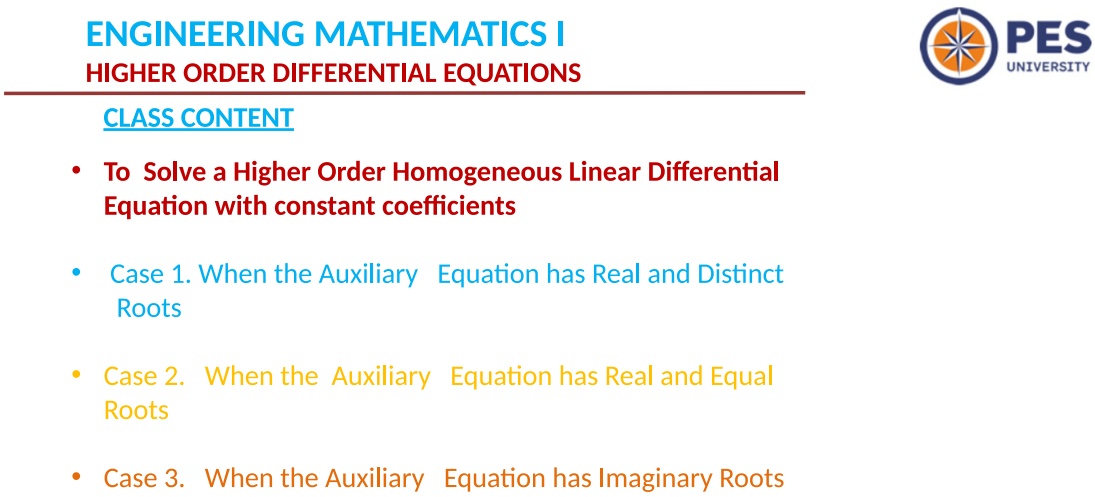

Figure G000021. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 3.

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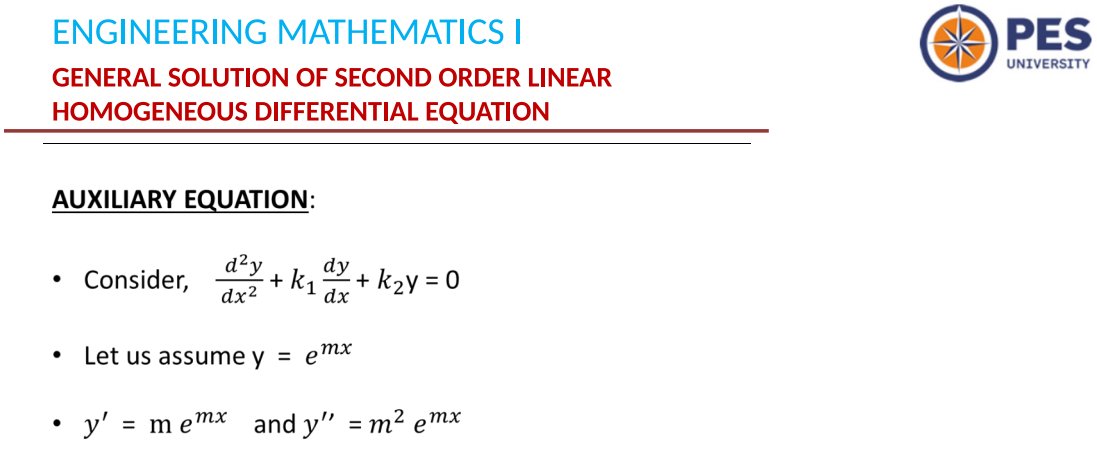

Figure G000022. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 4.

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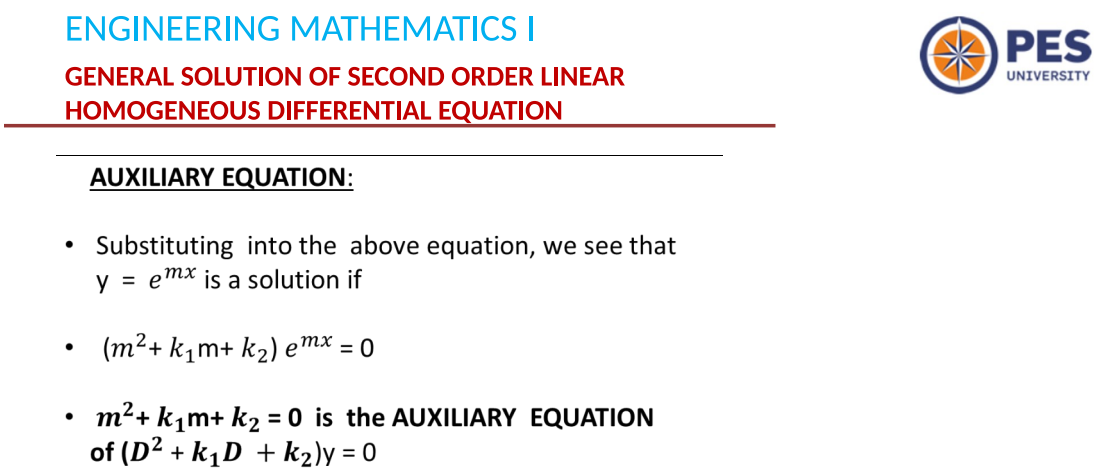

Figure G000023. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 5.

EV0000333

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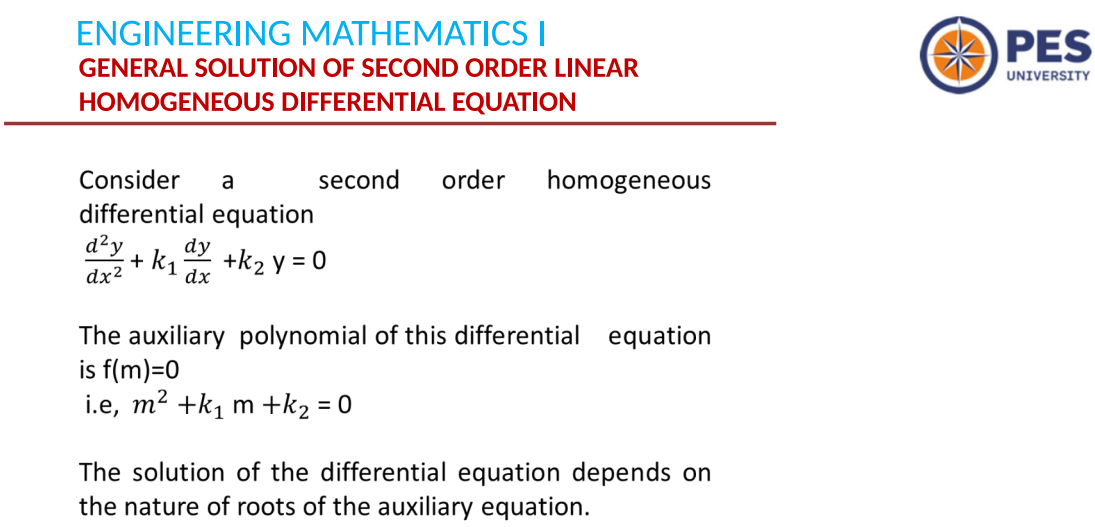

Figure G000024. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 6.

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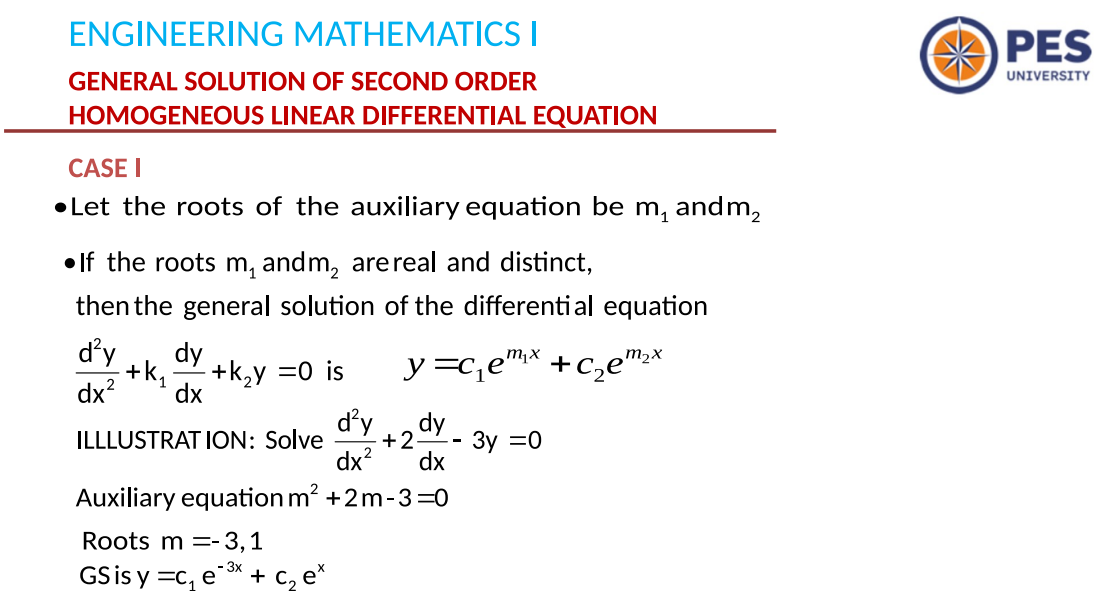

Figure G000025. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 7.

EV0000339

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ENGINEERING MATHEMATICS I

GENERAL SOLUTION OF SECOND ORDER LINEAR DIFFERENTIAL EQUATION

CASE II

- If the roots m_1 and m_2 are real and equal, then the general solution of the differential equation $\frac{d^2y}{dx^2} + k_1\frac{dy}{dx} + k_2 y = 0$ is $y = (c_1 + c_2 x)e^{mx}$

ILLLUSTRATION: Solve $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$

Auxiliary equation $m^2 + 4m + 4 = 0$

Roots $m = -2, -2$

GS is $y = (c_1 + c_2 x)e^{-2x}$

Figure G000026. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 8.

EV0000342

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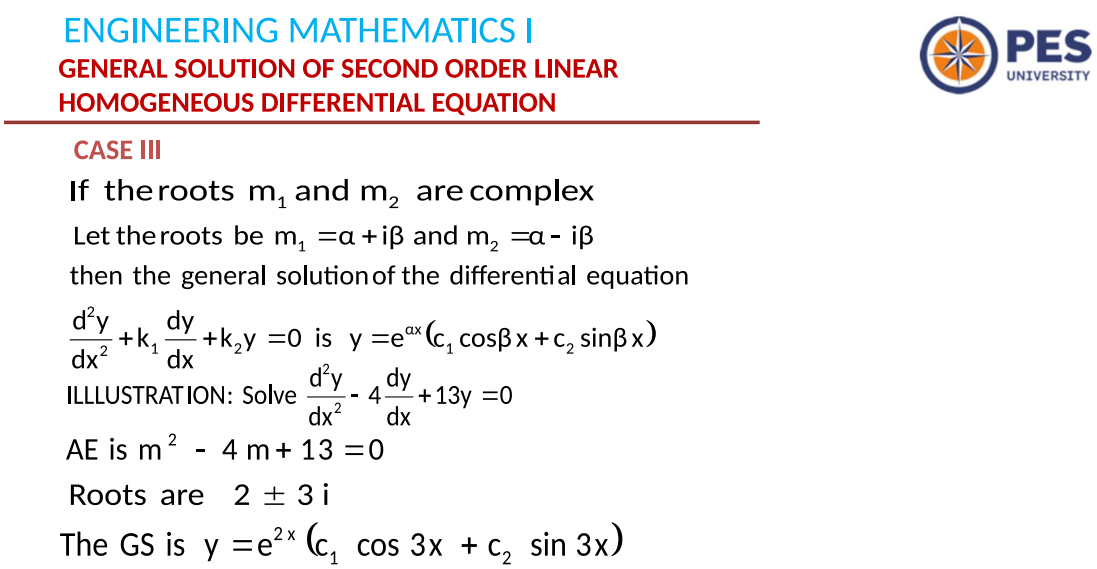

Figure G000027. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 9.

EV0000345

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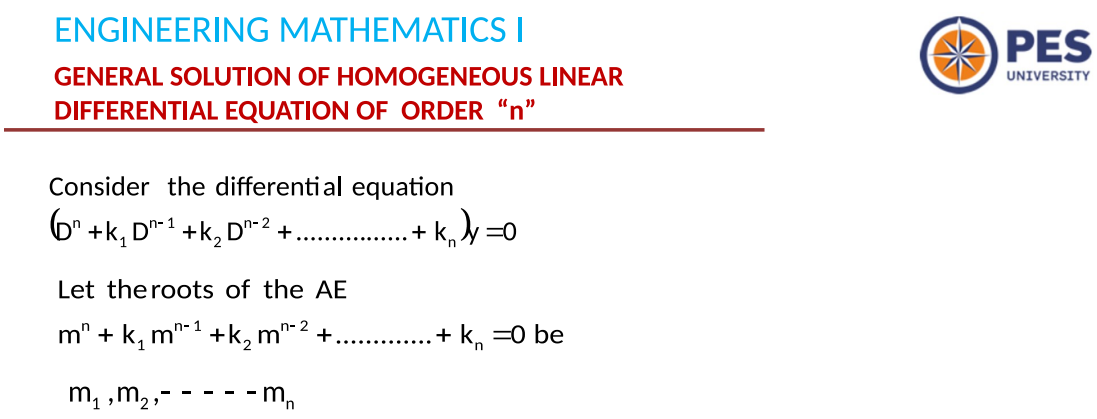

Figure G000028. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 10.

EV0000348

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

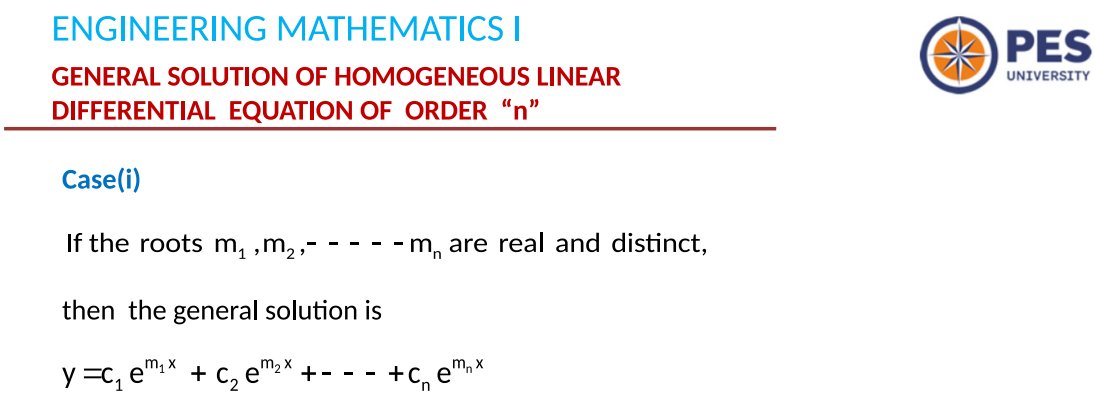

Figure G000029. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 11.

EV0000351

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

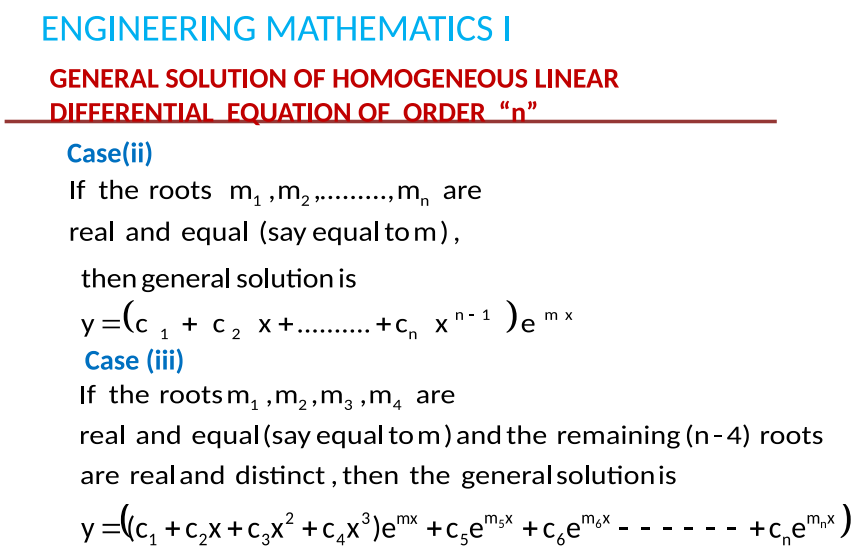

Figure G000030. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 12.

EV0000354

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

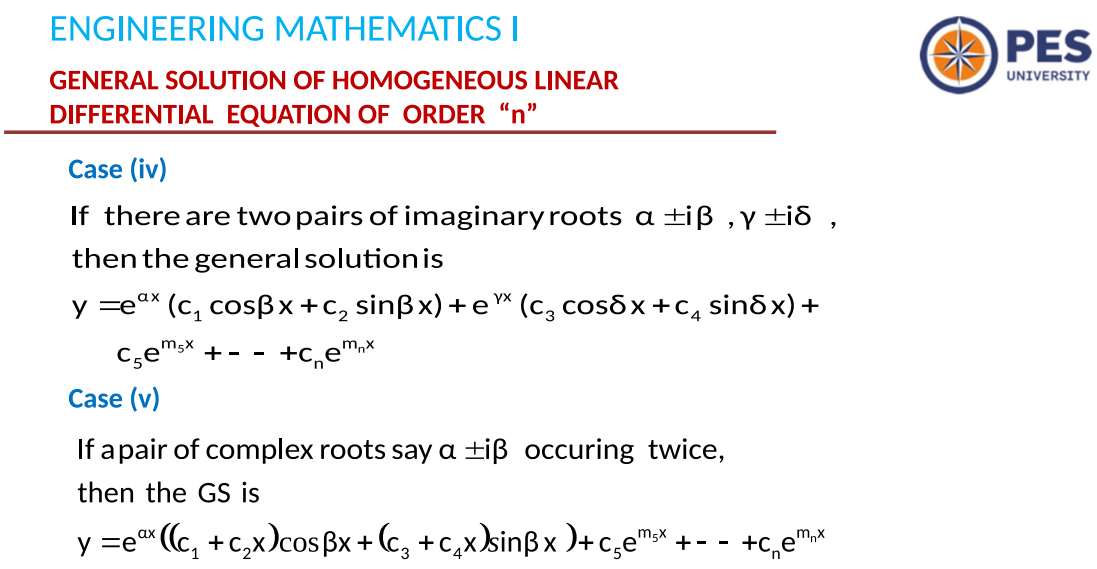

Figure G000031. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 13.

EV0000357

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS I

HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION OF ORDER "n"

PES UNIVERSITY

Solve: $\frac{d^3y}{dx^3} + 4\frac{dy}{dx} = 0$

$(D^3 + 4D)y = 0$

AE is $m^3 + 4m = 0$

Roots are $m = 0, \pm 2i$

$y = c_1 e^{0x} + e^{0x}(c_2 \cos 2x + c_3 \sin 2x)$

Thus, $y = c_1 + c_2 \cos 2x + c_3 \sin 2x$

Figure G000032. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 14.

EV0000360

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000033. Solutions of Homogeneous DE by introducing Complementary Functions · source slide page 15.

EV0000363

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000320 · EV0000321 · EV0000322
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS I UNIT 2: HIGHER ORDER DIFFERENTIAL EQUATIONS SESSION: 2 SUB-TOPIC: LDE with constant coefficients Department of Science and Humanities EV0000323 · EV0000324 · EV0000325
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS I HIGHER ORDER DIFFERENTIAL EQUATIONS CLASS CONTENT To Solve a Higher Order Homogeneous Linear Differential Equation with constant coefficients Case 1. When the Auxiliary Equation has Real and Distinct Roots Case 2. When the Auxiliary Equation has Real and Equal Roots Case 3. When the Auxiliary Equation has Imaginary Roots EV0000326 · EV0000327 · EV0000328
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS I GENERAL SOLUTION OF SECOND ORDER LINEAR HOMOGENEOUS DIFFERENTIAL EQUATION EV0000329 · EV0000330 · EV0000331
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS I GENERAL SOLUTION OF SECOND ORDER LINEAR HOMOGENEOUS DIFFERENTIAL EQUATION EV0000332 · EV0000333 · EV0000334

6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS I GENERAL SOLUTION OF SECOND ORDER LINEAR HOMOGENEOUS DIFFERENTIAL EQUATION EV0000335 · EV0000336 · EV0000337
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS I GENERAL SOLUTION OF SECOND ORDER HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION CASE I EV0000338 · EV0000339 · EV0000340
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS I GENERAL SOLUTION OF SECOND ORDER LINEAR DIFFERENTIAL EQUATION CASE II EV0000341 · EV0000342 · EV0000343
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS I GENERAL SOLUTION OF SECOND ORDER LINEAR HOMOGENEOUS DIFFERENTIAL EQUATION CASE III EV0000344 · EV0000345 · EV0000346
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS I GENERAL SOLUTION OF HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION OF ORDER "n" EV0000347 · EV0000348 · EV0000349
11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS I GENERAL SOLUTION OF HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION OF ORDER "n" Case(i) then the general solution is EV0000350 · EV0000351 · EV0000352
12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS I GENERAL SOLUTION OF HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION OF ORDER "n" Case(ii) Case (iii) EV0000353 · EV0000354 · EV0000355
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS I GENERAL SOLUTION OF HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION OF ORDER "n" Case (iv) Case (v) EV0000356 · EV0000357 · EV0000358
14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS I HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION OF ORDER "n" Solve: Thus, EV0000359 · EV0000360 · EV0000361
15. Source slide page 15 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000362 · EV0000363 · EV0000364

SHORT-NOTE RECAP

Source deck preserved: Solutions of Homogeneous DE by introducing Complementary Functions. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is score-based; this index does not claim a manual content review. EV0000320

19 Historical question-source pages · unit mapping candidates

This section preserves historical paper/question-bank pages whose extracted terms matched Engineering Mathematics - I, Unit_02 (Unit 2) above the dataset's candidate threshold. The association has not been manually confirmed, source text/OCR may be incomplete, and the pages are not rewritten or presented as solved questions. EV0000824 · EV0000825

Important points and why

1. Archive page 3 from Series Solution of Bessel__1GK7DgclHD.pdf; unit match candidate score 0.5; source text/OCR may be partial. Bessel Differential Equation ( ) 2 2 2 0 x y xy x n y ⊠⊠ ⊠ + + − = is known as Bessel differential equation. ENGINEERING MATHEMATICS I The second order linear ordinary differential equation in the form, (1) −−−− Here n is the order of the Bessel differential equation. EV0000824
2. Archive page 14 from Math_Unit2+Unit3+Unit4__1w-kU1CiDL.pdf; unit match candidate score 0.5; source text/OCR may be partial. W - 4 · Non-homogeneous PDE · Linear PDE but order of all P . D aren't equal en : (D2-(D'l 2 + D - D'S z = 0 · Working : · Finding CF , convert F(D , D') into linear factors of the form () D - m , D' - a ,) . (D - myD' -az) .... (D - mpD' - an))z = 0 (F = e* y , (y + m ,x) + ea2y(y + myn) +... + eannyn(y + mnn) a , 92, ... an & m , M2 ... Mn are constant · (D - mD - a) = = 0 p - my - az = 0 Pr + Qq = R After converting du =dy= into linear factors I directly write in -m(dn =dy J& Feder this form y + ma = C logz = <n + log(2 Ch z = 2 . 9 [= f(x) = J(y + mu) z = f(y +mu) . e L · Easy way to factorize non-homogeneous PDE Assume (a, D + b , D' +c)(a2D + b2D +x) en : (2D"-DD' - D + D - D') z coeft of D2 = a , az = ezn +3y coett of D' = b , b2 a, ac = 2 constant = <[2 b, bz = - a : 1 &2 : 2 coeft of D = a , + ac C , 2 = 0 b. : - 1 be : 1 coeft of D' = b , 2 + b 22 a ,x + a24 = 1 c : 0 22 : I b ,c + b24 = - inen, (D-D'S (2D - D' + 1) = (D - D')(D - +z) if same m and a EV0000825

SHORT-NOTE RECAP

2 historical archive pages are included as question-source references. Inspect the original paper before relying on OCR or unit placement; no answers are newly authored here.

EV0000824

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Introduction to higher order differential equations, complementary function and particular integral transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to higher order differential equations, complementary function and particular integral transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to higher order differential equations, complementary function and particular integral transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.
Introduction to higher order differential equations, complementary function and particular integral transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Rules for finding the Complementary Function, Problems transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Rules for finding the Complementary Function, Problems transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Rules for finding the Complementary Function, Problems transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9989; only the JSON segmentation supplies the displayed timestamps.
Rules for finding the Complementary Function, Problems transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 1 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of standard functions - Type 1 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 1 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9990; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 1 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral_Type 1 and Type 2 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral_Type 1 and Type 2 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral_Type 1 and Type 2 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9979; only the JSON segmentation supplies the displayed timestamps.
Particular integral_Type 1 and Type 2 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 2 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 2 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 2 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of standard functions - Type 2 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 3 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 3 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 3 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9983; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 3 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral -Type 3 and Type 4 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 4 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_of_standard_functions_-_Type_4__603379d034480436.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of standard functions - Type 4 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_of_standard_functions_-_Type_4__603379d034480436.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 4 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_of_standard_functions_-_Type_4__603379d034480436.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 4 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_of_standard_functions_-_Type_4__603379d034480436.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Particular Integral when X is any other form transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_Integral_when_X_is_any_other_form__653ae8f75a03fb1e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular Integral when X is any other form transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_Integral_when_X_is_any_other_form__653ae8f75a03fb1e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular Integral when X is any other form transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_Integral_when_X_is_any_other_form__653ae8f75a03fb1e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9975; only the JSON segmentation supplies the displayed timestamps.
Particular Integral when X is any other form transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_Integral_when_X_is_any_other_form__653ae8f75a03fb1e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral when X is any other form_Continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_integral_when_X_is_any_other_form_Continued__5c24666421028690.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral when X is any other form_Continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_integral_when_X_is_any_other_form_Continued__5c24666421028690.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral when X is any other form_Continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_integral_when_X_is_any_other_form_Continued__5c24666421028690.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9967; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral when X is any other form_Continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_integral_when_X_is_any_other_form_Continued__5c24666421028690.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equation__04e8c1e6bb4bea05.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equation__04e8c1e6bb4bea05.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equation__04e8c1e6bb4bea05.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equation__04e8c1e6bb4bea05.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equations_Continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equations_Continued__e286e5075a9d86f4.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Legendre differential equations_Continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equations_Continued__e286e5075a9d86f4.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equations_Continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equations_Continued__e286e5075a9d86f4.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equations_Continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equations_Continued__e286e5075a9d86f4.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method_Continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method_Continued__97963caf81f64a9c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method_Continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method_Continued__97963caf81f64a9c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method_Continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method_Continued__97963caf81f64a9c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Variation of Parameters method_Continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method_Continued__97963caf81f64a9c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method__f1407a459331b120.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method__f1407a459331b120.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method__f1407a459331b120.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
Variation of Parameters method transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Variation_of_Parameters_method__f1407a459331b120.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Applications of higher-order linear differential equations - RLC Circuits transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Applications of higher-order linear differential equations - RLC Circuits transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Applications of higher-order linear differential equations - RLC Circuits transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps.
Applications of higher-order linear differential equations - RLC Circuits transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/014_Applications_of_higher-order_linear_differential_equations_-_RLC_Circuits__2d0217a8ab77a9c8.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Application problems on Linear differential equations - Simple harmonic oscillation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Application problems on Linear differential equations - Simple harmonic oscillation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Application problems on Linear differential equations - Simple harmonic oscillation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9983; only the JSON segmentation supplies the displayed timestamps.
Application problems on Linear differential equations - Simple harmonic oscillation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Introduction to second and higher order DE with constant coefficients	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Introduction to second and higher order DE with constant coefficients	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Introduction to second and higher order DE with constant coefficients (converted PDF; original hash 136d74b8e9f37faca1033756c92ffe840c372acf9410d1f1aa49c4b0f1c7a135)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Solutions of Homogeneous DE by introducing Complementary Functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Solutions of Homogeneous DE by introducing Complementary Functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Solutions of Homogeneous DE by introducing Complementary Functions (converted PDF; original hash 080c29fbcaa2829cff08b42036ee05fea171c295f98bf51470eeb038eb890d8a)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of Type 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral of Type 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Particular integral of Type 1 (converted PDF; original hash 84f8f81caf6c9c435e20a3d93ba55b4f658236fe7202cd78cec415f091aaeea7)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of Type 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral of Type 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral of Type 2 (converted PDF; original hash e2030fff59bf8246db5254db42fc95be9243c269abf0afa18169d12af58417a7)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of Type 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral of Type 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Particular integral of Type 3 (converted PDF; original hash a4e3b73e36d4d5c9a4e485bf34097ac2c7093d9ad3f92951f157fd54ecda9e30)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of Type 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral of Type 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral of Type 4 (converted PDF; original hash acefecfd876a5d49e837f0aff6ec3f919809c4b92bd183f728c942051c82f0bc)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral when X is any other form	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral when X is any other form	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Particular integral when X is any other form (converted PDF; original hash d00a5ae64546151ed598e60717523153e6f0bf0d300c6850837c25310f4ff4fb)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Legendre s differential equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Legendre s differential equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Legendre s differential equation (converted PDF; original hash 38cc6e95d1aeca7f1625a42af8612f370d1be985622142a7a86be4b2b3c74f9f)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Variation of Parameters method	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Variation of Parameters method	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Variation of Parameters method (converted PDF; original hash 7cacd33b88c550da86eed1eba7748a51cfa85706f05516eb85d299dbfe05cca4)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Applications of DE LCR circuit	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Applications of DE LCR circuit	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Applications of DE LCR circuit (converted PDF; original hash e3adc11eb94d5de98210fd2f254c966dc6692cab0f8977b9f66d27597f90cc6b)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Applications of DE SHM	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Applications of DE SHM	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Applications of DE SHM (converted PDF; original hash 7b58c61a57ed91331a4fd457402244c66823d50388e69eb1aefecfae82bf6bfc)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Extracted historical pages: Series Solution of Bessel__1GK7DgclHD.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Math 1/Extra/Extra Questions/Series Solution of Bessel__1GK7DgclHD.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Math_Unit2+Unit3+Unit4__1w-kU1CiDL.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Math 1/Extra/Revision Notes/Math_Unit2+Unit3+Unit4__1w-kU1CiDL.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV00000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.

Evidence	Source unit	Status	Decision
EV000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000010	SRC00002/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000011	SRC00003/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000012	SRC00004/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000013	SRC00005/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000014	SRC00002/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000015	SRC00003/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000016	SRC00004/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000017	SRC00005/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000018	SRC00002/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000019	SRC00003/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000020	SRC00004/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0100. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000021	SRC00005/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000022	SRC00002/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0112. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000023	SRC00003/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0112. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000024	SRC00004/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0112. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000025	SRC00005/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0112. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000026	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000027	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000028	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000029	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000030	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000031	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000032	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000033	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000034	SRC00006/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000035	SRC00007/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000036	SRC00008/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000037	SRC00009/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000038	SRC00006/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000039	SRC00007/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000040	SRC00008/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0081. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000041	SRC00009/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000042	SRC00006/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0090. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000043	SRC00007/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0090. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000044	SRC00008/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0090. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000045	SRC00009/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0090. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000046	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000047	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000048	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9990; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000049	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000050	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000051	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000052	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9990; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000053	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000054	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000055	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000056	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 0.9990; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000057	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000058	SRC00010/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000059	SRC00011/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000060	SRC00012/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 0.9990; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000061	SRC00013/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000062	SRC00010/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0085. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000063	SRC00011/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0085. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000064	SRC00012/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0085. Sidecar variants compare at similarity 0.9990; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000065	SRC00013/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0085. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000066	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000067	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000068	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000069	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000070	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000071	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000072	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000073	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000074	SRC00014/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0046. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000075	SRC00015/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0046. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000076	SRC00016/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0046. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000077	SRC00017/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0046. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000078	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000079	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000080	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000081	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000082	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000083	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000084	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000085	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000086	SRC00018/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000087	SRC00019/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000088	SRC00020/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000089	SRC00021/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000090	SRC00018/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000091	SRC00019/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000092	SRC00020/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000093	SRC00021/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000094	SRC00018/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0097. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000095	SRC00019/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0097. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000096	SRC00020/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0097. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000097	SRC00021/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0097. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000098	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000099	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000100	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000101	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000102	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000103	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000104	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000105	SRC00025/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000106	SRC00022/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0050. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000107	SRC00023/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0050. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000108	SRC00024/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0050. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000109	SRC00025/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0050. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000110	SRC00026/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000111	SRC00027/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000112	SRC00028/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000113	SRC00029/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 14	SRC00026/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 15	SRC00027/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 16	SRC00028/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 17	SRC00029/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 18	SRC00026/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 19	SRC00027/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 20	SRC00028/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000121	SRC00029/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000122	SRC00026/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000123	SRC00027/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000124	SRC00028/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0055. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000125	SRC00029/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000126	SRC00030/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000127	SRC00031/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001128	SRC00032/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001129	SRC00033/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001130	SRC00030/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001131	SRC00031/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001132	SRC00032/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001133	SRC00033/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001134	SRC00030/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000135	SRC00031/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000136	SRC00032/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000137	SRC00033/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000138	SRC00030/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000139	SRC00031/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000140	SRC00032/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0071. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000141	SRC00033/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0071. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000142	SRC00034/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000143	SRC00035/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000144	SRC00036/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9975; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000145	SRC00037/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000146	SRC00034/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000147	SRC00035/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000148	SRC00036/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare at similarity 0.9975; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001149	SRC00037/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001150	SRC00034/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001151	SRC00035/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001152	SRC00036/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0054. Sidecar variants compare at similarity 0.9975; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001153	SRC00037/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001154	SRC00034/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001155	SRC00035/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000156	SRC00036/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0055. Sidecar variants compare at similarity 0.9975; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000157	SRC00037/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000158	SRC00038/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000159	SRC00039/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000160	SRC00040/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9967; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000161	SRC00041/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000162	SRC00038/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000163	SRC00039/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000164	SRC00040/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 0.9967; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000165	SRC00041/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000166	SRC00042/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000167	SRC00043/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000168	SRC00044/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000169	SRC00045/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001170	SRC00042/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001171	SRC00043/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001172	SRC00044/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001173	SRC00045/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001174	SRC00042/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001175	SRC00043/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001176	SRC00044/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000177	SRC00045/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000178	SRC00042/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000179	SRC00043/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000180	SRC00044/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000181	SRC00045/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000182	SRC00042/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0087. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000183	SRC00043/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0087. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001184	SRC00044/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0087. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001185	SRC00045/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0087. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001186	SRC00046/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001187	SRC00047/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001188	SRC00048/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001189	SRC00049/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001190	SRC00046/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001191	SRC00047/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001192	SRC00048/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001193	SRC00049/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001194	SRC00046/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001195	SRC00047/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001196	SRC00048/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001197	SRC00049/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 98	SRC00046/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 99	SRC00047/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 00	SRC00048/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0080. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 01	SRC00049/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 02	SRC00046/t00120 0-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0092. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 03	SRC00047/t00120 0-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0092. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 04	SRC00048/t00120 0-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0092. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000205	SRC00049/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0092. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000206	SRC00050/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000207	SRC00051/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000208	SRC00052/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000209	SRC00053/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000210	SRC00050/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000211	SRC00051/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000212	SRC00052/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000213	SRC00053/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000214	SRC00050/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000215	SRC00051/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000216	SRC00052/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000217	SRC00053/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000218	SRC00050/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00002219	SRC00051/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002220	SRC00052/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002221	SRC00053/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002222	SRC00050/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0098. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002223	SRC00051/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0098. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002224	SRC00052/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0098. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002225	SRC00053/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0098. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00002026	SRC00054/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002027	SRC00055/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002028	SRC00056/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002029	SRC00057/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002030	SRC00054/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002031	SRC00055/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002032	SRC00056/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000233	SRC00057/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000234	SRC00054/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000235	SRC00055/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000236	SRC00056/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000237	SRC00057/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000238	SRC00058/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000239	SRC00059/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000240	SRC00060/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000241	SRC00061/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000242	SRC00058/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000243	SRC00059/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000244	SRC00060/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000245	SRC00061/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000246	SRC00058/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000247	SRC00059/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000248	SRC00060/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000249	SRC00061/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0044. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000250	SRC00062/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000251	SRC00063/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000252	SRC00064/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000253	SRC00065/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00002254	SRC00062/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002255	SRC00063/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002256	SRC00064/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002257	SRC00065/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002258	SRC00062/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002259	SRC00063/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002260	SRC00064/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0056. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00002 61	SRC00065/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 62	SRC00062/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 63	SRC00063/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 64	SRC00064/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0060. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 65	SRC00065/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 66	SRC00066/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 67	SRC00068/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00002268	SRC00067/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00002269	SRC00066/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002270	SRC00068/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002271	SRC00067/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00002272	SRC00066/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002273	SRC00068/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002274	SRC00067/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 75	SRC00066/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 76	SRC00068/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 77	SRC00067/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 78	SRC00066/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 79	SRC00068/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 80	SRC00067/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 81	SRC00066/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 82	SRC00068/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 83	SRC00067/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 84	SRC00066/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 85	SRC00068/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 86	SRC00067/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000287	SRC00066/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000288	SRC00068/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000289	SRC00067/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000290	SRC00066/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000291	SRC00068/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000292	SRC00067/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000293	SRC00066/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000294	SRC00068/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000295	SRC00067/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000296	SRC00066/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000297	SRC00068/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000298	SRC00067/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 99	SRC00066/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 00	SRC00068/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 01	SRC00067/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 02	SRC00066/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 03	SRC00068/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 04	SRC00067/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 05	SRC00066/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 06	SRC00068/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 07	SRC00067/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 08	SRC00066/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 09	SRC00068/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 10	SRC00067/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 11	SRC00066/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 12	SRC00068/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 13	SRC00067/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 14	SRC00066/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 15	SRC00068/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 16	SRC00067/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000317	SRC00066/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000318	SRC00068/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/001_Introduction_to_second_and_higher_order_DE_with_constant_coefficients__8ef20ff90fd33088.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000319	SRC00067/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000320	SRC00069/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000321	SRC00071/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000322	SRC00070/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000323	SRC00069/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000324	SRC00071/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000325	SRC00070/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000326	SRC00069/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000327	SRC00071/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000328	SRC00070/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003329	SRC00069/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003330	SRC00071/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003331	SRC00070/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003332	SRC00069/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003333	SRC00071/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003334	SRC00070/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000335	SRC00069/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000336	SRC00071/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000337	SRC00070/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000338	SRC00069/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000339	SRC00071/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000340	SRC00070/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003341	SRC00069/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003342	SRC00071/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003343	SRC00070/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00003344	SRC00069/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003345	SRC00071/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003346	SRC00070/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000347	SRC00069/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000348	SRC00071/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000349	SRC00070/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000350	SRC00069/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000351	SRC00071/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000352	SRC00070/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000353	SRC00069/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000354	SRC00071/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000355	SRC00070/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000356	SRC00069/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000357	SRC00071/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000358	SRC00070/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 59	SRC00069/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 60	SRC00071/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 61	SRC00070/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 62	SRC00069/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 63	SRC00071/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 64	SRC00070/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 65	SRC00072/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 66	SRC00074/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 67	SRC00073/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 68	SRC00072/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 69	SRC00074/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 70	SRC00073/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 71	SRC00072/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 72	SRC00074/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 73	SRC00073/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 74	SRC00072/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 75	SRC00074/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 76	SRC00073/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003377	SRC00072/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003378	SRC00074/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003379	SRC00073/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00003380	SRC00072/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003381	SRC00074/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003382	SRC00073/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003383	SRC00072/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003384	SRC00074/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003385	SRC00073/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00003386	SRC00072/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003387	SRC00074/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003388	SRC00073/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003389	SRC00072/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003390	SRC00074/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003391	SRC00073/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00003392	SRC00072/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003393	SRC00074/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003394	SRC00073/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000395	SRC00072/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000396	SRC00074/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000397	SRC00073/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000398	SRC00072/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000399	SRC00074/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000400	SRC00073/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000401	SRC00072/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000402	SRC00074/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000403	SRC00073/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000404	SRC00072/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000405	SRC00074/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000406	SRC00073/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000407	SRC00072/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000408	SRC00074/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000409	SRC00073/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000410	SRC00072/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000411	SRC00074/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000412	SRC00073/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000413	SRC00072/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000414	SRC00074/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000415	SRC00073/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000416	SRC00072/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000417	SRC00074/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000418	SRC00073/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000419	SRC00075/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000420	SRC00077/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000421	SRC00076/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000422	SRC00075/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000423	SRC00077/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000424	SRC00076/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004025	SRC00075/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004026	SRC00077/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004027	SRC00076/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00004028	SRC00075/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004029	SRC00077/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004030	SRC00076/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000431	SRC00075/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000432	SRC00077/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000433	SRC00076/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000434	SRC00075/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000435	SRC00077/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000436	SRC00076/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000437	SRC00075/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000438	SRC00077/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000439	SRC00076/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000440	SRC00075/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000441	SRC00077/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000442	SRC00076/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000443	SRC00075/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000444	SRC00077/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000445	SRC00076/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000446	SRC00075/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000447	SRC00077/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000448	SRC00076/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000449	SRC00075/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000450	SRC00077/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000451	SRC00076/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000452	SRC00075/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000453	SRC00077/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000454	SRC00076/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000455	SRC00075/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000456	SRC00077/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000457	SRC00076/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000458	SRC00075/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000459	SRC00077/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000460	SRC00076/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000461	SRC00075/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000462	SRC00077/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000463	SRC00076/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000464	SRC00075/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000465	SRC00077/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000466	SRC00076/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000467	SRC00075/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000468	SRC00077/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000469	SRC00076/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000470	SRC00075/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000471	SRC00077/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000472	SRC00076/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000473	SRC00078/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000474	SRC00080/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000475	SRC00079/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000476	SRC00078/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000477	SRC00080/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000478	SRC00079/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 79	SRC00078/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 80	SRC00080/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 81	SRC00079/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 82	SRC00078/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 83	SRC00080/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 84	SRC00079/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000485	SRC00078/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000486	SRC00080/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000487	SRC00079/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000488	SRC00078/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000489	SRC00080/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000490	SRC00079/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000491	SRC00078/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000492	SRC00080/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000493	SRC00079/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000494	SRC00078/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000495	SRC00080/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000496	SRC00079/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 97	SRC00078/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 98	SRC00080/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 99	SRC00079/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 00	SRC00078/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 01	SRC00080/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 02	SRC00079/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000503	SRC00078/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000504	SRC00080/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000505	SRC00079/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000506	SRC00078/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000507	SRC00080/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000508	SRC00079/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000509	SRC00078/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000510	SRC00080/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000511	SRC00079/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000512	SRC00078/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000513	SRC00080/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000514	SRC00079/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000515	SRC00078/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000516	SRC00080/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000517	SRC00079/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000518	SRC00081/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000519	SRC00083/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000520	SRC00082/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000521	SRC00081/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000522	SRC00083/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000523	SRC00082/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000524	SRC00081/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000525	SRC00083/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000526	SRC00082/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 27	SRC00081/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 28	SRC00083/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 29	SRC00082/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 30	SRC00081/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 31	SRC00083/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 32	SRC00082/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000533	SRC00081/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000534	SRC00083/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000535	SRC00082/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000536	SRC00081/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000537	SRC00083/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000538	SRC00082/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000539	SRC00081/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000540	SRC00083/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000541	SRC00082/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000542	SRC00081/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000543	SRC00083/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000544	SRC00082/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000545	SRC00081/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000546	SRC00083/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000547	SRC00082/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000548	SRC00081/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000549	SRC00083/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000550	SRC00082/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000551	SRC00081/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000552	SRC00083/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000553	SRC00082/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000554	SRC00081/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000555	SRC00083/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000556	SRC00082/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000557	SRC00081/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000558	SRC00083/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000559	SRC00082/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000560	SRC00081/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000561	SRC00083/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/006_Particular_integral_of_Type_4__45b50704e800c4c6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000562	SRC00082/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000563	SRC00084/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000564	SRC00086/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000565	SRC00085/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000566	SRC00084/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000567	SRC00086/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000568	SRC00085/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000569	SRC00084/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000570	SRC00086/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000571	SRC00085/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000572	SRC00084/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000573	SRC00086/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000574	SRC00085/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000575	SRC00084/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000576	SRC00086/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000577	SRC00085/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000578	SRC00084/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000579	SRC00086/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000580	SRC00085/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000581	SRC00084/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000582	SRC00086/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000583	SRC00085/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000584	SRC00084/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000585	SRC00086/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000586	SRC00085/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000587	SRC00084/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000588	SRC00086/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000589	SRC00085/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000590	SRC00084/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000591	SRC00086/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000592	SRC00085/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000593	SRC00084/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000594	SRC00086/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000595	SRC00085/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000596	SRC00084/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000597	SRC00086/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000598	SRC00085/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 99	SRC00084/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006 00	SRC00086/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00006 01	SRC00085/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV00006 02	SRC00084/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006 03	SRC00086/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00006 04	SRC00085/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000605	SRC00084/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000606	SRC00086/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000607	SRC00085/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000608	SRC00084/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000609	SRC00086/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000610	SRC00085/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000611	SRC00084/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000612	SRC00086/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000613	SRC00085/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000614	SRC00084/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000615	SRC00086/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000616	SRC00085/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000617	SRC00087/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000618	SRC00089/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000619	SRC00088/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000620	SRC00087/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000621	SRC00089/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000622	SRC00088/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000623	SRC00087/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000624	SRC00089/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000625	SRC00088/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000626	SRC00087/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000627	SRC00089/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000628	SRC00088/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000629	SRC00087/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000630	SRC00089/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000631	SRC00088/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000632	SRC00087/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000633	SRC00089/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000634	SRC00088/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000635	SRC00087/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000636	SRC00089/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000637	SRC00088/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000638	SRC00087/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000639	SRC00089/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000640	SRC00088/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000641	SRC00087/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000642	SRC00089/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000643	SRC00088/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000644	SRC00087/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000645	SRC00089/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000646	SRC00088/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000647	SRC00087/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000648	SRC00089/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Legendre_s_differential_equation__0cbc3f8639965d66.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000649	SRC00088/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000650	SRC00090/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000651	SRC00092/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000652	SRC00091/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000653	SRC00090/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000654	SRC00092/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000655	SRC00091/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000656	SRC00090/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000657	SRC00092/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000658	SRC00091/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000659	SRC00090/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000660	SRC00092/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000661	SRC00091/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000662	SRC00090/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000663	SRC00092/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000664	SRC00091/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000665	SRC00090/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000666	SRC00092/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000667	SRC00091/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000668	SRC00090/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000669	SRC00092/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000670	SRC00091/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000671	SRC00090/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000672	SRC00092/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000673	SRC00091/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000674	SRC00090/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000675	SRC00092/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000676	SRC00091/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000677	SRC00090/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000678	SRC00092/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000679	SRC00091/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000680	SRC00090/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000681	SRC00092/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000682	SRC00091/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000683	SRC00090/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000684	SRC00092/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000685	SRC00091/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000686	SRC00090/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000687	SRC00092/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000688	SRC00091/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000689	SRC00090/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000690	SRC00092/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000691	SRC00091/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000692	SRC00090/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000693	SRC00092/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000694	SRC00091/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000695	SRC00090/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000696	SRC00092/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000697	SRC00091/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000698	SRC00090/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000699	SRC00092/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000700	SRC00091/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00007 01	SRC00090/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007 02	SRC00092/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/011_Variation_of_Parameters_method__56cafdd3e4db312b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 03	SRC00091/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV00007 04	SRC00093/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007 05	SRC00095/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 06	SRC00094/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000707	SRC00093/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000708	SRC00095/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000709	SRC00094/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000710	SRC00093/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000711	SRC00095/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000712	SRC00094/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000713	SRC00093/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000714	SRC00095/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000715	SRC00094/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000716	SRC00093/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000717	SRC00095/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000718	SRC00094/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000719	SRC00093/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000720	SRC00095/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000721	SRC00094/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000722	SRC00093/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000723	SRC00095/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000724	SRC00094/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00007 25	SRC00093/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007 26	SRC00095/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 27	SRC00094/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00007 28	SRC00093/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007 29	SRC00095/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 30	SRC00094/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000731	SRC00093/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000732	SRC00095/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000733	SRC00094/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000734	SRC00093/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000735	SRC00095/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000736	SRC00094/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000737	SRC00093/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000738	SRC00095/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000739	SRC00094/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000740	SRC00093/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000741	SRC00095/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000742	SRC00094/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000743	SRC00093/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000744	SRC00095/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000745	SRC00094/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000746	SRC00093/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000747	SRC00095/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000748	SRC00094/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000749	SRC00093/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000750	SRC00095/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000751	SRC00094/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000752	SRC00093/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000753	SRC00095/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/014_Applications_of_DE-LCR_circuit__1fb753421ad385f8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000754	SRC00094/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000755	SRC00096/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000756	SRC00098/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000757	SRC00097/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000758	SRC00096/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000759	SRC00098/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000760	SRC00097/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000761	SRC00096/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000762	SRC00098/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000763	SRC00097/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000764	SRC00096/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000765	SRC00098/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000766	SRC00097/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000767	SRC00096/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000768	SRC00098/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000769	SRC00097/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000770	SRC00096/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000771	SRC00098/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000772	SRC00097/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000773	SRC00096/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000774	SRC00098/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000775	SRC00097/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000776	SRC00096/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000777	SRC00098/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000778	SRC00097/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00007 79	SRC00096/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007 80	SRC00098/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 81	SRC00097/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00007 82	SRC00096/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007 83	SRC00098/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 84	SRC00097/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000785	SRC00096/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000786	SRC00098/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000787	SRC00097/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000788	SRC00096/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000789	SRC00098/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000790	SRC00097/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000791	SRC00096/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000792	SRC00098/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000793	SRC00097/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000794	SRC00096/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000795	SRC00098/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000796	SRC00097/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00007 97	SRC00096/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007 98	SRC00098/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 99	SRC00097/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV00008 00	SRC00096/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008 01	SRC00098/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008 02	SRC00097/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00008 03	SRC00096/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008 04	SRC00098/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008 05	SRC00097/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV00008 06	SRC00096/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008 07	SRC00098/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008 08	SRC00097/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000809	SRC00096/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000810	SRC00098/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000811	SRC00097/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000812	SRC00096/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000813	SRC00098/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000814	SRC00097/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00008 15	SRC00096/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008 16	SRC00098/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008 17	SRC00097/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.
EV00008 18	SRC00096/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008 19	SRC00098/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008 20	SRC00097/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00008021	SRC00096/p23	covered	Extracted slide page 23 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008022	SRC00098/p23	covered	Original slide page 23 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008023	SRC00097/p23	covered	Alternate extracted slide text page 23; normalized text matches the .slides.md page. Human comparison is pending.
EV00008024	SRC00099/q3	covered	Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
EV00008025	SRC00100/q14	covered	Original archive PDF page 14; candidate unit association is text-based and unconfirmed.

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